MATH 1300: Differential Calculus with Applications

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Preface

These are the first edition of these lecture notes for MATH 1300 (Differential Calculus with Applications). Consequently, there may be several typographical errors, missing exposition on necessary background, and more advanced topics for which there will not be time in class to cover. Future iterations of these notes will hopefully be fairly self-contained provided one has the necessary background. If you come across any typos, errors, omissions, or unclear expositions, please feel free to contact me so that I may continually improve these notes.

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Chapter 1

Limits

The concept of a limit is the foundation upon which differential calculus is built. In this chapter, we formalize the intuitive idea of a function's behavior as the input variable approaches a particular value. Rather than focusing solely on function values at a point, limits allow us to analyze what happens near that point, even when the function may be undefined or difficult to evaluate directly.

Limits arise naturally from two classical problems that motivated the development of calculus: the tangent problem (finding the slope of a curve at a point) and the velocity problem (determining instantaneous rate of change). Both problems require understanding how average quantities behave as an interval shrinks toward zero. The limit provides the mathematical language needed to make these ideas precise.

In this chapter, we begin by developing an intuitive understanding of limits using numerical tables, graphs, and real-world interpretations. We then introduce the formal definition of a limit and establish the limit laws, which allow us to compute limits algebraically. Special attention is given to situations where limits fail to exist, such as discontinuities, infinite behavior, and oscillations.

A key highlight of the chapter is the ε–δ (epsilon–delta) definition of a limit, which transforms intuition into rigorous proof. This definition plays a central role in theoretical calculus and is essential for understanding continuity and the logical structure of analysis. Students will be expected not only to compute limits, but also to justify them using precise arguments.

The chapter concludes by extending limits to infinity and using them to define continuity, a property that characterizes functions without "breaks" or "jumps." These ideas are indispensable for later chapters, where limits are used to define derivatives, analyze function behavior, and solve optimization problems.

1.1 The Limit of a Function

In this section, we introduce the central concept of a limit, which describes how a function behaves as the input variable approaches a particular value. Often in calculus, we are not interested in the value of a function at a specific point, but rather in what happens near that point. Limits allow us to make sense of quantities that cannot be evaluated directly and prepare the groundwork for defining derivatives and continuity.

We begin with an intuitive approach using numerical tables and graphs, emphasizing the idea of approach rather than attainment. This intuition will be refined and formalized in later sections, but it is essential to build a strong conceptual understanding first.

Definition 1.1. Let f be a function defined near (but not necessarily at) a number a. We say that

$$\lim_{x \to a} f(x) = L$$

if the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a, without requiring $x = a$.

The idea is that as x approaches a, $f(x)$ approaches L.

Example 1.2. One way to investigate limits is by constructing a table of values. Let $f(x) = x^2$. Consider the limit

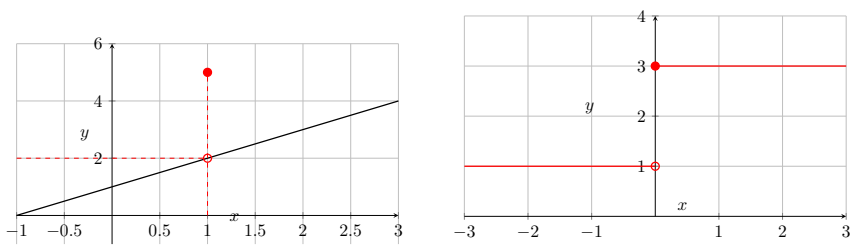
$$\lim_{x \to 2} x^2$$

$\boldsymbol{x}$	1.9	1.99	1.999	2	2.001	2.01	2.1
$\boldsymbol{f(x)}$	3.61	3.9601	3.996001	4	4.004001	4.0401	4.41

As $x \to 2$, $f(x) \to 4$. Thus,

$$\lim_{x \to 2} x^2 = 4.$$

Example 1.3. Limits can often be identified by examining the graph of a function.



From a graph, we observe the y-values that the function approaches. We do not focus on whether the function is defined at the point.

Sometimes a function behaves differently when approaching a point from the left or right.

Definition 1.4.

1. (Left-hand limit) Let f be a function defined near (but not necessarily at) a number a. We say that

$$\lim_{x \to a^-} f(x) = L$$

if the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a from the left, without requiring $x = a$.

2. (Right-hand limit) Let f be a function defined near (but not necessarily at) a number a. We say that

$$\lim_{x \to a^+} f(x) = L$$

if the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a from the right, without requiring $x = a$.

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One key observation for one-sided limits is the following.

Theorem 1.5. *Let f be a function defined near (but not necessarily at) a number a. Then $\lim_{x\to a} f(x) = L$ if and only if $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L$.*

Example 1.6. Let

$$f(x) = \begin{cases} x^2 & \text{if } x < 1 \\ 3 & \text{if } x \geq 1 \end{cases}$$

Compute $\lim_{x\to 1} f(x)$.

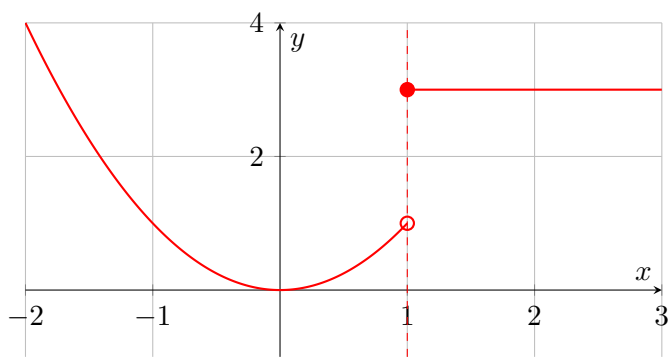
Solution. The left-hand limit is

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} x^2 = 1$$

and the right-hand limit is

$$\lim_{x\to 1^+} f(x) = 3$$

Since $1 \neq 3$, the limit *does not exist*.



It is crucial to distinguish between $\lim_{x\to a} f(x)$ and $f(a)$. They are independent concepts.

Example 1.7. Let

$$f(x) = \begin{cases} \frac{x^2-1}{x-1} & \text{if } x \neq 1 \\ 5 & \text{if } x = 1 \end{cases}$$

Simplify for $x \neq 1$

$$f(x) = \frac{(x+1)(x-1)}{x-1} = x + 1$$

Thus,

$$\lim_{x\to 1} f(x) = 2 \quad \text{but} \quad f(1) = 5$$

A limit may exist even if the function value is different or undefined.

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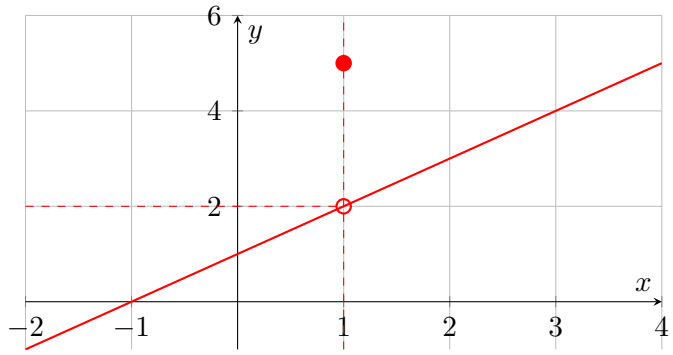

A limit may fail to exist due to

- Jump discontinuities,
- Infinite behaviour, or
- Oscillations.

Example 1.8. Let

$$f(x) = \frac{|x|}{x}$$

Recall that

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

We evaluate

$$\lim_{x \to 0} \frac{|x|}{x}$$

as follows. On the left-side, we have

$$\lim_{x \to 0^-} \frac{|x|}{x} = \lim_{x \to 0^-} \frac{-x}{x} = \lim_{x \to 0^-} (-1) = -1$$

On the other hand, on the right-side, we have

$$\lim_{x \to 0^+} \frac{|x|}{x} = \lim_{x \to 0^+} \frac{x}{x} = \lim_{x \to 0^+} 1 = 1$$

Therefore, since

$$\lim_{x \to 0^-} f(x) \neq \lim_{x \to 0^+} f(x)$$

The limit $\lim_{x \to 0} f(x)$ does not exist.

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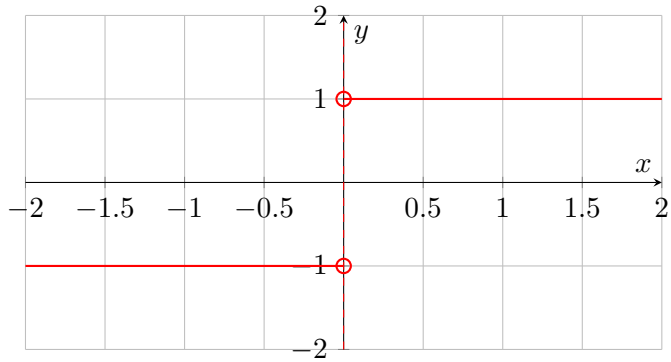
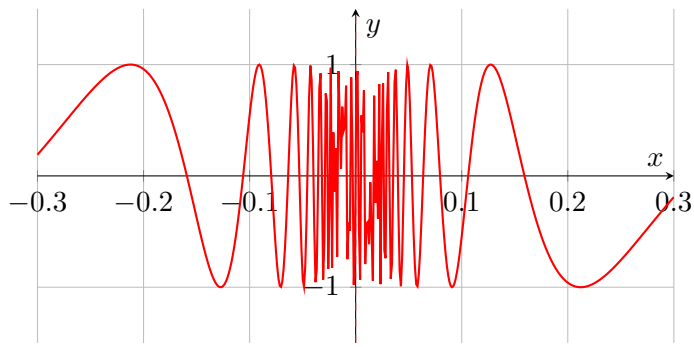


Example 1.9. Let

$$f(x) = \sin\left(\frac{1}{x}\right)$$

As $x \to 0$, the function oscillates infinitely and does not approach a single value. Therefore, $\lim_{x\to 0} \sin\left(\frac{1}{x}\right)$ does not exist.



A more careful (but still informal) phrasing: For every tolerance in the output, we can restrict the input so that the function values remain within that tolerance. This idea will be formalized rigorously in Section 2.3: The Precise Definition of a Limit.

1.2 Calculating Limits Using Limit Laws

In Section 2.1, limits were introduced conceptually through numerical tables and graphs. While these approaches are essential for understanding what a limit represents, they are not efficient for systematic calculation. In this section, we develop a collection of limit laws that allow us to compute limits algebraically, without relying on guesswork or graphical approximation.

Limit laws formalize the intuitive idea that limits behave well with respect to basic algebraic operations such as addition, multiplication, division, and powers. Using these laws, many limits can be evaluated simply by direct substitution. When direct substitution fails, the laws guide us toward appropriate algebraic simplifications.

This section emphasizes computation, but it also lays the groundwork for rigorous justification. Each law can be proved from the precise definition of a limit (ε–δ), which will be

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introduced in Section 2.3. Students should become fluent in applying limit laws accurately and recognizing when they do—and do not—apply.

Theorem 1.10 (**Limit Laws**). *Suppose that*

$$\lim_{x\to a} f(x) = L \quad \text{and} \quad \lim_{x\to a} g(x) = M$$

where L and M are finite real numbers. Then the following laws hold:

1. *Constant Law:* $\lim_{x\to a} c = c$ *for any constant* c.
2. *Identity Law:* $\lim_{x\to a} x = a$.
3. *Sum Law:* $\lim_{x\to a}[f(x) + g(x)] = L + M$.
4. *Constant Multiple Law:* $\lim_{x\to a}[cf(x)] = cL$
5. *Product Law:* $\lim_{x\to a}[f(x)g(x)] = LM$.
6. *Quotient Law: If* $M \neq 0$, *then* $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{L}{M}$.
7. *Power Law:* $\lim_{x\to a}[f(x)]^n = L^n$ *for all* $n \in \mathbb{N}$.
8. *Root Law:* $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{L}$ *if the root is defined.*

These laws are consequences of the precise definition of a limit, which we will prove rigorously later. For now, they are tools to simplify limits algebraically.

Theorem 1.11. *If $p(x)$ is a polynomial function, then*

$$\lim_{x\to a} p(x) = p(a).$$

If $r(x) = \frac{p(x)}{q(x)}$ is a rational function with $q(a) \neq 0$, then

$$\lim_{x\to a} r(x) = r(a).$$

Example 1.12. Compute

$$\lim_{x\to 3}(2x^2 - 5x + 1).$$

Solution. Since $2x^2 - 5x + 1$ is a polynomial, then using Theorem 1.11, we get

$$\lim_{x\to 3}(2x^2 - 5x + 1) = 2(3)^2 - 5(3) + 1 = 18 - 15 + 1 = 4$$

Polynomials are continuous everywhere, while rational functions are continuous wherever the denominator is zero. Thus, limits can be evaluated by direct substitution unless the denominator becomes zero.

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Example 1.13. Compute

$$\lim_{x\to 2} \frac{x^2+1}{x-1}$$

Solution. Substituting $x = 2$ gives

$$\frac{2^2+1}{2-1} = 5$$

Sometimes, direct substitution leads to expressions including

$$\frac{0}{0} \quad \text{or} \quad \infty - \infty$$

These are called indeterminate forms, and require algebraic simplification. Note that this does not mean that the limit does not exist!

Example 1.14. Compute

$$\lim_{x\to 1} \frac{x^2-1}{x-1}$$

Solution. By direct substitution, observe that

$$\lim_{x\to 1} \frac{x^2-1}{x-1} = \frac{1^2-1}{1-1} = \frac{0}{0}$$

So we obtain an indeterminate form. By *factoring*, observe that

$$\frac{x^2-1}{x-1} = \frac{(x-1)(x+1)}{x-1} = x+1$$

Now take the limit

$$\lim_{x\to 1} \frac{x^2-1}{x-1} = \lim_{x\to 1}(x+1) = 2$$

Example 1.15. Compute

$$\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4}$$

Solution. By direct substitution, observe that

$$\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4} = \frac{\sqrt{4}-2}{4-4} = \frac{0}{0}$$

So we obtain an indeterminate form. By *rationalizing the numerator*, we multiply the numerator and denominator by the conjugate

$$\frac{\sqrt{x}-2}{x-4} \cdot \frac{\sqrt{x}+2}{\sqrt{x}+2} = \frac{x-4}{(x-4)(\sqrt{x}+2)} = \frac{1}{\sqrt{x}+2}$$

Now take the limit

$$\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4} = \lim_{x\to 4} \frac{1}{\sqrt{x}+2} = \frac{1}{\sqrt{4}+2} = \frac{1}{4}$$

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Example 1.16. Compute

$$\lim_{x\to 1} \frac{x^{\frac{1}{3}} - 1}{x - 1}$$

Solution. Observe by direct substitution, we have

$$\lim_{x\to 1} \frac{x^{\frac{1}{3}} - 1}{x - 1} = \frac{1^{\frac{1}{3}} - 1}{1 - 1} = \frac{0}{0}$$

So we obtain an indeterminate form. In this case, it is easier to evaluate by *substitution*. Let $u = x^{\frac{1}{3}}$. Then $u^3 = x$. Observe that as $x \to 1$, then $u \to 1$ as well. Then we have

$$\frac{x^{\frac{1}{3}} - 1}{x - 1} = \frac{u - 1}{u^3 - 1}$$

Observe that the denominator can be factored and simplified

$$\frac{u - 1}{u^3 - 1} = \frac{u - 1}{(u - 1)(u^2 + u + 1)} = \frac{1}{u^2 + u + 1}$$

Therefore, taking the limit, we have

$$\lim_{x\to 1} \frac{x^{\frac{1}{3}} - 1}{x - 1} = \lim_{u\to 1} \frac{1}{u^2 + u + 1} = \frac{1}{1^2 + 1 + 1} = \frac{1}{3}$$

Example 1.17. Compute

$$\lim_{x\to 2} |x - 2|$$

Solution. As $x \to 2$, the expression inside the absolute value approaches 0, so

$$\lim_{x\to 2} |x - 2| = 0$$

However, extra care must be taken when absolute values appear in denominators or piecewise contexts.

Example 1.18. Compute

$$\lim_{x\to 2} \frac{|x - 2|}{x - 2}$$

Solution. The definition of the absolute value function is

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Thus,

$$|x - 2| = \begin{cases} x - 2 & \text{if } x - 2 \geq 0 \\ -(x - 2) & \text{if } x - 2 < 0 \end{cases} = \begin{cases} x - 2 & \text{if } x \geq 2 \\ -(x - 2) & \text{if } x < 2 \end{cases}$$

Thus, to evaluate

$$\lim_{x\to 2} \frac{|x - 2|}{x - 2}$$

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we need to evaluate the left-hand and right-hand limits. Indeed, for the left-hand limit, we have

$$\lim_{x\to 2^-} \frac{|x-2|}{x-2} = \lim_{x\to 2^-} \frac{-(x-2)}{x-2} = \lim_{x\to 2^-} (-1) = -1$$

On the other hand, for the right-hand limit, we have

$$\lim_{x\to 2^+} \frac{|x-2|}{x-2} = \lim_{x\to 2^+} \frac{x-2}{x-2} = \lim_{x\to 2^+} 1 = 1$$

Since the left-hand and right-hand limits do not agree, this implies that

$$\lim_{x\to 2} \frac{|x-2|}{x-2}$$

does not exist.

Some limits cannot be evaluated directly using algebraic manipulation or limit laws. In particular, functions that oscillate or involve trigonometric expressions may resist simplification. In such cases, we can determine a limit by bounding the function between two simpler functions whose limits are known.

The idea is formalized in the Squeeze Theorem (sometimes called the Sandwich Theorem).

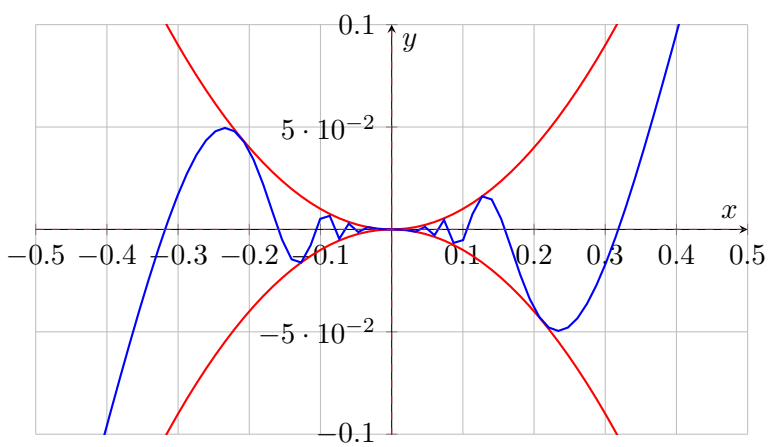
Theorem 1.19 (Squeeze Theorem). *Let f, g, and h be functions defined on an interval containing a, except possibly at a. Suppose that*

1. $f(x) \le g(x) \le h(x)$ *for all x near a with $x \ne a$,*
2. $\lim_{x\to a} f(x) = L$,
3. $\lim_{x\to a} h(x) = L$.

Then

$$\lim_{x\to a} g(x) = L.$$

If $g(x)$ is always trapped between $f(x)$ and $h(x)$, and both $f(x)$ and $h(x)$ approach the same value L, then $g(x)$ has no choice but to approach L as well.



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Example 1.20. Evaluate

$$\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right)$$

Solution. Recall that for all real numbers θ,

$$-1 \leq \sin(\theta) \leq 1$$

Thus,

$$-x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2$$

Now compute the limits of the bounding functions

$$\lim_{x\to 0}(-x^2) = 0, \quad \lim_{x\to 0} x^2 = 0$$

By Squeeze Theorem (Theorem 1.19),

$$\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) = 0.$$

Example 1.21. Evlauate

$$\lim_{x\to 0} |x| \cos\left(\frac{1}{x}\right)$$

Solution. Since

$$-1 \leq \cos\left(\frac{1}{x}\right) \leq 1$$

we have

$$-|x| \leq |x| \cos\left(\frac{1}{x}\right) \leq |x|$$

Both bounding functions approach 0 as $x \to 0$, so by Squeeze Theorem (Theorem 1.19)

$$\lim_{x\to 0} |x| \cos\left(\frac{1}{x}\right) = 0$$

1.3 The Precise Definition of a Limit

Up to this point, limits have been understood intuitively through graphs, tables, and algebraic computation using limit laws. While this intuition is extremely valuable, calculus requires a rigorous foundation to justify why these methods work and to handle subtle cases where intuition may fail. This section provides that foundation by introducing the precise (ε-δ) definition of a limit.

The ε-δ definition replaces informal language such as "x gets close to a" and "$f(x)$ gets close to L" with exact numerical conditions. It allows us to prove limit statements with complete logical certainty and is the standard definition used in real analysis. Mastery of this definition is essential not only for proofs involving limits, but also for understanding continuity, derivatives, and the logical structure of calculus.

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This section marks a transition from computational calculus to theoretical calculus. Students are expected to read, interpret, and construct rigorous arguments using inequalities and quantifiers.

Recall the informal statement

$$\lim_{x \to a} f(x) = L$$

means that $f(x)$ can be made as close to L as we like by taking x sufficiently close to a. The problem with this statement is that it is vague.

- What does "close" mean?
- How close is "sufficiently close"?
- How do we prove such a statement?

To resolve this, we introduce two positive numbers:

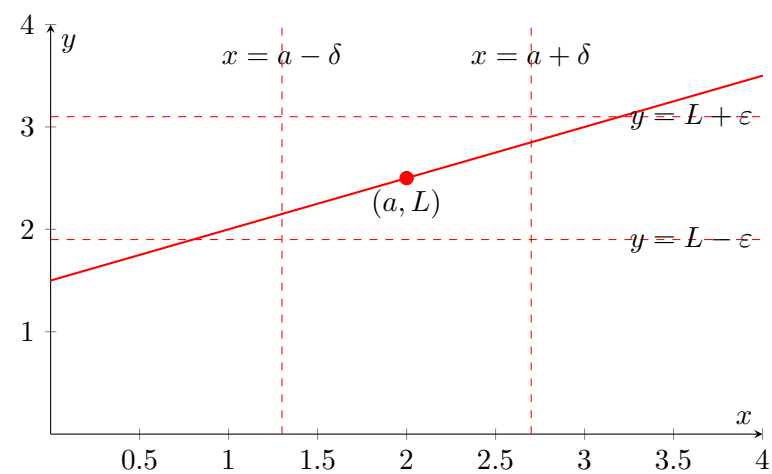
- ε (epsilon): controls closeness in the output.
- δ (delta): controls closeness in the input.

Definition 1.22. Let f be a function defined on an open interval containing a, except possibly at a itself. We say that

$$\lim_{x \to a} f(x) = L$$

if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that if $0 < |x - a| < \delta$, then $|f(x) - L| < \varepsilon$.

The hypothesis "$0 < |x - a| < \delta$" means that the value of x lies within a δ-neighbourhood of a, excluding a itself, and the conclusion "$|f(x) - L| < \varepsilon$" means that $f(x)$ lies within an ε-neighbourhood of L.



The horizontal dashed lines form the ε-neighbourhood, while the vertical dashed lines form the δ-neighourhood. The limit exists because whenever x lies in the δ-band, the graph lies entirely inside the ε-band. The point $x = a$ itself is not required to be on the graph.

The assumption "For every $\varepsilon > 0$, there exists a $\delta > 0$" means that no matter how small the tolerance input the output is, we can always find an iput restriction that works.

When working with the precise definition of the limit,

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- The value of $f(a)$ is irrelevant,
- The condition $0 < |x - a|$ explicitly excludes $x = a$,
- The same δ must work for all x satisfying the inequality.
- The definition does not tell us how to find δ; it only requires that one exists.

Example 1.23. Prove that

$$\lim_{x \to 2} (3x + 1) = 7$$

Proof. Let $\varepsilon > 0$ be arbitrary. We seek a $\delta > 0$ such that if $0 < |x - 2| < \delta$, we get $|(3x + 1) - 7| < \varepsilon$. Indeed, first simplify

$$|(3x + 1) - 7| = |3x - 6| = 3|x - 2|$$

Thus, it suffices to require

$$3|x - 2| < \varepsilon \implies |x - 2| < \frac{\varepsilon}{3}$$

Choose $\delta = \frac{\varepsilon}{3}$. Then whenever $0 < |x - 2| < \delta$, we have

$$|(3x + 1) - 7| = |3x - 6| = 3|x - 2| < 3 \cdot \frac{\varepsilon}{3} = \varepsilon$$

Therefore, as $\varepsilon > 0$ was arbitrary,

$$\lim_{x \to 2} (3x + 1) = 7$$

as desired. □

From the following example, to prove

$$\lim_{x \to a} f(x) = L$$

follow these steps.

1. Let $\varepsilon > 0$ be arbitrary.
2. Start with $|f(x) - L|$.
3. Simplify algebraically.
4. Bound the expression by something involving $|x - a|$.
5. Choose δ in terms of ε.
6. Conclude the proof.

Example 1.24. Prove that

$$\lim_{x \to 1} x^2 = 1$$

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Proof. Let $\varepsilon > 0$ be arbitrary. We seek a $\delta > 0$ such that if $0 < |x-1| < \delta$, then $|x^2 - 1| < \varepsilon$. Indeed, first factor

$$|x^2 - 1| = |x-1||x+1|$$

To control $|x+1|$, restrict x near 1. Assume $|x-1| < 1$, so $0 < x < 2$, and $|x+1| < 3$. Thus,

$$|x^2 - 1| = 3|x-1|$$

To ensure $|x^2 - 1| < \varepsilon$ it suffices to require

$$|x-1| < \frac{\varepsilon}{3}$$

Choose $\delta = \min\left\{1, \frac{\varepsilon}{3}\right\}$ so that

$$|x^2 - 1| = |x-1||x+1| < 3 \cdot \frac{\varepsilon}{3} = \varepsilon$$

Therefore, as $\varepsilon > 0$ was arbitrary,

$$\lim_{x \to 1} x^2 = 1$$

as desired. □

Example 1.25. Prove that

$$\lim_{x \to 4} (x^2 + x - 11) = 9$$

Proof. Let $\varepsilon > 0$ be arbitrary. We seek a $\delta > 0$ such that if $0 < |x-4| < \delta$, then $|(x^2 + x - 11) - 9| < \varepsilon$. First note that

$$|(x^2 + x - 11) - 9| = |x^2 + x - 20| = |x-4||x+5|$$

To control $|x+5|$, restrict x near 4. Assume $|x-4| < 1$, so $3 < x < 5$, and thus $|x+5| < 10$. To ensure $|(x^2 + x - 11) - 9| < \varepsilon$, it suffices to ensure that

$$|x-4| < \frac{\varepsilon}{10}$$

Choose $\delta = \min\left\{1, \frac{\delta}{10}\right\}$ so that

$$\begin{aligned} |(x^2 + x - 11) - 9| &= |x+5||x-4| \\ &< 10 \cdot \frac{\varepsilon}{10} \\ &= \varepsilon \end{aligned}$$

Therefore, as $\varepsilon > 0$ was arbitrary,

$$\lim_{x \to 4} (x^2 + x - 11) = 9$$

as desired. □

Example 1.26. Prove that

$$\lim_{x \to 2} x^3 = 8.$$

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Proof. Let $\varepsilon > 0$ be arbitrary. We seek a $\delta > 0$ such that if $0 < |x-2| < \delta$, then $|x^3 - 8| < \varepsilon$. First observe that

$$|x^3 - 8| = |x^3 - 2^3| = |x-2||x^2 + 2x + 4|$$

To control the second factor, restrict x near 2. Assume that $|x-2| < 1$, so $1 < x < 3$. Then

$$|x^2 + 2x + 4| \leq 3^2 + 2(3) + 4 = 19$$

Thus, whenever $|x-2| < 1$,

$$|x^3 - 8| \leq 19|x-2|$$

Choose $\delta = \min\left\{1, \frac{\varepsilon}{19}\right\}$. Then if $0 < |x-2| < \delta$, then

$$\begin{aligned} |x^3 - 8| &= |x-2||x^2 + 2x + 4| \\ &\leq 19 \cdot \frac{\varepsilon}{19} \\ &= \varepsilon \end{aligned}$$

□

Notice that from the step

$$|x^3 - 8| = |x-2||x^2 + 2x + 4|$$

some students try

$$|x^2 + 2x + 4| < \frac{\varepsilon}{|x-2|}$$

and then get stuck because the right-hand side depends on x, and even blows up as $x \to 2$. We cannot choose a single fixed δ from an inequality that has $|x-2|$ in the denominator like that.

In an ε-δ proof, you must pick *one number* δ that depends only on ε (and the point 2), not on x.

Instead, you bound the "extra factor" by forcing x to stay in a small interval around 2. For example, we assumed $|x-2| < 1$, so that $1 < x < 3$. Thus,

$$|x^2 + 2x + 4| \leq 3^2 + 2(3) + 4 = 19$$

which now we have a constant bound and we can finish

$$|x^3 - 8| \leq 19|x-2|$$

So if you choose $\delta \leq \frac{\varepsilon}{19}$, then we are done.

In Section 1.2, we saw functions whose behaviour differs depending on whether x approaches a point from the left or the right. To treat such cases rigorously, the ε-δ definition must be refined to distinguish directional approach.

One-sided limits are essential for piecewise-defined functions, endpoint behaviour, understanding jump discontinuities, and defining continuity on closed intervals.

Definition 1.27. We say that

$$\lim_{x \to a^+} f(x) = L$$

if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that if $0 < x - a < \delta$, then $|f(x) - L| < \varepsilon$.

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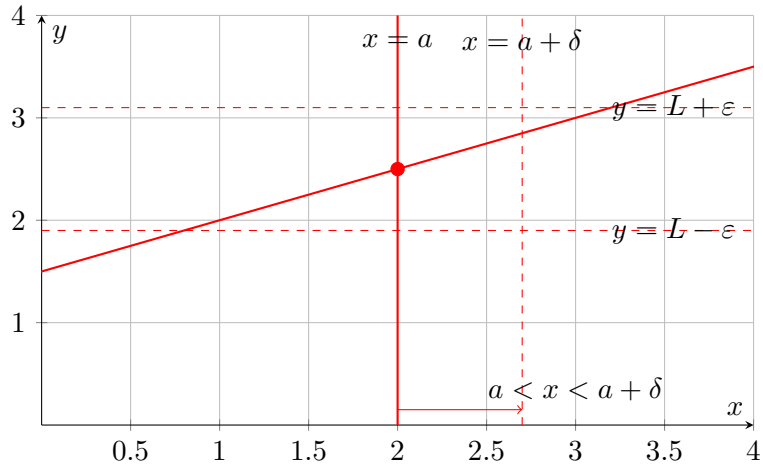
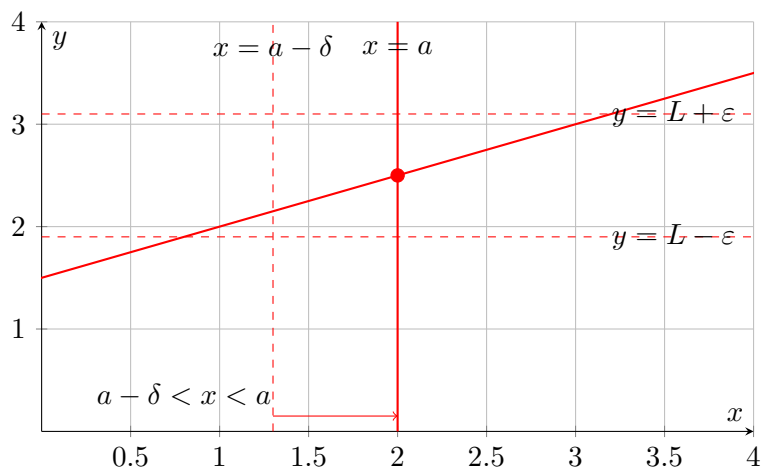


In this case, the values of x approaches a from the right. Only values with $x > a$ are considered. The function values must lie within an ε-neighbourhood of L.

Definition 1.28. We say that

$$\lim_{x \to a^-} f(x) = L$$

if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that if $0 < a - x < \delta$, then $|f(x) - L| < \varepsilon$.



In this case, the values of x approach a from the left, and only values with $x < a$ are considered.

A two-sided limit exists if and only if both one-sided limits exist and are equal.

Theorem 1.29. *Let f be defined on an interval about a (possibly not at a). Then $\lim_{x \to a} f(x) = L$ if and only if $\lim_{x \to a^-} f(x) = L$ and $\lim_{x \to a^+} f(x) = L$.*

Proof. ($\Rightarrow$) Assume $\lim_{x \to a} f(x) = L$. To see that $\lim_{x \to a^-} f(x) = L$ and $\lim_{x \to a^+} f(x) = L$, let $\varepsilon > 0$ be arbitrary. Since $\lim_{x \to a} f(x) = L$, there exists a $\delta > 0$ such that if $0 < |x - a| < \delta$, then $|f(x) - L| < \varepsilon$.

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To see that $\lim_{x\to a^+} f(x) = L$, if $0 < x-a < \delta$, then we have $0 < |x-a| < \delta$, so $|f(x)-L| < \varepsilon$. Thus, as $\varepsilon > 0$ was arbitrary,

$$\lim_{x\to a^+} f(x) = L$$

To see that $\lim_{x\to a^-} f(x) = L$, if $0 < a - x < \delta$, then $0 < |a-x| = |x-a| < \delta$, so $|f(x) - L| < \varepsilon$. Therefore, as $\varepsilon > 0$ was arbitrary,

$$\lim_{x\to a^-} f(x) = L$$

So both one-sided limits exist, and equals L.

($\Leftarrow$) Assume $\lim_{x\to a^-} f(x) = L$ and $\lim_{x\to a^+} f(x) = L$. To see that $\lim_{x\to a} f(x) = L$, let $\varepsilon > 0$ be arbitrary. Since $\lim_{x\to a^-} f(x) = L$, there exists a $\delta_- > 0$ such that if $0 < a - x < \delta_-$, then $|f(x) - L| < \varepsilon$. Similarly, since $\lim_{x\to a^+} f(x) = L$, there exists a $\delta_+ > 0$ such that if $0 < x - a < \delta_+$, then $|f(x) - L| < \varepsilon$.

Let $\delta = \min\{\delta_-, \delta_+\}$, and let x be such that $0 < |x - a| < \delta$. We consider the following cases:

1. If $x > a$, then $0 < x - a < \delta \leq \delta_+$, so $|f(x) - L| < \varepsilon$.
2. If $x < a$, then $0 < a - x < \delta \leq \delta_-$, so $|f(x) - L| < \varepsilon$.

In both cases, $0 < |x - a| < \delta$ implies $|f(x) - L| < \varepsilon$. Therefore, as $\varepsilon > 0$ was arbitrary,

$$\lim_{x\to a} f(x) = L$$

□

In earlier examples, limits failed to exist due to jumps or oscillations. We now show how such failures can be proved rigorously using the ε-δ definition.

Recall that to prove

$$\lim_{x\to a} f(x) \neq L$$

it suffices to show that there exists some $\varepsilon_0 > 0$ such that no matter how small $\delta > 0$ is chosen, there are values of x with

$$0 < |x - a| < \delta \quad \text{but} \quad |f(x) - L| \geq \varepsilon_0.$$

To put it simply, in quantifiers, the original definition of a limit is

$$\forall \varepsilon > 0, \exists \delta > 0 : \forall x \neq a, 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon$$

Thus, negating the statement gives

$$\exists \varepsilon > 0 : \forall \delta > 0, \exists x \neq a : 0 < |x - a| < \delta \wedge |f(x) - L| \geq \varepsilon$$

which is sufficient to prove that a limit does not exist.

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Example 1.30 (Heaviside Function). Let $H(x)$ be the Heaviside function defined by

$$H(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$$

Show that

$$\lim_{x \to 0} H(x)$$

does not exist.

Proof. Assume for a contradiction that

$$\lim_{x \to 0} H(x) = L$$

for some real number L. Let $\varepsilon_0 = \frac{1}{2}$, and let $\delta > 0$ be arbitrary. Choose

$$x_1 = -\frac{\delta}{2} < 0, \quad x_2 = \frac{\delta}{2} > 0$$

so that

$$H(x_1) = 0, \quad H(x_2) = 1$$

Both x_1 and x_2 satisfy $0 < |x| < \delta$. Now observe that for any real number L,

$$|0 - L| \geq \frac{1}{2} \quad \text{or} \quad |1 - L| \geq \frac{1}{2}$$

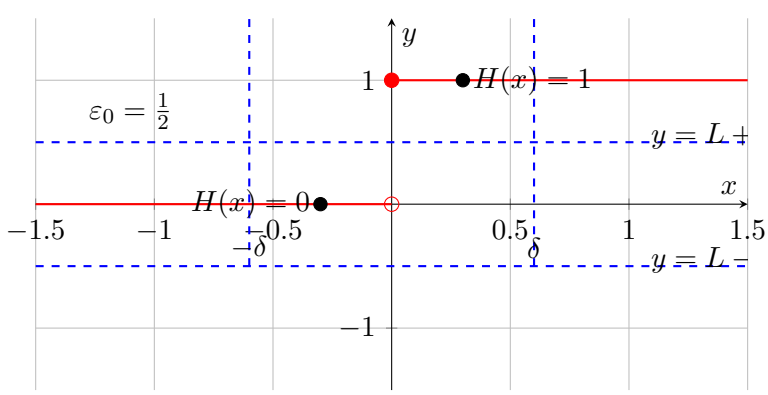
Therefore, at least one of the points x_1 or x_2 satisfies

$$|H(x) - L| \geq \varepsilon.$$

Since for the fixed tolerance $\varepsilon_0 = \frac{1}{2}$, no choice of $\delta > 0$ can force all values of $H(x)$ with $0 < |x| < \delta$ to remain within ε_0 of a single number L, the ε-δ condition for a limit fails. Therefore,

$$\lim_{x \to 0} H(x)$$

does not exist. □



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Here, the horizontal dashed lines represent the ε-band $|H(x) - L| < \varepsilon_0 = \frac{1}{2}$, and the vertical dashed lines represent an arbitrary δ-neighbourhood of 0.

No matter how small δ is, points just left of 0 give $H(x) = 0$, while point just right of 0 give $H(x) = 1$. Since 0 and 1 are more than ε_0 apart, they cannot both lie in the ε-band. Therefore, no δ can work for $\varepsilon_0 = \frac{1}{2}$.

Example 1.31. Show that

$$\lim_{x \to 0} \frac{1}{x^2}$$

does not exist.

Proof. Assume for a contradiction that there exists a real number L such that

$$\lim_{x \to 0} \frac{1}{x^2} = L$$

Let $\varepsilon_0 = 1$. By the ε-δ definition, there must exist $\delta > 0$ such that if $0 < |x| < \delta$, then

$$\left| \frac{1}{x} - L \right| < 1$$

In particular, this implies

$$\frac{1}{x^2} < L + 1$$

for all $0 < |x| < \delta$. Now choose

$$x = \min \left\{ \frac{\delta}{2}, \frac{1}{\sqrt{L+2}} \right\}$$

Then $0 < |x| < \delta$, so the above must hold. But with this choice,

$$|x| \leq \frac{1}{\sqrt{L+2}} \implies \frac{1}{x^2} \geq L + 2$$

Hence,

$$\left| \frac{1}{x^2} - L \right| \geq 2 > 1$$

contradicting the requirement $\left| \frac{1}{x^2} - L \right| < 1$.

Therefore, no such real number L exist, and the limit does not exist as a finite real number. □

Example 1.32. Show that

$$\lim_{x \to 0} \sin \left(\frac{1}{x} \right)$$

does not exist.

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Proof. Assume for a contradiction that the limit exists and equals some real number L

$$\lim_{x\to 0} \sin\left(\frac{1}{x}\right) = L$$

Let $\varepsilon_0 = \frac{1}{2}$. By the ε-δ definition of the limit, there must exist $\delta > 0$ such that whenever $0 < |x| < \delta$, we have

$$\left|\sin\left(\frac{1}{x}\right) - L\right| < \varepsilon_0$$

Now choose two sequences approaching 0: Let $(x_n)_{n=1}^{\infty}$ and $(y_n)_{n=1}^{\infty}$ be defined by

$$x_n = \frac{1}{\frac{\pi}{2} + 2\pi n}, \quad y_n = \frac{1}{\frac{3\pi}{2} + 2\pi n}$$

for $n \in \mathbb{N}$. Observe that as $n \to \infty$, $x_n, y_n \to 0$, and

$$\sin\left(\frac{1}{x_n}\right) = \sin\left(\frac{\pi}{2} + 2\pi n\right) = 1$$
$$\sin\left(\frac{1}{y_n}\right) = \sin\left(\frac{3\pi}{2} + 2\pi n\right) = -1$$

Since $x_n \to 0$, there exists $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$,

$$0 < |x_n| < \delta.$$

Similarly, since $y_n \to 0$, there exists $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$,

$$0 < |y_n| < \delta.$$

Let $N = \max\{N_1, N_2\}$. Then for all $n \geq N$, both x_n and y_n satisfy $0 < |x_n| < \delta$, so the definition demands

$$|1 - L| < \frac{1}{2} \quad \text{and} \quad |-1 - L| < \frac{1}{2}$$

But this is impossible: no real number L can be within $\frac{1}{2}$ of both 1 and -1 (since 1 and -1 are 2 units apart).

This contradiction shows the limit cannot exist. □

At an intuitive level, a limit describes a single value that a function approaches as the input variable gets closer and closer to a point. If a function could approach two different values at the same time, the notion of "approach" would lose all meaning. The uniqueness of limits formalizes the idea that approach must be unambiguous.

Proposition 1.33. *Assume*

$$\lim_{x\to a} f(x) = L \quad \textit{and} \quad \lim_{x\to a} f(x) = M$$

Then $L = M$.

In words, a function cannot have two different limits at the same point.

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Proof. Assume $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} f(x) = M$. Suppose for a contradiction that $L \neq M$. Let

$$\varepsilon_0 = \frac{|L-M|}{2} > 0$$

Since $\lim_{x\to a} f(x) = L$, there exists $\delta_1 > 0$ such that if $0 < |x-a| < \delta_1$, then $|f(x) - L| < \varepsilon_0$. Similarly, since $\lim_{x\to a} f(x) = M$, there exists a $\delta_2 > 0$ such that if $0 < |x-a| < \delta_2$, then $|f(x) - M| < \varepsilon_0$.

Let $\delta = \min\{\delta_1, \delta_2\}$. Then for any x with $0 < |x-a| < \delta$, both inequalities hold

$$|f(x) - L| < \varepsilon_0 \quad \text{and} \quad |f(x) - M| < \varepsilon_0$$

Using the triangle inequality,

$$\begin{aligned} |L-M| &= |L - f(x) + f(x) - M| \\ &\leq |L - f(x)| + |f(x) - M| \\ &< \varepsilon_0 + \varepsilon_0 \\ &= 2\varepsilon_0 \\ &= |L-M| \end{aligned}$$

However, this implies $1 < 1$, which is a contradiction. Therefore, our assumption $L \neq M$ is false, and hence, $L = M$. □

The Squeeze Theorem can be proved directly from the ε-δ definition of the limit. The idea is that if $|f(x) - L| < \varepsilon$ and $|h(x) - L| < \varepsilon$, then any function between them must also satisfy the same inequality.

Proof of Squeeze Theorem (Theorem 1.19). Let $\varepsilon > 0$ be arbitrary. Since $\lim_{x\to a} f(x) = L$, then there exists $\delta_1 > 0$ such that if $0 < |x-a| < \delta_1$, then $|f(x) - L| < \varepsilon$. In particular, $|f(x) - L| < \varepsilon$ implies

$$L - \varepsilon < f(x) < L + \varepsilon$$

so for $0 < |x-a| < \delta_1$, we have

$$f(x) > L - \varepsilon$$

Similarly, since $\lim_{x\to a} h(x) = L$, there exists $\delta_2 > 0$ such that if $0 < |x-a| < \delta_2$, then $|h(x) - L| < \varepsilon$. In particular, $|h(x) - L| < \varepsilon$ implies

$$L - \varepsilon < h(x) < L + \varepsilon$$

so for $0 < |x-a| < \delta_2$,

$$h(x) < L + \varepsilon$$

Now also assume the squeezing inequality $f(x) \leq g(x) \leq h(x)$ holds whenever $0 < |x-a| < \delta$ with $\delta = \min\{\delta_1, \delta_2\}$. Then for any x such that $0 < |x-a| < \delta$, we have

$$L - \varepsilon < f(x) \leq g(x) \leq h(x) < L + \varepsilon$$

so in particular,

$$L - \varepsilon < g(x) < L + \varepsilon$$

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which is equivalent to $|g(x) - L| < \varepsilon$. Since $\varepsilon > 0$ was arbitrary, by the ε-δ definition of a limit,

$$\lim_{x \to a} g(x) = L.$$

□

1.4 Continuity

In the study of limits, we focused on how a function behaves near a point. In this section, we examine a stronger and more intuitive idea: whether a function behaves smoothly at the point itself. This leads to the concept of continuity, which formalizes the notion that a graph can be drawn without lifting one's pencil.

Continuity connects limits directly to function values. Informally, a function is continuous at a point if:

- the function is defined there,
- the limit exists there,
- the limit equals the function value.

This section is a crucial bridge between limits and derivatives. The derivative is defined using a limit, and it can only exist at points where the function is continuous. Moreover, many important theorems in calculus—such as the Intermediate Value Theorem and the Extreme Value Theorem—require continuity as a hypothesis.

Students should expect to use ε-δ reasoning, limit laws, and algebraic arguments to determine continuity, classify discontinuities, and justify claims rigorously.

Definition 1.34. A function f is said to be *continuous at a point* a if the graph of f has no breaks, jumps, or holes at $x = a$, and the values of $f(x)$ close to $x = a$ are close to $f(a)$.

This intuition will now be made more precise using limits.

Definition 1.35. A function f is *continuous at a point* a if all three of the following conditions are satisfied.

1. $f(a)$ is defined,
2. $\lim_{x \to a} f(x)$ exists, and
3. $\lim_{x \to a} f(x) = f(a)$

If any one of these conditions fail, then f is not continuous at a.

Using the definition of a limit, continuity at a can be expressed as: for every $\varepsilon > 0$, there exists $\delta > 0$ such that if $|x - a| < \delta$, then $|f(x) - f(a)| < \varepsilon$. The inequality now includes $x = a$, because $f(a)$ must be defined.

Example 1.36. Let $f(x) = x^3 - 2x + 1$. Note that polynomials are defined for all real numbers, limits exist everywhere, and the limits equal function values everywhere. Therefore, f is continuous for all $x \in \mathbb{R}$.

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Example 1.37. Let

$$f(x) = \frac{x^2 - 1}{x - 1}$$

Observe that

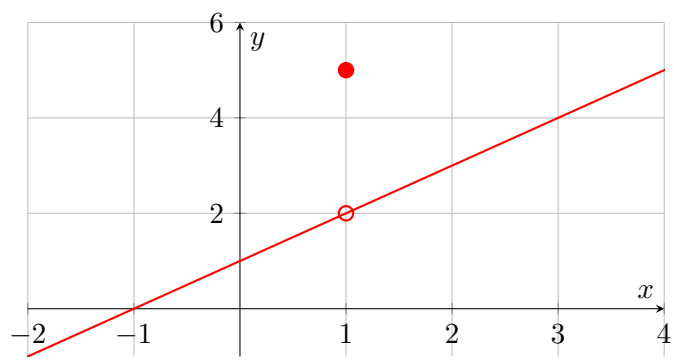
$$f(x) = \frac{(x-1)(x+1)}{x-1} = x + 1, \quad x \neq 1$$

The function f is not defined at $x = 1$, the limit exists at $x = 1$, but $f(1)$ does not exist. Therefore, f is not continuous at $x = 1$.

A function may fail to be continuous in different ways.

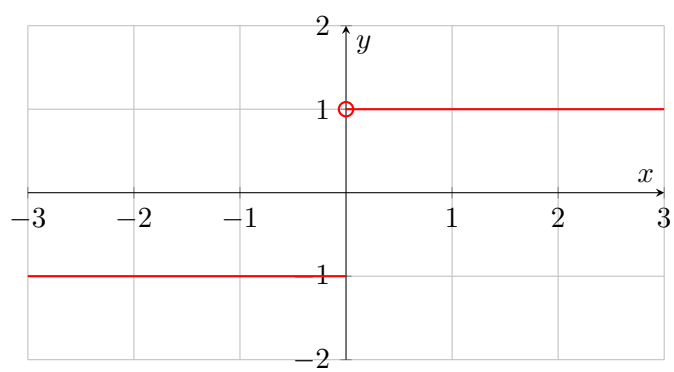
1. Removable discontinuities: the limit exists but the function value is missing or incorrect.

   Example 1.38. $f(x) = \begin{cases} \frac{x^2-1}{x-1} & \text{if } x \neq 1 \\ 5 & \text{if } x = 1 \end{cases}$



2. Jump discontinuity: one-sided limits exist, but are not equal.

   Example 1.39. $f(x) = \dfrac{|x|}{x}$



3. Infinite discontinuity: Function grows without bound near a point.

   Example 1.40. $f(x) = \dfrac{1}{x}$

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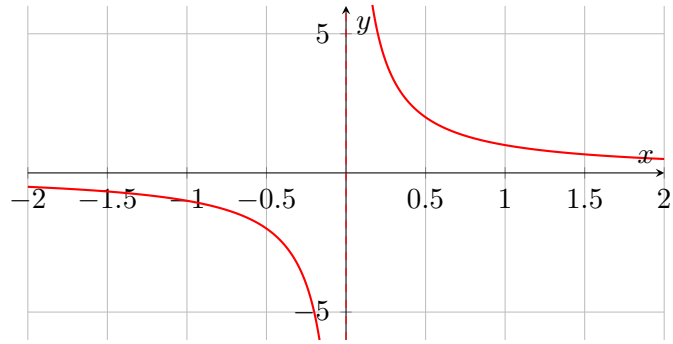

Definition 1.41. Let f be a function. Then we say that f is

1. Continuous on an open interval (a, b) if f is continuous at every point in (a, b).
2. Continuous on a closed interval $[a, b]$ if f is continuous on (a, b), right-continuous at a $\lim_{x \to a^+} f(x) = f(a)$, and left-continuous at b $\lim_{x \to b^-} f(x) = f(b)$.

The following functions are continuous on their domains:

- Polynomials
- Rational Functions
- Root Functions
- Exponential Functions
- Logarithmic Functions
- Trigonometric Functions

This allows us to evaluate many limits by direct substitution.

Theorem 1.42. *If f and g are continuous at a, then so are*

1. $f + g$
2. cf *for any constant* $c \in \mathbb{R}$,
3. fg
4. $\frac{f}{g}$, *provided that* $g(a) \neq 0$.

Proof. Follows directly from limit laws and the definition of continuity. □

Example 1.43. Prove that $f(x) = 3x$ is continuous at $x = 2$.

Proof. Let $\varepsilon > 0$ be arbitrary. We seek a $\delta > 0$ such that if $|x - 2| < \delta$, then $|f(x) - L| < \varepsilon$. Observe

$$|f(x) - f(2)| = |3x - 6| = 3|x - 2| < \varepsilon$$

Thus, choose $\delta = \frac{\varepsilon}{3}$. Then if $|x - 2| < \delta$, we have $|f(x) - f(2)| < \varepsilon$. Thus, f is continuous at $x = 2$. □

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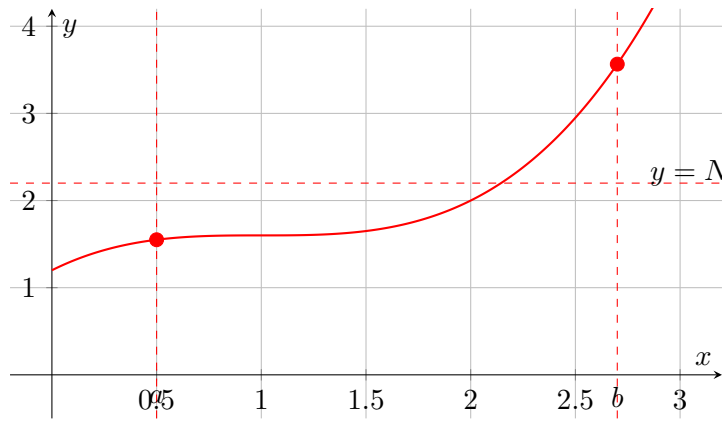
One of the most powerful consequences of continuity is that a continuous function cannot skip values. Intuitively, if you draw the graph of a continuous function without lifting your pencil, then moving from one point on the graph to another forces the function to pass through every intermediate height.

Theorem 1.44 (**Intermediate Value Theorem**). *Let f be a function that is continuous on a closed interval $[a, b]$. Let N be any real number between $f(a)$ and $f(b)$. Then there exists $c \in (a, b)$ such that $f(c) = N$.*



The theorem only guarantees the existence of a value, but not the uniqueness (as there may be more than one such c). The continuity assumption on the entire interval $[a, b]$ is essential.

Example 1.45. Suppose the temperature in a room is measured continuously over time. If the temperature is 18°C at 9:00 AM, and 25°C at noon, then at some time between 9:00 AM and noon, the temperature must have been exactly 20°C.

Example 1.46. Show that the equation

$$x^3 - x - 1 = 0$$

has at least one real solution.

Solution. Let $f(x) = x^3 - x - 1$. Then f is a polynomial function, thus, is continuous on $\mathbb{R}$. Computing $f(1) = -1$ and $f(2) = 5$. Since 0 lies between -1 and 5, the Intermediate Value Theorem (Theorem 1.44) guarantees that there exists a number $c \in (1, 2)$ such that $f(c) = 0$. Thus, the equation has at least one real root in $(1, 2)$.

Example 1.47. Show that $\cos(x) = x$ has a solution in the interval $(0, 1)$.

Solution. Let $f(x) = \cos(x) - x$. Then f is continuous on $[0, 1]$. Evaluate $f(0) = 1$ and $f(1) = \cos(1) - 1 < 0$. Since 0 lies between $f(0)$ and $f(1)$, the Intermediate Value Theorem (Theorem 1.44) implies there exists $c \in (0, 1)$ such that $f(c) = 0$, or equivalently, $\cos(c) = c$.

Example 1.48. Suppose f is continuous on $[1, 5]$ and the only solutions of the equation $f(x) = 6$ are $x = 1$ and $x = 4$. If $f(2) = 8$, we claim that $f(3) > 6$.

To see this, assume for a contradiction $f(3) \leq 6$. Since f is continuous on $[2, 3]$, by the Intermediate Value Theorem (Theorem 1.44), f takes every value between $f(2)$ and $f(3)$ on $[2, 3]$. We consider the following cases.

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1. If $f(3) < 6$, then 6 lies between $f(2) = 8$ and $f(3)$, so there exists some $c \in (2, 3)$ such that $f(c) = 6$.

2. If $f(3) = 6$, then $x = 3$ is a solution of $f(x) = 6$.

Either way, we obtain a solution $c \in (2, 3]$ to $f(x) = 6$, which contradicts the assumption that the only solutions of $f(x) = 6$ on $[1, 5]$ are $x = 1$ and $x = 4$.

Therefore, our assumption $f(3) \leq 6$ must be false, so $f(3) > 6$, as desired.

Example 1.49. If a and b are positive numbers, prove that the equation

$$\frac{a}{x^3 + 2x^2 - 1} + \frac{b}{x^3 + x - 2} = 0$$

has at least one solution in the interval $(-1, 1)$.

Solution. Let

$$f(x) = \frac{a}{x^3 + 2x^2 - 1} + \frac{b}{x^3 + x - 2}$$

Observe the denominator can be factored using long division as $x + 1$ is a factor

$$\begin{array}{r|rrrr}
 & & & x^2 & +x-1 \\ \cline{2-5}
x+1 & & x^3 & +2x^2 & -1 \\
 & & -x^3 & -x^2 & \\ \cline{2-4}
 & & & x^2 & \\
 & & & -x^2 & -x \\ \cline{3-5}
 & & & & -x-1 \\
 & & & & x+1 \\ \cline{4-5}
 & & & & 0
\end{array}$$

Thus,

$$x^3 + 2x^2 - 1 = (x + 1)(x^2 + x + 1)$$

So its real solutions are

$$x = -1, \quad x = \frac{-1 \pm \sqrt{5}}{2}$$

Among these, the one in $(-1, 1)$ is

$$r = \frac{-1 + \sqrt{5}}{2} \in (0, 1)$$

So $x^3 + 2x - 1 = 0$ at $x = r$, meaning f has a vertical asymptote at $x = r$ inside $(-1, 1)$.

For the second term, we can do long division again as $x - 1$ is a factor to get

$$x^3 + x - 2 = (x - 1)(x^2 + x + 2)$$

But $x^2 + x + 2$ has discriminant $1 - 8 = -7 < 0$ so it has no real roots. Hence, the only real zero is $x = 1$ which is an endpoint and not inside $(-1, 1)$.

Therefore, F is continuous on the open interval $(r, 1)$.

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Also note for every $x < 1$,

$$x^3 + x - 2 < 0$$

since it has only one real root at 1, and at $x = 0$ it equals -2. In particular $x^3 + x - 2 < 0$ for all $x \in (r, 1)$. Hence

$$\frac{b}{x^3 + x - 2} < 0$$

on $(r, 1)$.

Now observe,

- As $x \to r^+$, we have $x^3 + 2x^2 - 1 \to 0^+$ (as it changes sign from negative to positive at the root r), so

  $$\frac{a}{x^3 + 2x^2 - 1} \to \infty$$

  Meanwhile x^3+x-2 stays finite and negative near r, so $\frac{b}{x^3+x-2}$ stays bounded. Therefore

  $$\lim_{x \to r^+} f(x) = \infty$$

- As $x \to 1^-$, we have $x^3 + x - 2 \to 0^-$, so

  $$\frac{b}{x^3 + x - 2} \to -\infty$$

  and $x^3 + 2x^2 - 1 \to 2$, so $\frac{a}{x^3+2x^2-1} \to \frac{a}{2}$, a finite number. Therefore,

  $$\lim_{x \to 1^-} f(x) = -\infty$$

From the limits above, we can choose points c and d with $r < c < d < 1$, $f(c) > 0$ and $f(d) < 0$. Since f is continuous on $[c, d] \subseteq (r, 1)$, the Intermediate Value Theorem implies there exists $x_0 \in (c, d) \subseteq (r, 1) \subseteq (-1, 1)$ such that $f(x_0) = 0$.

So the equation

$$\frac{a}{x^3 + 2x^2 - 1} + \frac{b}{x^3 + x - 2} = 0$$

has at least one solution in $(-1, 1)$.

1.5 Limits at Infinity and Infinite Limits

In previous sections, limits described the behavior of a function as the input approaches a finite value. In many important applications, however, we are interested in what happens to a function as the input grows without bound or as the function values themselves become arbitrarily large. This leads to the study of limits at infinity and infinite limits.

Limits at infinity allow us to describe the long-term behavior of functions and to identify horizontal asymptotes, which model steady-state behavior in physical, biological, and economic systems. Infinite limits describe situations where a function grows without bound near a finite input value, giving rise to vertical asymptotes.

This section emphasizes careful interpretation of notation, precise language, and graphical understanding. While infinity is not a number, limits involving infinity can still be defined

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rigorously using inequalities. These ideas will play a central role later in curve sketching, optimization, and applied modeling.

A limit at infinity describes how a function behaves as x becomes very large (positive or negative). We write

$$\lim_{x\to\infty} f(x) = L \quad \text{or} \quad \lim_{x\to-\infty} f(x) = L$$

That is, as x increases (or decreases) without bound, the function values approach the number L.

Formally,

Definition 1.50. We say that

$$\lim_{x\to\infty} f(x) = L$$

if for every $\varepsilon > 0$, there exists a number $M > 0$ such that for all $x > M$, $|f(x) - L| < \varepsilon$.

Similarly, we say that

$$\lim_{x\to-\infty} f(x) = L$$

if for every $\varepsilon > 0$, there exists a number $M > 0$ such that for all $x < -M$, $|f(x) - L| < \varepsilon$.

The difference between Definition 1.50 and the ε-δ definition is we control how large x must be, not how close it is to a finite point.

Definition 1.51. The line $y = L$ is a horizontal asymptote of f if either

$$\lim_{x\to\infty} f(x) = L \quad \text{or} \quad \lim_{x\to-\infty} f(x) = L$$

A function may have

- no horizontal asymptotes,
- one horizontal asymptote, or
- two horizontal asymptotes as $x \to \pm\infty$.

Example 1.52. Let $f(x) = \frac{1}{x}$. Then

$$\lim_{x\to\infty} \frac{1}{x} = 0$$

As x grows larger, the fraction becomes arbitrarily small.

Example 1.53. Evaluate

$$\lim_{x\to\infty} \frac{2x+1}{x+3}$$

Solution. Divide the numerator and denominator by x:

$$\lim_{x\to\infty} \frac{2+\frac{1}{x}}{1+\frac{3}{x}} = \frac{2}{1} = 2$$

Thus, $y = 2$ is a horizontal asymptote.

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Let

$$f(x) = \frac{a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0}{b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0}$$

The key idea is to factor out the largest power of x from the numerator and denominator (equivalently, divide the top and bottom by x^m or x^n, whichever is larger). There are three cases

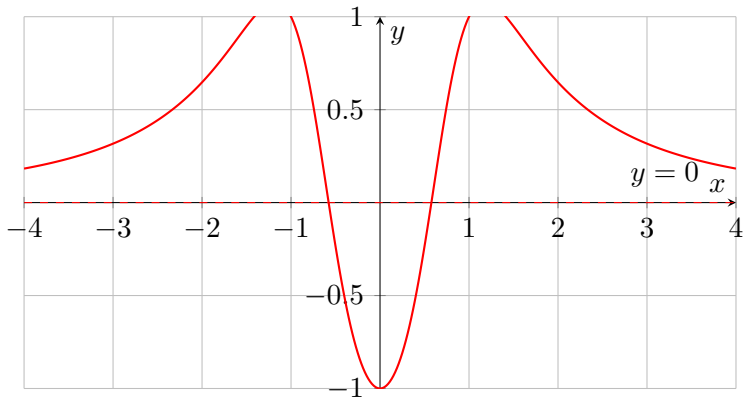
1. If $n < m$, then $x^n < x^m$. Divide the numerator and denominator by x^m to get

$$f(x) = \frac{a_n \frac{1}{x^{m-n}} + a_{n-1}\frac{1}{x^{m-(n-1)}} + \cdots + a_0 \frac{1}{x^m}}{b_m + b_{m-1}\frac{1}{x} + \cdots + b_0 \frac{1}{x^m}}$$

As $x \to \pm\infty$, every term $\frac{1}{x^k} \to 0$, so the numerator converges to zero, and the denominator converges to b_m, so

$$\lim_{x\to\pm\infty} f(x) = \frac{0}{b_m} = 0$$

Example 1.54. $\displaystyle\lim_{x\to\infty} \frac{3x^2 - 1}{5x^4 + 7} = 0.$



2. If $n = m$, then divide the numerator and denominator by $x^n = x^m$

$$f(x) = \frac{a_n + a_{n-1}\frac{1}{x} + a_{n-2}\frac{1}{x^2} + \cdots + a_0 \frac{1}{x^n}}{b_n + b_{n-1}\frac{1}{x} + b_{n-2}\frac{1}{x^2} + \cdots + b_0 \frac{1}{x^n}}$$

As $x \to \pm\infty$, all $\frac{1}{x^k} \to 0$, so

$$\lim_{x\to\pm\infty} f(x) = \frac{a_n}{b_n}$$

Example 1.55. $\displaystyle\lim_{x\to-\infty} \frac{4x^3 - x}{2x^3 + 7} = \frac{4}{2} = 2$

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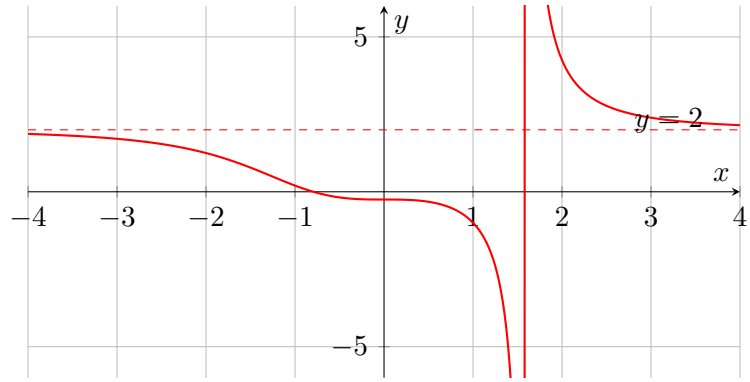

3. If $n > m$, the behaviour here is unbounded. The function typically has a polynomial (slant/oblique) asymptote. Divide by x^m (the highest power in the denominator)

$$f(x) = \frac{a_n x^{n-m} + a_{n-1}x^{n-1-m} + \cdots + a_0 x^{-m}}{b_m + b_{m-1}\frac{1}{x} + \cdots + b_0 \frac{1}{x^m}}$$

As $x \to \pm\infty$, the denominator approaches $b_m \neq 0$, while the numerator is dominated by $a_n x^{n-m}$. So the size of $f(x)$ behaves like

$$f(x) \sim \frac{a_n}{b_m} x^{n-m}$$

What happens depends on $n - m$

(a) If $n - m$ is odd, then $x^{n-m} \to \infty$ as $x \to \infty$, and $x^{n-m} \to -\infty$ as $x \to -\infty$. So two ends go in opposite directions, with signs determined by $\frac{a_n}{b_m}$.

Example 1.56. Let

$$f(x) = \frac{x^2}{x-1}$$

Applying long division gives

$$\begin{array}{r|l} & x + 1 \\ \hline x - 1 & x^2 \\ & -x^2 + x \\ \hline & \ x \\ & \ -x + 1 \\ \hline & \ 1 \end{array}$$

So then

$$\frac{x^2}{x-1} = x + 1 + \frac{1}{x-1}$$

So $f(x)$ approaches the slant asymptote $y = x + 1$, and since that line goes to $\pm\infty$, the limit is not finite.

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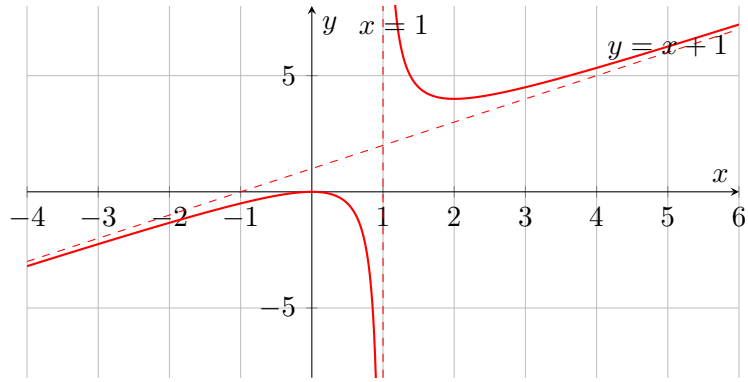


(b) If $n - m$ is even, then $x^{n-m} \to \infty$ as $x \to \infty$, and as $x \to -\infty$. So both ends go to the same direction (both up or both down) depending on the sign of $\frac{a_n}{b_m}$.

Example 1.57. Let

$$f(x) = \frac{-2x^4 + 1}{x^2 + 3}$$

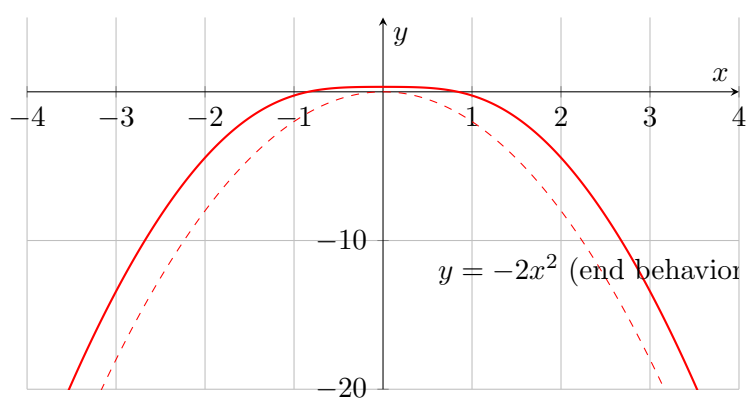
Then long division gives

$$\begin{array}{r|rrr} & & -2x^2 & +6 \\ \hline x^2+3 & -2x^4 & & +1 \\ & 2x^4 & +6x^2 & \\ \hline & & 6x^2 & +1 \\ & & -6x^2 & -18 \\ \hline & & & -17 \end{array}$$

Then

$$\frac{-2x^4 + 1}{x^2 + 3} = -2x^2 + 6 - \frac{17}{x^2 + 3}$$

So $f(x)$ approaches the quadratic asymptote $y = -2x^2 + 6$ and since this quadratic goes to $-\infty$ in both directions, the limit is not finite.



Limits at infinity involving square roots often appear complicated at first glance, but their behaviour is governed by the highest-degree term inside the square root. The key technique is to factor out the highest power of x from inside the radical.

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For large $|x|$,

$$\sqrt{ax^n + \text{lower-degree terms}} \quad \text{behaves like} \quad \sqrt{a}|x|^{\frac{n}{2}}$$

When comparing such expressions with polynomials, always

1. Factor out the highest power of x inside the square root.
2. Simplify using absolute vlaues.
3. Treat $x \to \infty$ and $x \to -\infty$ separately when necessary.

Example 1.58. Evaluate

$$\lim_{x\to\pm\infty} \frac{\sqrt{x^2+3x+1}}{x}$$

Solution. First we compute the limit as $x \to \infty$. Since the highest power of x inside the square root is x^2, then factor x^2 from inside the square root to get

$$\sqrt{x^2+3x+1} = \sqrt{x^2\left(1+\frac{3}{x}+\frac{1}{x^2}\right)} = |x|\sqrt{1+\frac{3}{x}+\frac{1}{x^2}}$$

Since $x \to \infty$, we have $|x| = x$, so

$$\frac{\sqrt{x^2+3x+1}}{x} = \frac{x\sqrt{1+\frac{3}{x}+\frac{1}{x^2}}}{x} = \sqrt{1+\frac{3}{x}+\frac{1}{x^2}}$$

Taking the limit

$$\lim_{x\to\infty} \sqrt{1+\frac{3}{x}+\frac{1}{x^2}} = 1$$

Therefore,

$$\lim_{x\to\infty} \frac{\sqrt{x^2+3x+1}}{x} = 1$$

Next we compute the limit as $x \to -\infty$. Again, factoring out the x^2 inside the square root gives us

$$\sqrt{x^2+3x+1} = |x|\sqrt{1+\frac{3}{x}+\frac{1}{x^2}}$$

Now since $x \to -\infty$, we have $|x| = -x$, and so

$$\frac{\sqrt{x^2+3x+1}}{x} = \frac{-x\sqrt{1+\frac{3}{x}+\frac{1}{x^2}}}{x} = -\sqrt{1+\frac{3}{x}+\frac{1}{x^2}}$$

Taking the limit

$$\lim_{x\to-\infty} \left(-\sqrt{1+\frac{3}{x}+\frac{1}{x^2}}\right) = -1$$

Therefore,

$$\lim_{x\to-\infty} \frac{\sqrt{x^2+3x+1}}{x} = -1$$

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Example 1.59. Evaluate

$$\lim_{x\to\infty}\left(\sqrt{x^2+4x}-x\right)$$

Solution. We multiply and divide by the conjugate

$$\sqrt{x^2+4x}-x=(\sqrt{x^2+4x}-x)\cdot\frac{\sqrt{x^2+4x}+x}{\sqrt{x^2+4x}+x}=\frac{(x^2+4x)-x^2}{\sqrt{x^2+4x}+x}=\frac{4x}{\sqrt{x^2+4x}+x}$$

Factoring x^2 inside the square root,

$$\frac{4x}{\sqrt{x^2\left(1+\frac{4}{x}\right)}+x}=\frac{4x}{|x|\sqrt{1+\frac{4}{x}}+x}$$

Since $x\to\infty$, we have $|x|=x$, so

$$\frac{4x}{|x|\sqrt{1+\frac{4}{x}}+x}=\frac{4x}{x\sqrt{1+\frac{4}{x}}+x}=\frac{4x}{x\left(\sqrt{1+\frac{4}{x}}+1\right)}=\frac{4}{\sqrt{1+\frac{4}{x}}+1}$$

Taking the limit gives

$$\lim_{x\to\infty}\frac{4}{\sqrt{1+\frac{4}{x}}+1}=\frac{4}{1+1}=2$$

Therefore,

$$\lim_{x\to\infty}\left(\sqrt{x^2+4x}-x\right)=2$$

Example 1.60. Evaluate

$$\lim_{x\to\infty}\frac{\sqrt{x^3+2x}}{x^{\frac{3}{2}}}$$

Solution. Factor x^3 inside the square root

$$\sqrt{x^3+2x}=\sqrt{x^3\left(1+\frac{2}{x^2}\right)}=x^{\frac{3}{2}}\sqrt{1+\frac{2}{x^2}}$$

with $x>0$. Then

$$\frac{\sqrt{x^3+2x}}{x^{\frac{3}{2}}}=\sqrt{1+\frac{2}{x^2}}$$

Taking the limit

$$\lim_{x\to\infty}\frac{\sqrt{x^3+2x}}{x^{\frac{3}{2}}}=1$$

For large negative x, note that $x^3+2x<0$, so $\sqrt{x^3+2x}$ is not defined, and thus,

$$\lim_{x\to-\infty}\frac{\sqrt{x^3+2x}}{x^{\frac{3}{2}}}$$

does not exist.

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Example 1.61. Evaluate

$$\lim_{x\to\infty}\left(\sqrt{x^3+x}-x^{\frac{3}{2}}\right)$$

Solution. Rationalize

$$\sqrt{x^3+x}-x^{\frac{3}{2}}=\left(\sqrt{x^3+x}-x^{\frac{3}{2}}\right)\cdot\frac{\sqrt{x^3+x}+x^{\frac{3}{2}}}{\sqrt{x^3+x}+x^{\frac{3}{2}}}=\frac{(x^3+x)-x^3}{\sqrt{x^3+x}+x^{\frac{3}{2}}}=\frac{x}{\sqrt{x^3+x}+x^{\frac{3}{2}}}$$

Factor $x^{\frac{3}{2}}$ from the denominator

$$\frac{x}{x^{\frac{3}{2}}\left(\sqrt{1+\frac{1}{x^2}}+1\right)}=\frac{1}{\sqrt{x}\left(\sqrt{1+\frac{1}{x^2}}+1\right)}$$

As $x\to\infty$, the denominator grows without bound, so

$$\lim_{x\to\infty}\left(\sqrt{x^3+x}-x^{\frac{3}{2}}\right)=0$$

An infinite limit occurs when function values grow without bound as x approaches a finite value. We write

$$\lim_{x\to a}f(x)=\infty\quad\text{or}\quad\lim_{x\to a}f(x)=-\infty$$

Definition 1.62. We say that

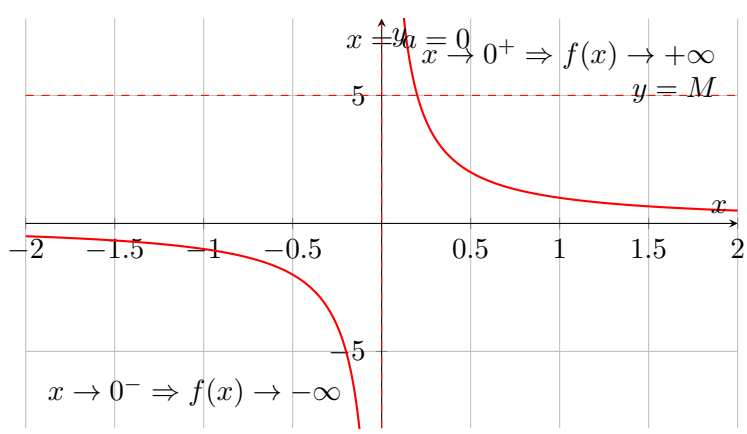
$$\lim_{x\to a}f(x)=\infty$$

if for every $M>0$, there exists a $\delta>0$ such that if $0<|x-a|<\delta$, then $f(x)>M$.

Similarly, we say that

$$\lim_{x\to a}f(x)=-\infty$$

if for every $M>0$, there exists a $\delta>0$ such that if $0<|x-a|<\delta$, then $f(x)<-M$.



Definition 1.63. The line $x=a$ is a *vertical asymptote* of f if at least one of the following holds

$$\lim_{x\to a^-}f(x)=\pm\infty\quad\text{or}\quad\lim_{x\to a^+}f(x)=\pm\infty$$

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Example 1.64. We have

$$\lim_{x\to 0^+} \frac{1}{x} = \infty \quad \text{and} \quad \lim_{x\to 0^-} \frac{1}{x} = -\infty$$

Thus, $x = 0$ is a vertical asymptote.

Example 1.65. We have

$$\lim_{x\to 2} \frac{1}{(x-2)^2} = \infty$$

Both sides approach ∞

$$\lim_{x\to 2^+} \frac{1}{(x-2)^2} = \lim_{x\to 2^-} \frac{1}{(x-2)^2} = \infty$$

It is important to note that ∞ is not a number. Statements such as $\lim_{x\to\infty} f(x) = \infty$ describe behaviour, not values. A vertical asymptote may exist on only one side. Asymptotes can be crossed.

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Chapter 2

Differentiation

Differentiation is one of the central ideas of calculus and provides a precise mathematical framework for describing change. Many quantities of interest in mathematics, science, economics, and engineering are not static: they vary with time, position, or other variables. The goal of this chapter is to develop the concept of the derivative, which measures how a function changes with respect to its input.

We begin with concrete motivating problems involving rates of change, such as velocity, growth rates, and slopes of curves. These problems naturally lead to the notion of the tangent line to a curve at a point and raise a fundamental question: how can we define and compute the slope of a curve at a single point when a curve is not linear?

The answer lies in the concept of a limit. By examining the behavior of secant lines as two points on a curve move closer together, we arrive at a precise definition of the derivative as a limit. This definition forms the foundation of differentiation and allows us to treat derivatives not just as numbers at specific points, but as functions in their own right.

Once the derivative is defined, the chapter develops systematic rules for computing derivatives efficiently, including the power rule, product rule, quotient rule, and chain rule, as well as derivatives of trigonometric, exponential, and logarithmic functions. Throughout, careful attention is paid to both computational skill and theoretical understanding. Proofs are included whenever possible to justify the differentiation rules and to emphasize the logical structure underlying calculus.

In addition to explicit differentiation, we study techniques such as implicit differentiation and related rates, which allow us to analyze situations where variables are linked by equations rather than explicit formulas. These methods are essential for modeling real-world problems.

2.1 Derivatives and Rates of Change

In many problems, we are interested not only in the value of a function, but in how that value changes. For example:

- How fast is a car moving at a specific instant?
- How quickly is a population growing at a given time?
- How steep is a curve at a particular point?

These questions cannot be answered by average changes alone. Instead, they require a notion of instantaneous rate of change, which leads to the central concept of this section: the derivative.

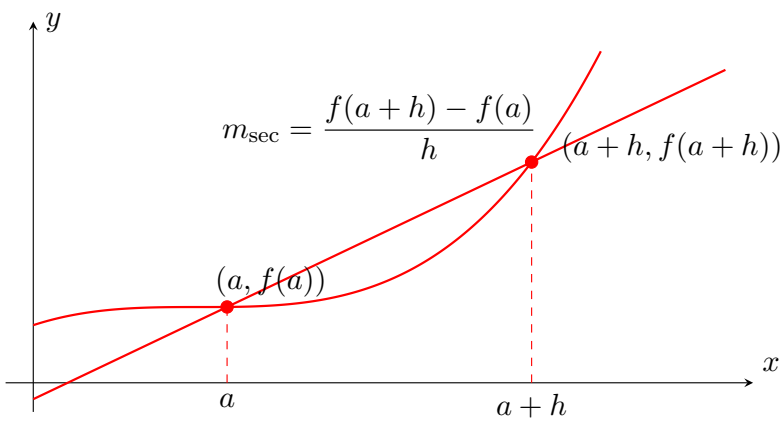
We begin by reviewing average rates of change, then show how instantaneous rates of change arise naturally by taking limits.

Definition 2.1. Let f be a function defined on an interval containing $x = a$ and $x = a + h$, with $h \neq 0$. The *average rate of change* of f from $x = a$ to $x = a + h$ is

$$\frac{f(a+h) - f(a)}{h}$$

This quantity represents the change in the output divided by the change in the input.

The average rate of change of f from a to $a + h$ is the slope of the secant line joining the points $(a, f(a))$ and $(a + h, f(a + h))$.



Example 2.2. Suppose the position of an object moving along a line is given by

$$s(t) = t^2$$

where s is measured in meters, and t is measured in seconds.

The average velocity from $t = 1$ and $t = 1 + h$ is

$$\frac{s(1+h) - s(1)}{h} = \frac{(1+h)^2 - 1}{h} = \frac{2h + h^2}{h} = 2 + h$$

This gives the average velocity over the time interval $[1, 1 + h]$.

Average rates of change depend on an interval. To measure how a quantity changes at a single point, we examine what happens as the interval becomes smaller and smaller.

Definition 2.3. The *instantaneous rate of change* of a function f at $x = a$ is defined as the limit of the average rate of change as $h \to 0$, provided the limit exists

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

This limit, when it exists, is called the *derivative of f at a*.

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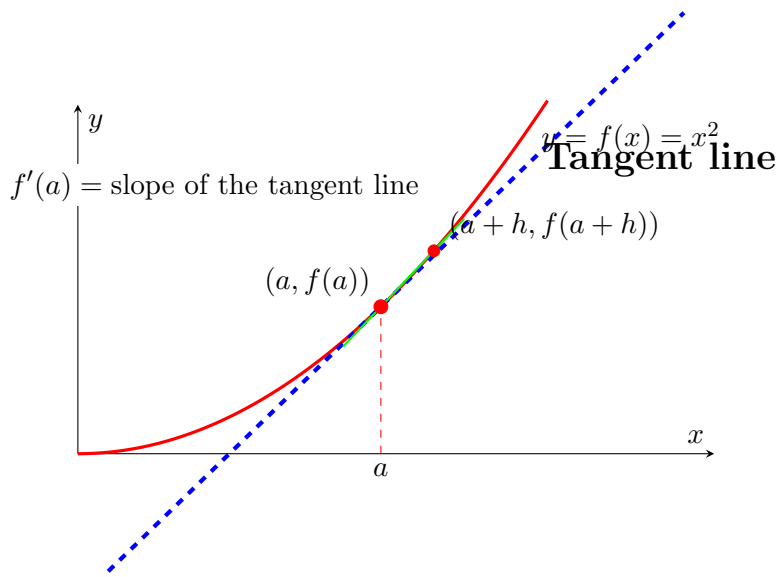
Definition 2.4. Let f be a function defined near $x = a$. The *derivative of f at a* is

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

if this limit exists.

The derivative $f'(a)$ can be interpreted in several equivalent ways:

1. Instantaneous rate of change of f with respect to x at $x = a$.
2. Slope of the tangent line to the graph of $y = f(x)$ at $x = a$.
3. Velocity, if f represents position as a function of time.



Example 2.5. Let $f(x) = x^2$. Compute $f'(a)$ using the definition.

Solution. Using the definition of the limit,

$$\begin{aligned} f'(a) &= \lim_{h \to 0} \frac{(a+h)^2 - a^2}{h} \\ &= \lim_{h \to 0} \frac{a^2 + 2ah + h^2 - a^2}{h} \\ &= \lim_{h \to 0} \frac{2ah + h^2}{h} \\ &= \lim_{h \to 0} (2a + h) \\ &= 2a \end{aligned}$$

Thus,

$$f'(a) = 2a$$

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Definition 2.6. The *tangent line* to the curve $y = f(x)$ at the point $(a, f(a))$ is the line passing through that point with slope $f'(a)$. Its equation is

$$y - f(a) = f'(a)(x - a)$$

Example 2.7. Let $f(x) = x^2$. Find the equation of the tangent line at $x = 1$.

Solution. At $x = 1$, we have $f(1) = 1$. Furthermore, from Example 2.5, $f'(1) = 2$. Thus,

$$\begin{aligned} y - 1 &= 2(x - 1) \\ y &= 2x - 1 \end{aligned}$$

One of the earliest applications of derivatives comes from physics.

Definition 2.8. Let $s(t)$ denote the position of an object at time t.

- The *average velocity* on $[t, t + h]$ is the quantity

$$\frac{s(t + h) - s(t)}{h}$$

- The *instantaneous velocity* at time t is

$$v(t) = \lim_{h \to 0} \frac{s(t + h) - s(t)}{h} = s'(t)$$

- The *acceleration* at time t is

$$a(t) = \lim_{h \to 0} \frac{v(t + h) - v(t)}{h} = v'(t) = s''(t)$$

Example 2.9. If $s(t) = x^2 - 6x$, then by the definition of the limit,

$$\begin{aligned} v(t) &= \lim_{h \to 0} \frac{(t + h)^2 - 6(t + h) - t^2 + 6t}{h} \\ &= \lim_{h \to 0} \frac{t^2 + 2th + h^2 - 6t - 6t - t^2 + 6t}{h} \\ &= \lim_{h \to 0} \frac{2th + h^2 - 6h}{h} \\ &= \lim_{h \to 0} (2t + h - 6) \\ &= 2t - 6 \end{aligned}$$

A function may fail to have a derivative at a point if

1. The graph has a corner or cusp,
2. The function is discontinuous, or
3. The tangent line is vertical.

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Example 2.10. Let $f(x) = |x|$. We show that $f(x)$ does not have a derivative at $x = 0$. Indeed, observe that from the left

$$\lim_{h\to 0^-} \frac{|h| - 0}{h} = \lim_{h\to 0^-} \frac{-h}{h} = \lim_{h\to 0^-} (-1) = -1$$

On the other hand,

$$\lim_{h\to 0^+} \frac{|h| - 0}{h} = \lim_{h\to 0^+} \frac{h}{h} = \lim_{h\to 0} 1 = 1$$

Since the one-sided limits are not equal, $f'(0)$ does not exist.

2.2 The Derivative as a Function

In Section 2.1, we defined the derivative of a function at a single point as an instantaneous rate of change and as the slope of the tangent line at that point. However, in most applications, we are not interested in the derivative at just one value of the variable, but rather in how the rate of change itself varies from point to point.

For example,

- How does the velocity of a moving object change over time?
- At which points is a curve increasing most rapidly?
- How does the slope of a curve depend on the input value?

To answer such questions, we extend the notion of a derivative from a number to a function. This leads to the concept of the derivative function, which assigns to each input value the derivative of the original function at that point.

Recall that the derivative of f at $x = a$ is defined by

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}$$

provided the limit exists. This definition gives a single number corresponding to the slope of the tangent at $x = a$.

Definition 2.11. Let f be a function defined on an open interval I. If the limit

$$\lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

exists for every $x \in I$, then the function defined by

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

is called the *derivative of* f, or the *derivative function*.

The derivative function $f'(x)$ represents the instantaneous rate of change of f at x, the slope of the tangent line to the graph of $y = f(x)$ at x, and a new function whose values describe how f is changing.

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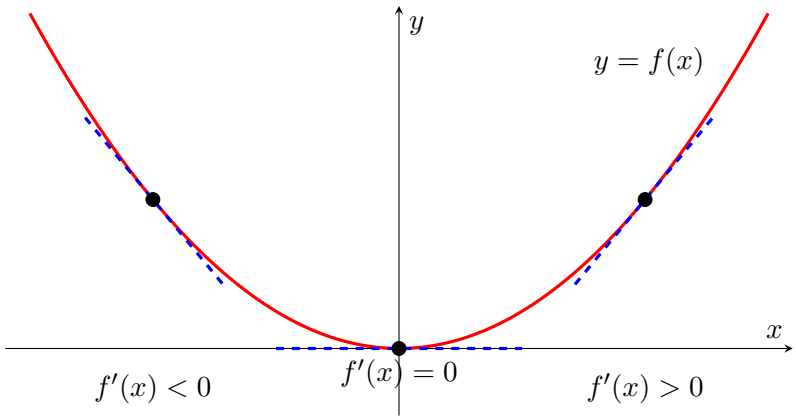

In this section, derivatives are computed directly from the limit definition. This emphasizes understanding and prepares us for proofs of differentiation rules. A lot of these examples to come, are simply limit laws and computation of limit laws that we have explored in Section 1.2, the examples that we did, are similar to the ones for the definition of the derivative.

Example 2.12. Use the definition of the derivative to compute the derivative of $f(x) = x^2$.

Solution. Using the definition,

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h\to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\ &= \lim_{h\to 0} \frac{2xh + h^2}{h} \\ &= \lim_{h\to 0} (2x + h) \\ &= 2x \end{aligned}$$

Therefore, the derivative function is $f'(x) = 2x$.

Example 2.13. Use the definition of the derivative to compute the derivative of $f(x) = c$, where c is a constant.

Solution. Using the definition,

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{c - c}{h} \\ &= \lim_{h\to 0} 0 \\ &= 0 \end{aligned}$$

Thus, the derivative of a constant function is zero everywhere.

Definition 2.14. A function f is said to be *differentiable at* x if the derivative $f'(x)$ exists. A function is *differentiable on an interval* if it is differentiable at every point in that interval.

An important observation that we can make about differentiability is the following.

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Theorem 2.15. *If f is differentiable, then f is continuous.*

Proof. Assume that f is differentiable at a. Then by definition,

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists. Write

$$f(a+h) - f(a) = h \cdot \frac{f(a+h) - f(a)}{h}$$

Taking limits as $h \to 0$, and using the fact that the limit of a product is the product of the limits,

$$\lim_{h \to 0} [f(a+h) - f(a)] = 0$$

which implies

$$\lim_{x \to a} f(x) = f(a)$$

Thus, f is continuous at a. □

Example 2.16. Let $f(x) = |x|$. Assume that $h \to 0^+$, then

$$f'(0) = \lim_{h \to 0^+} \frac{|h| - 0}{h} = \lim_{h \to 0^+} 1 = 1$$

and as $h \to 0^-$, we have

$$f'(0) = \lim_{h \to 0^-} \frac{|h| - 0}{h} = \lim_{h \to 0^-} (-1) = -1$$

Thus, since the left- and right-hand limits are not the same, $f'(0)$ does not exist. Thus, the derivative function of $f(x)$ is defined on $(-\infty, 0) \cup (0, \infty)$.

The graph of $f'(x)$ provides direct information about the behaviour of $f(x)$.

Sign of $f'(x)$	$f(x)$
$f'(x) > 0$	f is increasing
$f'(x) < 0$	f is decreasing
$f'(x) = 0$	horizontal tangent line

Example 2.17. If $f(x) = x^2$, then $f'(x) = 2x$. Note then

- $f'(x) = 2x > 0$ whenever $x > 0$, so $f(x)$ is increasing on $(0, \infty)$.
- $f'(x) = 2x < 0$ whenever $x < 0$, so $f(x)$ is decreasing on $(-\infty, 0)$.
- $f'(0) = 0$.

This explains the shape of the parabola and foreshadows the connection between derivatives and optimization. This will explained more later on in Chapter 3.

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Several notations are used for derivatives, depending on context.

$$\begin{aligned} f'(x) &\quad \text{(Lagrange notation)} \\ \frac{dy}{dx} &\quad \text{(Leibniz notation)} \\ y' &\quad \text{(prime notation)} \\ D_x f(x) &\quad \text{(operator notation)} \end{aligned}$$

If $y = f(x)$, then all notations above represent the same derivative.

Since $f'(x)$ is a function itself, it may also be differentiable.

Definition 2.18. If $f'(x)$ is differentiable, its derivative is called the second derivative of f, denoted by

$$f''(x) = \frac{d^2y}{dx^2}.$$

Example 2.19. If $f(x) = x^2$, then

$$f'(x) = 2x, \quad f''(x) = 2$$

Second derivatives will later be interpreted as acceleration and concavity.

2.3 Derivatives of Polynomial Functions

In Sections 2.1 and 2.2, derivatives were defined using limits and interpreted as instantaneous rates of change and slopes of tangent lines. While the limit definition is fundamental, it quickly becomes impractical for routine computations, especially for complicated functions.

Fortunately, many of the functions encountered in applications, such as those modeling in motion, cost, and growth, are polynomial functions. In this section, we derive simple and powerful rules that allow us to compute derivatives of polynomials efficiently, without returning to the limit definition each time.

The key result of this section is the power rule, which will be proved rigorously using limits. From this, we obtain derivatives of all polynomial functions by combining differentiation rules for constants and sums.

Theorem 2.20 (**Constant Rule for Derivatives**). *If $f(x) = c$, where c is a constant, then*

$$f'(x) = 0$$

for all x.

Proof. Using the definition of the derivative,

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h\to 0} \frac{c - c}{h} \\ &= \lim_{h\to 0} 0 \\ &= 0 \end{aligned}$$

□

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Theorem 2.21 (**Power Rule for Derivatives,** $\mathbb{N}$)**.** *If $f(x) = x^n$, where n is a positive integer, then*

$$f'(x) = nx^{n-1}$$

for all x.

Proof. Using the definition of the derivative,

$$f'(x) = \lim_{h \to 0} \frac{(x+h)^n - x^n}{h}$$

Using the binomial theorem,

$$(x+h)^n = x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + h^2$$

Subtracting x^n and factoring out h,

$$\frac{(x+h)^n - x^n}{h} = nx^{n-1} + \binom{n}{2}x^{n-2}h + \cdots + h^{n-1}$$

Taking the limit as $h \to 0$,

$$f'(x) = nx^{n-1}$$

as desired. □

Although the power rule works nicely for natural numbers, because we have the binomial theorem, we will require other tools later on to prove the power rule for any real number.

Theorem 2.22 (**Power Rule for Derivatives,** $\mathbb{R}$)**.** *If $f(x) = x^r$, where r is a real number, then*

$$f'(x) = rx^{r-1}$$

This will require us to prove that the power rule works for integers, then rational numbers, then real numbers.

Theorem 2.23. *If f and g are differentiable functions, and c is a constant, then*

1. *$f + g$ is differentiable with $(f+g)'(x) = f'(x) + g'(x)$, and*
2. *cf is differentiable with $(cf)'(x) = cf'(x)$.*

Recall that we define the functions $f + g$ and cf as

$$(f+g)(x) = f(x) + g(x) \quad \text{and} \quad (cf)(x) = cf(x)$$

Proof. Assume that f and g are differentiable. Then

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \quad \text{and} \quad g'(x) = \lim_{h \to 0} \frac{g(x+h) - g(x)}{h}$$

To see that (1) holds, we use the limit definition

$$(f+g)'(x) = \lim_{h \to 0} \frac{(f+g)(x+h) - (f+g)(x)}{h}$$

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$$
\begin{aligned}
&= \lim_{h\to 0} \frac{f(x+h)+g(x+h)-f(x)-g(x)}{h} \\
&= \lim_{h\to 0} \frac{f(x+h)-f(x)+g(x+h)-g(x)}{h} \\
&= \lim_{h\to 0} \left[\frac{f(x+h)-f(x)}{h} + \frac{g(x+h)-g(x)}{h}\right] \\
&= \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} + \lim_{h\to 0} \frac{g(x+h)-g(x)}{h} \\
&= f'(x)+g'(x)
\end{aligned}
$$

To see that (2) holds, we use the limit definition

$$
\begin{aligned}
(cf)'(x) &= \lim_{h\to 0} \frac{(cf)(x+h)-(cf)(x)}{h} \\
&= \lim_{h\to 0} \frac{cf(x+h)-cf(x)}{h} \\
&= \lim_{h\to 0} c\frac{f(x+h)-f(x)}{h} \\
&= c\lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \\
&= cf'(x)
\end{aligned}
$$

as desired. □

Theorem 2.23 tells us that the derivative operator is *linear*. As students will be taking MATH 1021 (Linear Algebra I), and then eventually MATH 2022 (Linear Algebra II), linearity comes up a lot.

Definition 2.24. A *polynomial function* is a function of the form

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

where $a_0, a_1, ..., a_n$ are real constants and n is a nonnegative integer.

Theorem 2.25. *Every polynomial function is differentiable for all real numbers, and its derivative is given by*

$$f'(x) = na_n x^{n-1} + (n-1)a_{n-1}x^{n-2} + \cdots + a_1$$

Proof. The proof follows from the Power Rule (Theorem 2.21) and linearity of the derivative (Theorem 2.23). □

Example 2.26. Differentiate $f(x) = 3x^4 - 5x^3 + 7x - 9$.

Solution. Using the Power Rule (Theorem 2.21) and Theorem 2.23, we have

$$
\begin{aligned}
f'(x) &= 3\cdot 4x^3 - 5\cdot 3x^2 + 7\cdot 1 - 0 \\
&= 12x^3 - 15x^2 + 7
\end{aligned}
$$

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Since the derivative of a polynomial is again a polynomial, we may differentiate repeatedly.

Example 2.27. Find $f'(x)$, $f''(x)$, and $f'''(x)$, where $f(x) = x^4 - 2x^2 + 3$.

Solution. Using the Power Rule (Theorem 2.21) and Theorem 2.23, we have

$$\begin{aligned} f'(x) &= 4x^3 - 4x, \\ f''(x) &= 12x^2 - 4, \\ f'''(x) &= 24x. \end{aligned}$$

Eventually, repeated differentiation yields the zero function.

If position is given by a polynomial $s(t)$, then the velocity $v(t)$ is the derivative of $s(t)$, and the acceleration $a(t)$ is the derivative of $v(t)$, or the second derivative of $s(t)$.

Example 2.28. Find the velocity $v(t)$ and acceleration functions $a(t)$, where $s(t) = t^3 - 6t^2 + 9t$.

Solution. Since the velocity is the derivative of the position,

$$v(t) = s'(t) = 3t^2 - 12t + 9$$

and since the acceleration is the derivative of the velocity,

$$a(t) = v'(t) = s''(t) = 6t - 12$$

In many applications, we are not interested in the derivative as a general function, but rather in the slope of the tangent line at a specific point, or equivalently, the equation of the tangent line to a curve at that point. Once the derivative $f'(x)$ is known, the slope $x = a$ is obtained by evaluation, and the tangent line is given by the point-slope form

$$y - y_0 = m(x - x_0)$$

where m is the slope and (x_0, y_0) is the point of tangency.

We use the following notations to denote the slope of the tangent at a point.

$$f'(a), \quad \left.\frac{dy}{dx}\right|_{x=a}$$

Example 2.29. Find the equation of the tangent line to

$$y = x^3 - 3x + 1$$

at $x = 1$.

Solution. Differentiating using the Power Rule (Theorem 2.21)

$$\frac{dy}{dx} = 3x^2 - 3$$

Then evaluate at $x = 1$

$$\left.\frac{dy}{dx}\right|_{x=1} = 3(1)^2 - 3 = 0$$

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and the point on the curve is

$$y = (1)^3 - 3(1) + 1 = -1$$

Therefore, the tangent line at $x = 1$ is

$$\begin{aligned} y - y_0 &= m(x - x_0) \\ y + 1 &= 0 \\ y &= -1 \end{aligned}$$

2.4 Product and Quotient Rule

In the previous section, we learned how to differentiate polynomial functions efficiently using the power rule and the linearity of differentiation. However, many functions of practical interest are not simple polynomial functions, but are formed by multiplying or dividing functions.

For example,

- the area of a rectangle with varying length and width,
- the product of position and velocity in physics,
- rational functions formed by ratios of polynomials.

A natural question arises: is the derivative of a product equal to the product of the derivatives?

The answer to this question is no. This section introduces the Product Rule and Quotient Rule, which provide correct and systematic methods for differentiating products and quotients of differentiable functions. These rules will be derived rigorously from the limit definition of the derivative.

Example 2.30. Consider the function $f(x) = x^2$, which can be written as $x^2 = x \cdot x$. Notice that

$$\frac{d}{dx}(x^2) = 2x \quad \text{and} \frac{d}{dx}(x)\frac{d}{dx}(x) = 1$$

Thus,

$$\frac{d}{dx}(x^2) \neq \frac{d}{dx}(x)\frac{d}{dx}(x)$$

In general,

$$\frac{d}{dx}[f(x)g(x)] \neq f'(x)g'(x)$$

This failure motivates the need for new differentiation rules.

Theorem 2.31 (**The Product Rule**). *If f and g are differentiable at x, then the product fg is differentiable at x, and*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x)$$

Recall we define the function fg at x as

$$(fg)(x) = f(x)g(x)$$

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Proof. Assume that f and g are differentiable. Then

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \quad \text{and} \quad g'(x) = \lim_{h\to 0} \frac{g(x+h) - g(x)}{h}$$

Using the limit definition of the derivative,

$$\begin{aligned}
(fg)'(x) &= \lim_{h\to 0} \frac{(fg)(x+h) - (fg)(x)}{h} \\
&= \lim_{h\to 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\
&= \lim_{h\to 0} \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h} \\
&= \lim_{h\to 0} \left[\frac{f(x+h) - f(x)}{h} g(x+h) + f(x) \frac{g(x+h) - g(x)}{h} \right] \\
&= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} g(x+h) + \lim_{h\to 0} f(x) \frac{g(x+h) - g(x)}{h}
\end{aligned}$$

Since f and g are differentiable, hence, continuous,

$$\lim_{h\to 0} g(x+h) = g(x)$$

and so

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x)$$

as desired. □

Example 2.32. Differentiate $f(x) = x^2(3x^4 - 5)$.

Solution. Let $u(x) = x^2$ and $v(x) = 3x^4 - 5$. Then $u'(x) = 2x$ and $v'(x) = 12x^3$, and so

$$\begin{aligned}
f'(x) &= u'(x)v(x) + u(x)v'(x) \\
&= 2x(3x^4 - 5) + x^2(12x^3) \\
&= 6x^5 - 10x + 12x^5 \\
&= 18x^5 - 10x
\end{aligned}$$

Example 2.33. Differentiate $f(x) = (2x^3 - x)(x^2 + 4)$.

Solution. Let $u(x) = 2x^3 - x$ and $v(x) = x^2 + 4$. Then $u'(x) = 6x^2 - 1$ and $v'(x) = 2x$, and so

$$\begin{aligned}
f'(x) &= u'(x)v(x) + u(x)v'(x) \\
&= (6x^2 - 1)(x^2 + 4) + (2x^3 - x)(2x)
\end{aligned}$$

Example 2.34. Find the equation of the tangent line to

$$y = x^2(x+1)$$

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at $x = 1$.

Solution. Differentiating using the product rule (Theorem 2.31)

$$\frac{dy}{dx} = 2x(x+1) + x^2$$

Then evaluate at $x = 1$

$$m = \left.\frac{dy}{dx}\right|_{x=1} = 2(1)(1+1) + 1^2 = 5$$

and the point the curve

$$y = 1^2(1+1) = 2$$

Therefore, the equation of the tangent line is

$$y - 2 = 5(x-1)$$

In general, the product rule can be expanded to more than two functions.

Theorem 2.35. *Let f, g, and h be differentiable functions. Then fgh is also differentiable, and*

$$(fgh)'(x) = f'(x)g(x)h(x) + f(x)g'(x)h(x) + f(x)g(x)h'(x)$$

Proof. We use the product rule with f and gh. Let $u(x) = f(x)$ and $v(x) = (gh)(x)$. Then

$$\begin{aligned} u'(x) &= f'(x) \\ v'(x) &= g'(x)h(x) + g(x)h'(x) \end{aligned}$$

and so

$$\begin{aligned} (fgh)'(x) &= u'(x)v(x) + u(x)v'(x) \\ &= f'(x)g(x)h(x) + f(x)(g'(x)h(x) + g(x)h'(x)) \\ &= f'(x)g(x)h(x) + f(x)g'(x)h(x) + f(x)g(x)h'(x) \end{aligned}$$

as desired. □

Example 2.36. Differentiate $f(x) = x(x+1)(x-2)$.

Solution. This is a product of three functions. You may either expand first, or apply the product rule twice.

Let $u(x) = x$ and $v(x) = v_1(x)v_2(x)$ where $v_1(x) = x+1$ and $v_2(x) = x-2$. Then $u'(x) = 1$, $v_1'(x) = 1$, $v_2'(x) = 1$, and

$$v'(x) = v_1'(x)v_2(x) + v_1(x)v_2'(x) = 1 \cdot x + x \cdot 1 = 2x$$

Therefore,

$$\begin{aligned} f'(x) &= u'(x)v(x) + u(x)v'(x) \\ &= (x+1)(x-2) + x(2x) \\ &= x^2 - x - 2 + 2x^2 \\ &= 3x^2 - x - 2 \end{aligned}$$

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Next, we will prove the quotient rule, where we have a quotient of two functions. If we consider the function $f(x) = \frac{x^2}{x}$. Observe that

$$\frac{d}{dx}\left(\frac{x^2}{x}\right) = \frac{d}{dx}(x) = 1$$

However,

$$\frac{\frac{d}{dx}(x^2)}{\frac{d}{dx}(x)} = \frac{2x}{1} = 2x$$

So clearly,

$$\frac{d}{dx}\left(\frac{x^2}{x}\right) \neq \frac{\frac{d}{dx}(x^2)}{\frac{d}{dx}(x)}$$

and in general,

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) \neq \frac{f'(x)}{g'(x)}$$

Theorem 2.37 (**The Quotient Rule**). *If f and g are differentiable at x with $g(x) \neq 0$, then $\frac{f}{g}$ is differentiable at x with*

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}$$

Recall we define the function $\frac{f}{g}$ at x as

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

Proof. Assume that f and g are differentiable with $g(x) \neq 0$. Then

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \quad \text{and} \quad g'(x) = \lim_{h\to 0} \frac{g(x+h) - g(x)}{h}$$

Using the limit definition of the derivative,

$$\begin{aligned}
\left(\frac{f}{g}\right)'(x) &= \lim_{h\to 0} \frac{\left(\frac{f}{g}\right)(x+h) - \left(\frac{f}{g}\right)(x)}{h} \\
&= \lim_{h\to 0} \frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h} \\
&= \lim_{h\to 0} \frac{f(x+h)g(x) - f(x)g(x+h)}{hg(x+h)g(x)} \\
&= \lim_{h\to 0} \frac{f(x+h)g(x) - f(x)g(x) + f(x)g(x) - f(x)g(x+h)}{hg(x+h)g(x)} \\
&= \lim_{h\to 0} \left[\frac{f(x+h) - f(x)}{h}\frac{g(x)}{g(x+h)g(x)} - \frac{f(x)}{g(x+h)g(x)}\frac{g(x+h) - g(x)}{h}\right]
\end{aligned}$$

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Since f and g are differentiable, hence, continuous,

$$\lim_{h\to 0} g(x+h) = g(x)$$

and so

$$\begin{aligned}\left(\frac{f}{g}\right)'(x) &= f'(x)\frac{g(x)}{g^2(x)} - \frac{f(x)}{g^2(x)}g'(x)\\ &= \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}\end{aligned}$$

as desired. □

Example 2.38. Differentiate

$$f(x) = \frac{x^3-1}{x^2+1}$$

Solution. Let $u(x) = x^3 - 1$ and $v(x) = x^2 + 1$. Then $u'(x) = 3x^2$ and $v'(x) = 2x$, so then

$$\begin{aligned}f'(x) &= \frac{u'(x)v(x) - u(x)v'(x)}{v^2(x)}\\ &= \frac{3x^2(x^2+1) - (x^3-1)(2x)}{(x^2+1)^2}\end{aligned}$$

Example 2.39. Differentiate

$$f(x) = \frac{3x^2-5x+1}{x^2+4}$$

Solution. Let $u(x) = 3x^2 - 5x + 1$ and $v(x) = x^2 + 4$. Then $u'(x) = 6x - 5$ and $v'(x) = 2x$, and so

$$\begin{aligned}f'(x) &= \frac{u'(x)v(x) - u(x)v'(x)}{v^2(x)}\\ &= \frac{(6x-5)(x^2+4) - (3x^2-5x+1)(2x)}{(x^2+4)^2}\end{aligned}$$

Example 2.40. Find the equation of the tangent line to

$$f(x) = \frac{x^3+2}{x-1}$$

at $x = 2$.

Solution. Differentiate using the quotient rule (Theorem 2.37)

$$\begin{aligned}f'(x) &= \frac{(3x^2)(x-1) - (x^3+2)(1)}{(x-1)^2}\\ &= \frac{3x^3 - 3x^2 - x^3 - 2}{(x-1)^2}\end{aligned}$$

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$$= \frac{2x^3 - 3x^2 - 2}{(x-1)^2}$$

Evaluate at $x = 2$ gives

$$m = f'(2) = \frac{2(2)^3 - 3(2)^2 - 2}{(2-1)^2} = 2$$

and the point on the curve at $x = 2$ is

$$f(1) = \frac{2^3 + 2}{2 - 1} = 10$$

Therefore, the equation of the tangent line to the curve at $x = 2$ is

$$y - 10 = 2(x - 2)$$

The following is a consequence that is often useful in computing derivatives of reciprocal functions.

Theorem 2.41 (**Derivative of Reciprocal Functions**). *Let $g(x)$ be a function such that $g(x) \neq 0$ for all $x \in \mathbb{R}$. If $f(x) = \frac{1}{g(x)}$, then f is differentiable for all $x \in \mathbb{R}$ such that $g(x) \neq 0$, and*

$$f'(x) = -\frac{g'(x)}{g^2(x)}$$

Proof. Using the quotient rule (Theorem 2.37), we have

$$f'(x) = \frac{(0)g(x) - (1)g'(x)}{g^2(x)} = -\frac{g'(x)}{g^2(x)}$$

□

To conclude this section, we will prove the power rule for all integers.

Theorem 2.42 (**Power Rule For Derivatives, $\mathbb{Z}$**). *If $f(x) = x^n$, where n is an integer, then*

$$f'(x) = nx^{n-1}$$

for all x.

Proof. Since we already proved the power rule holds for all positive integers, we simply need to show that the power rule holds for all negative integers. Indeed, if n is a negative integer, set $n = -m$, where m is a positive integer. Then we have

$$f(x) = x^{-m} = \frac{1}{x^m}$$

Here, we can use the Quotient Rule (Theorem 2.37) to find the derivative. Indeed, let $u(x) = 1$ and $v(x) = x^m$. Then $u'(x) = 0$ and $v'(x) = mx^{m-1}$ as m is a positive integer. Then

$$f'(x) = \frac{u'(x)v(x) - u(x)v'(x)}{v^2(x)}$$

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$$
\begin{aligned}
&= \frac{(0)(x^m) - (1)(mx^{m-1})}{x^{2m}} \\
&= -\frac{mx^{m-1}}{x^{2m}} \\
&= -mx^{-m-1}
\end{aligned}
$$

Since $n = -m$, we have

$$f'(x) = nx^{n-1}$$

Therefore the power rule for derivatives hold for all integers n. □

Example 2.43. Compute the derivative of $f(x) = 3x^{-4} + 6x^2 + 8x^{-1} + 3$, and write final answer with positive exponents.

Solution. Using the power rule for integers (Theorem 2.42),

$$
\begin{aligned}
f'(x) &= 3 \cdot (-4)x^{-5} + 6 \cdot 2x + 8 \cdot (-1)x^{-2} + 0 \\
&= -12x^{-5} + 12x - 8x^{-2} \\
&= -\frac{12}{x^5} + 12x - \frac{8}{x^2}
\end{aligned}
$$

2.5 The Chain Rule

Up to this point, we have learned how to differentiate sums, products, and quotients of functions. However, many functions encountered in applications are built not by combining functions side-by-side, but by composing them.

Some examples include

$$(3x^2 + 1)^5, \quad \sin(x^3), \quad e^{\sqrt{x}}$$

or temperature as a function of altitude, where altitude is a function of time.

In such cases, the input of one function is itself the output of another. The Chain Rule provides a systematic method for differentiating composite functions and is one of the most important tools in differential calculus.

Definition 2.44. If f and g are functions, the *composition of f and g* is the function $f \circ g$, where

$$(f \circ g)(x) = f(g(x))$$

for all x.

Example 2.45. Let $f(u) = u^2$ and $g(x) = 3x + 1$. Then

$$(f \circ g)(x) = f(g(x)) = f(3x + 1) = (3x + 1)^2$$

We now present the statement and proof of the chain rule. The method to prove the chain rule works differently than the proof of the product rule or chain rule.

Theorem 2.46 (**The Chain Rule**). *Let g be a differentiable at x, and let f be differentiable at $g(x)$. Then the composite function $f \circ g$ is differentiable at x and*

$$(f \circ g)'(x) = f'(g(x))g'(x)$$

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Alternatively, if $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

The second notation highlights the idea that rates of change multiply.

Proof. Let $u = g(x)$ and consider the difference quotient for $f \circ g$

$$\frac{(f \circ g)(x+h) - (f \circ g)(x)}{h} = \frac{f(g(x+h)) - f(g(x))}{h}$$

Set $\Delta x = h$ and $\Delta u = g(x+h) - g(x)$. Then

$$\frac{f(g(x+h)) - f(g(x))}{h} = \frac{f(u+\Delta u) - f(u)}{h} \cdot \frac{\Delta u}{\Delta x}$$

provided $\Delta u \neq 0$.

The issue is that it is possible for $\Delta u = 0$ for some small h. We handle this carefully by introducing an auxiliary function.

Since f is differentiable at u, by definition,

$$f'(u) = \lim_{\Delta u \to 0} \frac{f(u+\Delta u) - f(u)}{\Delta u}$$

Equivalently, define

$$\phi(\Delta u) = \begin{cases} \frac{f(u+\Delta u)-f(u)}{\Delta u} - f'(u) & \text{if } \Delta u \neq 0 \\ 0 & \text{if } \Delta u = 0 \end{cases}$$

so that

$$\lim_{\Delta u \to 0} \phi(\Delta u) = 0$$

Then for all Δu,

$$f(u+\Delta u) - f(u) = (f'(u) + \phi(\Delta u))\Delta u$$

Then substituting $\Delta u = g(x+h) - g(x)$, we have

$$f(g(x+h)) - f(g(x)) = (f'(u) + \phi(\Delta u(h)))\Delta u(h)$$

Dividing both sides by $h \neq 0$,

$$\frac{f(g(x+h)) - f(g(x))}{h} + (f'(u) + \phi(\Delta u(h)))\frac{\Delta u(h)}{h}$$

Taking the limit as $h \to 0$, since g is differentiable, then it is continuous, and thus,

$$\Delta u(h) = g(x+h) - g(x) \to 0$$

as $h \to 0$, thus,

$$\phi(\Delta u(h)) \to 0$$

as $h \to 0$. Also, by the definition of $g'(x)$,

$$\lim_{h \to 0} \frac{\Delta u(h)}{h} = \lim_{h \to 0} \frac{g(x+h) - g(x)}{h} = g'(x)$$

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Therefore,

$$\lim_{h\to 0}\frac{f(g(x+h))-f(g(x))}{h}=(f'(u)+0)g'(x)=f'(g(x))g'(x)$$

So $f\circ g$ is differentiable, and

$$(f\circ g)'(x)=f'(g(x))g'(x)$$

□

Example 2.47. Differentiate $y=(3x+1)^5$.

Solution. Let $u=3x+1$. Then

$$\frac{dy}{du}=5u^4 \quad \text{and} \quad \frac{du}{dx}=3$$

Therefore, by the Chain Rule (Theorem 2.46),

$$\begin{aligned}\frac{dy}{dx}&=\frac{dy}{du}\cdot\frac{du}{dx}\\&=5u^4\cdot 3\\&=15(3x+1)^4\end{aligned}$$

Example 2.48. Differentiate

$$y=(2x^3-5x+1)^4$$

Solution. Let $u=2x^3-5x+1$. Then

$$\frac{dy}{du}=4u^3 \quad \text{and} \quad \frac{du}{dx}=6x^2-5$$

Therefore, by the Chain Rule (Theorem 2.46), we have

$$\begin{aligned}\frac{dy}{dx}&=\frac{dy}{du}\cdot\frac{du}{dx}\\&=4(2x^3-5x+1)^3(6x^2-5)\end{aligned}$$

Example 2.49. Differentiate

$$y=(x^2-4x+7)^{-3}$$

Solution. Let $u=x^2-4x+7$. Then using the Power Rule for Derivatives for Integers (Theorem 2.42),

$$\frac{dy}{du}=-3u^{-4} \quad \text{and} \quad \frac{du}{dx}=2x-4$$

Therefore, by the Chain Rule (Theorem 2.46),

$$\begin{aligned}\frac{dy}{dx}&=\frac{dy}{du}\cdot\frac{du}{dx}\\&=-3(x^2-4x+7)^{-4}(2x-4)\\&=-\frac{3(2x-4)}{(x^2-4x+7)^4}\end{aligned}$$

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Example 2.50. Find the equation of the tangent line to

$$y = (2x^2 + 1)^3$$

at $x = 1$.

Solution. Differentiate using the Chain Rule (Theorem 2.46)

$$\frac{dy}{dx} = 3(2x^2 + 1)^2(4x) = 12x(2x^2 + 1)^2$$

Then evaluate at $x = 1$

$$m = \left.\frac{dy}{dx}\right|_{x=1} = 12(1)(2(1)^2 + 1)^2 = 108$$

and the point on the curve is

$$y = (2(1)^2 + 1)^3 = 27$$

Thus, the equation of the tangent line to the curve at $x = 1$ is

$$y - 27 = 108(x - 1)$$

Of course, there are examples where we can combine the chain rule with the product rule or the quotient rule.

A number of forthcoming examples will require algebraic simplification by factoring out common terms. Careful identification of shared factors can significantly simplify expressions and reduce the likelihood of computational errors.

Example 2.51. One may encounter

$$4x^2(3x^2 + 2)^3(4x + 1)^2 + 2x(3x^2 + 2)^4(4x + 1)^3$$

Observe that each term contains common factors. In particular,

- $2x$ is a common factor of $4x^2$ and $2x$,
- $(3x^2 + 2)^3$ is a common factor of $(3x^2 + 2)^3$ and $(3x^2 + 2)^4$, and
- $(4x + 1)^2$ is a common factor of $(4x^2 + 1)^2$ and $(4x + 1)^3$.

Factoring out these common terms allows the expression to be written in a simpler and more manageable form.

$$\begin{aligned}
&4x^2(3x^2 + 2)^3(4x + 1)^2 + 2x(3x^2 + 2)^4(4x + 1)^3 \\
&= 2x(3x^2 + 2)^3(4x + 1)^2[2x + (3x^2 + 2)(4x + 1)] \\
&= 2x(3x^2 + 2)^3(4x + 1)^2(2x + 12x^3 + 3x^2 + 8x + 2) \\
&= 2x(3x^2 + 2)^3(4x + 1)^2(12x^3 + 3x^2 + 10x + 2)
\end{aligned}$$

Such factorizations frequently arise after applying the product or quotient and chain rules, and recognizing common factors early can greatly simplify the final derivative.

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Example 2.52. Differentiate

$$y = x^2(3x^2+1)^4$$

Solution. We see that the function is a product, so product rule will be needed in this example. Let $u(x) = x^2$ and $v(x) = (3x^2+1)^4$. Then applying the product rule

$$\frac{dy}{dx} = u'(x)v(x) + u(x)v'(x)$$

We have $u'(x) = 2x$ and, by using the Chain Rule (Theorem 2.46),

$$v'(x) = 4(3x^2+1)^3(6x) = 24x(3x^2+1)^3$$

we have

$$\begin{aligned}\frac{dy}{dx} &= 2x(3x^2+1)^4 + x^2 \cdot 24x(3x^2+1)^3 \\ &= 2x(3x^2+1)^4 + 24x^3(3x^2+1)^3 \\ &= (3x^2+1)^3[2x(3x^2+1) + 24x^3] \\ &= (3x^2+1)^3(6x^3+2x+24x^3) \\ &= (3x^2+1)^3(30x^3+2x)\end{aligned}$$

Example 2.53. Differentiate

$$y = (x^2+1)^3(2x^3-5)^2$$

Solution. Let $u(x) = (x^2+1)^3$ and $v(x) = (2x^3-5)^2$. Then using the Chain Rule (Theorem 2.46) on each factor

$$\begin{aligned}u'(x) &= 3(x^2+1)^2(2x) = 6x(x^2+1)^2, \\ v'(x) &= 2(2x^3-5)(6x^2) = 12x^2(2x^3-5)\end{aligned}$$

Then using the product rule,

$$\begin{aligned}\frac{dy}{dx} &= u'(x)v(x) + u(x)v'(x) \\ &= 6x(x^2+1)^2(2x^3-5)^2 + (x^2+1)^3(12x^2(2x^3-5)) \\ &= 6x(x^2+1)^2(2x^3-5)^2 + 12x^2(x^2+1)^3(2x^3-5) \\ &= 6x(x^2+1)^2(2x^3-5)[(2x^3-5) + 2x(x^2+1)] \\ &= 6x(x^2+1)^2(2x^3-5)[2x^3-5+2x^3+2x] \\ &= 6x(x^2+1)^2(2x^3-5)(4x^3+2x-5)\end{aligned}$$

Example 2.54. Differentiate

$$y = \frac{(x^2+1)^5}{x^3}$$

Solution. Let $u(x) = (x^2+1)^5$ and $v(x) = x^3$. Then using the Chain Rule (Theorem 2.46) on $u(x)$, we have

$$u'(x) = 5(x^2+1)^4(2x) = 10x(x^2+1)^4$$

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$$v'(x) = 3x^2$$

Thus, using the Quotient Rule (Theorem 2.37), we have

$$\begin{aligned}
\frac{dy}{dx} &= \frac{u'(x)v(x) - u(x)v'(x)}{v^2(x)} \\
&= \frac{10x(x^2+1)^4(x^3) - (x^2+1)^5(3x^2)}{x^6} \\
&= \frac{10x^4(x^2+1)^4 - 3x^2(x^2+1)^5}{x^6} \\
&= \frac{x^2(x^2+1)^4[10x^2 - 3(x^2+1)]}{x^6} \\
&= \frac{(x^2+1)^4(10x^2 - 3x^2 - 3)}{x^4} \\
&= \frac{(x^2+1)^4(7x^2 - 3)}{x^4}
\end{aligned}$$

Example 2.55. Differentiate

$$y = \frac{x^2}{(3x-1)^4}$$

Solution. Let $u(x) = x^2$ and $v(x) = (3x-1)^4$. Then using the Chain Rule (Theorem 2.46) on $v(x)$, we have

$$\begin{aligned}
u'(x) &= 2x \\
v'(x) &= 4(3x-1)^3(3) = 12(3x-1)^3
\end{aligned}$$

Then by using the Quotient Rule (Theorem 2.37),

$$\begin{aligned}
\frac{dy}{dx} &= \frac{u'(x)v(x) - u(x)v'(x)}{v^2(x)} \\
&= \frac{2x(3x-1)^4 - x^2[12(3x-1)^3]}{(3x-1)^8} \\
&= \frac{2x(3x-1)^4 - 12x^2(3x-1)^3}{(3x-1)^8} \\
&= \frac{2x(3x-1)^3[(3x-1) - 6x]}{(3x-1)^8} \\
&= \frac{2x(-3x-1)}{(3x-1)^5} \\
&= -\frac{2x(3x+1)}{(3x-1)^5}
\end{aligned}$$

2.6 Implicit Differentiation

In the previous sections, we differentiated functions that were written explicitly in the form

$$y = f(x)$$

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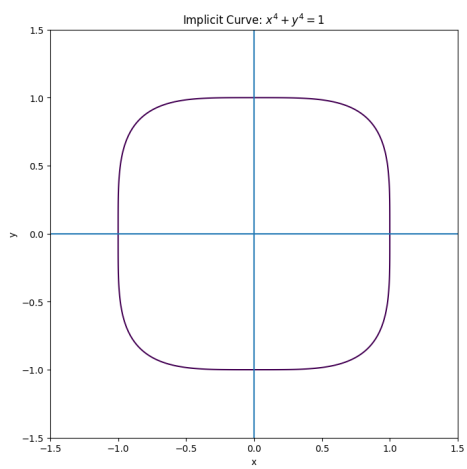
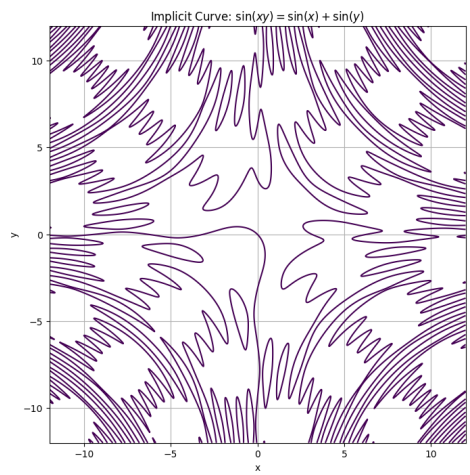
where one variable is expressed directly as a function of the other. However, many important relationships in mathematics, science, and engineering cannot be conveniently solved for one variable in terms of the other.

For example,

- The equation of a circle
$$x^2 + y^2 = 1$$
- Relations involving constraints, such as
$$x^2y + y^3 = 4$$
- Equations arising from geometry or physics where variables are interdependent.





In such cases, solving explicitly for y may be difficult or impossible. *Implicit differentiation* provides a systematic method for differentiating equations that define y implicitly as a function of x, without first solving for y.

Definition 2.56. A function is given *explicitly* if y is written directly in terms of x.

For example,
$$y = x^3 = 2x$$
or more in general, $y = f(x)$.

Definition 2.57. A relationship is *implicit* if x and y are related by an equation
$$x^2 + xy + y^2 = 3$$

Even when y is not explicitly isolated, it may still be possible to interpret y locally as a function of x.

When differentiating implicitly, we

1. Differentiate both sides of the equation with respect to x,

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2. Treat y as a function of x; that is, $y = y(x)$.
3. Apply the chain rule whenever a term involves y.

The key principle is

$$\frac{d}{dx}(y) = 1 \cdot \frac{dy}{dx}$$

and thus,

$$\frac{d}{dx}(y^n) = y^{n-1}\frac{dy}{dx}$$

Example 2.58. Differentiate implicitly

$$x^2 + y^2 = 1$$

Solution. Differentiate both sides with respect to x

$$2x + 2y\frac{dy}{dx} = 0$$

Solve for $\frac{dy}{dx}$

$$\frac{dy}{dx} = -\frac{x}{y}$$

Example 2.59. Differentiate

$$xy + y^3 = 4$$

Solution. Differentiate term by term

$$\frac{d}{dx}(xy) + \frac{d}{dx}(y^3) = 0$$

Then differentiate term-by-term

$$x\frac{dy}{dx} + y + 3y^3\frac{dy}{dx} = 0$$

Group terms involving $\frac{dy}{dx}$

$$(x + 3y^2)\frac{dy}{dx} = -y$$

Therefore,

$$\frac{dy}{dx} = -\frac{y}{x + 3y^2}$$

Example 2.60. Differentiate

$$x^3y^2 + y^5 = 7$$

Solution. Differentiate both sides and factor out $\frac{dy}{dx}$

$$\begin{aligned} 3x^2y^2 + 2x^3y\frac{dy}{dx} + 5y^4\frac{dy}{dx} &= 0 \\ (2x^3y + 5y^4)\frac{dy}{dx} &= -3x^2y^2 \\ \frac{dy}{dx} &= -\frac{3x^2y^2}{2x^3y + 5y^4} \end{aligned}$$

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Example 2.61. Differentiate

$$\frac{x}{y} + y^2 = 3$$

Solution. Rewrite if helpful

$$xy^{-1} + y^2 = 3$$

Then differentiate both sides, group and solve for $\frac{dy}{dx}$

$$\begin{aligned} y^{-1} + x(-y^{-2})\frac{dy}{dx} + 2y\frac{dy}{dx} &= 0 \\ \left(-\frac{x}{y^2} + 2y\right)\frac{dy}{dx} &= -\frac{1}{y} \\ \frac{dy}{dx} &= \frac{-\frac{1}{y}}{-\frac{x}{y^2} + 2y} \end{aligned}$$

Often we want the slope of the tangent line at a specific point, or even the equation of the tangent line to the curve at a specific point.

Example 2.62. Find the equation of the tangent line to

$$x^2 + xy + y^2 = 3$$

at the point $(1, 1)$.

Solution. First differentiate both sides and group terms involving $\frac{dy}{dx}$.

$$\begin{aligned} 2x + y + x\frac{dy}{dx} + 2y\frac{dy}{dx} &= 0 \\ (x + 2y)\frac{dy}{dx} &= -(2x + y) \\ \frac{dy}{dx} &= -\frac{2x + y}{x + 2y} \end{aligned}$$

Then at $(1, 1)$, the slope is

$$\left.\frac{dy}{dx}\right|_{(x,y)=(1,1)} = -\frac{2(1) + 1}{1 + 2(1)} = -1$$

Thus, the equation of the tangent line is

$$y - 1 = -(x - 1)$$

To conclude this section, we present another proof of the power rule for derivatives. After establishing the result for positive integer exponents and extending it to all integers, we use implicit differentiation to extend the power rule further to rational exponents.

Theorem 2.63 (**Power Rule for Derivatives,** $\mathbb{Q}$)**.** *If $f(x) = x^{\frac{a}{b}}$, where a and b are integers with $b \neq 0$, then*

$$f'(x) = \frac{a}{b}x^{\frac{a}{b}-1}$$

for all x.

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Proof. Let $y = x^{\frac{a}{b}}$. Then raising both sides to the power of b gives us

$$y^b = x^a$$

Then differentiating both sides using implicit differentiation,

$$by^{b-1}\frac{dy}{dx} = ax^{a-1}$$

Then solving for $\frac{dy}{dx}$ gives

$$\frac{dy}{dx} = \frac{ax^{a-1}}{by^{b-1}} = \frac{a}{b}\frac{x^{a-1}}{y^{b-1}}$$

But then since $y = x^{\frac{a}{b}}$, then

$$\begin{aligned}\frac{dy}{dx} &= \frac{a}{b}\frac{x^{a-1}}{(x^{\frac{a}{b}})^{b-1}}\\ &= \frac{a}{b}\frac{x^{a-1}}{x^{a-\frac{a}{b}}}\\ &= \frac{a}{b}x^{(a-1)-(a-\frac{a}{b})}\\ &= \frac{a}{b}x^{\frac{a}{b}-1}\end{aligned}$$

as desired. □

Example 2.64. Differentiate

$$y = \sqrt{x}$$

Solution. Rewrite using rational exponents

$$y = x^{\frac{1}{2}}$$

Then apply the power rule (Theorem 2.63)

$$\frac{dy}{dx} = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$$

Example 2.65. Differentiate

$$y = x^{\frac{5}{3}} + 2x^{\frac{1}{3}} - 4$$

Apply the power rule (Theorem 2.63) term-by-term

$$\frac{dy}{dx} = \frac{5}{3}x^{\frac{2}{3}} + \frac{2}{3}x^{-\frac{2}{3}}$$

Example 2.66. Differentiate

$$y = (3x+1)^{\frac{1}{2}}$$

Solution. Apply the Chain Rule (Theorem 2.46) and Power Rule (Theorem 2.63),

$$\frac{dy}{dx} = \frac{1}{3}(3x+1)^{-\frac{1}{2}}(3) = \frac{3}{2\sqrt{3x+1}}$$

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Example 2.67. Differentiate

$$y = (x^2 + 1)^{\frac{3}{2}}$$

Solution. Apply the Chain Rule (Theorem 2.46) and Power Rule (Theorem 2.63)

$$\frac{dy}{dx} = \frac{3}{2}(x^2 + 1)^{\frac{1}{2}}(2x) = 3x\sqrt{x^2 + 1}$$

2.7 Derivatives of Exponential Functions

Exponential functions arise naturally in a wide range of applications, including population growth, radioactive decay, compound interest, heat transfer, and differential equations. A defining feature of exponential behaviour is that the rate of change of a quantity is proportional to the quantity itself.

In this section, we develop the derivatives of exponential functions, beginning with the natural exponential function e^x. We will see that e^x is uniquely characterized by the property that its derivative is equal to the function itself. This remarkable fact makes e the most natural base for exponential growth and is the reason it plays a central role throughout calculus.

Recall that for a base $a > 0$, $a \neq 1$ the exponential function is defined by

$$f(x) = a^x$$

Special attention is given to the natural exponential function

$$f(x) = e^x$$

where $e \approx 2.71828$.

In order to find the derivative of the natural exponential function $f(x) = e^x$, we first require the following result.

Proposition 2.68. $\lim_{x \to 0} \frac{e^x - 1}{x} = 1$.

Lemma 2.69. *For all $x \in \mathbb{R}$,*

$$1 + x \leq e^x$$

This inequality can be proved using the definition of the exponential function.

Proof of Proposition 2.68. We prove the limit by checking the left- and right-hand limits

1. If $x > 0$, then

$$1 + x \leq e^x$$

Subtracting 1 and dividing both sides by $x > 0$,

$$1 \leq \frac{e^x - 1}{x}$$

Apply the same inequality to $-x$

$$1 - x \leq e^{-x}$$

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Multiply both sides by $e^x > 0$ to get

$$\begin{aligned} e^x - xe^x &\leq 1 \\ e^x - 1 &\leq xe^x \\ \frac{e^x - 1}{x} &\leq e^x \end{aligned}$$

For $h > 0$, we now have

$$1 \leq \frac{e^x - 1}{x} \leq e^x$$

Now we take the limit as $x \to 0^+$.

$$\lim_{x \to 0^+} 1 \leq \lim_{x \to 0^+} \frac{e^x - 1}{x} \leq \lim_{x \to 0^+} e^x = 1$$

Therefore, by Squeeze Theorem (Theorem 1.19)

$$\lim_{x \to 0^+} \frac{e^x - 1}{x} = 1$$

2. If $x < 0$, then $-x > 0$, so we apply (2) to $y = -x > 0$ to also obtain, similarly,

$$\lim_{x \to 0^-} \frac{e^x - 1}{x} = 1$$

Therefore, since both the left- and right-hand limits are equal,

$$\lim_{x \to 0} \frac{e^x - 1}{x} = 1$$

as desired. □

With this definition, we are ready to determine the derivative of the exponential function $f(x) = e^x$.

Theorem 2.70 (Derivative of Exponential Function, e^x). *For any real number x, if $f(x) = e^x$, then f is differentiable everywhere, and*

$$f'(x) = e^x.$$

Proof. We use the limit definition of the derivative and Proposition 2.68.

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{e^{x+h} - e^x}{h} \\ &= \lim_{h \to 0} \frac{e^x e^h - e^x}{h} \\ &= e^x \lim_{h \to 0} \frac{e^h - 1}{h} \\ &= e^x \cdot 1 && \text{(by Proposition 2.68)} \\ &= e^x \end{aligned}$$

as desired. □

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Now that we have the derivative of the exponential function at our disposal, we note that there could be functions in the exponential in which we may require the chain rule. For instance, we may have e^{x^2} as our function.

Theorem 2.71. *If $u(x)$ is differentiable, then*

$$\frac{d}{dx}(e^{u(x)}) = e^{u(x)}u'(x)$$

Proof. Let $y = e^u$, where $u = u(x)$. Then we have

$$\frac{dy}{du} = e^u \quad \text{and} \quad \frac{du}{dx} = u'(x)$$

Therefore, by the Chain Rule (Theorem 2.46),

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= e^{u(x)}u'(x)\end{aligned}$$

as desired. □

Example 2.72. Differentiate

$$y = e^{3x}$$

Solution. Let $u = 3x$. Then

$$\frac{dy}{du} = e^u \quad \text{and} \quad \frac{du}{dx} = 3$$

Therefore, by Theorem 2.71,

$$\frac{dy}{dx} = e^{3x} \cdot 3 = 3e^{3x}$$

Example 2.73. Differentiate

$$y = e^{x^2+1}$$

Solution. Let $u = x^2 + 1$. Then

$$\frac{dy}{du} = e^u \quad \text{and} \quad \frac{du}{dx} = 2x$$

Then by Theorem 2.71,

$$\frac{dy}{dx} = e^{x^2+1}(2x) = 2xe^{x^2+1}.$$

We can also find the derivative of a general exponential function, using the derivative of the natural exponential function e^x.

Theorem 2.74 (Derivative of Exponential Function, a^x). *For any real number x and $a > 0$ with $a \neq 1$, if $f(x) = a^x$, then f is differentiable everywhere and*

$$f'(x) = a^x \ln(a)$$

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Proof. If $f(x) = a^x$, then by using Proposition D.40,

$$f(x) = a^x = e^{x\ln(a)}$$

Let $u = x\ln(a)$. Then

$$\frac{dy}{du} = e^u \quad \text{and} \quad \frac{du}{dx} = \ln(a)$$

Thus, by Theorem 2.71,

$$\begin{aligned} f'(x) &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= e^{x\ln(a)}\ln(a) \end{aligned}$$

But since $e^{x\ln(a)} = a^x$,

$$f'(x) = a^x \ln(a)$$

as desired. □

Similar to the chain rule for e^x, we also have the chain rule for a^x.

Theorem 2.75. *If $u(x)$ is differentiable, then*

$$\frac{d}{dx}(a^{u(x)}) = a^{u(x)}u'(x)\ln(a)$$

Proof. Let $y = a^u$, where $u = u(x)$. Then

$$\frac{dy}{du} = a^u \ln(a) \quad \text{and} \quad \frac{du}{dx} = u'(x)$$

Then by the Chain Rule (Theorem 2.46)

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} \\ &= a^{u(x)}u'(x)\ln(a) \end{aligned}$$

as required. □

Example 2.76. Differentiate

$$f(x) = x^2 e^x$$

Solution. Using the product rule (Theorem 2.31), we have

$$f'(x) = 2xe^x + x^2e^x = e^x(x^2 + 2x)$$

Example 2.77. Differentiate

$$y = (3x - 1)2^x$$

Solution. Using the product rule (Theorem 2.31),

$$\frac{dy}{dx} = 3 \cdot 2^x + (3x - 1)2^x \ln(2) = 2^x[3 + (3x - 1)\ln(2)]$$

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Example 2.78. Differentiate

$$y = \frac{1}{x}e^{2x}$$

Solution. Rewrite exponents

$$y = x^{-1}e^{2x}$$

Then applying the product rule (Theorem 2.31),

$$\frac{dy}{dx} = -x^{-2}e^{2x} + x^{-1}(2e^{2x}) = -\frac{e^{2x}}{x^2} + \frac{2e^{2x}}{x} = e^{2x}\left(\frac{2}{x} - \frac{1}{x^2}\right)$$

Example 2.79. Differentiate

$$y = \frac{x^2}{e^x}$$

Solution. Using the quotient rule (Theorem 2.37)

$$\frac{dy}{dx} = \frac{2xe^x - x^2e^x}{e^{2x}} = \frac{e^x(2x - x^2)}{e^{2x}} = \frac{2x - x^2}{e^x}$$

Example 2.80. Differentiate

$$y = x^3e^{x^2}$$

Solution. Using the product rule and the chain rule (Theorem 2.71),

$$\begin{aligned}\frac{dy}{dx} &= 3x^2e^{x^2} + x^3e^{x^2}(2x)\\ &= 3x^2e^{x^2} + 2x^4e^{x^2}\\ &= x^2e^{x^2}(3 + 2x^2)\end{aligned}$$

Example 2.81. Find the equation of the tangent line to the curve

$$y = \frac{4^x}{(x^2+1)^4}$$

at the point $x = 1$.

Solution. At $x = 1$, we have

$$y = \frac{4^1}{(1^2+1)^4} = \frac{4}{16} = \frac{1}{4}$$

Now find the derivative using the quotient and chain rule

$$\begin{aligned}\frac{dy}{dx} &= \frac{4^x\ln(4)(x^2+1)^4 - 4^x(4(x^2+1)^3(2x))}{(x^2+1)^8}\\ &= \frac{4^x\ln(4)(x^2+1)^4 - 8x\cdot 4^x(x^2+1)^3}{(x^2+1)^8}\\ &= \frac{4^x(x^2+1)^3(\ln(4)(x^2+1) - 8x)}{(x^2+1)^8}\end{aligned}$$

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$$= \frac{4^x(\ln(4)(x^2+1) - 8x)}{(x^2+1)^5}$$

Thus, at $x = 1$, we have the slope

$$m = \left.\frac{dy}{dx}\right|_{x=1} = \frac{4^1(\ln(4)(2) - 8)}{(1^2+1)^5} = \frac{8\ln(4) - 8}{32} = \frac{2\ln(2)}{4} - \frac{1}{4} = \frac{\ln(2)}{2} - \frac{1}{4}$$

Therefore, the equation of the tangent line to the curve at $x = 1$ is

$$y - \frac{1}{4} = \left(\frac{\ln(2)}{2} - \frac{1}{4}\right)(x-1)$$

To conclude this section, we prove that the power rule works for any real number.

Theorem 2.82 (**Power Rule for Derivatives**). *For any real number r, if $f(x) = x^r$, then*

$$f(x) = rx^{r-1}$$

Proof. If $f(x) = x^r$, then using Proposition D.40, we have

$$f(x) = x^r = (e^{\ln(x)})^r = e^{r\ln(x)}$$

Then using the Chain Rule (Theorem 2.71),

$$f'(x) = e^{r\ln(x)} \cdot \frac{r}{x} = x^r\frac{r}{x} = r\frac{x^r}{x} = rx^{r-1}$$

as desired. □

2.8 Derivatives of Logarithmic Functions

Logarithmic functions arise naturally as inverses of exponential functions and play a central role in modeling phenomena involving relative change, scale, and growth. In many applications, such as population models, information theory, and elasticity, the rate of change of a logarithmic quantity is more meaningful than the rate of change of the original variable.

In this section, we develop the derivatives of logarithmic functions, beginning with the natural logarithm $\ln(x)$. We will see that logarithmic differentiation is closely connected to exponential functions and provides powerful techniques for differentiating complicated expressions involving products, quotients, and variable exponents.

Recall that the natural logarithm $\ln(x)$ is defined as the inverse function of e^x

$$y = \ln(x) \iff x = e^y, \quad x > 0$$

For a general base $a > 0$ with $a \neq 1$,

$$\log_a(x) = \frac{\ln(x)}{\ln(a)}$$

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Theorem 2.83 (**Derivative of Logarithmic Function, $\ln(x)$**). *For any real number $x > 0$, if $f(x) = \ln(x)$, then f is differentiable for all $x > 0$, and*

$$f'(x) = \frac{1}{x}$$

Proof. We will use implicit differentiation to differentiate $f(x) = \ln(x)$. Let $y = \ln(x)$. Then $x = e^y$, so then by implicit differentiation,

$$\begin{aligned} \frac{d}{dx}(x) &= \frac{d}{dx}(e^y) \\ 1 &= e^y \frac{dy}{dx} \\ \frac{dy}{dx} &= \frac{1}{e^y} \end{aligned}$$

since $x = e^y$, we ahve

$$\frac{dy}{dx} = \frac{1}{x}$$

as desired. □

The following theorem allows logarithmic differentiation to be applied to expressions involving negative values of x.

Theorem 2.84 (**Derivative of Logarithmic Function, $\ln|x|$**). *For any real number $x \neq 0$, if $f(x) = \ln|x|$, then f is differentiable for all $x \neq 0$, and*

$$f'(x) = \frac{1}{x}$$

We also wish to extend our differentiation results to logarithmic functions whose base is not e. Using the change-of-base formula (Proposition D.36), we can express such logarithms in terms of the natural logarithm and differentiate them accordingly.

Theorem 2.85 (**Derivative of Logarithmic Function, $\log_a(x)$**). *For any real number $x > 0$ and $a > 0$ with $a \neq 1$, if $f(x) = \log_a(x)$, then f is differentiable for all $x > 0$, and*

$$f'(x) = \frac{1}{x\ln(a)}$$

Proof. Using the change-of-base formula (Proposition D.36), write

$$\log_a(x) = \frac{\ln(x)}{\ln(a)}$$

Then using Theorem 2.83,

$$f'(x) = \frac{\frac{1}{x}}{\ln(a)} = \frac{1}{x\ln(a)}$$

as required. □

Using the chain rule, we can extend the derivatives of $\ln(x)$ and $\log_a(x)$ to compositions.

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Theorem 2.86. *For any real number $x > 0$ and $a > 0$ with $a \neq 1$, if $u = u(x)$ is differentiable, then*

$$\frac{d}{dx}(\ln(u(x)) = \frac{u'(x)}{u(x)} \quad \textit{and} \quad \frac{d}{dx}(\log_a(u(x)) = \frac{u'(x)}{u(x)\ln(a)}$$

Proof. Let $y = \ln(u)$. Then

$$\frac{dy}{du} = \frac{1}{u} \quad \text{and} \quad \frac{du}{dx} = u'(x)$$

Then by the Chain Rule (Theorem 2.46)

$$\frac{dy}{dx} = \frac{1}{u}u'(x) = \frac{u'(x)}{u(x)}$$

Similarly, if $y = \log_a(u)$, then

$$\frac{dy}{du} = \frac{1}{u\ln(a)} \quad \text{and} \quad \frac{du}{dx} = u'(x)$$

Then by the Chain Rule (Theorem 2.46),

$$\frac{dy}{dx} = \frac{1}{u\ln(a)}u'(x) = \frac{u'(x)}{u(x)\ln(a)}$$

□

Example 2.87. Differentiate

$$y = \ln(3x^2 + 1)$$

Solution. Apply the chain rule (Theorem 2.86)

$$\frac{dy}{dx} = \frac{1}{3x^2+1}(6x) = \frac{6x}{3x^2+1}$$

Example 2.88. Differentiate

$$y = \ln(x^{\frac{3}{2}})$$

Solution. We can rewrite using logarithmic properties

$$y = \ln(x^{\frac{3}{2}}) = \frac{3}{2}\ln(x)$$

Then differentiating

$$\frac{dy}{dx} = \frac{3}{2} \cdot \frac{1}{x} = \frac{3}{2x}$$

Example 2.89. Differentiate

$$y = x^2\ln(x)$$

Solution. Applying the product rule (Theorem 2.31),

$$\begin{aligned}\frac{dy}{dx} &= 2x\ln(x) + x^2\frac{1}{x} \\ &= 2x\ln(x) + x\end{aligned}$$

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Example 2.90. Differentiate

$$y = \frac{\ln(x^2+1)}{x^2}$$

Solution. Applying the quotient rule (Theorem 2.37) and chain rule (Theorem 2.86),

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{1}{x^2+1}(2x)\cdot x^2 - \ln(x^2+1)(2x)}{x^4} \\ &= \frac{\frac{2x^3}{x^2+1} - 2x\ln(x^2+1)}{x^4}\end{aligned}$$

Logarithmic differentiation is a powerful technique that combines implicit differentiation with the properties of logarithms. It is particularly effective for differentiating functions that involve complicated products, quotients, or variable exponents, where direct application of the product, quotient, or chain rules would be cumbersome or prone to algebraic error.

The central idea is to take the natural logarithm of both sides of an equation before differentiating. Doing so converts products into sums, quotients into differences, and powers into coefficients, greatly simplifying the differentiation process.

Logarithmic differentiation is especially useful when

- A function is a product of several factors,
- A function is a quotient of complicated expressions,
- the variable appears both in the base and the exponent,
- exponents are not constant.

Suppose $y = f(x)$ with $y > 0$. The steps are:

1. Take the natural logarithm of both sides

$$\ln(y) = \ln(f(x))$$

2. Use logarithmic properties to simplify the right-hand side.
3. Differentiate both sides with respect to x.
4. Solve for $\frac{dy}{dx}$.

Example 2.91. Differentiate

$$y = x^2(x^2+1)^3$$

Solution. For the purpose of this example, we will use logarithmic differentiation (product rule will also work here!). Indeed, taking the logarithm on both sides simplifies to

$$\begin{aligned}\ln(y) &= \ln(x^2(x^2+1)^3) \\ \ln(y) &= \ln(x^2) + \ln((x^2+1)^3) \\ \ln(y) &= 2\ln(x) + 3\ln(x^2+1)\end{aligned}$$

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Now differentiating both sides gives

$$\begin{aligned}\frac{d}{dx}(\ln(y)) &= \frac{d}{dx}(2\ln(x) + 3\ln(x^2+1))\\ \frac{1}{y}\frac{dy}{dx} &= 2\frac{1}{x} + 3\frac{2x}{x^2+1}\\ \frac{1}{y}\frac{dy}{dx} &= \frac{2}{x} + \frac{6x}{x^2+1}\\ \frac{dy}{dx} &= y\left(\frac{2}{x} + \frac{6x}{x^2+1}\right)\end{aligned}$$

Since $y = x^2(x^2+1)^3$, then

$$\begin{aligned}\frac{dy}{dx} &= x^2(x^2+1)^3\left(\frac{2}{x} + \frac{6x}{x^2+1}\right)\\ &= 2x(x^2+1)^3 + 6x^3(x^2+1)^2\end{aligned}$$

Example 2.92. Differentiate

$$y = \frac{x^3\sqrt{x+1}}{(x^2+1)^4}$$

Solution. Taking the logarithm on both sides simplifies to

$$\begin{aligned}\ln(y) &= \ln\left(\frac{x^3\sqrt{x+1}}{(x^2+1)^4}\right)\\ \ln(y) &= \ln(x^3\sqrt{x+1}) - \ln((x^2+1)^4)\\ \ln(y) &= \ln(x^3) + \ln((x+1)^{\frac{1}{2}}) - \ln((x^2+1)^4)\\ \ln(y) &= 3\ln(x) + \frac{1}{2}\ln(x+1) - 4\ln(x^2+1)\end{aligned}$$

Now, differentiating both sides gives

$$\begin{aligned}\frac{1}{y}\frac{dy}{dx} &= \frac{3}{x} + \frac{1}{2(x+1)} - \frac{4}{x^2+1}(2x)\\ \frac{dy}{dx} &= y\left(\frac{3}{x} + \frac{1}{2(x+1)} - \frac{8x}{x^2+1}\right)\end{aligned}$$

Since $y = \frac{x^3\sqrt{x+1}}{(x^2+1)^4}$, then

$$\begin{aligned}\frac{dy}{dx} &= \frac{x^3\sqrt{x+1}}{(x^2+1)^4}\left(\frac{3}{x} + \frac{1}{2(x+1)} - \frac{8x}{x^2+1}\right)\\ &= \frac{3x^2\sqrt{x+1}}{(x^2+1)^4} + \frac{x^3}{2(x^2+1)^4\sqrt{x+1}} - \frac{8x^4\sqrt{x+1}}{(x^2+1)^5}\end{aligned}$$

Example 2.93. Differentiate

$$y = x^x$$

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where $x > 0$.

Solution. This function cannot be differentiated using the standard power rule! Taking logarithms on both sides give

$$\begin{aligned} \ln(y) &= \ln(x^x) \\ \ln(y) &= x\ln(x) \end{aligned}$$

Then differentiating both sides

$$\begin{aligned} \frac{1}{y}\frac{dy}{dx} &= \ln(x) + 1 \\ \frac{dy}{dx} &= y(\ln(x) + 1) \end{aligned}$$

Since $y = x^x$, then

$$\frac{dy}{dx} = x^x(\ln(x) + 1)$$

The above example can be turned into a general statement.

Proposition 2.94. *Let $P(x)$ and $Q(x)$ be differentiable functions with $P(x) > 0$ for all x. Then the function*

$$y = P(x)^{Q(x)}$$

is differentiable and its derivative is given by

$$\frac{dy}{dx} = P(x)^{Q(x)}\left(Q'(x)\ln(P(x)) + Q(x)\frac{P'(x)}{P(x)}\right)$$

Proof. Take the natural logarithm of both sides gives

$$\ln(y) = Q(x)\ln(P(x))$$

Differentiating implicitly

$$\frac{1}{y}\frac{dy}{dx} = Q'(x)\ln(P(x)) + Q(x)\frac{P'(x)}{P(x)}$$

Multiplying both sides by $y = P(x)^{Q(x)}$ to obtain

$$\frac{dy}{dx} = P(x)^{Q(x)}\left(Q'(x)\ln(P(x)) + Q(x)\frac{P'(x)}{P(x)}\right)$$

as desired. □

This formula generalizes both the power rule and the exponential rule, and is especially useful when the base and exponent are both functions of x.

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Example 2.95. Differentiate

$$y = (x^2+1)^x$$

Solution. Note that for all x, $x^2+1>0$. Let $P(x)=x^2+1$ and $Q(x)=x$. Then $P'(x)=2x$ and $Q'(x)=1$. Applying the formula, we have

$$\begin{aligned}\frac{dy}{dx} &= (x^2+1)^x\left(1\cdot\ln(x^2+1)+x\cdot\frac{2x}{x^2+1}\right)\\ &= (x^2+1)^x\left(\ln(x^2+1)+\frac{2x^2}{x^2+1}\right)\end{aligned}$$

Example 2.96. Differentiate

$$y = \left(\frac{x+1}{x-1}\right)^{2x}$$

Solution. Note that $x>1$, so the past is also positive. Let $P(x)=\frac{x+1}{x-1}$ and $Q(x)=2x$. Then $Q'(x)=2$. For $\frac{P'}{P}$, compute efficiently using logarithms

$$\ln(P)=\ln(x+1)-\ln(x-1) \implies \frac{P'}{P}=\frac{1}{x+1}-\frac{1}{x-1}$$

Now applying the formula,

$$\frac{dy}{dx}=\left(\frac{x+1}{x-1}\right)^{2x}\left(2\ln\left(\frac{x+1}{x-1}\right)+2x\left(\frac{1}{x+1}-\frac{1}{x-1}\right)\right)$$

Example 2.97. Differentiate

$$y=(e^{x^2})^x$$

without using exponential properties.

Solution. Let $P(x)=e^{x^2}$ and $Q(x)=x$. Then $P'(x)=2xe^{x^2}$, $Q'(x)=1$, and $\frac{P'}{p}=2x$. Then applying the formula,

$$\begin{aligned}\frac{dy}{dx} &= (e^{x^2})^x\left(\ln(e^{x^2})+x\cdot 2x\right)\\ &= (e^{x^2})^x(x^2+2x^2)=3x^2(e^{x^2})^x\\ &= 3x^2e^{x^3}\end{aligned}$$

Example 2.98. Differentiate

$$y=(2x+3)^{\frac{1}{x}}$$

Solution. Let $P(x)=2x+3$ and $Q(x)=\frac{1}{x}$. Then $P'(x)=2$ and $Q'(x)=-\frac{1}{x^2}$, and so applying the formula gives

$$\begin{aligned}\frac{dy}{dx} &= (2x+3)^{\frac{1}{x}}\left(-\frac{1}{x^2}\ln(2x+3)+\frac{1}{x}\cdot\frac{2}{2x+3}\right)\\ &= (2x+3)^{\frac{1}{x}}\left(-\frac{\ln(2x+3)}{x^2}+\frac{2}{x(2x+3)}\right)\end{aligned}$$

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2.9 Derivatives of Trigonometric Functions

Trigonometric functions arise naturally in geometry, physics, engineering, and signal processing, where periodic behaviour plays a central role. Quantities such as displacement in oscillatory motion, alternating current, and wave propagation are often modeled using sine and cosine functions. To analyze how such quantities change, we must understand how to differentiate trigonometric functions.

In this section, we compute the derivative of the derivatives of basic trigonometric functions and develop techniques for differentiating more complicated expressions involving trigonometric functions together with polynomials, exponential functions, and exponential functions. These results will be essential in later applications, including curve sketching, optimization, and differential equations.

A review of trigonometric functions, identities, and their geometric interpretations is provided in Appendix A.9.

Recall the basic trigonometric functions

$$\sin(x), \quad \cos(x), \quad \tan(x), \quad \csc(x), \quad \sec(x), \quad \cot(x)$$

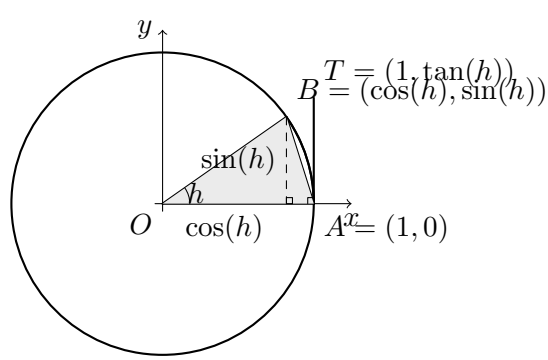
defined initially using right triangles and extend to all real numbers via the unit circle.

In order to use the definition of the derivative to compute the derivatives of $\sin(x)$ and $\cos(x)$, we require the following limit identities

Lemma 2.99. $\lim_{h\to 0} \dfrac{\sin(h)}{h} = 1$ *and* $\lim_{h\to 0} \dfrac{\cos(h)-1}{h} = 0$.

Proof. Assume that h is measured in radians, and consider $h \in \left(0, \frac{\pi}{2}\right)$. Using the unit circle compare areas:

- The area of a triangle with base $\sin(h)$ and height $\cos(h)$ is $\frac{1}{2}\sin(h)\cos(h)$.
- The area of the sector of angle h is $\frac{1}{2}$.
- The area of the triangle with base $\tan(h)$ and height 1 is $\frac{1}{2}\tan(h)$.



From the diagram, for $h \in \left(0, \frac{\pi}{2}\right)$,

$$\text{Area}(\Delta OAB) < \text{Area}(\text{sector } OAB) < \text{Area}(\Delta OAT)$$

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In which, gives

$$\sin(h) < h < \tan(h)$$

Divide through by $\sin(h) > 0$,

$$1 < \frac{h}{\sin(h)} < \frac{1}{\cos(h)}$$

Inverting the inequalities

$$\cos(h) < \frac{\sin(h)}{h} < 1$$

As $h \to 0$, we have $\cos(h) \to 0$. Therefore, by Squeeze Theorem (Theorem 1.19),

$$\lim_{h\to 0} \frac{\sin(h)}{h} = 1.$$

To see the second limit identity, use the trigonometric identity

$$\cos(h) - 1 = -2\sin^2\left(\frac{h}{2}\right)$$

so that

$$\frac{\cos(h)-1}{h} = -2\frac{\sin^2\left(\frac{h}{2}\right)}{h} = -2\frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}} \cdot \frac{\sin\left(\frac{h}{2}\right)}{2}$$

Take the limits term by term as $h \to 0$.

- From the above,

$$\lim_{h\to 0} \frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}} = 1$$

- $\lim_{h\to 0} \sin\left(\frac{h}{2}\right) = 0.$

Therefore,

$$\lim_{h\to 0} \frac{\cos(h)-1}{h} = -2 \cdot 1 \cdot 0 = 0$$

□

Theorem 2.100 (**Derivatives of Trigonometric Functions,** sin **and** cos)**.** *For all $x \in \mathbb{R}$,*

1. *If $f(x) = \sin(x)$, then $f'(x) = \cos(x)$.*
2. *If $f(x) = \cos(x)$, then $f'(x) = -\sin(x)$.*

Proof. To see that (1) holds, let $f(x) = \sin(x)$. Using the definition of the derivative, we have

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{\sin(x+h) - \sin(x)}{h}$$

Using the identity

$$\sin(x+h) = \sin(x)\cos(h) + \cos(x)\sin(h)$$

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we obtain

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{\sin(x)\cos(h) + \cos(x)\sin(h) - \sin(x)}{h} \\ &= \lim_{h\to 0} \left(\frac{\sin(x)(\cos(h)-1)}{h} + \frac{\cos(x)\sin(h)}{h} \right) \\ &= \sin(x) \lim_{h\to 0} \frac{\cos(h)-1}{h} + \cos(x) \lim_{h\to 0} \frac{\sin(h)}{h} \end{aligned}$$

Using Lemma 2.99, we have

$$f'(x) = \sin(x) \cdot 0 + \cos(x) \cdot 1 = \cos(x)$$

Therefore,

$$\frac{d}{dx}(\sin(x)) = \cos(x)$$

To see that (2) holds, let $f(x) = \cos(x)$. Using the definition of the derivative, we have

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h\to 0} \frac{\cos(x+h) - \cos(x)}{h} \end{aligned}$$

Using the identity

$$\cos(x+h) = \cos(x)\cos(h) - \sin(x)\sin(h)$$

we have

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{\cos(x)\cos(h) - \sin(x)\sin(h) - \cos(x)}{h} \\ &= \lim_{h\to 0} \left(\frac{\cos(x)(\cos(h)-1)}{h} - \frac{\sin(x)\sin(h)}{h} \right) \\ &= \cos(x) \lim_{h\to 0} \frac{\cos(h)-1}{h} - \sin(x) \lim_{h\to 0} \frac{\sin(h)}{h} \end{aligned}$$

Using Lemma 2.99, we have

$$f'(x) = \cos(x) \cdot 0 - \sin(x) \cdot 1 = -\sin(x)$$

□

Using the quotient identities and the chain rule, we obtain

Theorem 2.101 (Derivatives of Trigonometric Functions). *For any real number $x \in \mathbb{R}$, possibly except where one of the trigonometric functions may be undefined,*

$$\frac{d}{dx}(\sin(x)) = \cos(x), \quad \frac{d}{dx}(\cos(x)) = -\sin(x), \quad \frac{d}{dx}(\tan(x)) = \sec^2(x),$$
$$\frac{d}{dx}(\csc(x)) = -\csc(x)\cot(x), \quad \frac{d}{dx}(\sec(x)) = \sec(x)\cot(x), \quad \frac{d}{dx}(\cot(x)) = -\csc^2(x)$$

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Proof. The derivatives of $\sin(x)$ and $\cos(x)$ were proven in Theorem 2.100.

For the derivative of $\tan(x)$, we use the identity

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$

Therefore, taking the derivative and using the quotient rule (Theorem 2.37), we have

$$\begin{aligned} f'(x) &= \frac{\cos(x)\cos(x) - \sin(x)(-\sin(x))}{\cos^2(x)} \\ &= \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} \end{aligned}$$

We use the following identities

$$\sin^2(x) + \cos^2(x) = 1, \quad \frac{1}{\cos(x)} = \sec(x)$$

Therefore,

$$f'(x) = \sec^2(x)$$

as desired.

For the derivative of $\csc(x)$, we use the identity

$$\csc(x) = \frac{1}{\sin(x)}$$

Thus, taking the derivative using the derivative of a reciprocal function (Theorem 2.41), we have

$$f'(x) = -\frac{\cos(x)}{\sin^2(x)} = -\frac{1}{\sin(x)} \cdot \frac{\cos(x)}{\sin(x)}$$

Then using the identity above, and the identity

$$\cot(x) = \frac{\cos(x)}{\sin(x)}$$

we have

$$f'(x) = -\csc(x)\cot(x)$$

as desired.

For the derivative of $\sec(x)$, we use the identity

$$\sec(x) = \frac{1}{\cos(x)}$$

Thus, taking the derivative using the derivative of a reciprocal function (Theorem 2.41), we have

$$f'(x) = -\frac{-\sin(x)}{\cos^2(x)} = \frac{1}{\cos(x)} \cdot \frac{\sin(x)}{\cos(x)}$$

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Then using the identity above, and the identity

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$

we have

$$f'(x) = \sec(x)\tan(x)$$

as desired.

Finally, for the derivative of $\cot(x)$, we use the identity

$$\cot(x) = \frac{\cos(x)}{\sin(x)}$$

Then taking the derivative and using the quotient rule (Theorem 2.37), we have

$$\begin{aligned} f'(x) &= \frac{-\sin(x)\sin(x) - \cos(x)\cos(x)}{\sin^2(x)} \\ &= -\frac{\sin^2(x) + \cos^2(x)}{\sin^2(x)} \end{aligned}$$

We use the following identities

$$\sin^2(x) + \cos^2(x) = 1, \quad \frac{1}{\sin(x)} = \csc(x)$$

Therefore

$$f'(x) = -\csc^2(x).$$

□

Using the chain rule, we can extend the derivatives of all trigonometric functions to compositions.

Theorem 2.102. *For any real number $x \in \mathbb{R}$, if $u(x)$ is differentiable for all $x \in \mathbb{R}$, possibly except where one of the trigonometric functions may be undefined,*

$$\begin{aligned} &\frac{d}{dx}(\sin(u(x))) = u'(x)\cos(u(x)), \quad \frac{d}{dx}(\cos(u(x))) = -u'(x)\sin(u(x)), \\ &\frac{d}{dx}(\tan(u(x))) = u'(x)\sec^2(u(x)), \quad \frac{d}{dx}(\sec(u(x))) = u'(x)\sec(u(x))\tan(u(x)), \\ &\frac{d}{dx}(\csc(u(x))) = -u'(x)\csc^2(u(x)), \quad \frac{d}{dx}(\cot(u(x))) = -u'(x)\csc^2(u(x)). \end{aligned}$$

Proof. We will provide a proof for the derivative of $\sin(u(x))$, and the others will follow similarly. Let $y = \sin(u)$, where $u = u(x)$. Then we have

$$\frac{dy}{du} = \cos(u) \quad \text{and} \quad \frac{du}{dx} = u'(x)$$

Therefore, by the Chain Rule (Theorem 2.46),

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \cos(u(x))u'(x)$$

as desired. □

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Example 2.103. Differentiate

$$y = \sin(3x^2)$$

Solution. Apply the chain rule (Theorem 2.102)

$$\frac{dy}{dx} = \cos(3x^2)(6x) = 6x\cos(3x^2)$$

Example 2.104. Differentiate

$$y = \cos(\sqrt{x})$$

Solution. Rewrite $\sqrt{x} = x^{\frac{1}{2}}$ and so applying the power rule (Theorem 2.82)

$$\frac{dy}{dx} = -\sin(\sqrt{x}) \cdot \frac{1}{2}x^{-\frac{1}{2}} = -\frac{1}{2\sqrt{x}}\sin(\sqrt{x})$$

Example 2.105. Differentiate

$$y = x^2\sin(x)$$

Solution. Apply the product rule

$$\frac{dy}{dx} = 2x\sin(x) + x^2\cos(x)$$

Example 2.106. Differentiate

$$y = xe^x\cos(x)$$

Solution. Apply the product rule for three functions (Theorem 2.35)

$$\frac{dy}{dx} = e^x\cos(x) + xe^x\cos(x) - xe^x\sin(x)$$

Example 2.107. Differentiate

$$y = \frac{\sin(x)}{x}$$

Solution. Applying the quotient rule,

$$\frac{dy}{dx} = \frac{x\cos(x) - \sin(x)}{x^2}$$

Example 2.108. Differentiate

$$y = \frac{\ln(x)}{\cos(x)}$$

Solution. Note that

$$y = \ln(x)\sec(x)$$

Therefore, using the product rule, we have

$$\frac{dy}{dx} = \frac{1}{x}\sec(x) + \ln(x)\sec(x)\tan(x)$$

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Example 2.109. Differentiate implicitly

$$x^2 + x\sin(y^2) = 2y^3$$

Solution. Differentiate both sides

$$\begin{aligned} 2x + \sin(y^2) + x\cos(y^2)(2y)\frac{dy}{dx} &= 6y^2\frac{dy}{dx} \\ 2x + \sin(y^2) &= 6y^2\frac{dy}{dx} - 2xy\cos(y^2)\frac{dy}{dx} \\ 2x + \sin(y^2) &= \frac{dy}{dx}\left(6y^2 - 2xy\cos(y^2)\right) \\ \frac{dy}{dx} &= \frac{2x + \sin(y^2)}{6y^2 - 2xy\cos(y^2)} \end{aligned}$$

Example 2.110. Differentiate

$$y = (\sin(x))^x$$

with $\sin(x) > 0$.

Solution. Let $P(x) = \sin(x)$ and $Q(x) = x$. Then $P'(x) = \cos(x)$, $Q'(x) = 1$, and $\frac{P'(x)}{P(x)} = \frac{\cos(x)}{\sin(x)} = \cot(x)$. Then

$$\frac{dy}{dx} = (\sin(x))^x\left(1 \cdot \ln(\sin(x)) + x\cot(x)\right)$$

2.10 Derivatives of Inverse Functions

Many functions that arise in mathematics and its applications possess inverses. For example, exponential and logarithmic functions are inverses of one another, as are trigonometric functions (after appropriate domain restrictions) and thier inverse trigonometric counterparts.

A natural question arises: *If a function f is differentiable and has an inverse f^{-1}, how can we compute the derivative of f^{-1}?*

Rather than differentiating the inverse directly, we will derive a general formula expressing the derivative of the inverse function in terms of the derivative of the original function. This result will provide a powerful tool that allows us to compute derivatives of inverse trigonometric functions, logarithmic functions, and many others in a systematic way.

Recall the definition of an inverse function.

Definition 2.111. A function f has an inverse f^{-1} if and only if f is one-to-one, i.e.

$$f(x_1) = f(x_2) \implies x_1 = x_2$$

If f is invertible, then

$$f(f^{-1}(x)) = x, \quad f^{-1}(f(x)) = x$$

The graph of f^{-1} is obtained by reflecting the graph of f across the line $y = x$. Slopes of corresponding tangent lines are reciprocals, when defined.

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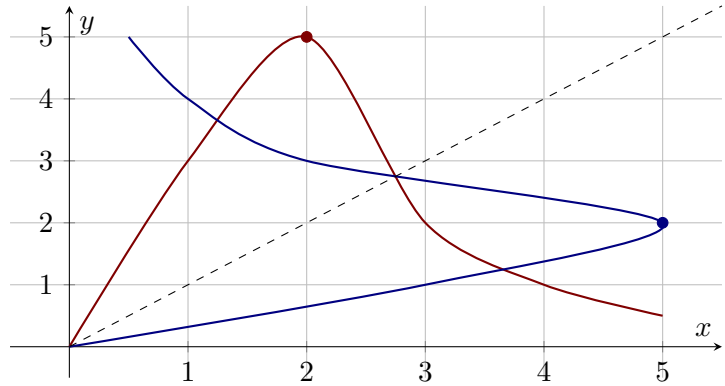

This geometric observation suggests the formula we are about to prove.

Theorem 2.112 (Derivative of an Inverse Function). *Let f be differentiable and one-to-one on an interval I, and suppose $f'(a) \neq 0$. Then f^{-1} is differentiable at $b = f(a)$, and*

$$(f^{-1})'(b) = \frac{1}{f'(a)}$$

Equivalently,

$$\frac{d}{dx}(f^{-1}(x)) = \frac{1}{f'(f^{-1}(x))}$$

Proof. Let $y = f^{-1}(x)$. Then $f(y) = 1$. Differentiating both sides with respect to x gives

$$f'(y)\frac{dy}{dx} = 1$$

Solving for $\frac{dy}{dx}$ gives

$$\frac{dy}{dx} = \frac{1}{f'(y)}$$

But since $y = f^{-1}(x)$, we obtain

$$\frac{d}{dx}(f^{-1}(x)) = \frac{1}{f'(f^{-1}(x))}.$$

□

Indeed, if $f'(a) = m$ and $b = f(a)$, then

$$(f^{-1})'(b) = \frac{1}{m}$$

Thus, the slope of the inverse at b is the reciprocal of the slope of the original function at a. This matches the reflection symmetry across $y = x$.

Theorem 2.112 requires three conditions:

1. f must be differentiable,
2. f must be one-to-one, and

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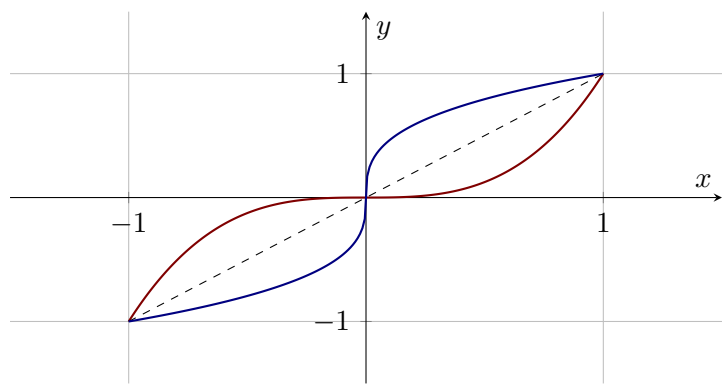
3. $f'(a) \neq 0$

If $f'(a) = 0$, then the inverse typically has a vertical tangent.

Example 2.113. Let $f(x) = x^3$. Then $f'(0) = 0$. The inverse is $f^{-1}(x) = \sqrt[3]{x}$, but then

$$\frac{d}{dx}\sqrt[3]{x} = \frac{1}{3x^{\frac{2}{3}}}$$

which is undefined at $x = 0$.



Thus, the condition $f'(a) \neq 0$ is essential.
The derivative of inverse functions can always be derived via

1. Write $y = f^{-1}(x)$.
2. Rewrite as $f(y) = x$.
3. Apply implicit differentiation.

2.11 Derivatives of Inverse Trigonometric Functions

Trigonometric functions such as $\sin(x)$, $\cos(x)$, and $\tan(x)$ are not one-to-one on $\mathbb{R}$, and therefore, do not possess global inverses. However, by restricting their domains appropriately, we obtain the *inverse trigonometric functions*

$$\arcsin(x), \quad \arccos(x), \quad \arctan(x), \quad \operatorname{arccsc}(x), \quad \operatorname{arcsec}(x), \quad \operatorname{arccot}(x)$$

These functions arise naturally when solving trigonometric equations, in geometry, and in applications involving angles determined by algebraic relationships. To differentiate expressions involving inverse trigonometric functions, we must combine

- Implicit differentiation,
- Knowledge from trigonometric identities,
- Careful attention to domain restrictions.

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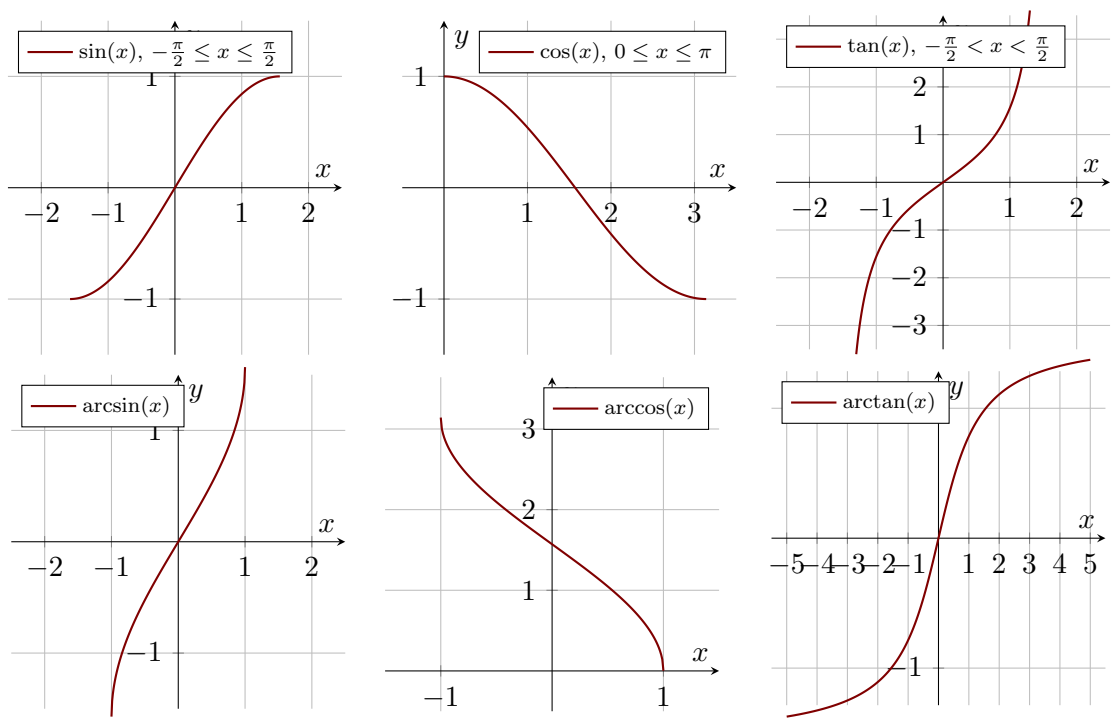
In this section, we derive the formulas for the derivatives of inverse trigonometric functions and apply them to a variety of problems.

To define inverses, we restrict domains

$$\begin{aligned}\arcsin(x) &\in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right],\\ \arccos(x) &\in [0, \pi],\\ \arctan(x) &\in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)\end{aligned}$$

These restrictions determine the sign of square roots that appear in their derivatives.



Theorem 2.114 (**Derivatives of Inverse Trigonometric Functions**). *The six derivatives of inverse trigonometric functions are given as follows.*

1. $\frac{d}{dx}(\arcsin(x)) = \frac{1}{\sqrt{1-x^2}}$, $-1 < x < 1$
2. $\frac{d}{dx}(\arccos(x)) = -\frac{1}{\sqrt{1-x^2}}$, $-1 < x < 1$
3. $\frac{d}{dx}(\arctan(x)) = \frac{1}{1+x^2}$, $x \in \mathbb{R}$
4. $\frac{d}{dx}(\operatorname{arccsc}(x)) = -\frac{1}{|x|\sqrt{x^2-1}}$, $|x| > 1$
5. $\frac{d}{dx}(\operatorname{arcsec}(x)) = \frac{1}{|x|\sqrt{x^2-1}}$, $|x| > 1$
6. $\frac{d}{dx}(\operatorname{arccot}(x)) = -\frac{1}{1+x^2}$, $x \in \mathbb{R}$

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Proof. To prove the derivative for $\arcsin(x)$, let $y = \arcsin(x)$. Then $x = \sin(y)$. Using implicit differentiation and solving for $\frac{dy}{dx}$, we have

$$1 = \cos(y)\frac{dy}{dx} \implies \frac{dy}{dx} = \frac{1}{\cos(y)}$$

Since $\sin(y) = x$, use the identity

$$\cos(y) = \sqrt{1 - \sin^2(y)} = \sqrt{1 - x^2}$$

since $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ implies $\cos(y) \geq 0$. Hence,

$$\frac{dy}{dx} = \frac{1}{\sqrt{1 - x^2}}$$

as desired.

To prove the derivative for $\arccos(x)$, let $y = \arccos(x)$. Then $x = \cos(y)$. Using implicit differentiation and solving for $\frac{dy}{dx}$ we have

$$1 = -\sin(y)\frac{dy}{dx} \implies \frac{dy}{dx} = -\frac{1}{\sin(y)}$$

Since $\cos(y) = x$, use the identity

$$\sin(y) = \sqrt{1 - \cos^2(y)} = \sqrt{1 - x^2}$$

since $y \in [0, \pi]$ implies $\sin(y) \geq 0$. Hence,

$$\frac{dy}{dx} = -\frac{1}{\sqrt{1 - x^2}}$$

as desired.

To prove the derivative for $\arctan(x)$, let $y = \arctan(x)$. Then $x = \tan(y)$. Using implicit differentiation and solving for $\frac{dy}{dx}$ we have

$$1 = \sec^2(y)\frac{dy}{dx} \implies \frac{dy}{dx} = \frac{1}{\sec^2(y)}$$

Since $\tan(y) = x$, use the identity

$$\sec^2(y) = 1 + \tan^2(y) = 1 + x^2$$

Therefore,

$$\frac{dy}{dx} = \frac{1}{1 + x^2}$$

as desired.

To prove the derivative for $\operatorname{arccsc}(x)$, let $y = \operatorname{arccsc}(x)$. Then $x = \csc(y)$. Using implicit differentiation and solving for $\frac{dy}{dx}$, we have

$$1 = -\csc(y)\cot(y)\frac{dy}{dx} \implies \frac{dy}{dx} = -\frac{1}{\csc(y)\cot(y)}$$

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Since $\csc(y) = x$,

$$\frac{dy}{dx} = -\frac{1}{x\cot(y)}$$

Using the identity

$$\cot^2(y) = \csc^2(y) - 1 = x^2 - 1$$

Then

$$\cot(y) = \pm\sqrt{x^2 - 1}$$

Careful sign analysis (using the principal range of $\operatorname{arccsc}(x)$) yields

$$x\cot(y) = |x|\sqrt{x^2 - 1}$$

Therefore

$$\frac{dy}{dx} = -\frac{1}{|x|\sqrt{x^2 - 1}}$$

as desired.

To prove the derivative for $\operatorname{arcsec}(x)$, let $y = \operatorname{arcsec}(x)$. Then $x = \sec(y)$. Using implicit differentiation and solving for $\frac{dy}{dx}$, we have

$$1 = \sec(y)\tan(y)\frac{dy}{dx} \implies \frac{dy}{dx} = \frac{1}{\sec(y)\tan(y)}$$

Using $\sec(y) = x$,

$$\frac{dy}{dx} = \frac{1}{x\tan(y)}$$

Using the identity

$$\tan^2(y) = \sec^2(y) - 1 = x^2 - 1$$

Then

$$\tan(y) = \pm\sqrt{x^2 - 1}$$

Careful about sign analysis (using the principal range of $\operatorname{arcsec}(x)$) yields

$$\frac{dy}{dx} = \frac{1}{|x|\sqrt{x^2 - 1}}$$

as desired.

Finally, to prove the derivative for $\operatorname{arccot}(x)$, let $y = \operatorname{arccot}(x)$. Then $x = \cot(y)$. Using implicit differentiation and solving for $\frac{dy}{dx}$, we have

$$1 = -\csc^2(y)\frac{dy}{dx} \implies \frac{dy}{dx} = -\frac{1}{\csc^2(y)}$$

Since $\cot(y) = x$, use the identity

$$\csc^2(y) = \cot^2(y) + 1 = x^2 + 1$$

Then

$$\frac{dy}{dx} = -\frac{1}{1 + x^2}$$

as desired. □

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We extend the derivatives of inverse trigonometric functions to involve the chain rule.

Theorem 2.115. *Let $u = u(x)$ be a differentiable function. Then*

1. $\frac{d}{dx}(\arcsin(u(x))) = \frac{u'(x)}{\sqrt{1-u^2(x)}}$, $-1 < u(x) < 1$
2. $\frac{d}{dx}(\arccos(x)) = -\frac{u'(x)}{\sqrt{1-u^2(x)}}$, $-1 < u(x) < 1$
3. $\frac{d}{dx}(\arctan(x)) = \frac{u'(x)}{1+u^2(x)}$, $u(x) \in \mathbb{R}$
4. $\frac{d}{dx}(\operatorname{arccsc}(u(x))) = -\frac{u'(x)}{|u(x)|\sqrt{u^2(x)-1}}$, $|u(x)| > 1$
5. $\frac{d}{dx}(\operatorname{arcsec}(x)) = \frac{u'(x)}{|u(x)|\sqrt{u^2(x)-1}}$, $|u(x)| > 1$
6. $\frac{d}{dx}(\operatorname{arccot}(x)) = -\frac{u'(x)}{1+u^2(x)}$, $u(x) \in \mathbb{R}$

Example 2.116. Differentiate

$$y = \sin^{-1}(3x)$$

Solution. Using Theorem 2.115

$$\frac{dy}{dx} = \frac{3}{\sqrt{1-(3x)^2}} = \frac{3}{\sqrt{1-9x^2}}$$

Example 2.117. Differentiate

$$y = \cos^{-1}(2x-1)$$

Solution. Using Theorem 2.115

$$\frac{dy}{dx} = -\frac{2}{\sqrt{1-(2x-1)^2}}$$

Example 2.118. Differentiate

$$y = \arctan(x^2)$$

Solution. Using Theorem 2.115

$$\frac{dy}{dx} = \frac{2x}{1+(x^2)^2} = \frac{2x}{1+x^4}$$

Example 2.119. Differentiate

$$y = \operatorname{arcsec}(2x)$$

Solution. Using Theorem 2.115

$$\frac{dy}{dx} = \frac{2}{|2x|\sqrt{(2x)^2-1}} = \frac{2}{|2x|\sqrt{4x^2-1}}$$

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Example 2.120. Differentiate

$$y = \operatorname{arccsc}(x^3)$$

Solution. Using Theorem 2.115

$$\frac{dy}{dx} = -\frac{3x^2}{|x^3|\sqrt{(x^3)^2 - 1}} = -\frac{3x^2}{|x^3|\sqrt{x^6 - 1}}$$

Example 2.121. Differentiate

$$y = \operatorname{arccot}(5x)$$

Solution. Using Theorem 2.115

$$\frac{dy}{dx} = -\frac{5}{1 + (5x)^2} = -\frac{5}{1 + 25x^2}$$

Example 2.122. Differentiate

$$y = x \arcsin(x)$$

Solution. Using the Product Rule (Theorem 2.31) and Theorem 2.114,

$$\frac{dy}{dx} = \arcsin(x) + x\frac{1}{\sqrt{1 - x^2}} = \arcsin(x) + \frac{x}{\sqrt{1 - x^2}}$$

Example 2.123. Differentiate

$$y = e^x \arctan(x^5)$$

Solution. Using the Product Rule (Theorem 2.31) and Theorem 2.115

$$\frac{dy}{dx} = e^x \arctan(x^5) + e^x \frac{5x^4}{1 + (x^5)^2} = e^x \arctan(x^5) + \frac{5x^4 e^x}{1 + x^{10}}$$

Example 2.124. Differentiate

$$y = \frac{\arcsin(x)}{x}$$

Solution. Using the Quotient Rule (Theorem 2.37) and Theorem 2.114,

$$\frac{dy}{dx} = \frac{\frac{x}{\sqrt{x^2-1}} - \arcsin(x)}{x^2}$$

Example 2.125. Differentiate

$$\arctan(xy^2) = x^2 + e^{xy} + y$$

Solution. Using implicit differentiation, the Product Rule (Theorem 2.31) and the Chain Rule (Theorem 2.46), we have

$$\frac{d}{dx}(\arctan(xy^2)) = \frac{d}{dx}(x^2 + e^{xy} + y)$$

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$$\begin{aligned}
\frac{\frac{d}{dx}(xy^2)}{1+(xy^2)^2} &= 2x + e^{xy}\left(y + \frac{dy}{dx}\right) + \frac{dy}{dx} \\
\frac{y^2 + 2xy\frac{dy}{dx}}{1+x^2y^4} &= 2x + ye^{xy} + e^{xy}\frac{dy}{dx} + \frac{dy}{dx} \\
\frac{2xy}{1+x^2y^4}\frac{dy}{dx} - e^{xy}\frac{dy}{dx} - \frac{dy}{dx} &= 2x + ye^{xy} - \frac{y^2}{1+x^2y^4} \\
\frac{dy}{dx}\left(\frac{2xy}{1+x^2y^4} - e^{xy} - 1\right) &= 2x + ye^{xy} - \frac{y^2}{1+x^2y^4} \\
\frac{dy}{dx} &= \frac{2x + ye^{xy} - \frac{y^2}{1+x^2y^4}}{\frac{2xy}{1+x^2y^4} - e^{xy} - 1}
\end{aligned}$$

Example 2.126. Find the equation of the tangent line to

$$y = \arctan(x)$$

at $x = 1$.

Solution. The slope at $x = 1$ is

$$m = \frac{1}{1+1^2} = \frac{1}{2}$$

and at $x = 1$, we have $y = \frac{\pi}{4}$. Therefore, the equation of the tangent line to the curve at the point $(1, \frac{\pi}{4})$ is

$$y - \frac{\pi}{4} = \frac{1}{2}(x-1)$$

Example 2.127. Differentiate

$$y = \ln(\operatorname{arcsec}(x))$$

Solution. Using the Chain Rule (Theorem 2.46)

$$\frac{dy}{dx} = \frac{1}{\operatorname{arcsec}(x)} \cdot \frac{1}{|x|\sqrt{x^2-1}} = \frac{1}{\operatorname{arcsec}(x)|x|\sqrt{x^2-1}}$$

2.12 Related Rates

In many applications, quantities do not change in isolation. Instead, several variables are connected by a geometric, physical, or algebraic relationship, and a change in one quantity induces changes in others. For example, as the radius of a balloon increases, its volume increases; as a ladder slides down a wall, the distance of its base from the wall changes as well.

Related rates problems study how the rates of change of different variables are connected through such relationships. The key idea is that although the variables may be related algebraically, their rates of change with respect to time are related through differentiation. Solving these problems requires implicit differentiation together with careful attention to units, given information, and evaluation at a specific instant.

Suppose two or more variables x, y, z,... depend on time t and satisfy an equation

$$F(x, y, z, ...) = 0$$

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Even if this equation does not involve t explicitly, the variables themselves do.

Differentiating both sides with respect to t using the chain rule produces an equation relating the rates

$$\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}, \ldots$$

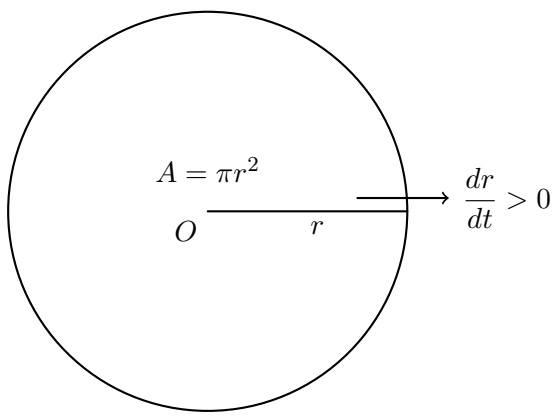
The key idea is to differentiate with respect to time first, then substitute numerical values last.

To solve a related rates problem,

1. Draw a diagram and introduce variables.
2. Write an equation relating the variables (geometry, physics, etc.)
3. Differentiate implicitly with respect to time t.
4. Substitute known values (including the specific instant).
5. Solve for the desired rate.

Skipping steps often leads to sign errors or incorrect results.

Example 2.128. The radius r of a circle increases at a rate of 2 cm/s. How fast is the area increasing when $r = 5$ cm?



Solution. The relationship we require is the area of the circle, which is given by

$$A = \pi r^2$$

Differentiating with respect to t gives

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

Since $\frac{dr}{dt} = 2$ cm/s, then

$$\frac{dA}{dt} = 2\pi(5)(2) = 20\pi \text{ cm}^2/\text{s}$$

Therefore, the area is increasing at a rate of 20π cm^2/s when $r = 5$.

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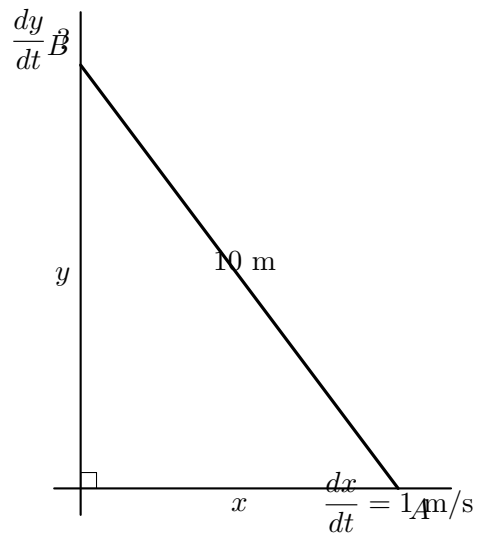
Example 2.129. A 10-m ladder leans against a vertical wall. The bottom slides away from the wall at 1 m/s. How fast is the top sliding down when the bottom is 6 m from the wall?



Solution. Let x be the distance from the wall, and y be the height of the ladder. Then by Pythagorean Theorem, our relationship becomes

$$x^2 + y^2 = 100$$

Differentiating both sides with respect to t

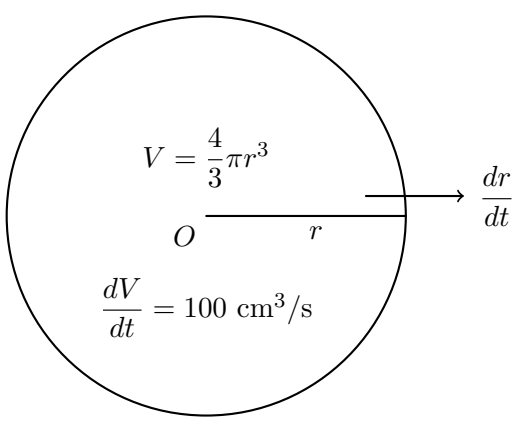
$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

When $x = 6$, $y = 8$, and $\frac{dx}{dt} = 1$, we have

$$2(6)(1) + 2(8)\frac{dy}{dt} = 0 \implies \frac{dy}{dt} = -\frac{3}{4} \text{ m/s}$$

The negative sign indicates downward motion.

Example 2.130. Air is being pumped into a spherical balloon at a rate of 100 cm^3/s. How fast is the radius increasing when the radius is 5 cm?



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Solution. The relationship we require is the volume of a sphere

$$V = \frac{4}{3}\pi r^3$$

Differentiating both sides give

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

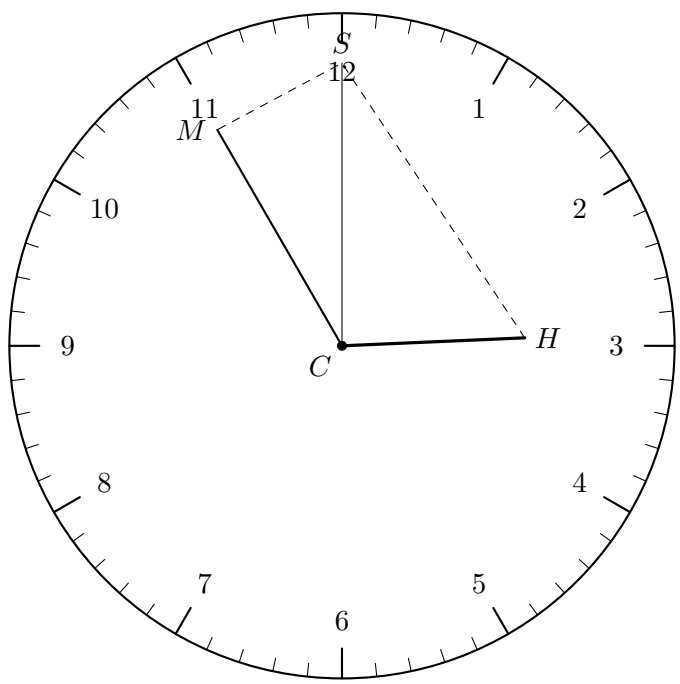
Substituting the values give

$$100 = 4\pi(5)^2 \frac{dr}{dt}$$

Thus,

$$\frac{dr}{dt} = \frac{1}{\pi} \text{ cm/s}$$

Example 2.131. A clock has an hour hand of length 4 cm, a minute hand of length 5.5 cm, and a second hand of length 6 cm. How quickly is the area of the quadrilateral formed by the centre of the clock and the tips of the three clock hands changing at exactly at 2:55?



Solution. Let C be the center of the clock, and let H, M, and S be the tips of the hour, minute, and second hands, respectively.

We define the quadrilateral $CHSM$ and split it into two triangles

$$A = A_1 + A_2$$

where

$$A_1 = \text{Area}(\Delta CHS), \quad A_2 = \text{Area}(\Delta CSM)$$

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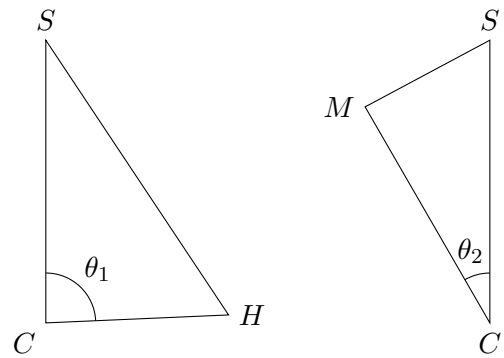

Let θ_1 be the angle between the hour and second hands, and let θ_2 be the angle between the minute and angle hands. For two sides a and b with included angle θ, the area relationship is given as

$$\text{Area} = \frac{1}{2}ab\sin(\theta)$$

Thus,

$$A_1 = \text{Area}(\Delta CHS) = \frac{1}{2}(4)(6)\sin(\theta_1) = 12\sin(\theta_1)$$

and

$$A_2 = \text{Area}(\Delta CSM) = \frac{1}{2}(6)(5.5)\sin(\theta_2) = 16.5\sin(\theta_2)$$

Using the angle measurements at the instant 2:55:00,

$$\theta_S = \frac{\pi}{2}, \quad \theta_H = \frac{\pi}{72}, \quad \theta_M = \frac{2\pi}{3}$$

the included angles are

$$\theta_1 = \theta_S - \theta_H = \frac{\pi}{2} - \frac{\pi}{72} = \frac{35\pi}{72}$$

and

$$\theta_2 = \theta_M - \theta_S = \frac{2\pi}{3} - \frac{\pi}{2} = \frac{\pi}{6}$$

So at 2:55:00,

$$A_1 = 12\sin\left(\frac{35\pi}{72}\right), \quad A_2 = 16.5\sin\left(\frac{\pi}{6}\right)$$

The hands rotate at constant angular speeds (in radians per second)

$$\omega_S = \frac{2\pi}{60}, \quad \omega_M = \frac{2\pi}{3600}, \quad \omega_H = \frac{2\pi}{12 \cdot 3600}$$

Since $\theta_1 = \theta_S - \theta_H$ and $\theta_2 = \theta_M - \theta_S$, then

$$\frac{d\theta_1}{dt} = \omega_S - \omega_H, \quad \frac{d\theta_2}{dt} = \omega_M - \omega_S$$

Differentiating A_1 and A_2 with respect to time t gives

$$\frac{dA_1}{dt} = 12\cos(\theta_1)\frac{d\theta_1}{dt}, \quad \frac{dA_2}{dt} = 16.5\cos(\theta_2)\frac{d\theta_2}{dt}$$

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Substituting the values, we get

$$\frac{dA_1}{dt} = 12\cos\left(\frac{35\pi}{72}\right)\left(\frac{2\pi}{60} - \frac{2\pi}{12\cdot 3600}\right) \approx -0.0547 \text{ cm}^2/\text{s}$$

and

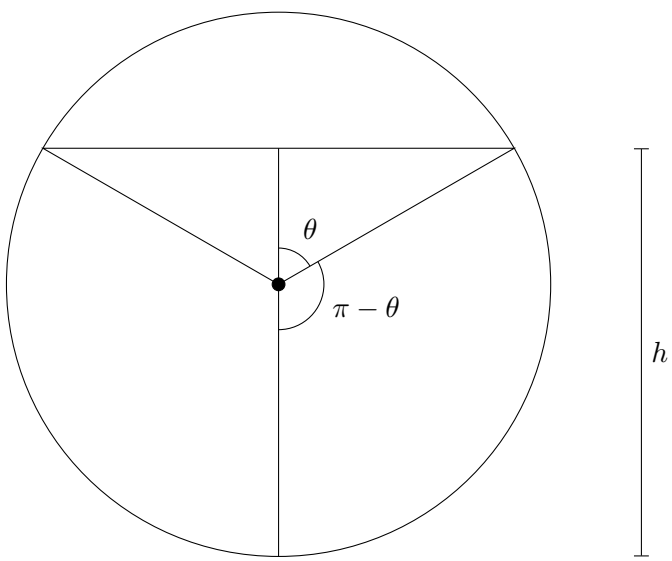
$$\frac{dA_2}{dt} = 16.5\cos\left(\frac{\pi}{6}\right)\left(\frac{2\pi}{3600} - \frac{2\pi}{60}\right) \approx 1.4714 \text{ cm}^2/\text{s}$$

Hence,

$$\frac{dA}{dt} = \frac{dA_1}{dt} + \frac{dA_2}{dt} \approx -0.0547 + 1.4714 = 1.4167 \text{ cm}^2/\text{s}.$$

Therefore, at 2:55, the area of the quadrilateral formed by the centre of the clock and the tips of the three hands is increasing at approximately 1.4167 cm^2/s.

Example 2.132. A water pipe with radius 10 cm and length 20 m is filling with water at the rate of 70 cm^3/min. If both ends are sealed off and the pipe is parallel to the ground, how quickly is the water level in the pipe rising when it is 3/4 full?



Solution. Let $h(t)$ denote the height of the water in the pipe. Since the pipe is horizontal, the volume of water is

$$V = LA$$

where A is the cross-sectional area of the water. Differentiating with respect to time,

$$\frac{dV}{dt} = L\frac{dA}{dt} \implies \frac{dA}{dt} = \frac{70}{2000} = \frac{7}{200} \text{ cm}^2/\text{min}$$

Let θ be defined as in the diagram, where the water surface subtends to an angle 2θ at the center of the pipe. The water-filled region consists of

- a circular sector of angle $2(\pi - \theta)$
- minus an isosceles triangle of vertex angle 2θ.

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Hence,

$$A = (\text{Area of sector}) - (\text{Area of triangle}) = A_S - A_T$$

The sector area is

$$A_S = \frac{2(\pi - \theta)}{2\pi} \cdot \pi r^2 = 100(\pi - \theta)$$

and the area of the triangle is

$$A_T = \frac{1}{2}(2r\sin(\theta))(r\cos(\theta)) = 50\sin(2\theta)$$

Thus,

$$A = 100(\pi - \theta) - 50\sin(2\theta)$$

Differentiating implicitly,

$$\frac{dA}{dt} = (-100 - 100\cos(2\theta))\frac{d\theta}{dt}$$

Substituting the values give

$$\frac{7}{200} = (-100 - 100\cos(2\theta))\frac{d\theta}{dt}$$

When the pipe is 3/4 full, the empty region is 1/4 of the circle, which corresponds to

$$\theta = \frac{\pi}{3}$$

Thus,

$$\cos(2\theta) = \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2}$$

Substitute

$$\frac{7}{200} = (-100 + 50)\frac{d\theta}{dt} = -50\frac{d\theta}{dt}$$

so

$$\frac{d\theta}{dt} = -\frac{7}{10000}$$

From the geometry

$$h = r + r\cos(\theta) = 10 + 10\cos(\theta)$$

Differentiate

$$\frac{dh}{dt} = -10\sin(\theta)\frac{d\theta}{dt}$$

At $\theta = \frac{\pi}{3}$,

$$\sin(\theta) = \frac{\sqrt{3}}{2}$$

Thus,

$$\frac{dh}{Dt} = -10 \cdot \frac{\sqrt{3}}{2}\left(-\frac{7}{10000}\right) = \frac{7\sqrt{3}}{2000} \approx 0.00202 \text{ cm/min}$$

The water level is rising at approximately 0.00202 cm/min when the pipe is 3/4 full.

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2.13 Linear Approximations and Differentials

In many applications, it is difficult—or unnecessary—to compute the exact value of a function. Instead, we often seek a good approximation that is easier to evaluate and interpret. Near a point where a function is differentiable, the graph of the function is well-approximated by its tangent line. This observation leads to the idea of linear approximation, which replaces a nonlinear function with a linear function that closely matches it near a chosen point.

Closely related is the concept of differentials, which provide a systematic way to estimate small changes in a quantity and to analyze how errors propagate through calculations. Linear approximations and differentials form the foundation for numerical methods, scientific computation, and many applications in physics, engineering, and economics.

Let f be differentiable at a point $x = a$. The equation of the tangent line to the graph of $y = f(x)$ at $x = a$ is

$$y = f(a) + f'(a)(x - a)$$

Definition 2.133. The *linear approximation* (or *linearization*) of f at $x = a$ is the function

$$L(x) = f(a) + f'(a)(x - a)$$

For values of x close to a,

$$f(x) \approx L(x)$$

The graph of $y = L(x)$ is the tangent line to $y = f(x)$ at $x = a$. Near $x = a$, the tangent line and the curve are nearly indistinguishable. The approximation becomes more accurate as $x \to a$. The idea is sometimes summarized as: a differentiable function is locally linear.

Example 2.134. Approximate $f(x) = x^3$ near $x = 2$.

Solution. Compute $f(2) = 8$ and its derivative at 2

$$f'(x) = 3x^2 \implies f'(2) = 12$$

Therefore,

$$L(x) = 8 + 12(x - 2)$$

To approximate 2.01^3,

$$2.01^3 \approx 8 + 12(0.01) = 8.12$$

The exact value is 8.120601, so the approximation is quite accurate.

Example 2.135. Approximate $e^{0.02}$.

Solution. Let $f(x) = e^x$ and choose $a = 0$. Then

$$f(0) = 1 \quad \text{and} \quad f'(0) = 1$$

So then

$$L(x) = 1 + x$$

Thus,

$$e^{0.02} \approx 1.02$$

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Example 2.136. Approximate $\ln(1.05)$.

Solution. Let $f(x) = \ln(x)$ and choose $a = 1$. Then

$$f(1) = 0 \quad \text{and} \quad f'(1) = 1$$

So then

$$\ln(1.05) \approx 0 + 1(0.05) = 0.05$$

Definition 2.137. Let $y = f(x)$ be a differentiable function. The *differential of* y is

$$dy = f'(x)\ dx$$

where dx is an independent variable representing a small change in x.

Here, dx is a small input change, while dy is the corresponding linear approximation of the change in y. For small dx, we have

$$\Delta y = f(x + dx) - f(x) \approx dy$$

Example 2.138. Let $y = \sqrt{x}$. Estimate the change in y when x changes from 16 to 16.1.

Solution. First note that the derivative is given as

$$f'(x) = \frac{1}{2\sqrt{x}}$$

At $x = 16$,

$$\frac{dy}{dx} = \frac{1}{8} \implies dy = \frac{1}{8}\ dx$$

With $dx = 0.1$, we have

$$dy \approx \frac{1}{8}(0.1) = 0.0125$$

Thus,

$$\sqrt{16.1} \approx 4.0125$$

Using the linear approximation

$$f(a + dx) \approx f(a) + f'(a)\ dx$$

we see that

$$\Delta y \approx dy$$

Thus, linear approximation and differentials are two viewpoints of the same idea.

Differentials are particularly useful for estimating how measurement errors affects computed quantities.

Example 2.139. A sphere has a radius $r = 10$ cm with a possible measurement error of ± 0.1 cm. Estimate the resulting error in the volume.

Solution. The volume of the sphere is given as

$$V = \frac{4}{3}\pi r^3$$

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and the differential of V is given by

$$dV = 4\pi r^2 \, dr$$

At $r = 10$,

$$dV \approx 4\pi(10)^2(0.1) = 40\pi \approx 126 \text{ cm}^3$$

Example 2.140. Find the linear approximation to $f(x) = \ln(x)$ at $x = e$ and use it to approximate $\ln(2.7)$.

Solution. Note that $f(e) = 1$ and $f'(e) = \frac{1}{e}$. Then the linear approximation is

$$L(x) = 1 + \frac{1}{e}(x - e)$$

Then

$$\ln(2.7) \approx 1 + \frac{1}{e}(2.7 - e)$$

Linear approximation is most accurate when x is close to a, $f''(x)$ is small near a, or the function is smooth and slowly varying. For larger deviations, higher-order approximations (Taylor polynomials) are needed.

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Chapter 3

Applications of Derivatives

In Chapter **??**, we developed the derivative as a limit of difference quotients and interpreted it as a rate of change and slope of the tangent line. In this chapter, we move from computation to application. The derivative is no longer just an algebraic formula, it becomes a powerful analytical tool for understanding the behaviour of functions.

The central theme of this chapter is: *The derivative reveals qualitative information about a function.*

- From the sign of $f'(x)$, we determine whether the function is increasing or decreasing.
- From the zeros of $f'(x)$, we locate potential minima and maxima.
- From $f''(x)$, we detect concavity and inflection points.
- From linearization, we approximate values.
- From iterative derivative-based methods we solve nonlinear equations.

This chapter builds a bridge between pure differentiation and real-world modeling.

Many natural and economic problems ask questions such as

- When is a quantity largest or smallest?
- How can we optimize cost, revenue, or efficiency?
- How can we approximate complicated functions?
- How can we find solutions to equations that cannot be solved algebraically?

These questions cannot be answered by algebra alone. They require a deeper understanding of how functions behave, and the derivative provides exactly that information.

3.1 Maximum and Minimum Values

In the previous chapter, we developed the derivative as a measure of instantaneous rate of change and as the slope of the tangent. We now begin one of the central applications of derivatives: maximum and minimum values of functions.

In many practical situations, we are interested in questions such as

- When is profit maximized?
- What is the smallest possible cost?
- What is the highest or lowest point of a curve?
- When does a quantity stop increasing and begin decreasing.

These questions naturally lead to the study of *extrema* (maximum and minimum values).

The key idea of this section is that extreme values are closely connected to the behaviour of the derivative. In particular,

Interior extrema occur only where $f'(x) = 0$ or $f'(x)$ does not exist.

This section introduces the precise definitions of local and absolute extrema, defines critical numbers, and proves the foundational theorem that links extrema and derivatives.

Definition 3.1. Let f be a function defined on a set D and let $c \in D$.

1. We say that $f(c)$ is an *absolute maximum value* of f on D if for all $x \in D$,
$$f(x) \leq f(c)$$
2. We say that $f(c)$ is an *absolute minimum value* of f on D if for all $x \in D$,
$$f(x) \geq f(c)$$

The point c is called a *point where the absolute minimum (or maximum) occurs.*

Definition 3.2. Let f be defined on an interval containing c.

1. We say that $f(c)$ is a *local maximum value* if there exists a $\delta > 0$ such that
$$f(x) \leq f(c)$$
for all x satisfying $|x - c| < \delta$.
2. We say that $f(c)$ is a *local minimum value* if there exists a $\delta > 0$ such that
$$f(x) \geq f(c)$$
for all x satisfying $|x - c| < \delta$.

Absolute extrema compare values on the entire domain, while local extrema compare values only near the point. An absolute maximum is always a local minimum (if interior), but a local maximum need not be absolute.

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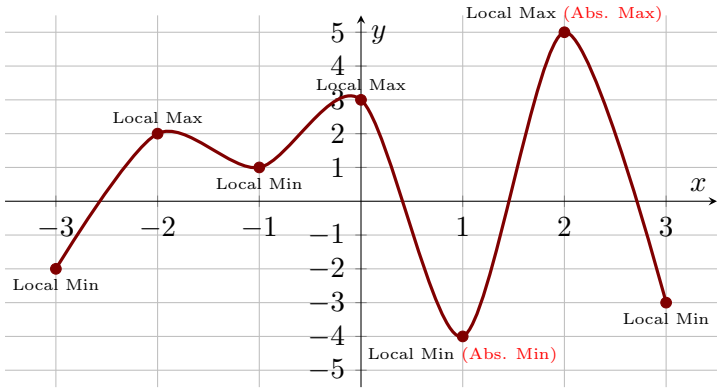

Definition 3.3. Let f be defined on an interval. A number c in the domain of f is called a *critical number* of f if either

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ does not exist}$$

The corresponding point $(c, f(c))$ is called a *critical point.*

Theorem 3.4 (**Fermat's Theorem**)**.** *Let f be differentiable at c. If f has a local maximum or local minimum at c, then*

$$f'(c) = 0$$

Proof. Suppose f has a local maximum at c. Then there exists a $\delta > 0$ such that

$$f(x) \leq f(c)$$

for all x with $|x - c| < \delta$.

Consider $h > 0$ sufficiently small. Then $c + h$ lies within the δ-neighbourhood, so

$$f(c + h) \leq f(c)$$

Thus,

$$\frac{f(c + h) - f(c)}{h} \leq 0$$

Now let $h < 0$ with small magnitude. Again,

$$f(c + h) \leq f(c)$$

so that

$$\frac{f(c + h) - f(c)}{h} \geq 0$$

since dividing by a negative number reverses the inequality.

Taking limits as $h \to 0^+$ and $h \to 0^-$, we obtain

$$\lim_{h \to 0^+} \frac{f(c + h) - f(c)}{h} \leq 0$$

and

$$\lim_{h \to 0^-} \frac{f(c + h) - f(c)}{h} \geq 0$$

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If $f'(c)$ exists, these one-sided limits must be equal. Therefore,

$$f'(c) = 0.$$

The proof for a local minimum is similar. □

This theorem gives a necessary condition, and not a sufficient one. Indeed, $f'(c) = 0$ does not guarantee an extrema, as the following example demonstrates.

Example 3.5. Let $f(x) = x^3$. Then $f'(x) = 3x^2$. At $x = 0$, $f'(0) = 0$. However, $f(x) = x^3$ is increasing everywhere, so $x = 0$ is not a maximum or minimum. Thus, $f'(c) = 0$ does not imply an extrema.

Theorem 3.6. *If f has a local maximum or minimum at c, then c is a critical number of f.*

Proof. If f is differentiable at c, then by Fermat's Theorem (Theorem 3.4), $f'(c) = 0$. If f is not differentiable at c, then by definition c is still a critical number. Therefore, every local extremum occurs at a critical number. □

Let f be a continuous function on $[a, b]$. To find absolute extrema on a closed interval,

1. Find the critical numbers in (a, b).
2. Evaluate f at each critical number.
3. Evaluate $f(a)$ and $f(b)$.
4. Compare all values.

The largest value is the absolute maximum, and the smallest value is the absolute minimum.

Example 3.7. Find the absolute extrema of $f(x) = x^3 - 3x^2 + 1$ on the interval $[-1, 3]$.

Solution. Compute the derivative, we get

$$f'(x) = 3x^2 - 6x = 3x(x - 2)$$

To find the critical numbers, set $f'(x) = 0$. Then we solve

$$3x(x - 2) = 0$$

Thus, either $x = 0$ or $x = 2$.

Evaluating at the endpoints and at the critical numbers,

x	$f(x)$
-1	-3
0	1
2	-3
3	1

Thus, the absolute maximum value is 1 (at $x = 0$ and $x = 3$) and absolute minimum value is -3 (at $x = -1$ and $x = 2$).

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Example 3.8. Let $f(x) = |x|$. Then

$$f'(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$$

The derivative does not exist at $x = 0$.

Since $f(0) = 0$ is less than nearby values, $x = 0$ is a local (and absolute) minimum.

Thus, nondifferentiable points must also be considered.

The difference between finding absolute extrema and local extrema is that you only want to know where the local extrema occur, and not the value of the function at that point.

Example 3.9. Find all local extrema of

$$f(x) = x^2 e^{-x}$$

Solution. We compute the derivative using the product rule (Theorem 2.31)

$$\begin{aligned} f'(x) &= (x^2)' e^{-x} + x^2 (e^{-x})' \\ &= 2xe^{-x} - x^2 e^{-x} \\ &= e^{-x}(2x - x^2) \\ &= xe^{-x}(2 - x) \end{aligned}$$

Since $e^{-x} > 0$ for all x, critical numbers occur when

$$x(2 - x) = 0$$

Thus, $x = 0$ or $x = 2$ are the critical numbers where local extrema occur.

Example 3.10. Find all local extrema of

$$f(x) = \frac{x}{x^2 + 1}$$

Solution. We compute the derivative using the quotient rule (Theorem 2.37)

$$\begin{aligned} f'(x) &= \frac{(x)'(x^2 + 1) - (x)(x^2 + 1)'}{(x^2 + 1)^2} \\ &= \frac{(x^2 + 1) - x(2x)}{(x^2 + 1)^2} \\ &= \frac{-x^2 + 1}{(x^2 + 1)^2} \end{aligned}$$

Critical numbers occur whenever

$$\frac{-x^2 + 1}{(x^2 + 1)^2} = 0$$

Since $x^2 + 1 > 0$ for all x, there are no x values that make $f'(x)$ undefined. Thus,

$$-x^2 + 1 = 0 \implies (1 - x)(1 + x) = 0$$

and so local extrema occur at $x = 1$ and $x = -1$.

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Example 3.11. Find the local extrema of

$$f(x) = \sin(x^2)$$

Solution. We compute derivative using the chain rule (Theorem 2.46)

$$\begin{aligned} f'(x) &= \sin'(x^2)(x^2)' \\ &= \cos(x^2)(2x) \\ &= 2x\cos(x^2) \end{aligned}$$

Critical numbers occur whenever

$$2x\cos(x^2) = 0$$

Thus, either $x = 0$ or $\cos(x^2) = 0$. Equivalently,

$$x^2 = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}$$

and thus,

$$x = \pm\sqrt{\frac{\pi}{2} + k\pi}, \quad k \in \mathbb{Z}$$

Example 3.12. Find all local extrema of

$$f(x) = x\ln(x)$$

for $x > 0$.

Solution. We compute the derivative using the product rule (Theorem 2.31)

$$\begin{aligned} f'(x) &= (x)'\ln(x) + x\ln'(x) \\ &= \ln(x) + x \cdot \frac{1}{x} \\ &= \ln(x) + 1 \end{aligned}$$

Critical numbers occur whenever

$$\ln(x) + 1 = 0$$

Thus,

$$\ln(x) = -1 \implies x = e^{-1}$$

Example 3.13. Find all local extrema of

$$f(x) = e^x\sin(x)$$

Solution. We compute the derivative using the product rule (Theorem 2.31)

$$\begin{aligned} f'(x) &= (e^x)'\sin(x) + e^x\sin'(x) \\ &= e^x\sin(x) + e^x\cos(x) \\ &= e^x(\sin(x) + \cos(x)) \end{aligned}$$

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Since $e^x \neq 0$, we solve

$$\begin{aligned} \sin(x) + \cos(x) &= 0 \\ \sin(x) &= -\cos(x) \\ \tan(x) &= -1 \end{aligned}$$

Thus,

$$x = -\frac{\pi}{4} + k\pi, \quad k \in \mathbb{Z}$$

3.2 The Extreme Value Theorem

In Section 3.1, we learned that if a function has a local maximum or minimum at an interior point and is differentiable there, then the derivative must be zero. However, an important question still remains: *Do extrema necessarily exist?*

We have seen examples where functions fail to attain maximum or minimum values. For instance, the function $f(x) = \frac{1}{x}$ on $(0, 1)$ is continuous but has no absolute maximum or minimum on that open interval.

Thus, differentiability alone does not guarantee existence of extrema. The key missing ingredient is the combination of

- continuity,
- and a closed, bounded interval.

The Extreme Value Theorem provides a powerful existence result: a continuous function on a closed interval must attain both an absolute maximum and an absolute minimum.

This theorem is foundational for optimization, the First and Second Derivative Tests, and many deeper results in analysis.

Definition 3.14. Let $S \subseteq \mathbb{R}$.

1. A number M is an *upper bound* of S if

$$x \leq M$$

   for all $x \in S$.

2. A number M is a *lower bound* of S if

$$x \geq M$$

   for all $x \in S$

Definition 3.15. Let $S \subseteq \mathbb{R}$ be nonempty.

1. If S is bounded above, the *supremum of* S, denoted $\sup(S)$ is the smallest upper bound of S.
2. If S is bounded below, the *infimum of* S, denoted $\inf(S)$, is the largest lower bound of S.

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The supremum or infimum may or may not belong to S. If $\sup(S) \in S$, then it is the maximum of S, while if $\inf(S) \in S$, then it is the minimum of S.

Example 3.16. Let $S = (0, 1)$. Then

$$\sup(S) = 1, \quad \inf(S) = 0$$

However, neither 0 nor 1 belong to S. Thus, S has no maximum and no minimum.

Example 3.17. Let $S = [0, 1]$. Then

$$\sup(S) = 1, \quad \inf(S) = 0$$

and both are elements of S. Thus, S has both a maximum and a minimum.

Definition 3.18. Let f be defined on a set D. The function f is *bounded* on D, if there exists a real number $M > 0$ such that

$$|f(x)| \leq M$$

for all $x \in D$. That is, the set $\{f(x) : x \in D\}$ is bounded above and below.

Theorem 3.19 (**Extreme Value Theorem**). *Let f be continuous on a closed interval $[a, b]$. Then*

1. *f is bounded on $[a, b]$.*
2. *There exist numbers $c, d \in [a, b]$ such that*

$$f(c) \leq f(x) \leq f(d)$$

for all $x \in [a, b]$.

Thus, f attains both an absolute minimum and an absolute maximum on $[a, b]$.

The theorem contains three essential hypotheses

1. Continuity,
2. Closed interval,
3. Bounded interval.

If any one of these fail, the conclusion may fail.

Note that we do not prove this theorem in this course. Its proof relies on deeper properties of the real numbers studied in real analysis (MATH 2001). However, we will illustrate why each hypothesis is necessary.

Example 3.20. Consider

$$f(x) = x$$

on the interval $(0, 1)$. The function is continuous. However, as $x \to 1$, $f(x) \to 1$, but then $1 \notin (0, 1)$. Thus, f does not attain an absolute maximum. The interval is not closed, so EVT does not apply.

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Example 3.21. Define

$$f(x) = \begin{cases} x & \text{if } x \neq \frac{1}{2} \\ 2 & \text{if } x = \frac{1}{2} \end{cases}$$

on $[0, 1]$. The function is defined on a closed interval but is not continuous at $x = \frac{1}{2}$. The supremum of the function values is 1, but the value 1 is never attained. Therefore, there is no absolute maximum.

Corollary 3.22 (**Closed Interval Method**). *Let f be continuous on $[a, b]$. Then absolute extrema exist, and they can occur only at*

1. *Critical numbers in (a, b),*
2. *Endpoints a and b.*

Thus, to find absolute extrema,

1. Compute $f'(x)$.
2. Find the critical numbers in (a, b).
3. Evaluate f at critical numbers.
4. Evaluate $f(a)$ and $f(b)$.
5. Compare all values.

Example 3.23. Find the absolute extrema of

$$f(x) = \ln(x) - x$$

on $[1, 4]$.

Solution. We compute the derivative using differentiation rules

$$f'(x) = \frac{1}{x} - 1$$

Critical numbers occur when

$$\frac{1}{x} - 1 = 0 \implies \frac{1}{x} = 1 \implies x = 1$$

Evaluating at the endpoints and critical numbers,

x	$f(x)$
1	-1
4	$\ln(4) - 4 \approx -2.6137$

Therefore, the absolute maximum is -1 (occurs at $x = 1$), and the absolute minimum is $\ln(4) - 4$ (occurs at $x = 4$).

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Example 3.24. Find all absolute extrema of

$$f(x) = (x^2+2)^3(3x^4+4)^2$$

on $[-2, 2]$.

Solution. We compute the derivative using the product rule (Theorem 2.31) and the chain rule (Theorem 2.46).

$$\begin{aligned} f'(x) &= 3(x^2+2)^2(2x)(3x^4+4)^2 + (x^2+2)^3(2)(3x^4+4)(12x^3) \\ &= 6x(x^2+2)^2(3x^4+4)^2 + 24x^3(x^2+2)^3(3x^4+4) \end{aligned}$$

Simplifying by common factoring

$$\begin{aligned} f'(x) &= 6x(x^2+2)^2(3x^4+4)[(3x^4+4) + 4x^2(x^2+2)] \\ &= 6x(x^2+2)^2(3x^4+4)(3x^4+4+4x^4+2) \\ &= 6x(x^2+2)^2(3x^4+4)(7x^4+6) \end{aligned}$$

Critical numbers occur when

$$6x(x^2+2)^2(3x^4+4)(7x^4+6) = 0$$

So either $6x = 0$ or $x^2+2 = 0$, or $3x^4+4 = 0$ or $7x^4+6 = 0$. However, the equations $x^2+2 = 0$, $3x^4+4 = 0$, and $7x^4+6 = 0$ have no real solutions. Thus, the only critical number is $x = 0$.

Evaluating at the endpoints and the critical number, we have

x	$f(x)$
-2	584064
0	0
2	584064

Therefore, the absolute minimum is 0 (occurs at $x = 0$) and the absolute maximum is 584064 (occurs at $x = -2$ and $x = 2$).

Example 3.25. Find all absolute extrema of

$$f(x) = \frac{x^2-2}{3x+1}$$

on $[1, 5]$.

Solution. We compute the derivative using the quotient rule (Theorem 2.37)

$$\begin{aligned} f'(x) &= \frac{2x(3x+1) - (x^2-2)(3)}{(3x+1)^2} \\ &= \frac{6x^2+2x-3x^2+6}{(3x+1)^2} \\ &= \frac{3x^2+2x+6}{(3x+1)^2} \end{aligned}$$

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Critical numbers occur whenever

$$\frac{3x^2+2x+6}{(3x+1)^2}=0 \implies 3x^2+2x+6=0$$

However, checking the discriminant

$$2^2-4(3)(6)<0$$

Therefore, the quadratic has no real solutions, and thus, there are no critical numbers. Evaluating at the endpoints then

x	$f(x)$
1	$-\frac{1}{4}$
5	$\frac{23}{16}$

Therefore, the absolute maximum is $\frac{23}{16}$ (occurs at $x=5$) and the absolute minimum is $-\frac{1}{4}$ (occurs at $x=1$).

3.3 The Mean Value Theorem

In Section 3.2, the Extreme Value Theorem (Theorem 3.19) guaranteed that a continuous function on a closed interval attains its maximum and minimum values. We now move from existence of extrema to a deeper connection between derivatives and global behaviour.

Recall

- The derivative represents an instantaneous rate of change.
- The difference quotient
$$\frac{f(b)-f(a)}{b-a}$$
represents the average rate of change on $[a,b]$.

A natural and profound question arises: *Must the instantaneous rate of change equal the rate of change at some point?*

The answer is yes, under the appropriate hypotheses. This is the result of the Mean Value Theorem (MVT), one of the most important theorems in calculus.

The Mean Value Theorem

- Justifies why derivative control global behaviour,
- Forms the theoretical basis for increasing/decreasing tests,
- Underlies error estimates and approximation theory,
- Is fundamental in advanced analysis.

We first prove a special case known as Rolle's Theorem, which is fully provable using the Extreme Value Theorem (Theorem 3.19) and Fermat's Theorem (Theorem 3.4).

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Definition 3.26. Let f be defined on $[a, b]$ with $a < b$. The *average rate of change* of f on $[a, b]$ is

$$\frac{f(b) - f(a)}{b - a}$$

Geometrically, this is the slope of the secant line connecting $(a, f(a))$ and $(b, f(b))$.

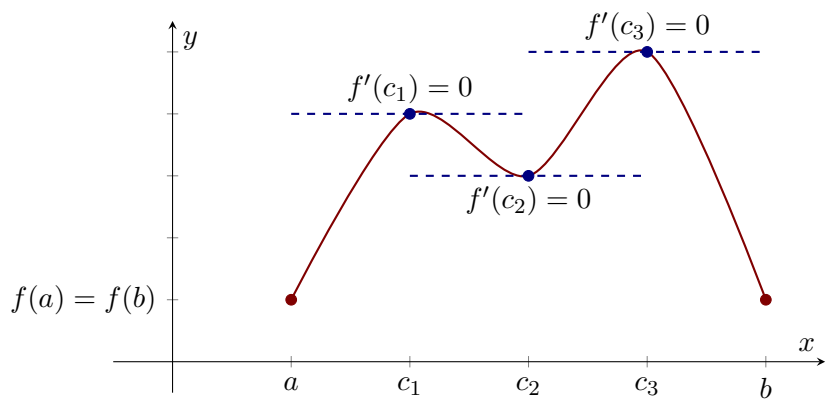
Theorem 3.27 (Rolle's Theorem). *Let f be a function on $[a, b]$ such that*

1. *f is continuous on $[a, b]$,*
2. *f is differentiable on (a, b),*
3. *$f(a) = f(b)$.*

Then there exists $c \in (a, b)$ such that

$$f'(c) = 0$$

In other words, if a differentiable curve begins and ends at the same height, then somewhere in between it must have a horizontal tangent.



Proof. Since f is continuous on $[a, b]$, the Extreme Value Theorem (Theorem 3.19) guarantees that f attains both a maximum and minimum on $[a, b]$. Let M and m denote the maximum and minimum values. Then we consider the following cases.

Case 1: If f is constant. If $f(x) = k$ for all $x \in [a, b]$, then

$$f'(x) = 0$$

for all $x \in (a, b)$, so the theorem holds.

Case 2: If f is not constant. Then either the maximum or minimum occur at some point $c \in (a, b)$. Indeed, if both maximum and minimum occurred only at endpoints, then since $f(a) = f(b)$, both extrema would equal the same value, forcing the function to be constant, contradicting our assumption.

Thus, there exists $c \in (a, b)$ where f has a local maximum or minimum. Since f is differentiable at c, Fermat's Theorem (Theorem 3.4) implies

$$f'(c) = 0$$

as desired. □

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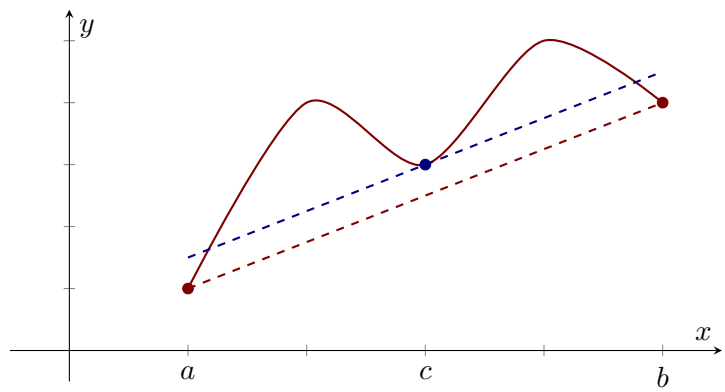
Theorem 3.28 (Mean Value Theorem). *Let f be a function on $[a, b]$ such that*

1. *f is continuous on $[a, b]$,*
2. *f is differentiable on (a, b).*

Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

In this version, there exists a point where the instantaneous rate of change equals the average rate of change. Geometrically, there is a tangent line parallel to the secant line.



Proof. Define the function

$$g(x) = f(x) - \left[f(a) + \frac{f(b) - f(a)}{b - a}(x - a)\right]$$

Observe that the expression in the brackets is the equation of the secant line, and thus, $g(x)$ measures the vertical distance between $f(x)$ and the secant line.

Since f is continuous on $[a, b]$ and differentiable on (a, b), and linear functions are continuous and differentiable everywhere, it follows that g is continuous on $[a, b]$ and differentiable on (a, b). Observe further that

$$g(a) = f(a) - \left[f(a) + \frac{f(b) - f(a)}{b - a}(a - a)\right] = 0$$

and

$$g(b) = f(b) - \left[f(a) + \frac{f(b) - f(a)}{b - a}(b - a)\right] = 0$$

Thus, $g(a) = g(b)$.

Using Rolle's Theorem (Theorem 3.27), there exists $c \in (a, b)$ such that

$$g'(c) = 0$$

In particular,

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}$$

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and evaluating at $x = c$ gives

$$0 = f'(x) - \frac{f(b) - f(a)}{b - a}$$

and thus,

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

□

With the Mean Value Theorem (Theorem 3.28) available to us, we are able to prove the following consequences.

Corollary 3.29. *If $f'(x) = 0$ for all x in an interval I, then f is constant on I.*

Proof. Let $a, b \in I$. By the Mean Value Theorem (Theorem 3.28), there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Since $f'(c) = 0$, we have

$$f(b) - f(a) = 0$$

Thus, $f(a) = f(b)$ for all $a, b \in I$, so f is constant. □

Corollary 3.30. *If $f'(x) = g'(x)$ for all x in an interval I, then*

$$f(x) = g(x) + C$$

for some constant C.

Proof. Consider $h(x) = f(x) - g(x)$. Then $h'(x) = 0$. By Corollary 3.29, h is constant, or

$$f(x) - g(x) = C \implies f(x) = g(x) + C$$

□

Some applications of the Mean Value Theorem is justifying the number of solutions of a function on an interval.

Example 3.31. Show that the equation

$$x^3 - 3x + 1 = 0$$

has at most one real solution in $(0, 1)$.

Solution. Let

$$f(x) = x^3 - 3x + 1$$

Compute the derivative

$$f'(x) = 3x^2 - 3 = 3(x^2 - 1)$$

On $(0, 1)$, the function $x^2 - 1 < 0$, so $f'(x) < 0$. A strictly decreasing function can cross the x-axis at most once. Hence, there is at most one solution in $(0, 1)$.

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Example 3.32. Verify the Mean Value Theorem for $f(x) = x^2$ on $[1, 3]$.

Solution. The average rate of change is

$$\frac{f(3) - f(1)}{3 - 1} = \frac{9 - 1}{2} = 4$$

Computing the derivative gives

$$f'(x) = 2x$$

and set equal to 4.

$$2c = 4 \implies c = 2$$

Since $2 \in (1, 3)$, the theorem is verified.

The following result is a generalization of the above example.

Theorem 3.33. *Let f be continuous on $[a, b]$ and differentiable on (a, b). Suppose that*

$$f'(x) > 0$$

for all $x \in (a, b)$. Then f can have at most one root in $[a, b]$.

Proof. Assume for a contradiction that f has two distinct roots in $[a, b]$. Then there exist $x_1, x_2 \in [a, b]$ with $x_1 < x_2$ such that $f(x_1) = 0$ and $f(x_2) = 0$.

Apply the Mean Value Theorem to f on $[x_1, x_2]$. Since f is continuous on $[x_1, x_2]$ and differentiable on (x_1, x_2), there exists $c \in (x_1, x_2)$ such that

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

But then $f(x_2) = f(x_1)$, so

$$f'(c) = \frac{0 - 0}{x_2 - x_1} = 0$$

This contradicts the assumption that $f'(x) > 0$ for all $x \in (a, b)$. Therefore, f cannot have two distinct roots in $[a, b]$, and hence has at most one root. □

The Mean Value Theorem can also be used to justify bounded derivatives. Before we mention about this application, we need to introduce the concept of Lipschitz functions.

Definition 3.34. A function f is said to be *Lipschitz* on an interval I if for all $x, y \in I$,

$$|f(x) - f(y)| \leq M|x - y|$$

Theorem 3.35. *Assume f is continuous on $[a, b]$ and differentiable on (a, b). Assume further that there exists $M > 0$ such that*

$$|f'(x)| \leq M$$

for all $x \in (a, b)$. Then f is Lipschitz, i.e. for any $x, y \in [a, b]$,

$$|f(x) - f(y)| \leq M|x - y|$$

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Proof. Let $x, y \in [a, b]$ with $x \neq y$. Without loss of generality, assume $x < y$.

Since f is continuous on $[x, y]$ and differentiable on (x, y), the Mean Value Theorem (Theorem 3.28) applies. Therefore, there exists $c \in (x, y)$ such that

$$f'(c) = \frac{f(y) - f(x)}{y - x}$$

Taking absolute value gives

$$|f'(c)| = \left| \frac{f(y) - f(x)}{y - x} \right| = \frac{|f(y) - f(x)|}{|y - x|}$$

By the assumption $|f'(c)| \leq M$, we obtain

$$\frac{|f(y) - f(x)|}{|y - x|} \leq M$$

and thus, multiplying both sides by $|y - x|$,

$$|f(y) - f(x)| \leq M|y - x|$$

Since $|f(y) - f(x)| = |f(x) - f(y)|$ and $|y - x| = |x - y|$, the desired inequality follows for all $x, y \in [a, b]$. □

This result shows that if a derivative is bounded, the function cannot change too fast. It is a rigorous way to justify maximum slope controls maximum change.

3.4 The First Derivative Test

In Section 3.1, we learned that if a function has a local maximum or minimum at an interior point and is differentiable there, then the derivative must be zero. However, we also saw that the condition

$$f'(c) = 0$$

is only a necessary condition, not a sufficient one.

In Section 3.3, the Mean Value Theorem revealed something deeper: derivatives control the global behaviour of functions. In particular, we proved that if the derivative is always positive on an interval, then the function cannot have more than one root there. This suggests a more general principle: the sign of $f'(x)$ determines whether f is increasing or decreasing.

This section makes the precise idea. We prove that

- If $f'(x) > 0$ on an interval, then f is increasing.
- If $f'(x) < 0$ on an interval, then f is decreasing.

Using these results, we classify critical points by analyzing how the sign of the derivative changes around them. This leads to the First Derivative Test.

Definition 3.36. Let f be defined on an interval I.

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1. f is *increasing* on I if for all $x_1, x_2 \in I$ with $x_1 < x_2$,
$$f(x_1) \leq f(x_2)$$
2. f is *strictly increasing* if for all $x_1, x_2 \in I$ with $x_1 < x_2$,
$$f(x_1) < f(x_2)$$
3. f is *decreasing* if for all $x_1, x_2 \in I$ with $x_1 < x_2$,
$$f(x_1) \geq f(x_2)$$
4. f is *strictly decreasing* if for all $x_1, x_2 \in I$ with $x_1 < x_2$,
$$f(x_1) > f(x_2)$$

Theorem 3.37. *Let f be continuous on $[a, b]$ and differentiable on (a, b).*

1. *If $f'(x) > 0$ for all $x \in (a, b)$, then f is strictly increasing on $[a, b]$.*
2. *If $f'(x) < 0$ for all $x \in (a, b)$, then f is strictly decreasing on $[a, b]$.*

Proof. We prove part (1), since the proof of part (2) is analogous.

Let $x_1, x_2 \in [a, b]$ with $x_1 < x_2$. Since f is continuous on $[x_1, x_2]$ and differentiable on (x_1, x_2), the Mean Value Theorem (Theorem 3.28) applies. Thus, there exists $c \in (x_1, x_2)$ such that
$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$
By the hypothesis $f'(c) > 0$ and $x_2 - x_1 > 0$. Therefore,
$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} > 0$$
which implies
$$f(x_2) - f(x_1) > 0$$
Thus,
$$f(x_2) > f(x_1)$$
Hence, f is strictly increasing. □

Theorem 3.38 (**The First Derivative Test**). *Let f be continuous near c and differentiable on an interval around c except possibly at c. Suppose c is a critical number of f.*

1. *If $f'(x)$ changes from positive to negative as x passes through c, then f has a local maximum at c.*
2. *If $f'(x)$ changes from negative to positive as x passes through c, then f has a local minimum at c.*
3. *If $f'(x)$ does not change sign, then f has no local extremum at c.*

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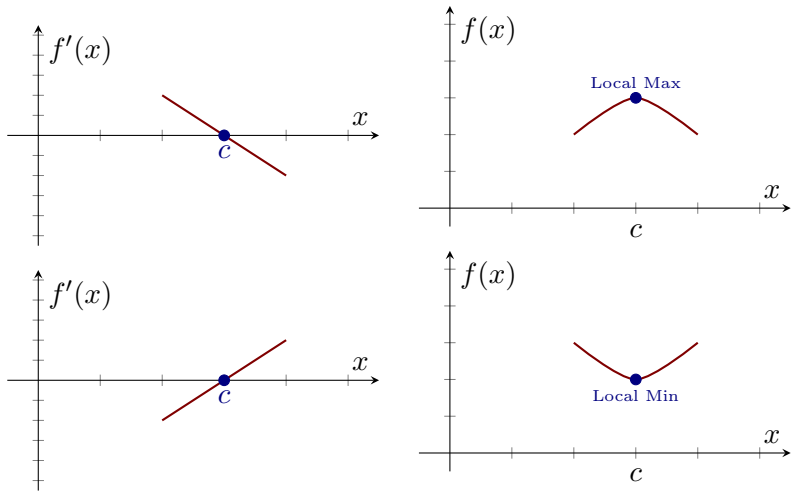

Proof. We will prove (1) as (2) is analogous.

Suppose $f'(x) > 0$ for x near c with $x < c$, and $f'(x) < 0$ for x near c with $x > c$. Then by Theorem 3.37, f is increasing to the left of c, and f is decreasing to the right of c. Thus, for x sufficiently close to c, $f(x) \leq f(c)$. Therefore, $f(c)$ is a local maximum.

Part (3) follows directly from Theorem 3.37. □

Let f be a function and let c be a critical number. We provide a procedure for classifying the critical points using the First Derivative Test (Theorem 3.38).

1. Find the critical numbers.
   - Compute $f'(x)$.
   - Solve $f'(x) = 0$ and also include any x in the domain where $f'(x)$ does not exist.
   - Keep only those values that are in the domain.
2. Set up a sign chart for $f'(x)$.
   - Mark the critical numbers on a number line or chart.
   - These points split the number line into intervals.
3. Determine the sign of $f'(x)$ on each interval.
   - For each interval, choose a test point x and compute the sign of $f'(x)$. Use Theorem 3.38 to classify the critical point.

Example 3.39. Find and classify the critical points of

$$f(x) = x^3 - 3x^2 + 2$$

Solution. First compute

$$f'(x) = 3x^2 - 6x = 3x(x - 2)$$

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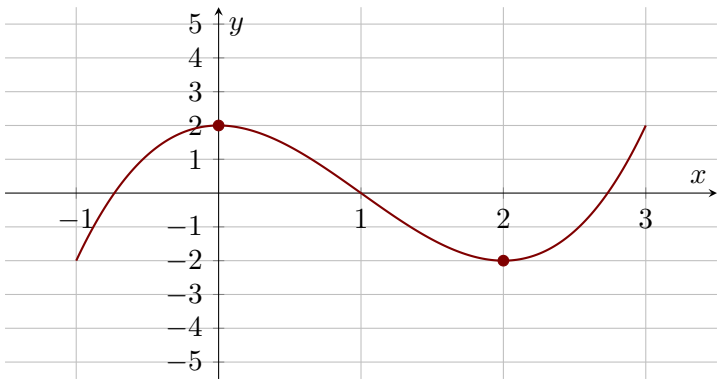
Solving $f'(x) = 0$, we obtain critical numbers $x = 0$ and $x = 2$. We now perform sign analysis using an interval chart. We need intervals $(-\infty, 0)$, $(0, 2)$, and $(2, \infty)$.

	$(-\infty, 0)$	0	$(0, 2)$	2	$(2, \infty)$
x	$-$	0	$+$		$+$
$x - 2$	$-$		$-$	0	$+$
$f'(x)$	$+$	0	$-$	0	$+$
$f(x)$	INC	MAX	DEC	MIN	INC



Example 3.40. Classify the critical points of

$$f(x) = x^3$$

Solution. Computing the derivative gives $f'(x) = 3x^2$ and solving for x gives only $x = 0$. Since $3x^2 \geq 0$ for all x, and does not change sign, there is no local extrumum at $x = 0$.

Example 3.41. Find intervals where

$$f(x) = xe^{-x}$$

is increasing or decreasing.

Solution. Computing the derivative gives

$$f'(x) = e^{-x} - xe^{-x} = e^{-x}(1 - x)$$

Since $e^{-x} > 0$ for all x, the sign of $f'(x)$ depends on $1 - x$, and thus, the critical number is $x = 1$. We now perform sign analysis using an interval chart.

	$(-\infty, 1)$	1	$(1, \infty)$
$1 - x$	$+$	0	$-$
$f'(x)$	$+$	0	$-$
$f(x)$	INC	MAX	DEC

Therefore, the function is increasing on $(-\infty, 1)$ and decreasing on $(1, \infty)$.

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Example 3.42. Classify the critical points of

$$f(x) = (x^3+1)^2(x^2-1)^3$$

Solution. We compute the derivative

$$\begin{aligned} f'(x) &= 2(x^3+1)(3x^2)(x^2-1)^3 + (x^3+1)^2(3)(x^2-1)^2(2x) \\ &= 6x^2(x^3+1)(x^2-1)^3 + 6x(x^3+1)^2(x^2-1)^2 \end{aligned}$$

This is where simplifying further and common factoring becomes very useful. Indeed,

$$\begin{aligned} f'(x) &= 6x(x^3+1)(x^2-1)^2[x(x^2-1)+(x^3+1)] \\ &= 6x(x^3+1)(x^2-1)^2[x^3-x+x^3+1] \\ &= 6x(x^3+1)(x^2-1)^2(2x^3-x+1) \\ &= 6x(x+1)(x^2-x+1)(x+1)^2(x-1)^2(x+1)(2x^2-2x+1) \\ &= 6x(x+1)^4(x-1)^2(x^2-x+1)(2x^2-2x+1) \end{aligned}$$

Solving $f'(x)=0$, we get either $x=0$, $x=-1$, or $x=1$. Furthermore, note that $x^2-x+1>0$ and $2x^2-2x+1>0$ for all x, so the sign of $f'(x)$ depends on $x(x+1)^4(x-1)^2$. We now perform sign analysis using an interval chart.

	$(-\infty,-1)$	-1	$(-1,0)$	0	$(0,1)$	1	$(1,\infty)$
x	$-$		$-$	0	$+$		$+$
$(x+1)^4$	$+$	0	$+$		$+$		$+$
$(x-1)^2$	$+$		$+$		$+$	0	$+$
$f'(x)$	$-$	0	$-$	0	$+$	0	$+$
$f(x)$	DEC	$-$	DEC	MAX	INC	$-$	INC

Therefore, the function is increasing on $(-\infty,-1)\cup(-1,0)$ and decreasing on $(0,1)\cup(1,\infty)$. The critical number $x=-1$ and $x=1$ do not change sign, so there is no local extrema at $x=-1$ and $x=1$. The critical number around $x=0$ changes from decreasing to increasing so $x=0$ is a local maximum.

Example 3.43. Classify the critical points of

$$f(x) = (x^2-1)e^x$$

Solution. Using the product rule,

$$f'(x) = 2xe^x + (x^2-1)e^x = e^x(x^2+2x-1)$$

Since $e^x>0$ for all x, set

$$x^2+2x-1=0$$

Then using the quadratic formula,

$$x = \frac{-2\pm\sqrt{4+4}}{2} = -1\pm\sqrt{2}$$

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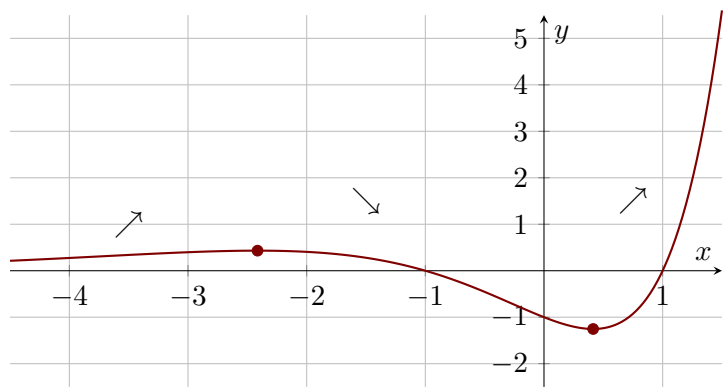
Because $e^x > 0$, the sign depends only on the quadratic. Rewrite the quadratic using the roots found

$$x^2 + 2x - 1 = (x + 1 - \sqrt{2})(x + 1 + \sqrt{2})$$

We now perform sign analysis using an interval chart.

	$(-\infty, -1-\sqrt{2})$	$-1-\sqrt{2}$	$(-1-\sqrt{2}, -1+\sqrt{2})$	$-1+\sqrt{2}$	$(-1+\sqrt{2}, \infty)$
$x+1-\sqrt{2}$	$-$	0	$-$		$+$
$x+1+\sqrt{2}$	$-$		$+$	0	$+$
$f'(x)$	$+$	0	$-$	0	$+$
$f(x)$	$\nearrow$	MAX	$\searrow$	MIN	$\nearrow$

Therefore, the intervals of increase are $(-\infty, -1-\sqrt{2}) \cup (-1+\sqrt{2}, \infty$ and the intervals of decrease is $(-1-\sqrt{2}, -1+\sqrt{2})$. Thus, since around $x = -1-\sqrt{2}$ goes from positive to negative, then $x = -1-\sqrt{2}$ is a local maximum, while since around $x = -1+\sqrt{2}$ goes from negative to positive, then $x = -1+\sqrt{2}$ is a local minimum.



Example 3.44. Classify the critical points of

$$f(x) = \frac{x^2}{x+1}, \quad x \neq -1$$

Solution. Using the quotient rule, we have

$$\begin{aligned} f'(x) &= \frac{2x(x+1) - x^2}{(x+1)^2} \\ &= \frac{2x^2 + 2x - x^2}{(x+1)^2} \\ &= \frac{x^2 + 2x}{(x+1)^2} \\ &= \frac{x(x+2)}{(x+1)^2} \end{aligned}$$

The denominator is never zero except at -1 (which is not in the domain). Thus, set the numerator equal to zero. Then the critical numbers are $x = 0$ or $x = -2$.

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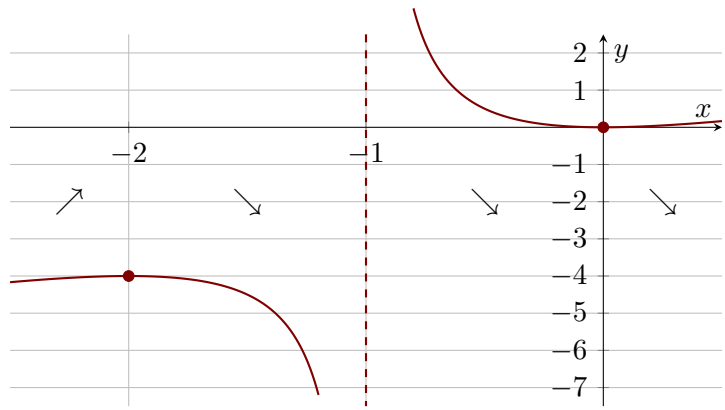
We now perform sign analysis using an interval chart.

	$(-\infty,-2)$	-2	$(-2,-1)$	-1	$(-1,0)$	0	$(0,\infty)$
x	$-$		$-$		$-$	0	$+$
$x+2$	$-$	0	$+$		$+$		$+$
$(x+1)^2$	$+$		$+$	0	$+$		$+$
$f'(x)$	$+$	0	$-$	DNE	$-$	0	$+$
$f(x)$	$\nearrow$	MAX	$\searrow$	VA	$\searrow$	MIN	$\nearrow$

Therefore, the intervals of increase are $(-\infty,-2)\cup(0,\infty)$ and the intervals of decrease are $(-2,-1)\cup(-1,0)$. The local maximum occurs at $x=-2$ and the local minimum occurs at $x=0$.



Example 3.45. Classify the critical points of

$$f(x)=\cos(x^2)$$

on $[0,\pi]$.

Solution. Using the chain rule, we have

$$f(x)=-2x\sin(x^2)$$

Finding the critical numbers, either $x=0$ or $\sin(x^2)=0$. In particular,

$$x^2=k\pi \implies x=\pm\sqrt{k\pi}$$

for $k\in\mathbb{Z}$. Since $0\le x\le\pi$, then we only need $x=\sqrt{k\pi}$ for $k=0,1,2,3$. We now perform sign analysis using an interval chart.

	0	$(0,\sqrt{\pi})$	$\sqrt{\pi}$	$(\sqrt{\pi},\sqrt{2\pi})$	$\sqrt{2\pi}$	$(\sqrt{2\pi},\sqrt{3\pi})$	$\sqrt{3\pi}$	$(\sqrt{3\pi},\pi)$	π
$-x$	0	$-$		$-$		$-$		$-$	
$\sin(x^2)$	0	$+$	0	$-$	0	$+$	0	$-$	0
$f'(x)$	0	$-$	0	$+$	0	$-$	0	$+$	0
$f(x)$	MAX	$\searrow$	MIN	$\nearrow$	MAX	$\searrow$	MIN	$\nearrow$	MAX

Therefore, the intervals of increase are $(\sqrt{\pi},\sqrt{2\pi})\cup(\sqrt{3\pi},\pi)$ and the intervals of decrease are $(0,\sqrt{\pi})\cup(\sqrt{2\pi},\sqrt{3\pi})$. The local maximum occurs at $x=0,\sqrt{2\pi},\pi$ and the local minimum occurs $x=\sqrt{\pi},\sqrt{3\pi}$.

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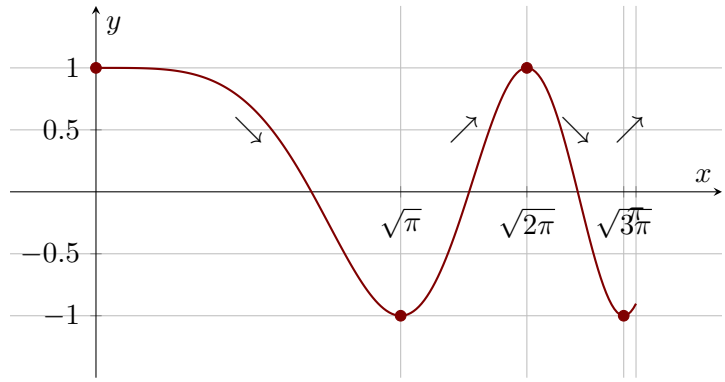


Example 3.46. Classify the critical points of

$$f(x) = x\ln(x), \quad x > 0$$

Solution. Using the product rule,

$$f'(x) = \ln(x) + 1$$

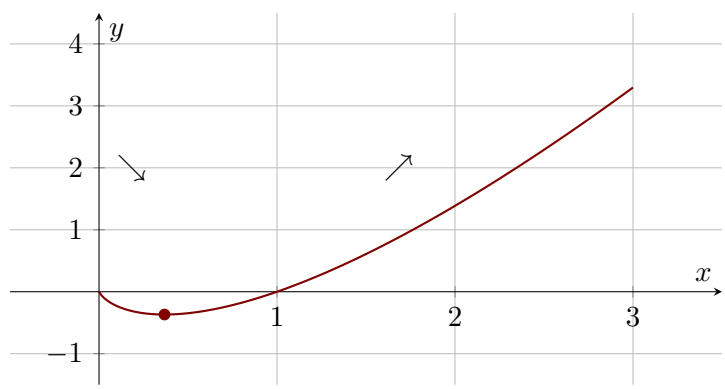
Finding the critical numbers, we solve

$$\ln(x) + 1 = 0 \implies \ln(x) = -1 \implies x = e^{-1}$$

We now perform sign analysis using an interval chart.

	$(0, e^{-1})$	e^{-1}	(e^{-1}, ∞)
$\ln(x) + 1$	$-$	0	$+$
$f'(x)$	$-$	0	$+$
$f(x)$	$\searrow$	MIN	$\nearrow$

Therefore, the interval of increase is (e^{-1}, ∞) and the interval of decrease is $(0, e^{-1})$. The local minimum occurs at $x = e^{-1}$.



3.5 The Second Derivative Test

In Section 3.4, we developed the First Derivative Test, which classifies critical points by analyzing the sign changes of $f'(x)$. While powerful, that method requires constructing a sign chart and testing intervals around each critical number.

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A natural question arises: *Can we classify critical points more directly?* The answer is yes, provided the function is *twice differentiable*. The key idea is geometric:

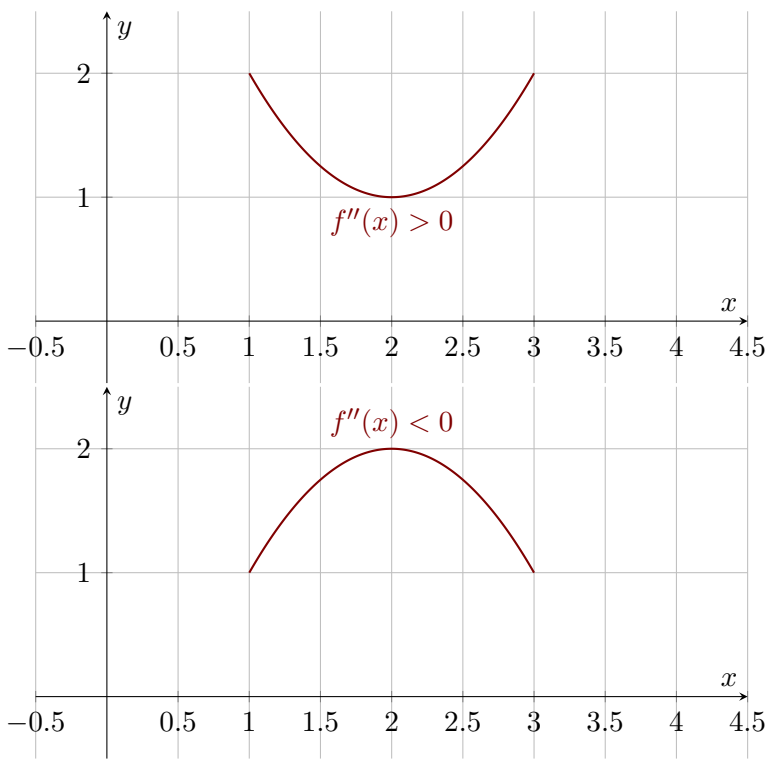
- If a graph bends upward (concave up), it resembles a bowl and tends to have a local minimum.
- If a graph bends downward (concave down), it resembles an upside-down bowl and tends to have a local maximum.

The second derivative measures concavity. The Second Derivative Test uses this information to classify critical points without constructing a full sign chart.

However, like the First Derivative Test, it is not always conclusive. Understanding when and why it works is essential.

Definition 3.47. Let f be twice differentiable on an interval I.

1. f is *concave up* on I if $f''(x) > 0$ for all $x \in I$.
2. f is *concave down* on I if $f''(x) < 0$ for all $x \in I$.



Theorem 3.48. *Let f be twice differentiable on an interval I.*

1. *If $f''(x) > 0$ on I, then $f'(x)$ is increasing on I.*
2. *If $f''(x) < 0$ on I, then $f'(x)$ is decreasing on I.*

Proof. Since $f''(x) = (f'(x))'$, the result follows directly from Theorem 3.37 applied to f'. □

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Theorem 3.49 (**Second Derivative Test**). *Let f be twice differentiable near c, and suppose*

$$f'(c) = 0$$

1. *If $f''(c) > 0$, then f has a local maximum at c.*
2. *If $f''(c) < 0$, then f has a local maximum at c.*
3. *If $f''(c) = 0$, the test is inconclusive.*

Proof. Suppose $f'(c) = 0$. We consider the following cases.

Case 1: If $f''(c) > 0$. By continuity of f'', there exists an interval around c where $f''(x) > 0$. By Theorem 3.48, $f'(x)$ is increasing near c. Since $f'(c) = 0$ and f' is increasing,

- For $x < c$ sufficiently close to c, $f'(x) < 0$.
- For $x > c$ sufficiently close to c, $f'(x) > 0$.

Thus, the derivative changes from negative to positive. By the First Derivative Test (Theorem 3.38), f has a local minimum at c.

Case 2: If $f''(c) < 0$. Similarly, $f''(x) < 0$ near c, so $f'(x)$ is decreasing. Since $f'(c) = 0$

- For $x < c$, $f'(x) > 0$.
- For $x > c$, $f'(x) < 0$.

Thus, the derivative changes from positive to negative. Hence, f has a local maximum at c.

Case 3: If $f''(c) = 0$. No conclusion can be drawn. □

Let f be twice differentiable. To apply the Second Derivative Test,

1. Compute $f'(x)$.
2. Solve $f'(x) = 0$ to find critical numbers.
3. Compute $f''(x)$.
4. Evaluate $f''(c)$ at each critical number. Then decide
   - If $f''(c) > 0$, then f has a local minimum at $x = c$.
   - If $f''(c) < 0$, then f has a local maximum at $x = c$.
   - If $f''(c) = 0$, then no conclusion can be made.

Example 3.50. Classify the critical points of

$$f(x) = x^3 - 3x^2 + 2$$

Solution. We computed the derivative to be

$$f'(x) = 3x(x - 2)$$

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and the critical numbers to be $x = 0$ or $x = 2$. Computing the second derivative

$$f''(x) = 6x - 6$$

Evaluating at each critical number,

x	$f''(x)$	$f(x)$
0	$-$	MAX
2	$+$	MIN

This example confirms our sign analysis from the previous section, where we have determined that if a function goes from $\nearrow$ to $\searrow$, then the critical number is a local maximum, or if a function goes from $\searrow$ to $\nearrow$, then the critical number is a local minimum. The second derivative test is a shortcut, if we simply care about what the local extrema actually becomes.

Example 3.51. Classify the critical point of

$$f(x) = x\ln(x)$$

Solution. We have found in a previous example that $f'(x) = \ln(x) + 1$ and the critical number occur at $x = e^{-1}$. Computing the second derivative, we have

$$f''(x) = \frac{1}{x}$$

Since

$$f''(e^{-1}) = e > 0$$

then there exists a local minimum at $x = e^{-1}$.

Example 3.52. Classify the critical point of

$$f(x) = x^4$$

Solution. Computing the first derivative, we have

$$f'(x) = 4x^3$$

Thus, the critical number is $x = 0$. Computing the second derivative, we get

$$f''(x) = 12x^2$$

However, evaluating at the critical number

$$f''(x) = 0$$

Thus, the Second Derivative Test is inconclusive.

Instead, using the First Derivative Test,

- If $x < 0$, we have $f'(x) < 0$, and
- If $x > 0$, we have $f'(x) > 0$.

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Thus, a local minimum exists at $x = 0$.

Up to this point, we used the second derivative to classify critical points. We now pause to understand the geometric meaning of $f''(x)$. The second derivative describes how the first derivative changes, and therefore how the slope of the tangent line behaves.

Proposition 3.53. *Let f be twice differentiable on an interval I.*

1. *If $f''(x) > 0$ on I, then the graph of f lies above its tangent on I.*
2. *If $f''x) < 0$ on I, then the graph of f lies below its tangent lies on I.*

Concavity can change from up to down or down to up.

Definition 3.54. A point c is called an *infection point* of f if

1. f is continuous at c, and
2. The concavity of f changes at c.

If f has an inflection point at c and f'' exists near c, then either $f''(c) = 0$ or $f''(c)$ does not exist. However, this condition is not sufficient.

Example 3.55. Determine the intervals of concavity and inflection points for

$$f(x) = x^3$$

Solution. The second derivative is

$$f''(x) = 6x$$

and thus, the inflection points are located at $x = 0$. We now use an interval chart to perform sign analysis.

	$(-\infty, 0)$	0	$(0, \infty)$
x	$-$	0	$+$
$f''(x)$	$-$	0	$+$
$f(x)$	$\frown$	INF	$\smile$

Therefore, the interval of concave up is $(0, \infty)$ and the interval of concave down is $(-\infty, 0)$. The point of infection occurs at $x = 0$.

Example 3.56. Determine whether $x = 0$ is an inflection point for

$$f(x) = x^4$$

Solution. We computed the second derivative earlier to be

$$f''(x) = 12x^2$$

However, $12x^2 \geq 0$ for all x, so the function is concave up everywhere. Thus, concavity does not change at 0, so $x = 0$ is not an inflection point.

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Up to this point, we have developed several powerful tools

- The First Derivative Test (Theorem 3.38), which determines where a function is increasing or decreasing and classifies critical points.
- The Second Derivative Test (Theorem 3.49), which uses concavity to classify critical points.
- The concept of inflection points, where concavity is changing.

Each of these ideas describe a different geometric aspect of a function.

- $f'(x)$ controls slope and monotonicity.
- $f''(x)$ controls concavity and curvature.

However in practice, we rarely use these tools separately. Instead, we combine them to obtain a complete qualitative understanding of the graph.

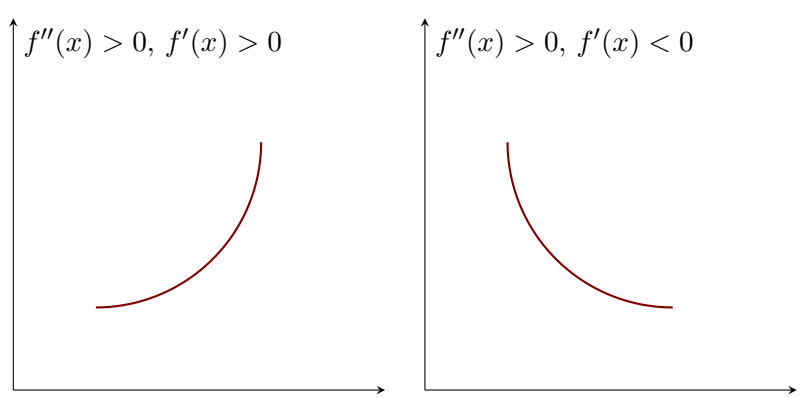
Before moving to full curve sketching in Section 3.6, we now analyze a function using

- Critical numbers,
- The First Derivative Test,
- The Second Derivative Test,
- Intervals of increase and decrease,
- Intervals of concavity,
- Points of inflection.

The goal is not yet to draw the graph, but to understand its shape purely from derivative information. We do the steps in the following order: Given a twice differentiable function f,

1. Compute $f'(x)$ and determine its critical numbers.
2. Compute $f''(x)$ and determine its points of inflection.
3. Create a big sign chart that covers all critical points, inflection points, and includes $f''(x)$, $f'(x)$, and $f(x)$.

On each subinterval, the graph should look like one of the four shapes



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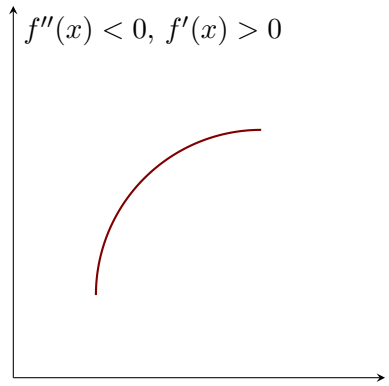

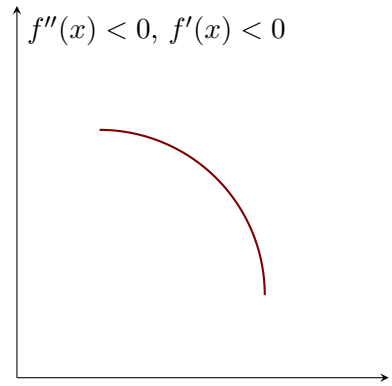

We will adapt the following notation for our sign analysis

$f'(x)$	$f''(x)$	$f(x)$	Meaning
$+$	$+$	$\nearrow \smile$	Increasing, Concave Up
$+$	$-$	$\nearrow \frown$	Increasing, Concave Down
$-$	$+$	$\searrow \smile$	Decreasing, Concave Up
$-$	$-$	$\searrow \frown$	Decreasing, Concave Down

Example 3.57. Analyze the function

$$f(x) = x^4 - 4x^3$$

by finding its critical numbers, points of inflection, intervals of increase/decrease, and intervals of concavity.

Solution. Computing the first derivative of f we get

$$f'(x) = 4x^3 - 12x^2 = 4x^2(x-3)$$

Finding the critical numbers we have $x = 0$ and $x = 3$.

Computing the second derivative of f, we get

$$f''(x) = 12x^2 - 24x = 12x(x-2)$$

Finding the points of inflection we have $x = 0$ and $x = 2$.

We now analyze the function by performing sign analysis. We need the intervals $(-\infty, 0)$, $(0, 2)$, $(2, 3)$, and $(3, \infty)$. This covers both the critical numbers and points of inflection.

	$(-\infty, 0)$	0	$(0, 2)$	2	$(2, 3)$	3	$(3, \infty)$
x^2	$+$	0	$+$		$+$		$+$
$x - 3$	$-$		$-$		$-$	0	$+$
$f'(x)$	$-$	0	$-$		$-$	0	$+$
x	$-$	0	$+$		$+$		$+$
$x - 2$	$-$		$-$	0	$+$		$+$
$f''(x)$	$+$	0	$-$	0	$+$		$+$
$f(x)$	$\searrow \smile$	INF	$\searrow \frown$	INF	$\searrow \smile$	MIN	$\nearrow \smile$

Therefore

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- Interval(s) of increase: $(3, \infty)$
- Interval(s) of decrease: $(-\infty, 3)$
- Interval(s) of concave up: $(-\infty, 0) \cup (2, \infty)$
- Interval(s) of concave down: $(0, 2)$
- The minimum occurs at $(3, f(3))$, where $f(3) = -27$.
- The points of inflection occur at $(0, 0)$ and $(2, f(2))$, where $f(2) = -16$.

Example 3.58. Analyze the function

$$f(x) = x^{\frac{1}{3}} - 2x^{\frac{2}{3}}$$

by finding its critical numbers, points of inflection, intervals of increase/decrease, and intervals of concavity.

Solution. Computing the first derivative of f, we get

$$f'(x) = \frac{1}{3}x^{-\frac{2}{3}} - \frac{4}{3}x^{-\frac{1}{3}} = \frac{1}{3}x^{-\frac{2}{3}}(1 - 4x^{\frac{1}{3}})$$

Since $\frac{1}{3}x^{-\frac{2}{3}} \neq 0$ (but it is undefined at $x = 0$), the zeros come from

$$1 - 4x^{\frac{1}{3}} = 0 \implies x^{\frac{1}{3}} = \frac{1}{4} \implies x = \frac{1}{64}$$

Thus, the critical numbers are $x = 0$ and $x = \frac{1}{64}$.

Computing the second derivative of f, we get

$$f''(x) = -\frac{2}{9}x^{-\frac{5}{3}} + \frac{4}{9}x^{-\frac{4}{3}} = -\frac{2}{9}x^{-\frac{5}{3}}(1 - 2x^{\frac{1}{3}})$$

Similarly, since $-\frac{2}{9}x^{-\frac{5}{3}} \neq 0$ (but it is undefined at $x = 0$), the zeros come from

$$1 - 2x^{\frac{1}{3}} = 0 \implies x^{\frac{1}{3}} = \frac{1}{2} \implies x = \frac{1}{8}$$

Thus, the points of inflection are $x = 0$ and $x = \frac{1}{8}$.

We now analyze the function by performing sign analysis. We need the intervals $(-\infty, 0)$, $(0, \frac{1}{64})$, $(\frac{1}{64}, \frac{1}{8})$ and $(\frac{1}{8}, \infty)$. This covers both the critical numbers and points of inflection.

	$(-\infty, 0)$	0	$(0, \frac{1}{64})$	$\frac{1}{64}$	$(\frac{1}{64}, \frac{1}{8})$	$\frac{1}{8}$	$(\frac{1}{8}, \infty)$
$x^{-\frac{2}{3}}$	$+$	DNE	$+$		$+$		$+$
$1 - 4x^{\frac{1}{3}}$	$+$		$+$	0	$-$		$-$
$f'(x)$	$+$	DNE	$+$	0	$-$		$-$
$-x^{-\frac{5}{3}}$	$+$	DNE	$-$		$-$		$-$
$1 - 2x^{\frac{1}{3}}$	$+$		$+$		$+$	0	$-$
$f''(x)$	$+$	DNE	$-$		$-$	0	$+$
$f(x)$	$\nearrow \smile$	INF	$\nearrow \frown$	MAX	$\searrow \frown$	INF	$\searrow \smile$

Therefore,

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- Interval(s) of increase: $(-\infty, \frac{1}{64})$
- Interval(s) of decrease: $(\frac{1}{64}, \infty)$
- Interval(s) of concave up: $(-\infty, 0) \cup (\frac{1}{8}, \infty)$
- Intervals of concave down: $(0, \frac{1}{8})$
- The local maximum occurs at and $(\frac{1}{64}, f(\frac{1}{64}))$, where $f(\frac{1}{64}) = \frac{1}{8}$
- The points of inflection: $(0, 0)$ and $(\frac{1}{8}, f(\frac{1}{8}))$ where $f(\frac{1}{8}) = 0$.

3.6 Curve Sketching

Up to this point, we developed several tools for analyzing functions:

- First derivatives determine intervals of increase and decrease and allow us to classify critical points.
- Second derivatives determine concavity and identify points of inflection.
- Limits allow us to study asymptotic behaviour of functions.

In practice, these ideas can be combined to obtain a detailed understanding of the shape of a graph, without relying on graphing technology. This process is called curve sketching.

The goal of curve sketching is not to compute many points, but rather to use calculus to determine the qualitative features of a function. In particular, we want to identify:

- Intercepts,
- Asymptotes,
- Critical points,
- Points of infection,
- Intervals of increase and decrease,
- Intervals of concavity.

These features together determine the overall behaviour of the graph.

Rational functions provide particularly useful examples because their graphs exhibit a rich variety of behaviours, including asymptotes and multiple extrema. In this section, we develop a systematic method for sketching such graphs.

The following procedure provides a structured method for sketching the graph of a rational function.

Let $f(x) = \frac{p(x)}{q(x)}$ be a rational function.

1. Find the x-intercepts:
   - For x-intercepts, solve $p(x) = 0$, provided $q(x) \neq 0$.
   - For y-intercept, compute $f(0)$, if defined.

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2. Find the vertical asymptotes: Vertical asymptotes occur at values where $q(x) = 0$, but $p(x) \neq 0$. For each vertical asymptote $x = a$, evaluate

$$\lim_{x \to a^-} f(x) \quad \text{and} \quad \lim_{x \to a^+} f(x)$$

to determine the behaviour of the graph near the asymptote.

3. Find the horizontal asymptote: Compute the limits

$$\lim_{x \to \infty} f(x), \quad \lim_{x \to -\infty} f(x)$$

These determine the horizontal asymptotes. After identifying a horizontal asymptote $y = L$, check whether the graph crosses it by solving $f(x) = L$.

4. Compute the first derivative: Find $f'(x)$ and determine the critical numbers by solving $f'(x) = 0$ or identifying where $f'(x)$ is undefined while $f(x)$ is defined.

5. Compute the second derivative: Find $f''(x)$ and determine the candidate points of inflection by solving $f''(x) = 0$ or where $f''(x)$ is undefined.

6. Construct a sign chart: Combine the information from f' and f'' to determine:

   - increasing/decreasing behaviour,
   - concavity,
   - extrema,
   - inflection points.

7. Sketch the graph: Use the information obtained from (1)-(6) to draw a graph that reflects the qualitative behaviour of the function.

Example 3.59. Sketch the graph of

$$f(x) = \frac{x}{x^2 - 1}$$

using curve sketching techniques.

Solution. We follow steps (1)-(7) as outlined above.

1. To find the x-intercepts, we solve the numerator equal to zero, in which case, we get $x = 0$.

   To find the y-intercepts, compute

$$f(0) = \frac{0}{0^2 - 1} = 0$$

2. The vertical asymptotes occur whenever

$$x^2 - 1 = 0$$

   In this case, $x = \pm 1$ are the vertical asymptotes. Note by factoring the denominator, we get

$$f(x) = \frac{x}{(x-1)(x+1)}$$

   So we analyze the behaviour as $x \to \pm 1^{\pm}$

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- As $x \to 1^+$,

$$\lim_{x\to 1^+} \frac{x}{(x-1)(x+1)} = \frac{1^+}{(1^+-1)(1^++1)} = \frac{(+)}{(+)(+)} = \infty$$

- As $x \to 1^-$

$$\lim_{x\to 1^-} \frac{x}{(x-1)(x+1)} = \frac{1^-}{(1^--1)(1^++1)} = \frac{(+)}{(-)(+)} = -\infty$$

- As $x \to -1^+$,

$$\lim_{x\to -1^+} \frac{x}{(x-1)(x+1)} = \frac{-1^+}{(-1^+-1)(-1^++1)} = \frac{(-)}{(-)(+)} = \infty$$

- As $x \to -1^-$,

$$\lim_{x\to -1^-} \frac{x}{(x-1)(x+1)} = \frac{-1^-}{(-1^--1)(-1^++1)} = \frac{(-)}{(-)(-)} = -\infty$$

3. Since the degree of the numerator is smaller than the degree of the numerator

$$\lim_{x\to\pm\infty} f(x) = 0$$

Thus, $y = 0$ is a horizontal asymptote. To determine crossing, we solve $f(x) = 0$, but $x = 0$ is the only value.

We analyze the behaviour as $x \to \pm\infty$.

- As $x \to \infty$, say we take $x = 1000$, then

$$\frac{1000}{1000^2 - 1} = 0^+$$

so as $x \to \infty$, $f(x) \to 0^+$.

- As $x \to -\infty$, say we take $x = -1000$, then

$$\frac{-1000}{(-1000)^2 - 1} = 0^-$$

so as $x \to -\infty$, $f(x) \to 0^-$.

4. Using the quotient rule, we compute the first derivative

$$f'(x) = \frac{(x^2-1) - 2x^2}{(x^2-1)^2} = \frac{-x^2-1}{(x^2-1)} = -\frac{x^2+1}{(x^2-1)^2}$$

The critical numbers are found by solving the equation $x^2 + 1 = 0$, which has no real solutions. So there are no critical numbers. However, also note that there are vertical asymptotes at $x = \pm 1$.

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5. Differentiating again gives

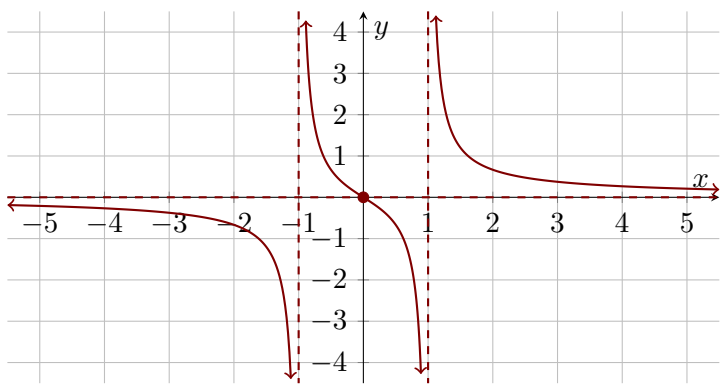
$$f''(x) = \frac{2x(x^2+3)}{(x^2-1)^3}$$

The points of inflection are found by solving the equation $2x(x^2+3) = 0$ which the only solution is $x = 0$.

6. The key points are $x = \pm 1$ $x = 0$, so we perform sign analysis of f' and f'' on the intervals $(-\infty, -1)$ $(-1, 0)$, $(0, 1)$, and $(1, \infty)$.

	$(-\infty,-1)$	-1	$(-1,0)$	0	$(0,1)$	1	$(1,\infty)$
$-(x^2+1)$	$-$		$-$		$-$		$-$
$(x^2-1)^2$	$+$	VA	$+$		$+$	VA	$+$
$f'(x)$	$-$	VA	$-$		$-$	VA	$-$
$2x$	$-$		$-$	0	$+$		$+$
x^2+3	$+$		$+$		$+$		$+$
$(x^2-1)^3$	$+$	VA	$-$		$-$	VA	$+$
$f''(x)$	$-$	VA	$+$	0	$-$	VA	$+$
$f(x)$	$\searrow \frown$	VA	$\searrow \smile$	INF	$\searrow \frown$	VA	$\searrow \smile$

7. Using the information from Steps 1-6, we obtain the following sketch of the graph.



3.7 Indeterminate Forms and L'Hôpital's Rule

In previous sections, we learned several techniques for evaluating limits, including laws, algebraic simplification, and trigonometric limit identities. In many cases, these methods allow us to compute limits directly. However, some limits arise in which direct substitution produces expressions that appear meaningless or ambiguous.

For example, consider the limit

$$\lim_{x \to 0}$$

Direct substitution yields

$$\frac{0}{0}$$

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which is not a defined number. Expressions of this type are called *indeterminate forms* because the expression does not determine the value of the limit. In some cases, the limit may exist and be finite, while in other cases it may diverge.

Indeterminate forms commonly arise when studying limits involving rational functions, exponential functions, logarithms, and trigonometric functions. To evaluate such limits efficiently, we introduce a powerful method known as L'Hôpital's Rule. This rule allows us to replace a difficult limit with a simpler one involving derivatives.

In this section, we first study the concept of indeterminate forms, and then develop L'Hôpital's Rule as a practical technique for evaluating limits.

Definition 3.60. An expression obtained from direct substitution in a limit is called an *indeterminate form* if it does not determine the value of the limit.

The most common indeterminate forms are

$$\frac{0}{0}, \quad \frac{\infty}{\infty}$$

These forms indicate that further analysis is required to determine the limit.

Example 3.61. Evaluate

$$\lim_{x\to 0} \frac{x^2}{x}$$

Solution. Direct substitution gives

$$\frac{0}{0}$$

which is an indeterminate form. However,

$$\frac{x^2}{x} = x, \quad x \neq 0$$

Thus,

$$\lim_{x\to 0} \frac{x^2}{x} = \lim_{x\to 0} x = 0$$

This example shows that the form $\frac{0}{0}$ does not determine the value of the limit.

Example 3.62. Evaluate

$$\lim_{x\to 0} \frac{x}{x}$$

Solution. Direct substitution again gives

$$\frac{0}{0}$$

However,

$$\frac{x}{x} = 1, \quad x \neq 0$$

Thus,

$$\lim_{x\to 0} \frac{x}{x} = 1$$

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Thus, the same indeterminate form can produce different limit values.

L'Hôpital's Rule provides a systematic method for evaluating limits that yield the forms $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

Theorem 3.63 (L'Hôpital's Rule). *Let f and g be differentiable functions on an open interval containing a, except possibly at a, and suppose*

$$\lim_{x\to a} f(x) = 0, \quad \lim_{x\to a} g(x) = 0$$

or

$$\lim_{x\to a} f(x) = \pm\infty, \quad \lim_{x\to a} g(x) = \pm\infty$$

If $g'(x) \neq 0$ near a and

$$\lim_{x\to a} \frac{f'(x)}{g'(x)}$$

exists or is infinite, then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

L'Hôpital's Rule replaces the original limit with a new limit involving derivatives. Open the new limit is easier to compute.

The proof relies on the Mean Value Theorem (Theorem 3.28).

Lemma 3.64 (Cauchy Mean Value Theorem). *Let f and g be continuous on $[a, b]$ and differentiable on (a, b). Assume $g(b) \neq g(a)$. Then there exists $c \in (a, b)$ such that*

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c)$$

If in addition, $g'(c) \neq 0$, then

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

Proof. Define an auxiliary function

$$H(x) = (f(b) - f(a))g(x) - (g(b) - g(a))f(x)$$

Since f and g are continuous on $[a, b]$ and differentiable on (a, b), so is H. Compute $H(a)$ and $H(b)$

$$H(a) = (f(b) - f(a))g(a) - (g(b) - g(a))f(a)$$
$$H(b) = (f(b) - f(a))g(b) - (g(b) - g(a))f(b)$$

Now expand $H(b) - H(a)$

$$H(b) - H(a) = (f(b) - f(a))g(b) - (g(b) - g(a))f(b) - (f(b) - f(a))g(a) + (g(b) - g(a))f(a) = 0$$

Thus, $H(a) = H(b)$. By Rolle's Theorem (Theorem 3.27), there exists $c \in (a, b)$ such that $H'(c) = 0$.

Differentiate H to get

$$H'(x) = (f(b) - f(a))g'(x) - (g(b) - g(a))f'(x)$$

So $H'(c) = 0$ gives

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c)$$

which is exactly the desired conclusion. □

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Proof of Theorem 3.63. We prove the theorem in two cases.

Case 1: If $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{0}{0}$. We prove the right-hand limit; the left-hand limit is analogous. Assume $x \to a^+$. Let $x > a$ sufficiently close to a. Apply the Cauchy Mean Value Theorem to f and g on the interval $[a, x]$ (note that f and g are continuous on $[a, x]$ and differentiable on (a, x)). Since $g'(t) \neq 0$ for $t \in (a, x)$, there exists $c \in (a, x)$ such that

$$\frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c)}{g'(c)}$$

Because $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$, we may (and do) define $f(a) = 0$ and $g(a) = 0$ without changing the limit. Then the previous equation becomes

$$\frac{f(x)}{g(x)} = \frac{f'(c)}{g'(c)}$$

Now let $x \to a^+$. Since $c \in (a, x)$, we necessarily have $c \to a^+$ as $x \to a^+$. Therefore,

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = \lim_{x\to a^+} \frac{f'(c)}{g'(c)} = \lim_{t\to a^+} \frac{f'(t)}{g'(t)}$$

This is exactly the desired conclusion.

Case 2: If $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{\infty}{\infty}$. Fix $x \in (a, b)$. Choose $t \in (a, x)$ so that $a < t < x$. Apply the Cauchy Mean Value Theorem (Lemma 3.64) to f and g on the interval $[t, x]$. Since f and g are differentiable on (a, b), they are continuous on $[t, x]$ and differentiable on (t, x), and $g'(u) \neq 0$ on (t, x). Hence, there exists $c \in (t, x)$ such that

$$\frac{f(x) - f(t)}{g(x) - g(t)} = \frac{f'(c)}{g'(c)} \tag{1}$$

Now we let $t \to a^+$ while keeping x fixed.

Since $g(t) \to \pm\infty$ as $x \to a^+$, we have $|g(t)| \to \infty$. Divide the numerator and denominator of the left-side of (1) by $g(t)$

$$\frac{f(x) - f(t)}{g(x) - g(t)} = \frac{\frac{f(x)}{g(t)} - \frac{f(t)}{g(t)}}{\frac{g(x)}{g(t)} - 1}$$

As $t \to a^+$, we have

$$\frac{f(x)}{g(t)} \to 0, \quad \frac{g(x)}{g(t)} \to 0$$

because $f(x)$ and $g(x)$ are fixed constants while $g(t) \to \infty$.

Thus, if we define

$$Q(t) = \frac{f(x) - f(t)}{g(x) - g(t)}$$

then

$$\lim_{t\to a^+} Q(t) = \lim_{t\to a^+} \frac{0 - \frac{f(t)}{g(t)}}{0 - 1} = \lim_{t\to a^+} \frac{f(t)}{g(t)}$$

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So we have shown that

$$\lim_{t\to a^+} \frac{f(x)-f(t)}{g(x)-g(t)} = \lim_{t\to a^+} \frac{f(t)}{g(t)} \tag{2}$$

Return to (1)

$$\frac{f(x)-f(t)}{g(x)-f(t)} = \frac{f'(c)}{g'(c)}$$

for some $c \in (t, x)$. Now send $t \to a^+$. Since $c \in (t, x)$ and $t \to a^+$ with x fixed, we have $c \to a^+$. Therefore,

$$\lim_{t\to a^+} \frac{f'(c)}{g'(c)} = L$$

Combining with (2) gives

$$\lim_{t\to a^+} \frac{f(t)}{g(t)} = \lim_{t\to a^+} \frac{f(x)-f(t)}{g(x)-g(t)} = \lim_{t\to a^+} \frac{f'(c)}{g'(c)} = L$$

Since the variable t is just a dummy variable approaching a^+, this proves

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = L$$

as desired. □

Example 3.65. Evaluate

$$\lim_{x\to 0} \frac{\sin(x)}{x}$$

Solution. Direct substitution gives

$$\frac{0}{0}$$

Applying L'Hôpital's Rule (Theorem 3.63)

$$\lim_{x\to 0} \frac{\sin(x)}{x} = \lim_{x\to 0} \frac{\cos(x)}{1} = 1$$

Example 3.66. Evaluate

$$\lim_{x\to 0} \frac{1-\cos(x)}{x^2}$$

Solution. Substitution gives

$$\frac{0}{0}$$

Apply L'Hôpital's Rule (Theorem 3.63)

$$\lim_{x\to 0} \frac{\sin(x)}{2x}$$

This is still indeterminate. Apply L'Hôpital's Rule (Theorem 3.63) again

$$\lim_{x\to 0} \frac{\cos(x)}{2} = \frac{1}{2}$$

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Example 3.67. Evaluate

$$\lim_{x\to\infty} \frac{\ln(x)}{x}$$

Solution. Direct substitution gives the form

$$\frac{\infty}{\infty}$$

Applying L'Hôpital's Rule (Theorem 3.63)

$$\lim_{x\to\infty} \frac{1/x}{1} = \lim_{x\to\infty} \frac{1}{x} = 0$$

Example 3.68. Evaluate

$$\lim_{x\to\infty} \frac{x}{e^x}$$

Solution. Substitution gives

$$\frac{\infty}{\infty}$$

Apply L'Hôpital's Rule (Theorem 3.63)

$$\lim_{x\to\infty} \frac{1}{e^x} = 0$$

L'Hôpital's Rule (Theorem 3.63) can only be used for limits producing

$$\frac{0}{0} \quad \text{or} \quad \frac{\infty}{\infty}$$

If the limit produces a different form, the rule cannot be applied directly.

Example 3.69. Evaluate

$$\lim_{x\to 0} \frac{1}{x}$$

Solution. The expression does not produce an indeterminate form. The limit diverges. Thus, the limit does not exist.

Example 3.70. Evaluate

$$\lim_{x\to 0^+} x\ln(x)$$

Solution. Note that direct substitution does give us an indeterminate form, but as

$$0 \cdot (-\infty)$$

L'Hôpital's Rule (Theorem 3.63) does not work on products. So we need to change the limit expression so that we have a quotient. Indeed, observe that

$$x\ln(x) = \frac{\ln(x)}{\frac{1}{x}}$$

Now observe by direct substitution, we get

$$\frac{\infty}{\infty}$$

So by L'Hôpital's Rule,

$$\lim_{x\to 0^+} \frac{\ln(x)}{\frac{1}{x}} = \lim_{x\to 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x\to 0^+} (-x) = 0$$

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3.8 Optimization Problems

In many applications in calculus, we are interested in determining the best possible value of a quantity. For example, we may want to

- maximize the area of a region,
- minimize the cost of production,
- determine the maximum height of a projectile,
- find the dimensions of a container with minimal material.

Problems of this type are called *optimization problems*. In such problems, we seek either the largest or smallest possible value of a quantity subject to certain constraints.

The derivative plays a fundamental role in solving optimization problems. Recall from previous sections that local extrema occur at critical numbers, where the derivative is zero or undefined. By expressing the quantity we wish to optimize as a function and analyzing its critical numbers, we can determine the optimal solution.

However, many optimization problems involve several variables related by constraints. Therefore, the first step is to reduce the problem to a single-variable function whose maximum or minimum can be determined using calculus.

In this section we develop a systematic approach for solving optimization problems. We will follow the mnemonic

D C O S T C T C

which outlines the logical sequence of steps used to translate a word problem into a calculus problem and solve it.

Definition 3.71. An *optimization problem* is a problem in which one seeks to determine the maximum or minimum value of a quantity subject to one or more constraints. The quantity being maximized or minimized is called the *objective function*, while the relationships restricting the variables are called *constraints*.

Example 3.72. Suppose a rectangle has a fixed perimeter P. If x and y denote the side lengths, then

$$2x + 2y = P$$

is the constraint, while

$$A = xy$$

is the objective function represent the area.

We now present a systematic method for solving optimization problems. To solve an optimization problem, follow the steps:

(D) Diagram: Draw a diagram representing the situation whenever possible. Clearly label all relevant quantities and variables.

A diagram helps visualize the relationships between variables and often reveals the constraint equation.

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(C) **Constraint:** Determine the constraint equation, which expresses a relationship between the variables in the problem.

The constraint typically arises from physical limitations, geometric relationships, or fixed quantities.

(O) **Optimization Function:** Identify the quantity to be optimized (maximized or minimized). Write an equation for this quantity in terms of the variables.

This equation defines the objective function.

(S) **Substitute:** Use the constraint to eliminate variables and express the objective function as a function of a single variable.

This step converts the problem into a standard calculus problem.

(D) **Derivative:** Compute the derivative of the function.

(C) **Critical Numbers:** Find the critical numbers by solving $f'(x) = 0$ or identifying where $f'(x)$ does not exist.

(T) **Test:** Use an appropriate test to determine whether the critical points produce maxima or minima.

Possible methods include

- First Derivative Test (Theorem 3.38)
- Second Derivative Test (Theorem 3.49)
- Comparison of values at endpoints.

(C) **Conclusion:** State the final answer clearly in the context of the problem.

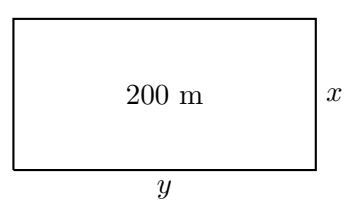
Be sure to include appropriate units and interpret the meaning of the result.

The DCOSDCTC procedure works because extrema of differentiable functions occur at critical numbers. By expressing the quantity we wish to optimize as a single-variable function, we can apply the techniques developed earlier in this chapter.

Example 3.73. A farmer wishes to fence a rectangular field of 200 m of fencing. What dimensions maximize the area of the field?

Solution. We solve the optimization problem using the procedure outlined above.

(D) Let x represent the length of the rectangular fence, and let y represent the width of the rectangular fence.



The rectangle has area $A = xy$.

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(C) The total fencing gives the perimeter constraint

$$2x + 2y = 200$$

(O) We wish to maximize the area

$$A = xy$$

(S) Solve for the constraint y

$$y = 100 - x$$

Substitute into the area function

$$A(x) = x(100 - x) = 100x - x^2$$

(D) Computing the derivative

$$A'(x) = 100 - 2x$$

(C) Finding the critical numbers by setting $A'(x) = 0$, we get

$$\begin{aligned} 100 - 2x &= 0 \\ x &= 50 \end{aligned}$$

(T) Compute the second derivative

$$A''(x) = -2 < 0$$

Thus, the critical point $x = 50$ corresponds to a maximum.

(C) Therefore, if $x = 50$, then $y = 50$. The rectangle with maximum area is a square with side lengths of 50 m. The maximum area is

$$A = 2500 \text{ m}^2$$

Example 3.74. Find two positive numbers whose sum is 20 and whose product is maximal.

Solution. We solve the optimization problem using the procedure outlined above.

(D) Let x and y be positive numbers. (Note that a visual diagram is not necessary for a problem like this)

(C) The sum of the two positive numbers is 20, so

$$x + y = 20$$

(O) We wish to maximize the product

$$P = xy$$

(S) Solve for the constraint y

$$y = 20 - x$$

Substitute into the objective function

$$P(x) = x(20 - x) = 20x - x^2$$

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(D) Computing the derivative,

$$P'(x) = 20 - 2x$$

(C) Finding the critical numbers by setting $P'(x) = 0$, we get

$$\begin{aligned} 20 - 2x &= 0 \\ x &= 10 \end{aligned}$$

(T) Compute the second derivative

$$P''(x) = -2 < 0$$

Therefore, the critical point $x = 10$ corresponds to a maximum.

(C) Therefore, if $x = 10$, then $y = 10$. The maximum product occurs when the two positive numbers are 10. The largest product is

$$P = 100$$

Many optimization problems share a common structure: when a quantity depending on two variables is optimized subject to a linear constraint, the optimal solution often occurs when the variables are equal. This phenomenon appears frequently in geometry, economics, and engineering.

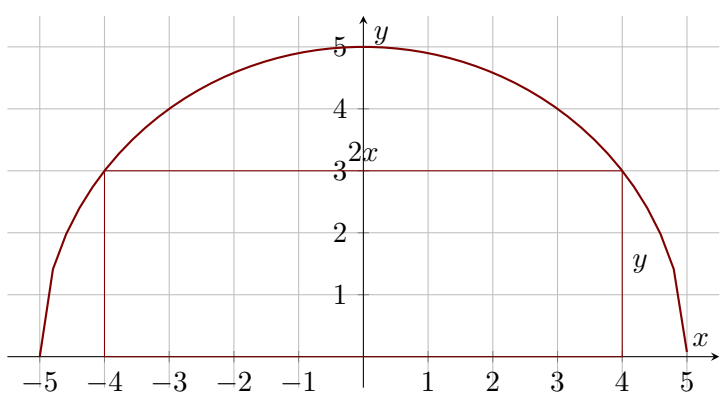
Example 3.75. A rectangle is inscribed inside the semicircle

$$x^2 + y^2 = r^2, \quad y \geq 0$$

Find the dimensions of the rectangle with maximum area.

Solution. We solve the optimization problem using the procedure outlined above.

(D) Consider the rectangle centered on the y-axis with its base on the x-axis and its top corners touching the semicircle.



Let x be half of the width of the rectangle, and let y be the height of the rectangle. Thus, the width of the rectangle is $2x$. The area of the rectangle is

$$A = 2xy$$

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(C) Since the top corners lie on the semicircle,

$$x^2 + y^2 = r^2$$

(O) We wish to maximize the area

$$A = 2xy$$

(S) Solve for the constraint

$$y = \sqrt{r^2 - x^2}$$

Substitute into the objective function

$$A(x) = 2x\sqrt{r^2 - x^2}$$

(D) Differentiate using the product and chain rules

$$A'(x) = 2\sqrt{r^2 - x^2} + 2x\left(-\frac{x}{\sqrt{r^2 - x^2}}\right) = \frac{2(r^2 - 2x^2)}{\sqrt{r^2 - x^2}}$$

(C) Solve for the critical numbers

$$r^2 - 2x^2 = 0$$
$$x = \frac{r}{\sqrt{2}}$$

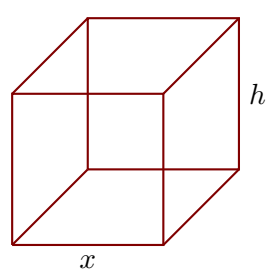
(T) Since the area increases before this value and decreases after it, this critical number gives a maximum.

(C) Thus, if $x = \frac{r}{\sqrt{2}}$, then $y = \frac{r}{\sqrt{2}}$. Thus, the rectangle of maximum area has width $\sqrt{2}r$ and height $\frac{r}{\sqrt{2}}$.

Example 3.76. A rectangular box with a square base is to have volume 32 m^3. Find the dimensions that minimize the surface area.

Solution. We solve the optimization problem using the procedure outlined above.

(D) Let x be the side length of the base and let h be the height.



(C) The volume condition is

$$x^2 h = 32$$

(O) Surface area of the box is

$$S = x^2 + 4xh$$

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(S) Solve the constraint

$$h = \frac{32}{x^2}$$

Substitute into the objective function

$$S(x) = x^2 + \frac{128}{x}$$

(D) Finding the derivative

$$S'(x) = 2x - \frac{128}{x^2}$$

(C) Solve

$$\begin{aligned} 2x - \frac{128}{x^2} &= 0 \\ 2x^3 - 128 &= 0 \\ x &= 4 \end{aligned}$$

(T) Computing the second derivative,

$$S''(x) = 2 + \frac{256}{x^3} > 0$$

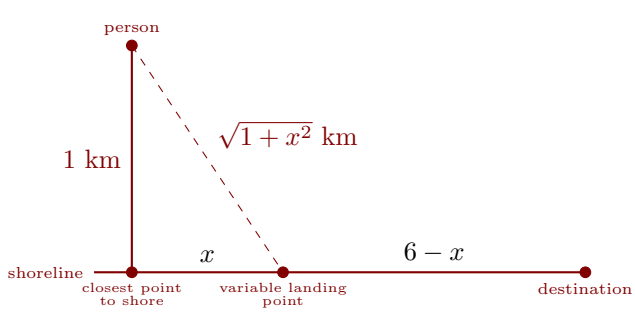
Thus, the critical number $x = 4$ is a minimum.

(C) Therefore, if $x = 4$, then $h = 2$. The box with minimal surface area has dimensions of side length 4 m and height of 2 m.

Example 3.77. A person is walking along a straight road at 5 km/h and is currently 1 km from a straight shoreline. A boat located on the shore is 6 km away from the shoreline. The person can walk to any point along the shore and then take the boat directly to the destination. If the boat travels at 15 km/h, where should the person enter the boat to minimize travel time?

Solution. We solve the optimization problem using the procedure outlined above.

(D) Let x be the distance walked along the shoreline. The person walks distance $\sqrt{1+x^2}$ and the boat travels $6 - x$.



(C) The total travel time is the sum of walking and boating time.

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(O) The optimization function is

$$T(x) = \frac{\sqrt{1+x^2}}{15} + \frac{6-x}{15}$$

(S) This step is not required since we do not have any constraint equation, so we can directly skip this step.

(D) Computing the derivative gives

$$T'(x) = \frac{x}{5\sqrt{1+x^2}} - \frac{1}{15}$$

(C) Solving for the critical numbers

$$\begin{aligned} \frac{x}{5\sqrt{1+x^2}} &= \frac{1}{15} \\ 3x &= \sqrt{1+x^2} \\ 9x^2 &= 1+x^2 \\ 8x^2 &= 1 \\ x &= \frac{1}{\sqrt{8}} \end{aligned}$$

(T) Since time decreases then increases, this point gives the minimum time.

(C) The person should walk $\frac{1}{\sqrt{8}}$ km along the shoreline before taking the boat.

Definition 3.78. Let q denote the number of units of a product produced or sold. If $p(q)$ represents the price per unit, then the *revenue function* is

$$R(q) = qp(q)$$

Definition 3.79. The *cost function* $C(q)$ represents the total cost of producing q units of a product. This may include fixed costs or variable costs.

Definition 3.80. The *profit function* is defined by

$$P(q) = R(q) - C(q)$$

Maximizing profit is one of the most important optimization problems in economics.

Proposition 3.81. *If $P(q)$ is differentiable, then the profit is maximized at a critical number q satisfying*

$$P'(q) = 0$$

Proof. By Section 3.1, a differentiable function can attain a local maximum only at a critical number where the derivative is zero or undefined. Since the profit function is typically differentiable for $q > 0$, a maximum must occur at a critical number. □

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Example 3.82. A company determines that the price per unit of a product is given by

$$p(q) = 100 - 2q$$

The cost of producing q units is

$$C(q) = 20q + 100$$

Find the level of production that maximizes profit.

Solution.

(D) Let q represent the number of units produced.

(C) No additional constraint is required since profit can be expressed directly in terms of q.

(O) The revenue function is

$$R(q) = q(100 - 2q) = 100q - 2q^2$$

and the profit equation is

$$P(q) = R(q) - C(q) = (100q - 20q^2) - (20q + 100) = -2q^2 + 80q - 100$$

(S) There are no substitution necessary for this question.

(D) Derivative is given by

$$P'(q) = -4q + 80$$

(C) Solving

$$\begin{aligned} -4q + 80 &= 0 \\ q &= 20 \end{aligned}$$

(T) Since

$$P''(q) = -4 < 0$$

The profit is maximized at the critical number.

(C) Therefore, the maximum profit occurs when $q = 20$.

Many optimization problems arise from physical principles involving distance, time, energy, or geometry. The following is one of the most classical calculus applications in physics

$$R(\theta) = \frac{v_0^2}{g} \sin(2\theta)$$

Example 3.83. A projectile is launched with initial speed v_0. Determine the launch angle that maximizes the horizontal range.

Solution.

(D) Let θ be the launch angle, let g be the acceleration due to gravity, and let v_0 represent the initial speed.

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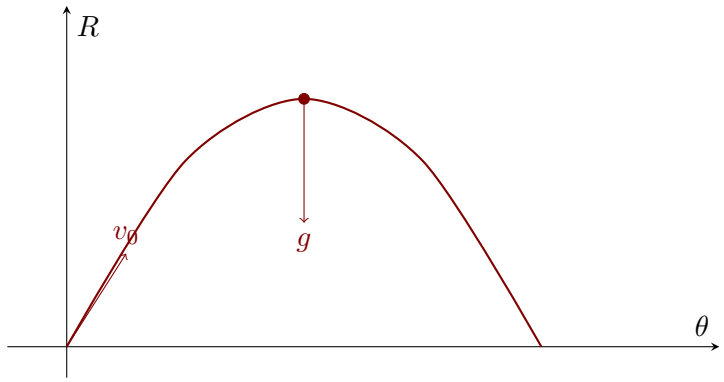

(C) There are no constraints involved in this problem.

(O) The objective function is the range of the projectile

$$R(\theta) = \frac{v_0^2}{g}\sin(2\theta)$$

(S) Since there are no constraints involved, we skip the part.

(D) The derivative is

$$R'(\theta) = \frac{v_0^2}{g}2\cos(2\theta)$$

(C) Solving

$$\begin{aligned}\cos(2\theta) &= 0\\ 2\theta &= \frac{\pi}{2}\\ \theta &= \frac{\pi}{4}\end{aligned}$$

(T) Since the function increases then decreases, the critical number is the maximum.

(C) The maximum range occurs at $\theta = \frac{\pi}{4}$.

Example 3.84. The yield of a chemical reaction at temperature T is modeled by

$$Y(T) = Te^{-\frac{T}{40}}$$

for $T > 0$. Find the temperature that maximizes the yield.

Solution.

(D) No geometric diagram is required in this case, but we note that T represents temperature, and $Y(T)$ represents the reaction yield. We seek the value of T that gives the greatest possible yield.

(C) The physical constraint is that temperature must be positive.

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(O) The quantity to be maximized is the yield

$$Y(T) = Te^{-\frac{T}{40}}$$

(S) No substitution is needed, since the objective function is already written in terms of a single variable T.

(D) Differentiating using the product rule,

$$Y'(T) = e^{-\frac{T}{40}} + T\left(-\frac{1}{40}e^{-\frac{T}{40}}\right)$$

Thus,

$$Y'(T) = e^{-\frac{T}{40}}\left(1 - \frac{T}{40}\right)$$

(C) Setting $Y'(T) = 0$,

$$e^{-\frac{T}{40}}\left(1 - \frac{T}{40}\right) = 0$$

Since $e^{-\frac{T}{40}} > 0$ for all T, the only way $Y'(T) = 0$ is if

$$1 - \frac{T}{40} = 0$$

Hence $T = 40$.

(T) Using the First Derivative Test,

- If $0 < T < 40$, then

$$1 - \frac{T}{40} > 0$$

  so $Y'(T) > 0$ and Y is increasing.
- If $T > 40$, then

$$1 - \frac{T}{40} < 0$$

  so $Y'(T) < 0$, and Y is decreasing.

Thus, the function increases before $T = 40$ and decreases after $T = 40$. Therefore, $T = 40$ gives a maximum number.

(C) The yield is maximized when the temperature is $T = 40$. The maximum yield is

$$Y(40) = 40e^{-1}$$

Example 3.85. A fence 10 ft tall runs parallel to a tall building at a distance of 2 ft from the building. Find the length of the shortest ladder that will reach from the ground over the fence to the wall of the wall of the building.

Solution.

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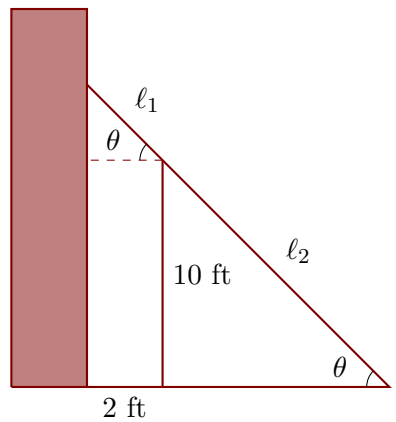
(D) Let ℓ_1 represent the ladder from the ground to the fence, and let ℓ_2 represent the ladder from the fence to the wall of the building. Then . Let θ be the angle formed between the ground and the ladder.



(C) From the small right triangle near the fence, the horizontal distance is 2 ft and the corresponding hypotenuse is ℓ_1, so

$$\cos(\theta) = \frac{2}{\ell_1}.$$

Hence

$$\ell_1 = \frac{2}{\cos(\theta)} = 2\sec(\theta).$$

From the triangle formed by the fence, the fence height is 10 ft and the corresponding hypotenuse is ℓ_2, so

$$\sin(\theta) = \frac{10}{\ell_2}.$$

Hence

$$\ell_2 = \frac{10}{\sin(\theta)} = 10\csc(\theta).$$

(O) The total length of the ladder is

$$L = \ell_1 + \ell_2$$

(S) Write

$$\ell_1 = \frac{2}{\cos(\theta)} = 2\sec(\theta), \quad \ell_2 = \frac{10}{\sin(\theta)} = 10\csc(\theta)$$

and substitute to the objective function

$$L(\theta) = 2\sec(\theta) + 10\csc(\theta)$$

(D) Differentiating gives

$$L'(\theta) = 2\sec(\theta)\tan(\theta) - 10\csc(\theta)\cot(\theta)$$

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(C) Setting $L'(\theta) = 0$,

$$\begin{aligned} 2\sec(\theta)\tan(\theta) &= 10\csc(\theta)\cot(\theta) \\ 2\frac{\sin(\theta)}{\cos^2(\theta)} &= 10\frac{\cos(\theta)}{\sin^2(\theta)} \\ \frac{\sin^3(\theta)}{\cos^3(\theta)} &= 5 \\ \tan^3(\theta) &= 5 \\ \tan(\theta) &= \sqrt[3]{5} \\ \theta &= \arctan(\sqrt[3]{5}) \end{aligned}$$

(T) Since

$$\lim_{\theta\to 0^+} L(\theta) = \infty \qquad \text{and} \qquad \lim_{\theta\to(\pi/2)^-} L(\theta) = \infty,$$

and there is only one critical number in $0 < \theta < \frac{\pi}{2}$, this critical number must give the absolute minimum value of L.

(C) Now compute the minimum ladder length. Since $\tan(\theta) = \sqrt[3]{5}$, let

$$t = \sqrt[3]{5}.$$

Then

$$\sec(\theta) = \sqrt{1+t^2}, \qquad \csc(\theta) = \frac{\sqrt{1+t^2}}{t}.$$

So

$$\begin{aligned} L_{\min} &= 2\sqrt{1+t^2} + 10\frac{\sqrt{1+t^2}}{t} \\ &= \sqrt{1+t^2}\left(2 + \frac{10}{t}\right). \end{aligned}$$

Since $t^3 = 5$, we have $\frac{10}{t} = 2t^2$, and thus

$$\begin{aligned} L_{\min} &= \sqrt{1+t^2}\,(2+2t^2) \\ &= 2(1+t^2)^{3/2}. \end{aligned}$$

Therefore,

$$L_{\min} = 2\left(1 + 5^{2/3}\right)^{3/2} \text{ ft.}$$

Numerically,

$$L_{\min} \approx 14.03 \text{ ft.}$$

Thus, the length of the shortest ladder is 14.03 ft.

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3.9 Newton's Method

In many mathematical and scientific problems, we are interested in solving equations of the form

$$f(x) = 0$$

Such equations arise naturally in numerous contexts, including

- determining equilibrium points in physics,
- solving nonlinear equations in engineering,
- computing optimal values in economics,
- modeling chemical reaction equilibria.

However, not all equations can be solved algebraically. For example, equations such as

$$x = \cos(x), \quad e^x = x^2$$

cannot be solved using elementary algebraic techniques. In such cases we must rely on numerical methods to approximate the solutions.

One of the most powerful, and widely used, numerical techniques is called Newton's Method, also known as the Newton–Raphson Method. This method uses derivatives to construct successive approximations that converge to a solution of the equation.

The key idea behind Newton's Method is to approximate the function near a point by its tangent line, and then use the point where the tangent line intersects the x-axis as a better approximation to the root.

In this section we derive Newton's Method, study its geometric interpretation, and apply it to approximate solutions of nonlinear equations.

Suppose we want to find a root of $f(x) = 0$, and suppose we begin with an initial approximation x_0.

Near the point $(x_0, f(x_0))$, the function can be approximated by its tangent line. The equation of the tangent line is

$$y = f(x_0) + f'(x_0)(x - x_0)$$

Newton's method uses the point where this tangent line intersects the x-axis as a new approximation to the root. To find where the tangent line intersects the x-axis, set $y = 0$.

$$0 = f(x_0) + f'(x_0)(x - x_0)$$

Solving for x, we get

$$x = x_0 - fracf(x_0)f'(x_0)$$

This value is denoted x_1, the next approximation. Repeating this process produces a sequence $x_0, x_1, x_2, ...$ that ideally converges to the root.

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Definition 3.86. Let f be a differentiable function and suppose x_0 is an initial approximation to a root of $f(x) = 0$. The sequence defined by

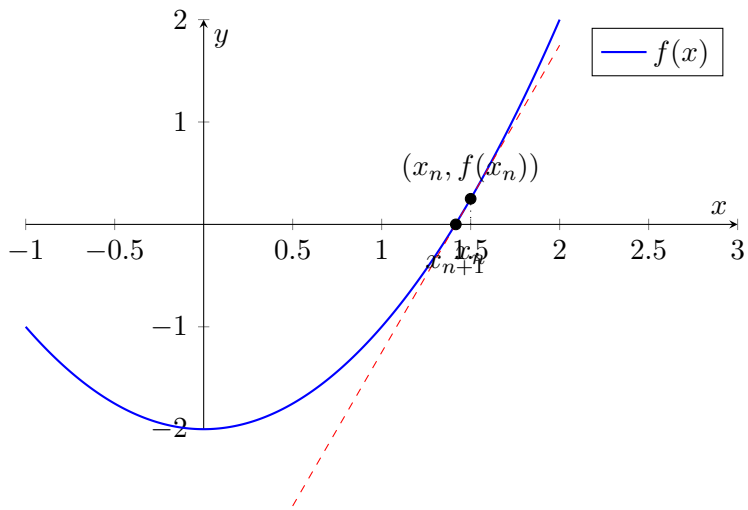
$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

is called *Newton's Method*. The numbers x_n are called *Newton iterates*.

Newton's method has a clear and geometric meaning.

1. Start at a point x_n on the graph of f.
2. Draw a tangent line at $(x_n, f(x_n))$.
3. The point where the tangent line intersects the x-axis becomes the next approximation x_{n+1}.

This process is repeated until the approximation stabilize.



To approximate a root of $f(x) = 0$.

1. Choose an initial guess x_0.
2. Compute
$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$
3. Repeat until successive approximations differ by less than a desired tolerance.

Example 3.87. Use Newton's Method to approximate $\sqrt{2}$.

Solution. We solve

$$x^2 - 2 = 0$$

Let $f(x) = x^2 - 2$. Then $f'(x) = 2x$. Newton's iteration becomes

$$x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n}$$

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Simplifying

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right)$$

Let $x_0 = 1$ be our initial guess. Then computing x_1

$$x_1 = \frac{1}{2}\left(1 + \frac{2}{1}\right) = 1.5$$

Now computing x_2,

$$x_2 = \frac{1}{2}\left(1.5 + \frac{1}{1.5}\right) = 1.4167$$

Computing x_3,

$$x_3 = \frac{1}{2}\left(1.4167 + \frac{2}{1.4167}\right) \approx 1.4142$$

Thus,

$$\sqrt{2} \approx 1.4142.$$

Example 3.88. Use Newton's Method to approximate a solution of

$$x^3 - x - 1 = 0$$

Solution. Let $f(x) = x^3 - x - 1$. Then $f'(x) = 3x^2 - 1$. Newton's iteration becomes

$$x_{n+1} = x_n - \frac{x_n^3 - x_n - 1}{3x_n^2 - 1}$$

Since $f(1) = -1$ and $f(2) = 5$, the root lies between 1 and 2. Choose $x_0 = 1.5$ as our initial guess. Compute the first iteration

$$x_1 = 1.5 - \frac{1.5^3 - 1.5 - 1}{3(1.5)^2 - 1} \approx 1.3478$$

Compute the second iteration

$$x_2 \approx 1.3252$$

The solution is approximately

$$x \approx 1.3247$$

When Newton's Method works well, the following result holds.

Proposition 3.89. *If f is differentiable, the root r satisfies $f'(r) \neq 0$, and the initial guess x_0 is sufficiently close to r, then the Newton sequence often converges rapidly to r.*

On the other hand, Newton's method may fail if $f'(x_n) = 0$, or the initial guess is far from the root, or the tangent line moves away from the root. Thus, careful selection of the initial approximation is important.

A very nice final example for Newton's Method is one where the iterations converge quickly and the results can be neatly summarized in a table of iterates. A classical choice is solving

$$\cos(x) = x$$

This equation cannot be solved algebraically and arises in fixed point problems, oscillations, and numerical analysis.

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Example 3.90. Use Newton's Method to approximate a solution of the equation

$$\cos(x) = x$$

Construct a table of successive approximations.

Solution. First rewrite the equation in the form

$$f(x) = \cos(x) - x$$

Then

$$f'(x) = -\sin(x) - 1$$

Newton's Method gives

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{\cos(x_n) - x_n}{-\sin(x_n) - 1}$$

Since $\cos(0) = 1 > 0$ and $\cos(1) - 1 < 0$, the root lies between 0 and 1. We choose $x_0 = 0.5$.

We compute successive approximations using Newton's formula

n	x_n	$f(x_n)$
0	0.500000	0.377583
1	0.755222	−0.027103
2	0.739142	−0.000095
3	0.739085	−0.000000

The approximation converges rapidly to $x = 0.739085$. Thus, the solution of $\cos(x) = x$ is approximately $x = 0.739085$.

This example illustrates an important feature of Newton's Method: when the initial guess is sufficiently close to the root, the method typically converges very rapidly, often doubling the number of correct digits at each iteration.

For those that are interested, we are able to create a program in Python that visualizes Newton's Method dynamically by showing how each tangent line produces the next approximation. Students are encouraged to experiment with different functions and initial guesses to observe how the method behaves. Furthermore, students interested may be interested in taking MATH 3241 and MATH 3242 (Numerical Methods I, II) in their upper year course. Access the code here.

3.10 Antiderivatives

Throughout this chapter we have studied the derivative of a function and learned how derivatives describe rates of change, slopes of tangent lines, and the behavior of functions. In many situations, however, we encounter the inverse problem: given the derivative of a function, can we determine the original function?

For example, suppose we know that a function F satisfies

$$F'(x) = 3x^2$$

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One possible function with this derivative is

$$F(x) = x^3$$

However, this is not the only such function. In fact

$$F(x) = x^3 + 5$$

and

$$F(x) = x^3 - 2$$

also have derivative $3x^2$. Thus, the derivative does not determine a unique function; instead it determines a family of functions differing by a constant.

The problem of finding a function whose derivative is known leads to the concept of an antiderivative. Antiderivatives form the foundation of the subject of integration, which will be studied more extensively in MATH 1310 (Integral Calculus with Applications).

In this section, we introduce the concept of antiderivatives, study their basic properties, and develop techniques for finding them.

In physics, the motion of an object is often described using three physical quantities: position, velocity, and acceleration. These quantities are related through derivatives.

Definition 3.91. If $s(t)$ represents the *position* of an object at time t, then the velocity is defined by

$$v(t) = s'(t)$$

Velocity represents the rate of change of position.

The *acceleration* is the derivative of velocity

$$a(t) = v'(t)$$

Thus, acceleration measures the rate of change of velocity.

These relationships can be summarized as

$$s(t) \xrightarrow{\frac{d}{dt}} v(t) \xrightarrow{\frac{d}{dt}} a(t)$$

That is, $v(t) = s'(t)$ and $a(t) = v'(t)$.

Suppose we know the acceleration of an object. Can we determine the velocity? And if we know the velocity, can we determine the position? That is exactly the reverse process of differentiation.

Example 3.92. Suppose $a(t) = 4$. This means the velocity satisfies

$$v'(t) = 4$$

We ask: what function has derivative 4? One possible answer is

$$v(t) = 4t$$

But this is not the only possibility. In fact,

$$v(t) = 4t + C$$

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also satisfies

$$v'(t) = 4$$

Thus, there are infinitely many functions whose derivative is 4. These functions are called antiderivatives of 4.

Now suppose $v(t) = 4t + C$. Since

$$v(t) = s'(t)$$

we seek a function whose derivative is $4t + C$.

One such function is

$$s(t) = 2t^2 + Ct + D$$

Thus,

$$s(t) \to v(t) \to a(t)$$

corresponds to

$$2t^2 + Ct + D \to 4t + C \to 4$$

This exmaple illustrates an important principle:

- Acceleration determines velocity up to a constant.
- Velocity determines position up to a constant.

The constants correspond to physical quantities: C is the initial velocity, and D is the initial position.

This motivates the concept of an antiderivative: An antiderivative reverses differentiation and allows us to recover a function from its rate of change.

Definition 3.93. Let f be a function defined on an interval I. A function F is called an *antiderivative* of f on I if

$$F'(x) = f(x)$$

for all $x \in I$.

Example 3.94. Find an antiderivative of $f(x) = 2x$.

Solution. We seek a function F such that

$$F'(x) = 2x$$

Since

$$\frac{d}{dx}(x^2) = 2x$$

one antiderivative is

$$F(x) = x^2$$

As observed from earlier, antiderivatives are not unique.

Theorem 3.95. *If F and G are both antiderivatives of f on an interval I, then*

$$F(x) - G(x) = C$$

for some constant C.

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Proof. Since F and G are antiderivatives of f, then $F'(x) = f(x)$ and $G'(x) = f(x)$. Thus,

$$(F - G)' = F' - G' = f(x) - f(x) = 0$$

A function whose derivative is zero on an interval must be constant (Corollary 3.29). Therefore,

$$F(x) - G(x) = C$$

for some constant C. □

Corollary 3.96. *If F is one antiderivative of f, then the set of all antiderivatives of f is*

$$F(x) + C$$

where C is an arbitrary constant.

Proposition 3.97 (Power Rule for Antiderivatives). *If $n \neq -1$, then*

$$\frac{d}{dx}\left(\frac{x^{n+1}}{n+1}\right) = x^n$$

Thus, an antiderivative of x^n is

$$\frac{x^{n+1}}{n+1}$$

Proof. By the power rule for differentiation,

$$\frac{d}{dx}\left(\frac{x^{n+1}}{n+1}\right) = \frac{n+1}{n+1}x^n = x^n$$

□

The reason for the restriction is that the formula would require dividing by $n + 1$. When $n = -1$, $n + 1 = 0$, so the formula breaks down. We therefore need a special case.

Proposition 3.98 (Antiderivative of $\frac{1}{x}$). *An antiderivative of $\frac{1}{x}$ is $\ln|x|$.*

$$\frac{d}{dx}(\ln|x|) = \frac{1}{x}, \quad x \neq 0$$

Proof. Recall that

$$\frac{d}{dx}(\ln(x)) = \frac{1}{x}$$

if $x > 0$ and

$$\frac{d}{dx}(\ln(-x)) = \frac{1}{x}$$

if $x < 0$. Thus, for $x \neq 0$,

$$\frac{d}{dx}(\ln|x|) = \frac{1}{x}$$

Therefore, $\ln|x|$ is an antiderivative of $\frac{1}{x}$. □

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Definition 3.99. Let f be a function defined on an interval I. The *indefinite integral* of f is the family of all antiderivatives of f, and is denoted by

$$\int f(x)\ dx$$

Thus,

$$\int f(x)\ dx = F(x) + C$$

where F is the antiderivative of f and C is an arbitrary constant.

The symbol

$$\int$$

is called the *integral sign*, and the expression

$$dx$$

indicates that the integration is performed with respect to the variable x.

The notation $\int f(x)\ dx$ does not represent a single function but rather an entire family of functions that differ by a constant. This is called the constant of integration.

Example 3.100. Evaluate

$$\int 2x\ dx$$

Solution. We seek a function whose derivative is $2x$. Since

$$\frac{d}{dx}(x^2) = 2x$$

an antiderivative of $2x$ is x^2. Therefore,

$$\int 2x\ dx = x^2 + C$$

The notation dx indicates that the integration is taken with respect to the variable x. If another variable appears, the notation changes accordingly

$$\int f(t)\ dt, \quad \int f(y)\ dy$$

Example 3.101. Find the general antiderivative of $f(x) = x^4$.

Solution. Using the power rule (Proposition 3.97),

$$\int x^4\ dx = \frac{x^5}{5} + C$$

Thus, the general antiderivative is

$$F(x) = \frac{x^5}{5} + C$$

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Example 3.102. Find the general antiderivative of $f(x) = \frac{1}{x}$.

Solution. Since

$$\frac{d}{dx}(\ln|x|) = \frac{1}{x}$$

an antiderivative is

$$F(x) = \ln|x|$$

Therefore, the general antiderivative is

$$F(x) = \frac{1}{x} + C$$

Proposition 3.103 (**Antiderivatives of Exponential Functions**). *An antiderivative of e^x is e^x.*

Proof. Since

$$\frac{d}{dx}(e^x) = e^x$$

the function e^x is its own antiderivative. □

Example 3.104. Find the general antiderivative of $f(x) = e^x$.

Solution. Since

$$\frac{d}{dx}(e^x) = e^x$$

the general antiderivative is

$$F(x) = e^x + C$$

Proposition 3.105 (**Antiderivatives of Trigonometric Functions**). *The following antiderivatives hold*

$$\frac{d}{dx}(\sin(x)) = \cos(x), \quad \frac{d}{dx}(-\cos(x)) = \sin(x)$$

Example 3.106. Find the general antiderivative of $f(x) = \cos(x)$.

Solution. Since

$$\frac{d}{dx}(\sin(x)) = \cos(x)$$

an antiderivative is

$$F(x) = \sin(x)$$

Thus the general antiderivative is

$$F(x) = \sin(x) + C$$

Proposition 3.107. *If F is an antiderivative of f and G is an antiderivative of g, then*

$$aF + bG$$

is an antiderivative of

$$af + bg$$

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Proof. Differentiate

$$(aF + bG)' = aF + bG' = af + bg$$

Thus, the result follows. □

Example 3.108. Find the general antiderivative of

$$f(x) = 3x^2 + 4e^x$$

Solution. Integrate term-by-term

$$\int 3x^2 \, dx = x^3, \quad \int 4e^x \, dx = 4e^x$$

Thus,

$$F(x) = x^3 + 4e^x + C$$

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Appendix A

MHF4U Review: Functions

Functions are the central objects of study in advanced mathematics. Nearly every concept encountered in calculus—limits, derivatives, integrals, and mathematical modelling—is built upon a precise understanding of what a function is and how it behaves. Appendix A provides a structured review of functions, with emphasis on both algebraic understanding and graphical interpretation, as expected in MHF4U.

In this appendix, we revisit the formal definition of a function, along with essential ideas such as domain, range, function notation, and common families of functions. These foundational concepts are not merely technical details; they determine whether a mathematical model is valid and whether subsequent operations—such as composition, inversion, or differentiation—are well-defined.

A major focus of this review is the graphical behaviour of functions. Understanding how graphs change under transformations allows complex functions to be analyzed quickly and accurately, without relying solely on computation. This skill is critical in both pre-calculus and calculus, where qualitative reasoning about graphs often precedes algebraic work.

The appendix concludes with inverse functions, a topic that unifies algebraic reasoning, graphical interpretation, and logical precision. Inverse functions introduce important ideas such as one-to-one behaviour and restricted domains, which will reappear throughout calculus, particularly in the study of exponential and logarithmic functions.

By the end of Appendix A, students should be comfortable moving between symbolic, graphical, and verbal descriptions of functions, and should be prepared to use functions confidently as the fundamental language of calculus.

A.1 Functions

Functions provide the formal language used to describe relationships between quantities. In advanced secondary mathematics and calculus, a clear and precise understanding of functions is essential: every graph, equation, and model represents a function or a relationship closely related to one. This section develops the foundational definition of a function and introduces the core concepts—domain, range, function notation, and basic classifications—that will be used throughout the remainder of this course and beyond.

Our goal is not only to perform computations with functions, but also to understand what functions are, what makes a mathematical rule a function, and how different representations

(algebraic, graphical, numerical, and verbal) describe the same underlying object.

Definition A.1. A *function* is a rule that assigns exactly one output to each input from a specified set.

More formally, a function f consists of

- a domain D, the set of allowable inputs, and
- a rule that assigns to each $x \in D$ a unique real number $f(x)$.

We write

$$f : D \to \mathbb{R}, \quad x \mapsto f(x)$$

A relation is a function if and only if no input corresponds to more than one output.

Example A.2. Let $f(x) = 2x + 3$. For each real number x, the rule produces exactly one output. Therefore, f is a function with domain $\mathbb{R}$.

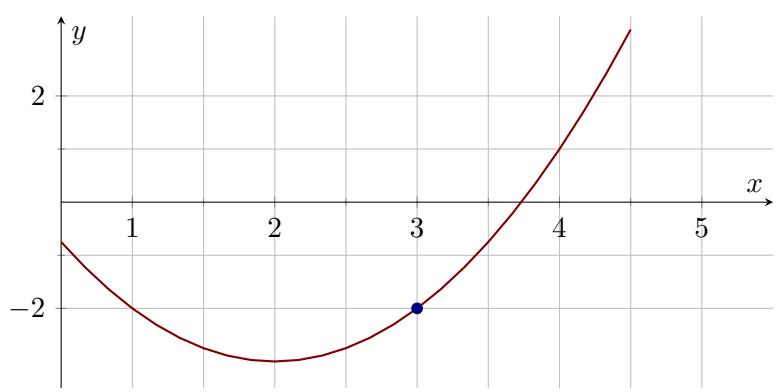
Example A.3. Suppose a relation assigns $x = 2 \mapsto y = 3$ and $y = 5$. This relation is not a function, since the input $x = 2$ corresponds to two different outputs.

Definition A.4. If f is a function and x is in its domain, then $f(x)$ denotes the output of the function corresponding to input x.

The symbol $f(x)$ is read as "f of x" and does not represent multiplication.

Example A.5. If $f(x) = x^2 - 4x + 1$, then

$$f(3) = 3^2 - 4(3) + 1 = 9 - 12 + 1 = -2$$



Example A.6. Let $f(x) = 2x - 1$. Find $f(a + h)$.

Solution. $f(a + h) = 2(a + h) - 1 = 2a + 2h - 1$

Definition A.7. The *domain* of a function is the set of all input values for which the function is defined.

Unless stated otherwise, the domain is assumed to be the largest possible set of real numbers for which the formula makes sense.

A value of x is excluded from the domain if it causes

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1. Division by zero,
2. An even root of a negative number, or
3. A logarithm of a non-positive number.

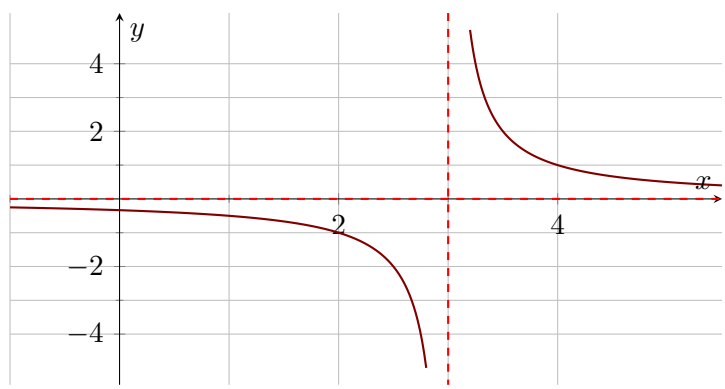
Example A.8. Find the domain of

$$f(x) = \frac{1}{x-3}$$

Solution. The denominator cannot be zero. Thus,

$$x - 3 \neq 0 \implies x \neq 3$$

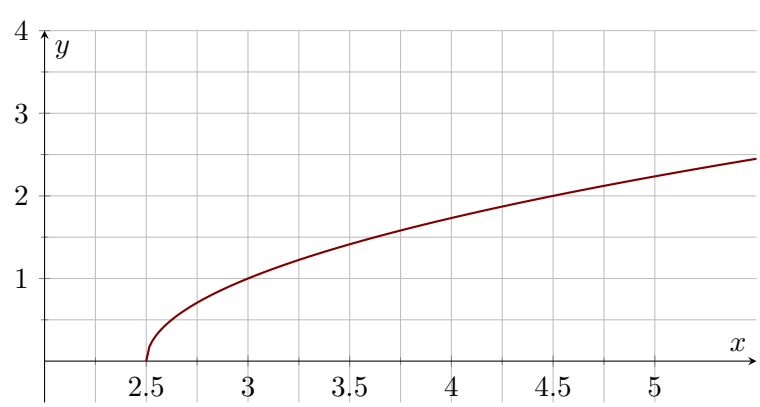
The domain of the function is $\mathbb{R} \setminus \{3\}$.



Example A.9. Find the domain of

$$f(x) = \sqrt{2x-5}$$

Solution. We require $2x - 5 \geq 0$, or $x \geq \frac{5}{2}$. The domain of the function is $\left[\frac{5}{2}, \infty\right)$.



Definition A.10. The *range* of a function is the set of all possible output values $f(x)$ as x varies over the domain.

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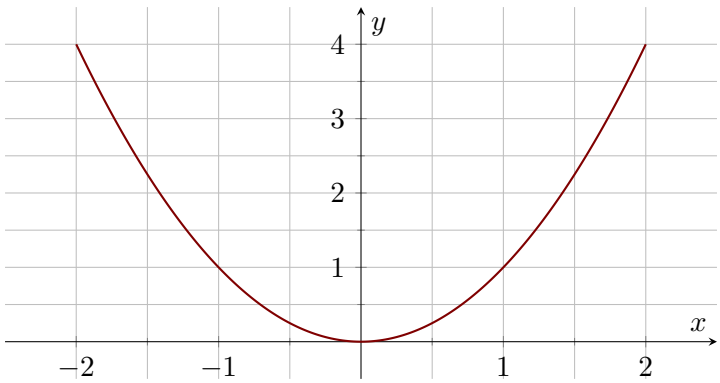
Finding the range is often more difficult than finding the domain and may require

- algebraic analysis,
- graphing, or
- reasoning about function behaviour.

Example A.11. Let $f(x) = x^2$. Since squares are always non-negative, $f(x) \geq 0$ for all $x \in \mathbb{R}$, so the range is $[0, \infty)$.

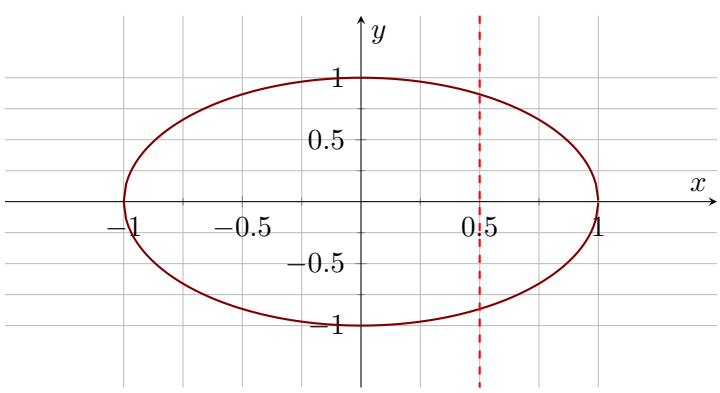


A function can be represented in several equivalent ways:

1. Algebraically: $f(x) = x^2 - 1$.
2. Graphically: a curve in the Cartesian plane.
3. Numerically: a table of values.
4. Verbally: a description of the relationship between quantities.

Theorem A.12 (The Vertical Line Test). *A graph in the Cartesian plane represents a function if and only if no vertical line intersects the graph more than once.*

Example A.13. The graph of a circle fails the vertical line test and therefore does not represent a function.



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Definition A.14. Two functions f and g are equal if they have the same domain, and $f(x) = g(x)$ for all x in the domain.

Example A.15. The expressions

$$f(x) = \frac{x^2 - 1}{x - 1}, \quad g(x) = x + 1$$

are not equal as functions, since f is undefined at $x = 1$, while g is defined for all real numbers.

A.2 Graphing Functions Using Transformations

Many complicated-looking functions can be understood quickly by recognizing how they are built from simpler, well-known graphs. Rather than plotting points one at a time, we can start with a parent function and apply a sequence of transformations—such as translations, reflections, and stretches—to obtain the graph of a new function. This approach emphasizes structure, efficiency, and conceptual understanding, all of which are essential skills in MHF4U and in calculus.

In this section, we develop a systematic method for graphing functions using transformations. We begin by identifying common parent functions, then study how algebraic changes in a function's formula affect its graph. Careful attention is paid to the order and meaning of transformations, since horizontal and vertical changes behave differently. These ideas will be used extensively in later topics, including curve sketching, inverse functions, and graphical analysis of limits and derivatives.

Definition A.16. A *parent function* is a basic function whose graph serves as a reference from which more complicated functions can be obtained using transformations.

Some common parent functions include the following.

Function	**Formula**	**Key Features**
Linear Function	$f(x) = x$	Straight line through origin
Quadratic Function	$f(x) = x^2$	Parabola opening upward
Absolute Value Function	$f(x) = \|x\|$	Distance from the x-axis
Square Root Function	$f(x) = \sqrt{x}$	Defined for $x \geq 0$
Reciprocal Function	$f(x) = \frac{1}{x}$	Asymptote at axes
Sine Function	$f(x) = \sin(x)$	Trigonometric function crossing the origin
Cosine Function	$f(x) = \cos(x)$	Trigonometric function crossing $(0, 1)$

Definition A.17. Let $f(x)$ be a function. The graph of $y = f(x) + k$ is obtained by shifting the graph of $y = f(x)$ *upward* by k units if $k > 0$, and *downward* by $|k|$ units if $k < 0$.

To visualize vertical translations, we may use table of values with the mapping rule $y \mapsto y + k$

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x	y	$\Rightarrow$	x	$y+k$
x_1	y_1		x_1	y_1+k
x_2	y_2		x_2	y_2+k
x_3	y_3		x_3	y_3+k

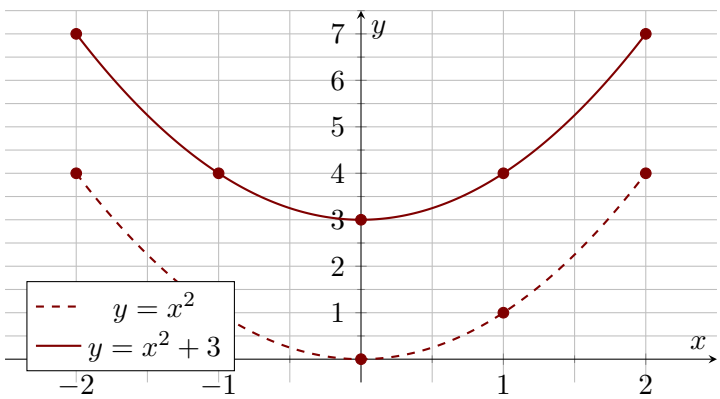
where the two columns on the right represent the transformed function $y = f(x) + k$.

Example A.18. Graph $y = x^2 + 3$.

Solution. Start with the parent function $y = x^2$ and shift the entire graph 3 units up.

Using a table of values,

x	y	$\Rightarrow$	x	$y+3$
-2	4		-2	7
-1	1		-1	4
0	0		0	3
1	1		1	4
2	4		2	7



Definition A.19. Let $f(x)$ be a function. The graph of $y = f(x - h)$ is obtained by shifting the graph of $y = f(x)$ *to the right* by h units, and the graph of $y = f(x + h)$ is obtained by shifting the graph of $y = f(x)$ *to the left* by h units.

Horizontal shifts act in the opposite direction of the sign inside the function.

To visualize horizontal translations, we may use table of values with the mapping rule $x \mapsto x + h$

x	y	$\Rightarrow$	$x+h$	y
x_1	y_1		x_1+h	y_1
x_2	y_2		x_2+h	y_2
x_3	y_3		x_3+h	y_3

where the two columns on the right represent the transformed function $y = f(x - h)$.

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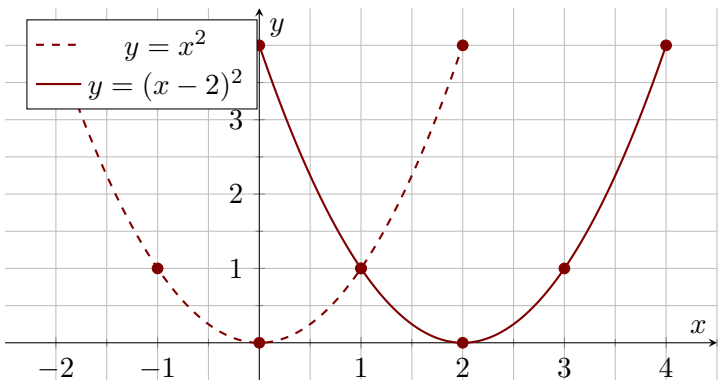
Example A.20. Graph $y = (x-2)^2$.

Solution. This is the graph $y = x^2$ shifted right 2 units.

Using a table of values,

x	y	$\Rightarrow$	$x+2$	y
-2	4		0	4
-1	1		1	1
0	0		2	0
1	1		3	1
2	4		4	4

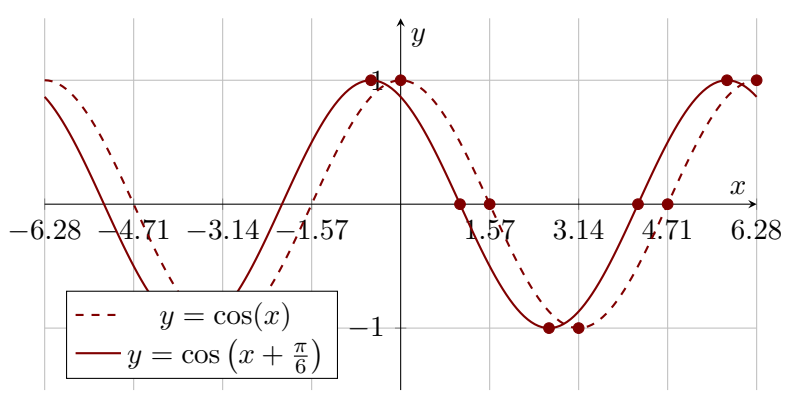


Example A.21. Graph $y = \cos\left(x + \frac{\pi}{6}\right)$.

Solution. This is the graph of $y = \cos(x)$ shifted $\frac{\pi}{6}$ units to the left.

Using a table of values,

x	y	$\Rightarrow$	$x - \frac{\pi}{6}$	y
0	1		$-\frac{\pi}{6}$	1
$\frac{\pi}{2}$	0		$\frac{\pi}{3}$	0
π	-1		$\frac{5\pi}{6}$	-1
$\frac{3\pi}{2}$	0		$\frac{4\pi}{3}$	0
2π	1		$\frac{11\pi}{6}$	1



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Definition A.22. Let $f(x)$ be a function and $a \neq 0$. The graph of $y = af(x)$ is obtained by multiplying all y-values of f by a. If

- $|a| > 1$, then the function is *stretched vertically.*
- If $0 < |a| < 1$, then the function is *compressed vertically.*

To visualize vertical stretches or compressions, we may use table of values with the mapping rule $y \mapsto ay$

x	y	$\Rightarrow$	x	ay
x_1	y_1		x_1	ay_1
x_2	y_2		x_2	ay_2
x_3	y_3		x_3	ay_3

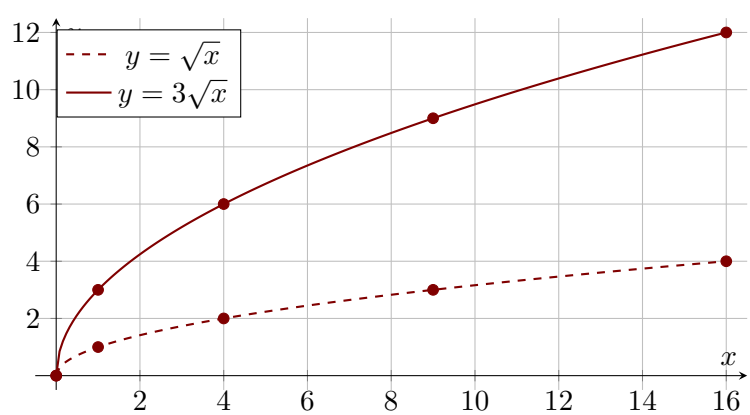
where the two columns on the right represent the transformed function $y = af(x)$.

Example A.23. Graph $y = 3\sqrt{x}$.

Solution. This is the graph of $y = \sqrt{x}$ vertically stretched by a factor of 3.

Using a table of values,

x	y	$\Rightarrow$	x	$3y$
0	0		0	0
1	1		1	3
4	2		4	6
9	3		9	9
16	4		16	12



Example A.24. Graph $y = \frac{1}{2}x^2$.

Solution. This is the graph of $y = x^2$ vertically compressed by a factor of $\frac{1}{2}$.

Using a table of values,

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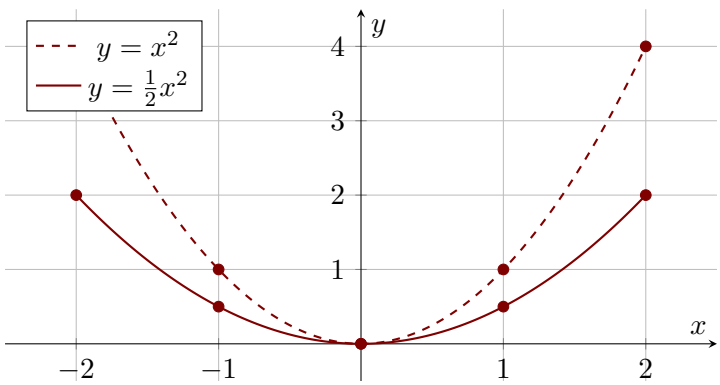
x	y	$\Rightarrow$	x	$\frac{1}{2}y$
-2	4		-2	2
-1	1		-1	$\frac{1}{2}$
0	0		0	0
1	1		1	$\frac{1}{2}$
2	4		2	2



Definition A.25. The graph of $y = -f(x)$ is the reflection of the graph of $y = f(x)$ across the x-axis.

To visualize reflections in the x-axis, we may use table of values with the mapping rule $y \mapsto -y$

x	y	$\Rightarrow$	x	$-y$
x_1	y_1		x_1	$-y_1$
x_2	y_2		x_2	$-y_2$
x_3	y_3		x_3	$-y_3$

where the two columns on the right represent the transformed function $y = -f(x)$.

Definition A.26. The graph of $y = f(-x)$ is the reflection of the graph $y = f(x)$ across the y-axis.

To visualize reflections in the x-axis, we may use table of values with the mapping rule $x \mapsto -x$

x	y	$\Rightarrow$	$-x$	y
x_1	y_1		$-x_1$	y_1
x_2	y_2		$-x_2$	y_2
x_3	y_3		$-x_3$	y_3

where the two columns on the right represent the transformed function $y = f(-x)$.

Example A.27. Graph $y = -|x|$.

Solution. Reflect the parent function $y = |x|$ across the x-axis.

Using a table of values

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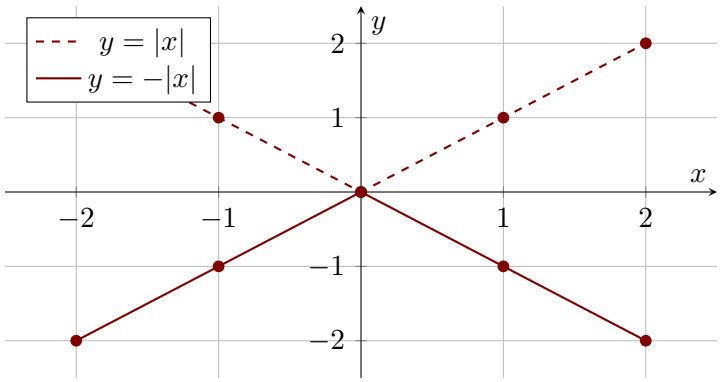
x	y	$\Rightarrow$	x	$-y$
-2	2		-2	-2
-1	1		-1	-1
0	0		0	0
1	1		1	-1
2	2		2	-2



Combining all the transformations we have available, we have the following result.

Theorem A.28. *For a function of the form*

$$y = af(x - h) + h$$

the transformations are applied in the following order

1. *Horizontal translation by h*
2. *Vertical stretch/compression by factor $|a|$*
3. *Reflection if $a < 0$*
4. *Vertical translation by k.*

To visualize combined transformations, we may use table of values with the mapping rule $x \mapsto x + h$ and $y \mapsto ay + k$

x	y	$\Rightarrow$	$x + h$	$ay + k$
x_1	y_1		$x_1 + h$	$ay_1 + k$
x_2	y_2		$x_2 + h$	$ay_2 + k$
x_3	y_3		$x_3 + h$	$ay_3 + k$

where the two columns on the right represent the transformed function $y = af(x - h) + k$.

Example A.29. Graph $y = -2(x - 1)^2 + 3$.

Solution. Starting with $y = x^2$,

1. Shift right 1 unit

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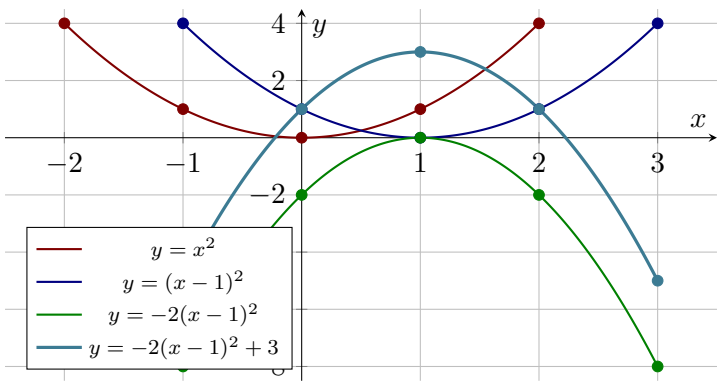
2. Stretch vertically by factor 2
3. Reflect across the x-axis
4. Shift up 3 units.

Using a table of values

x	y	$\Rightarrow$	$x+1$	y	$\Rightarrow$	$x+1$	$-2y$	$\Rightarrow$	$x+1$	$-2y+3$
-2	4		-1	4		-1	-8		-1	-5
-1	1		0	1		0	-2		0	1
0	0		1	0		1	0		1	3
1	1		2	1		2	-2		2	1
2	4		3	4		3	-8		3	-5



By applying transformations to a parent function, the domain and range will be affected, as the following theorem notes.

Theorem A.30. *Applying vertical translations affect the range, but not the domain. Similarly, applying horizontal translations affect the domain, but not the range.*

Example A.31. Let $f(x) = \sqrt{x}$. Graph the function $y = \sqrt{x-2} + 1$.

Solution. The domain of the transformed function is $x \geq 2$, and the range of the transformed function is $y \geq 1$. Using a table of values,

x	y	$\Rightarrow$	$x+2$	y	$\Rightarrow$	$x+2$	$y+1$
0	0		2	0		2	1
1	1		3	1		3	2
4	2		6	2		6	3
9	3		11	3		11	4
16	4		18	4		18	5

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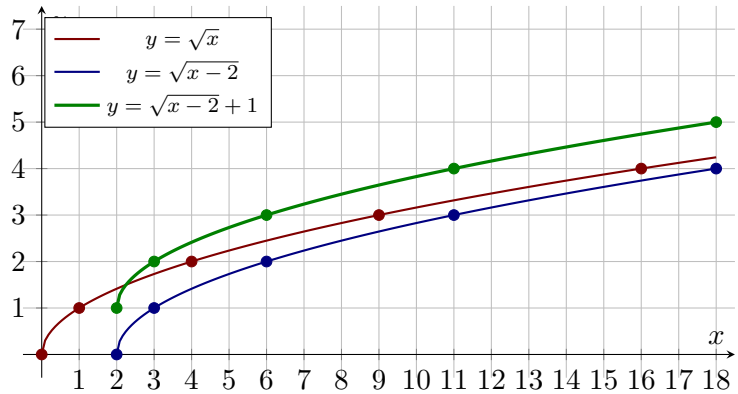

A.3 Inverse Functions

Many mathematical relationships are *reversible*: knowing the output allows us to recover the original input. Inverse functions formalize this idea. They allow us to "undo" a function's action and play a central role in advanced mathematics, particularly in calculus, where exponential and logarithmic functions are defined as inverses of one another.

In this section, we study inverse functions from algebraic, graphical, and logical perspectives. We develop the precise definition of an inverse function, explain why not every function has an inverse, and introduce the concept of one-to-one behaviour. We also learn how to find inverses algebraically and how to interpret them graphically as reflections across the line $y = x$.

Definition A.32. Let f be a function with domain D and range R. The *inverse function* of f, denoted f^{-1}, is the relation obtained by interchanging the roles of input and output. That is, if

$$f(x) = y$$

then

$$f^{-1}(y) = x$$

The notation f^{-1} does not mean $\frac{1}{f}$. It represents the inverse function, when it exists.

Example A.33. If $f(x) = 2x + 1$, then the inverse relation satisfies

$$y = 2x + 1 \iff x = \frac{y-1}{2}$$

Thus,

$$f^{-1}(x) = \frac{x-1}{2}$$

Definition A.34. A function f is said to be *one-to-one* (or *injective*) if distinct inputs produce distinct outputs. Formally, if x_1, x_2 are such that $f(x_1) = f(x_2)$, then $x_1 = x_2$.

A function is one-to-one of no horizontal line intersects its graph more than once.

Theorem A.35 (The Horizontal Line Test). *A function is one-to-one if and only if every horizontal line intersects its graph at most once.*

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Example A.36. The function

$$f(x) = x^3$$

is one-to-one, since it is strictly increasing on $\mathbb{R}$.

Example A.37. The function

$$f(x) = x^2$$

is not one-to-one on $\mathbb{R}$ since

$$f(2) = f(-2)$$

but clearly, $2 \neq -2$.

Theorem A.38. *A function has an inverse if and only if it is one-to-one.*

Consequently, if a function is not one-to-one on its natural domain, its inverse relation will fail to be a function.

Definition A.39. If a function is not one-to-one on its entire domain, it may become one-to-one when its domain is restricted.

Example A.40. Consider the function

$$f(x) = x^2$$

If we restrict the domain to $x \geq 0$, then f becomes one-to-one and has an inverse

$$f^{-1}(x) = \sqrt{x}$$

To find the inverse of a one-to-one function f,

1. Write $y = f(x)$
2. Interchange x and y
3. Solve for y
4. Replace y with $f^{-1}(x)$

Example A.41. Find the inverse of

$$f(x) = 3x - 5$$

Solution. Write $y = 3x - 5$. Interchanging x and y gives

$$x = 3y - 5$$

Solving for y gives

$$y = \frac{x+5}{3} \implies f^{-1}(x) = \frac{x+5}{3}$$

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Example A.42. Find the inverse of

$$f(x) = \frac{x-1}{2}$$

Solution. Write $y = \frac{x-1}{2}$. Interchanging x and y gives

$$x = \frac{y-1}{2}$$

Solving for y gives

$$2x = y - 1 \implies y = 2x + 1$$

and thus,

$$f^{-1}(x) = 2x + 1$$

Theorem A.43. *Functions f and g are inverses if and only if*

$$f(g(x)) = x \quad \text{and} \quad g(f(x)) = x$$

for all x in the appropriate domains.

Example A.44. Let $f(x) = 2x + 3$ and $g(x) = \frac{x-3}{2}$. Then

$$f(g(x)) = 2\left(\frac{x-3}{2}\right) + 3 = x$$

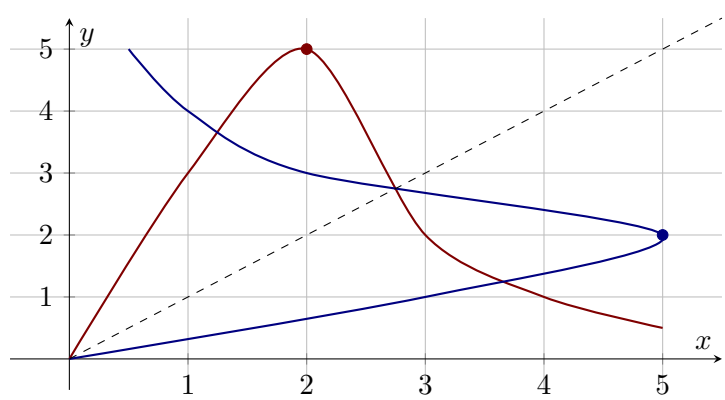
and

$$g(f(x)) = \frac{(2x+3)-3}{2} = x$$

Therefore, f and g are inverses.

Theorem A.45. *The graphs of a function and its inverse are reflections of each other across the line $y = x$.*

Example A.46. If the point $(2, 5)$ lies on the graph of f, then the point $(5, 2)$ lies on the graph of f^{-1}.



The domain and range of an inverse function f^{-1} is the range and domain of a function f, respectively.

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Theorem A.47. *If f has inverse f^{-1}, then*

- *The domain of f^{-1} is the range of f, and*
- *The range of f^{-1} is the domain of f.*

Example A.48. Let $f(x) = e^x$. Then the domain of e^x is $x \in \mathbb{R}$, and the range of e^x is $y > 0$. On the other hand, the inverse function

$$f^{-1}(x) = \ln(x)$$

has domain $x > 0$ and range $y \in \mathbb{R}$.

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Appendix B

MHF4U Review: Polynomial Functions and Rational Functions

Polynomial and rational functions form the algebraic backbone of advanced secondary mathematics and provide essential preparation for calculus. Their structure, behaviour, and solution methods appear repeatedly in limits, derivatives, curve sketching, and applied modelling problems. A strong understanding of these functions allows students to analyze graphs systematically, solve equations rigorously, and interpret algebraic expressions with confidence.

In this appendix, we first study polynomial functions, which are among the most fundamental and well-behaved classes of functions. Their continuity, end behaviour, and zero structure make them ideal for developing graphing intuition and algebraic fluency. We then turn to solving polynomial equations and inequalities, emphasizing factoring techniques, multiplicity of zeros, and sign analysis.

We then extend our study to rational functions, which are ratios of polynomials. Unlike polynomials, rational functions may have discontinuities, vertical asymptotes, and holes, introducing richer and more subtle behaviour. Understanding these features is crucial not only for graphing but also for later topics in calculus, such as limits and asymptotic analysis.

B.1 Polynomial Functions

Polynomial functions are among the most fundamental objects in algebra. They are algebraically simple, defined for all real numbers, and exhibit predictable large-scale behaviour. Because of their smoothness and structural clarity, polynomial functions play a central role in pre-calculus and form the foundation for many ideas in calculus, including derivatives, curve sketching, and approximation theory.

In this section, we study the definition and structure of polynomial functions, analyze their degree and leading coefficient, investigate zeros and multiplicity, and determine their end behaviour. We emphasize both algebraic reasoning and graphical interpretation, as these two perspectives reinforce one another.

Definition B.1. A *polynomial function* is a function of the form

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

where n is a nonnegative integer, $a_0, a_1, ..., a_n \in \mathbb{R}$, and $a_n \neq 0$. The integer n is called the *degree* of the polynomial and a_n is called the *leading coefficient*

Example B.2. Determine the degree and leading coefficient of the polynomial function

$$f(x) = 3x^4 - 2x^2 + 5x - 1$$

Solution. The degree of the polynomial is 4, and the leading coefficient is 3.

Theorem B.3. *Every polynomial function is defined for all real numbers.*

Theorem B.4. *Polynomial functions are continuous for all real numbers.*

This means their graphs have no breaks, holes, or asymptotes.

Definition B.5. A number r is called a *zero* (or *root*) of f if

$$f(r) = 0$$

Theorem B.6 (The Factor Theorem). *Let $f(x)$ be a polynomial. Then r is a zero of f if and only if $(x - r)$ is a factor of $f(x)$.*

Example B.7. Find the zeros of the function

$$f(x) = x^2 - 5x + 6$$

Solution. Factoring gives

$$f(x) = (x - 2)(x - 3)$$

Thus, since $(x - 2)$ and $(x - 3)$ are factors of $f(x)$, then $x = 2$ and $x = 3$ are zeros of $f(x)$.

Definition B.8. If

$$f(x) = (x - r)^k g(x)$$

where $g(r) \neq 0$, then r is zero of *multiplicity* k.

Theorem B.9. *Let r be a zero of multiplicity k.*

1. *If k is odd, the graph crosses the x-axis at $x = r$.*
2. *If k is even, the graph touches and turns at r.*

Example B.10. Classify the roots of the function

$$f(x) = (x - 1)^2(x + 2)$$

Solution. Note that $(x - 1)$ and $(x + 2)$ are factors of $f(x)$, so the roots of $f(x)$ are $x = 1$ with multiplicity 2, and $x = -2$ with multiplicity 1.

The large-scale behaviour of a polynomial is determined entirely by its leading term $a_n x^n$.

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Theorem B.11. *Let*

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

be a polynomial of degree n. As $x \to \pm\infty$, the graph behaves like $y = a_n x^n$.

There are two different cases, depending on the degree of the polynomial, and the leading coefficient.

Degree	**Sign**	**Graph**
Even	Positive	Both ends rise
Even	Negative	Both ends fall
Odd	Positive	Left end down, right end up
Odd	Negative	Left end up, right end down

Example B.12. Determine the end behaviour of the function

$$f(x) = -2x^3 + x$$

Solution. Since the degree of $f(x)$ is 3, and the leading coefficient is negative, then

- As $x \to -\infty$, $f(x) \to +\infty$, and
- As $x \to +\infty$, $f(x) \to -\infty$

Theorem B.13. *A polynomial of degree n has at most n real roots.*

Example B.14. A polynomial of degree 4 may have

- 4 real roots, or
- 2 real roots, but
- never more than 4 total roots.

To sketch the graph of a polynomial,

1. Determine the degree and leading coefficient.
2. Identify end behaviour.
3. Find real zeros and multiplicities.
4. Determine the y-intercept.
5. Sketch using behaviour at zeros and end behaviour.

Example B.15. Sketch the graph of

$$f(x) = x^3 - 4x$$

Solution. First we factor the polynomial

$$f(x) = x(x^2 - 4) = x(x-2)(x+2)$$

Thus, the zeros are $x = -2$, $x = 0$, and $x = 2$, all of which have multiplicity at 1. Furthermore, the y-intercept is 0.

Since the degree of the polynomial is 3, and the leading coefficient is positive,

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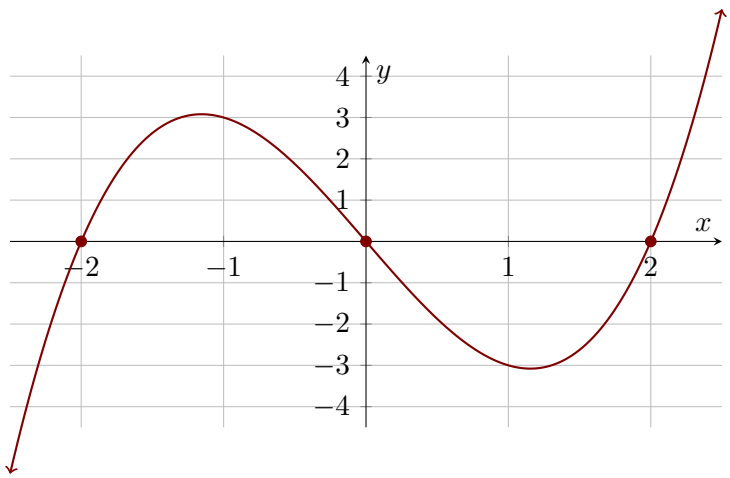
- As $x \to -\infty$, $f(x) \to -\infty$, and
- As $x \to +\infty$, $f(x) \to +\infty$.

Using the information above, we have the following sketch.



Polynomial functions provide a structured and predictable class of functions that serve as a foundation for rational functions and for calculus topics such as derivatives and curve sketching.

B.2 Solving Polynomials and Inequalities

Solving polynomial equations and inequalities is one of the most important algebraic skills developed in MHF4U. These techniques are foundational not only for pre-calculus, but also for calculus, where solving equations determines critical points, intercepts, and regions of increase/decrease.

Unlike linear equations, higher-degree polynomial equations require structural understanding: factoring, recognizing multiplicities, and interpreting sign changes. When solving inequalities, we go beyond finding zeros—we must determine where a polynomial is positive or negative. This requires careful reasoning and attention to logical structure.

Definition B.16. A *polynomial equation* is an equation of the form

$$f(x) = 0$$

where $f(x)$ is a polynomial.

Solving the equation means finding all real numbers x such that

$$f(x) = 0$$

To solve a polynomial equation, we must express the polynomial in factored form whenever possible.

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Theorem B.17 (**Zero Product Property**). *If*

$$ab = 0$$

then either $a = 0$ *or* $b = 0$.

This property allows us to solve polynomial equations once the polynomial is factored.

Example B.18. Solve

$$x^2 - 5x + 6 = 0$$

Solution. Factoring gives

$$(x-2)(x-3) = 0$$

Then by the Zero Product Property, either $x = 2$ or $x = 3$.

Factoring is not optional, and in fact, it is central. If the polynomial is not factored, the Zero Product Property cannot be applied.

Example B.19. Solve

$$x^3 - 4x = 0$$

Solution. Factor the left-hand side gives

$$x^3 - 4x = x(x^2 - 4) = x(x-2)(x+2) = 0$$

Therefore, we have either $x = -2$ or $x = 0$ or $x = 2$.

Always factor completely before starting solutions. As we have mentioned in Section B, whenever we have a factor $(x-r)^k$ appearing in a polynomial, then $x = r$ is a solution of multiplicity k.

Example B.20. Solve

$$(x-1)^2(x+3) = 0$$

Solution. The solutions are $x = 1$ (multiplicity 2) or $x = -3$.

Multiplicity affects graph behaviour but does not change the solution set. A repeated root is still a single solution, though counted multiple times in algebraic theory.

When solving polynomial equations of degree 4 or higher, we cannot rely on simple factoring. Instead, we proceed systematically.

Theorem B.21 (**Rational Root Theorem**). *Let*

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

be a polynomial with integer coefficients. If $\frac{p}{q}$ *(in lowest terms) is a rational root, then*

- p *divides the constant term* a_0*, and*
- q *divides the leading coefficient* a_n.

This theorem does not guarantee a rational root exist. It only tells us where to look.

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Definition B.22. *Synthetic division* is a simplified method for dividing a polynomial $f(x)$ by a linear factor of the form $x - r$.

Is is based on the Remainder Theorem.

Theorem B.23 (**The Remainder Theorem**). *If a polynomial $f(x)$ is divided by $x - r$, then the remainder is equal to $f(r)$. In particular, if $f(r) = 0$, then $x - r$ is a factor of $f(x)$.*

Instead of performing full polynomial long division, synthetic division efficiently computes the value of $f(r)$ and the coefficients of the reduced polynomial, if r is a root. It only works when dividing by linear factors of the form $x - r$.

To divide $f(x)$ by $x - r$,

1. Write only the coefficients of $f(x)$.
2. Place r on the left.
3. Bring down the first coefficient.
4. Multiply by r, add to the next coefficient.
5. Repeat.
6. The final number is the remainder.

If the final number is r, you have found a root. The remaining numbers form the coefficients of the reduced polynomial. Synthetic division is faster and cleaner than long division when testing possible rational roots.

Example B.24. Solve

$$x^3 - 4x^2 - 7x + 10 = 0$$

Solution. We first list the possible rational roots. Since the constant term is 10 and the leading coefficient is 1, the possible rational roots are ± 1, ± 2, ± 5.

Testing $x = 1$ gives

$$\begin{array}{c|cccc} 1 & 1 & -4 & -7 & 10 \\ & & 1 & -3 & -10 \\ \hline & 1 & -3 & -10 & 0 \end{array}$$

Thus, since the remainder is 0, then $x = 1$ is a root.

We now have

$$f(x) = (x - 1)(x^2 - 3x - 10)$$

Factoring the quadratic gives

$$f(x) = (x - 1)(x - 5)(x + 2) = 0$$

Therefore, the solutions are $x = 1$ or $x = 5$ or $x = -2$.

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Example B.25. Solve

$$f(x) = x^4 - 5x^3 + 6x^2 + 4x - 8 = 0$$

Solution. We first list the possible rational roots. Since the leading coefficient is 1 and the constant term is -8, the possible rational roots are ± 1, ± 2, ± 4, or ± 8.

Note if we test $x = 1$, we have

$$\begin{array}{c|ccccc} 1 & 1 & -5 & 6 & 4 & -8 \\ & & 1 & -4 & 2 & 6 \\ \hline & 1 & -4 & 2 & 6 & -2 \end{array}$$

Thus, the remainder is not zero, so $x = 1$ is not a root.

If we test $x = 2$, we have

$$\begin{array}{c|ccccc} 2 & 1 & -5 & 6 & 4 & -8 \\ & & 2 & -6 & 0 & 8 \\ \hline & 1 & -3 & 0 & 4 & 0 \end{array}$$

Thus, since the remainder is zero, $x = 2$ is a root. Thus,

$$f(x) = (x - 2)(x^3 - 3x^2 + 4)$$

Now continue factoring the cubic, in the same process.

Definition B.26. A *polynomial inequality* is an equality of the form

$$f(x) > 0, \quad f(x) \geq 0, \quad f(x) < 0, \quad \text{or} \quad f(x) \leq 0$$

where $f(x)$ is a polynomial.

Unlike equations, inequalities require determining where the polynomial is positive or negative.

To solve a polynomial inequality,

1. Rewrite the inequality so that one side equals zero.
2. Factor completely.
3. Determine the zeros.
4. Create a number line.
5. Test each sign of each interval.
6. Write the solution in interval notation.

Theorem B.27. *If a factor $(x - r)^k$ has*

- *odd multiplicity, then the sign changes at r.*
- *even multiplicity, then the sign does not change at r.*

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Example B.28. Solve

$$x^2 - 5x + 6 > 0$$

Solution. Factoring the left hand side gives

$$(x-2)(x-3) > 0$$

and the critical points at $x = 2$ and $x = 3$. Creating the sign chart

Interval	$(-\infty, 2)$	$(2, 3)$	$(3, \infty)$
$(x-2)$	$-$	$+$	$+$
$(x-3)$	$-$	$-$	$+$
$f(x)$	$+$	$-$	$+$

Therefore, $f(x) > 0$ whenever $x \in (-\infty, 2) \cup (3, \infty)$.

Notice that since the inequality is strict, the endpoints are not included.

Example B.29. Solve

$$(x-1)^2(x+2) \geq 0$$

Solution. The zeros of the above are $x = 1$ (multiplicity 2), and $x = -2$. Note that the sign changes at $x = -2$, but not at $x = 1$. Creating the sign chart

Interval	$(-\infty, -2)$	$(-2, 1)$	$(1, \infty)$
$(x-1)^2$	$+$	$+$	$+$
$(x+2)$	$-$	$+$	$+$
$f(x)$	$-$	$+$	$+$

Therefore, $f(x) \geq 0$ whenever $[-2, 1] \cup [1, \infty) = [-2, \infty)$.

B.3 Rational Functions

Rational functions extend polynomial functions by allowing division by another polynomial. While polynomial graphs are continuous and smooth everywhere, rational functions introduce new and richer behaviour: discontinuities, vertical asymptotes, horizontal asymptotes, and possible holes. These features make rational functions especially important in pre-calculus and in calculus, where limits and asymptotic behaviour are studied formally.

Definition B.30. A *rational function* is a function of the form

$$f(x) = \frac{p(x)}{q(x)}$$

where $p(x)$ and $q(x)$ are polynomials with $q(x) \neq 0$.

Definition B.31. The domain consists of all real numbers for which the denominator is nonzero $q(x) \neq 0$.

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Example B.32. Let

$$f(x) = \frac{x+1}{x-2}$$

Then the domain of this function is $x \neq 2$.

Unlike polynomials, rational functions may not be defined everywhere. The denominator controls the "danger points".

Since a fraction equals zero when the numerator equals zero, to find the x-intercepts, it suffices to determine when $p(x) = 0$, while $q(x) \neq 0$.

Definition B.33. Find the x-intercepts and y-intercept of the function

$$f(x) = \frac{x^2-4}{x-1}$$

Solution. Factoring the numerator gives

$$\frac{(x-2)(x+2)}{x-1}$$

Thus, the x-intercepts occur when the numerator is zero, so when

$$(x-2)(x+2) = 0$$

Therefore, $x = 2$ or $x = -2$.

The y-intercept occurs when $x = 0$. Indeed,

$$f(0) = \frac{-4}{-1} = 4$$

Definition B.34. The line $x = a$ is a *vertical asymptote* if as $x \to a^{\pm}$, $f(x) \to \pm\infty$. Algebraically, vertical asymptotes occur where $q(a) = 0$, and the factor does not cancel.

Example B.35. Find the vertical asymptote of the function

$$f(x) = \frac{x+1}{x-3}$$

Solution. The denominator is zero at $x = 3$, so the vertical asymptote is at $x = 3$.

Note that if a factor cancels, we get a hole, not an asymptote.

Definition B.36. If a factor appears in both numerator and denominator and cancels, the graph has a *hole* at that value.

Example B.37. Find the hole of the function

$$f(x) = \frac{(x-1)(x+2)}{x-1}$$

Solution. Simplifying the rational function gives

$$f(x) = x+2, \quad x \neq 1$$

Thus, there is a hole at $(1, 3)$.

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Theorem B.38. *Let*

$$f(x) = \frac{a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0}{b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0}$$

1. *If $n < m$, then there is a horizontal asymptote at $y = 0$.*
2. *If $n = m$, then there is a horizontal asymptote at $y = \frac{a_n}{b_m}$.*
3. *If $n > m$, then there is no horizontal asymptote.*

Example B.39. Find the horizontal asymptote of the function

$$f(x) = \frac{3x^2 + 1}{2x^2 - 5}$$

Solution. Since the degree of the numerator is equal to the degree of the denominator,

$$y = \frac{3}{2}$$

is a horizontal asymptote.

Horizontal asymptotes describe behaviour as $x \to \pm\infty$, not near vertical asymptotes. A rational function may cross its horizontal asymptote. To find crossing points, we solve

$$f(x) = \text{horizontal asymptote}$$

Theorem B.40. *To determine where a rational function is positive or negative,*

1. *Factor numerator and denominator.*
2. *Identify zeros and vertical asymptote.*
3. *Create sign chart.*
4. *Determine sign in each interval.*

Note that sign may change at either the zeros or at the vertical asymptotes.

Example B.41. Sketch the function

$$f(x) = \frac{(x-2)(x+2)}{(x-1)(x+1)}$$

Solution. First note that the domain is $x \neq \pm 1$. The x-intercepts occur when $y = 0$, and thus, when the numerator is zero

$$(x-2)(x+2) = 0 \implies x = \pm 2$$

and the y-intercept occurs when $x = 0$,

$$f(x) = \frac{-4}{-1} = 4$$

To find the vertical asymptotes, note that the denominator is zero whenever $x = \pm 1$, so the vertical asymptotes occur at $x = \pm 1$. Furthermore, observe that

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- As $x \to -1^-$, $f(x) \to -\infty$
- As $x \to -1^+$, $f(x) \to +\infty$
- As $x \to 1^-$, $f(x) \to +\infty$
- As $x \to 1^+$, $f(x) \to -\infty$

To find the horizontal asymptote, note that the degree of the numerator is equal to the degree of the denominator, so $y = 1$ is the horizontal asymptote. Furthermore, taking for example, $x = 1000$, we get

$$f(1000) \approx 0.999$$

this would imply that as $x \to \infty$, $f(x) \to 1^-$, and similarly, if $x = -1000$, we also get

$$f(-1000) \approx 0.999$$

so as $x \to -\infty$, $f(x) \to 1^-$.

To determine whether the function crosses the horizontal asymptote, we solve

$$\frac{(x+2)(x-2)}{(x-1)(x+1)} = 1$$

in which case we get

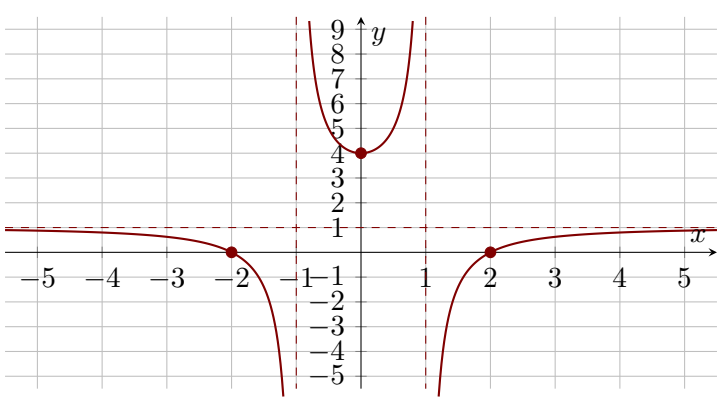
$$x^2 - 4 = x^2 - 1 \implies -4 = -1$$

which is impossible, so there is no crossing.

The critical values of $f(x)$ are $x = \pm 1, \pm 2$, so we create a sign chart for $f(x)$.

Interval	$(-\infty, -2)$	$(-2, -1)$	$(-1, 1)$	$(1, 2)$	$(2, \infty)$
$(x+2)$	$-$	$+$	$+$	$+$	$+$
$(x+1)$	$-$	$-$	$+$	$+$	$+$
$(x-1)$	$-$	$-$	$-$	$+$	$+$
$(x-2)$	$-$	$-$	$-$	$-$	$+$
$f(x)$	$+$	$-$	$+$	$-$	$+$

Now we sketch the function.



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B.4 Solving Rational Equations and Inequalities

Rational equations and inequalities extend the algebra of polynomial equations by introducing denominators that contain variables. This seemingly small change dramatically increases the need for logical care: solutions may be excluded from the domain, extraneous solutions may appear during manipulation, and sign behaviour becomes more intricate.

Definition B.42. A *rational equation* is an equation that contains one or more rational expressions, i.e. fractions whose numerator and/or denominator are polynomials.

For example,

$$\frac{p(x)}{q(x)} = \frac{r(x)}{s(x)}$$

Values that make any denominator equal to zero are excluded from the domain. These values can never be solutions. This is the most common source of errors. Students often "solve correctly", but forget to exclude values that make denominators zero. Always state the restrictions first.

To solve rational equations,

1. Determine domain restrictions.
2. Multiply both sides by the least common denominator.
3. Solve the resulting polynomial equation.
4. Check solutions against restriction.
5. State final solution set.

Theorem B.43. *If two rational expressions are equal and denominators are nonzero, multiply both sides by the LCD preserves equivalence.*

Example B.44. Solve

$$\frac{2}{x} = \frac{3}{x-1}$$

Solution. First note that $x \neq 0$ and $x \neq 1$. Multiplying by the least common denominator $x(x-1)$, we have

$$2(x-1) = 3x$$

Thus, solving the equation gives

$$\begin{aligned} 2x - 2 &= 3x \\ x &= -2 \end{aligned}$$

Note that $x = -2$ does not violate any restrictions, so $x = -2$ is the solution.

Example B.45. Solve

$$\frac{x}{x-2} = 1$$

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Solution. Note that $x \neq 2$. Multiplying by the least common denominator $x - 2$ gives

$$x = x - 2$$

Solving the equation gives

$$0 = -2$$

which is impossible. Therefore, there are no solutions.

This example shows that multiplying through may produce contradictions. Not every rational equation has a solution.

Definition B.46. A *rational inequality* is an inequality involving a rational expression, such as

$$\frac{p(x)}{q(x)} > 0$$

The key difference from equations is that we cannot multiply both sides by denominators, without knowing their sign. Thus, we must use sign analysis.

To solve a rational inequality,

1. Move all terms to one side.
2. Write as a single rational expression.
3. Factor numerator and denominator.
4. Identify critical numbers:
   - Zeros of numerator,
   - Zeros of denominator.
5. Create a sign chart.
6. Select intervals satisfying inequality.
7. Exclude denominator zeros.

Similar to polynomial inequalities and sign charts, we have the following.

Theorem B.47. *The sign of a rational function may change at*

- *Zeros of the numerator.*
- *Zeros of the denominator.*

Multiplicity determines whether the sign changes.

Denominator zeros are never included in the solution as they are not in the domain.

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Example B.48. Solve

$$\frac{x-1}{x+2} > 0$$

Solution. The numerator zero is $x = 1$ and the denominator zero is $x = -2$. The intervals for sign chart are then $(-\infty, -2)$, $(-2, 1)$, and $(1, \infty)$. Creating the sign chart,

Interval	$(-\infty, -2)$	$(-2, 1)$	$(1, \infty)$
$x+2$	$-$	$+$	$+$
$x-1$	$-$	$-$	$+$
$f(x)$	$+$	$-$	$+$

Therefore, the solution is $(-\infty, -2) \cup (1, \infty)$. Note that it is important to exclude $x = -2$, as it is undefined, and exclude $x = 1$ since inequality is strict.

Example B.49. Solve

$$\frac{(x-2)(x+1)}{x-3} \leq 0$$

Solution. The critical numbers are $x = -1$, $x = 2$, and $x = 3$. Creating the sign chart,

Interval	$(-\infty, -1)$	$(-1, 2)$	$(2, 3)$	$(3, \infty)$
$x+1$	$-$	$+$	$+$	$+$
$x-2$	$-$	$-$	$+$	$+$
$x-3$	$-$	$-$	$-$	$+$
$f(x)$	$-$	$+$	$-$	$+$

Therefore, the solution is $(-\infty, -1) \cup (2, 3)$. Since the inequality is ≤ 0, we include $x = -1$ and $x = 2$, but exclude $x = 3$. Thus, $(-\infty, -1] \cup [2, 3)$.

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Appendix C

MHF4U Review: Trigonometry

Trigonometry provides the mathematical framework for studying periodic behaviour, circular motion, and oscillatory phenomena. In advanced secondary mathematics, it connects geometry, algebra, and functions in a unified way. In calculus, trigonometric functions become essential tools for modeling motion, waves, growth cycles, and many natural processes.

This appendix reviews the foundational ideas of trigonometry with an emphasis on both geometric understanding and functional interpretation. We begin with angle measurement—especially radians, which are fundamental in calculus—and the unit circle, which provides the conceptual basis for defining sine, cosine, and tangent. From there, we study the graphs and transformations of trigonometric functions, emphasizing amplitude, period, phase shifts, and vertical shifts. Finally, we develop systematic methods for solving trigonometric equations, paying close attention to general solutions and interval restrictions.

C.1 Trigonometry

Trigonometry begins with the study of triangles, but quickly develops into the study of angles, circular motion, and periodic relationships. In earlier mathematics, trigonometric ratios were introduced using right triangles. In advanced mathematics, however, trigonometry is best understood using the unit circle, which allows us to define trigonometric functions for all real numbers, not just acute angles.

Definition C.1. A full revolution around a circle is divided into $360°$. One *degree* represents $\frac{1}{360}$ of a full rotation.

Definition C.2. An angle has measure 1 *radian* if it subtends an arc equal in length to the radius of the circle. More generally,

$$\theta = \frac{\text{arc length}}{\text{radius}}$$

Theorem C.3. *π radians is $180°$. Thus,*

$$\theta^\circ = \theta \cdot \frac{\pi}{180} \quad \textit{and } \theta \textit{ rad} = \theta^\circ \cdot \frac{180}{\pi}$$

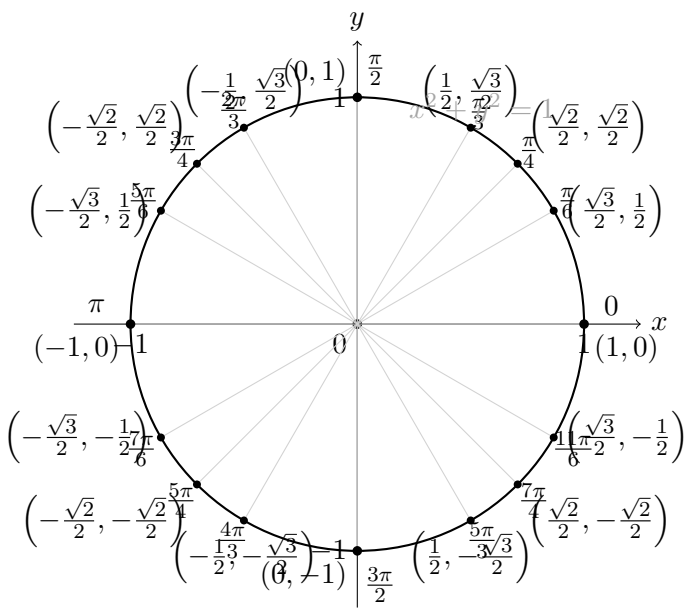
Example C.4. Convert 60° to radians. **Solution.** We have

$$60^\circ \cdot \frac{\pi}{180} = \frac{\pi}{3}$$

Radians are not just another unit. They are the natural unit of angle measure in calculus. All derivative formulas involving trigonometric functions require radians.

Definition C.5. The *unit circle* is the circle centered at the origin with radius 1

$$x^2 + y^2 = 1$$



Definition C.6. Let θ be an angle in standard position. If the terminal side intersects the unit circle at point (x, y), then

$$\cos(\theta) = x, \quad \sin(\theta) = y$$

Definition C.7. The *tangent* ratio is defined as

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$$

provided that $\cos(\theta) \neq 0$.

This definition extends sine and cosine to all real numbers, not just angles in triangles. The key angles that one should know are

$$0, \quad \frac{\pi}{6}, \quad \frac{\pi}{4}, \quad \frac{\pi}{3}, \quad \frac{\pi}{2}$$

These give the special values

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Angle	$\sin(\theta)$	$\cos(\theta)$
0	0	1
$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$
$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$
$\frac{\pi}{2}$	1	0

Example C.8. Find $\sin\left(\frac{5\pi}{6}\right)$.

Solution. Our reference angle is $\frac{\pi}{6}$. Since $\frac{5\pi}{6}$ is in the second quadrant, sine is positive, so

$$\sin\left(\frac{5\pi}{6}\right) = \frac{1}{2}$$

Theorem C.9.

- *Quadrant I: All positive.*
- *Quadrant II: Sine positive.*
- *Quadrant III: Tangent positive.*
- *Quadrant IV: Cosine positive.*

A useful mnemonic: "**A**ll **S**tudents **T**ake **C**alculus".

Theorem C.10. *From the unit circle equation*

$$x^2 + y^2 = 1$$

we obtain

$$\cos^2(\theta) + \sin^2(\theta) = 1$$

Dividing by $\cos^2(\theta)$ *gives*

$$1 + \tan^2(\theta) = \sec^2(\theta)$$

Dividing by $\sin^2(\theta)$ *gives*

$$\cot^2(\theta) + 1 = \csc^2(\theta)$$

Example C.11. If $\sin(\theta) = \frac{3}{5}$ and θ is in Quadrant I, find $\cos(\theta)$.

Solution. Using

$$\cos^2(\theta) = 1 - \sin^2(\theta) = 1 - \left(\frac{3}{5}\right)^2 = 1 - \frac{9}{25} = \frac{16}{25}$$

Therefore,

$$\cos(\theta) = \frac{4}{5}$$

Definition C.12. A function f is *T-periodic* if there exists $T > 0$ such that

$$f(x + T) = f(x)$$

for all x.

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Theorem C.13. $\sin(x+2\pi) = \sin(x)$ *and* $\cos(x+2\pi) = \cos(x)$. *Thus, sine and cosine have period* 2π. *For tangent,* $\tan(x+\pi) = \tan(x)$, *so tangent has period* π.

Periodicity explains oscillatory motion including waves, seasons, sound, or rotations. These become fundamental in later function analysis.

Example C.14. Find $\cos\left(\frac{13\pi}{6}\right)$.

Solution. Rewrite

$$\frac{13\pi}{6} = 2\pi + \frac{\pi}{6}$$

Since cosine has period 2π,

$$\cos\left(\frac{13\pi}{6}\right) = \cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

Theorem C.15 (**Compound Angle Formulas**). *For any real numbers* α *and* β,

$$\begin{aligned}
\sin(\alpha+\beta) &= \sin(\alpha)\cos(\beta) + \cos(\alpha)\sin(\beta) \\
\sin(\alpha-\beta) &= \sin(\alpha)\cos(\beta) - \cos(\alpha)\sin(\beta) \\
\cos(\alpha+\beta) &= \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta) \\
\cos(\alpha-\beta) &= \cos(\alpha)\cos(\beta) + \sin(\alpha)\sin(\beta)
\end{aligned}$$

These formulas are extremely important. They allow us to

- Evaluate trigonometric expressions without a calculator
- Derive double angle formulas
- Solve trigonometric equations
- Develop identities
- Eventually prove derivative formulas in calculus.

These identities can be derived using rotation matrices, or coordinates of rotated points on the unit circle. For example, rotating a point $(\cos(\alpha), \sin(\alpha))$ by angle β produces $(\cos(\alpha+\beta), \sin(\alpha+\beta))$. Expanding this rotation algebraically leads directly to the formulas above.

Example C.16. Evaluate $\sin\left(\frac{5\pi}{12}\right)$.

Solution. Write

$$\frac{5\pi}{12} = \frac{\pi}{4} + \frac{\pi}{6}$$

and apply compound formula

$$\begin{aligned}
\sin\left(\frac{\pi}{4} + \frac{\pi}{6}\right) &= \sin\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{6}\right) + \cos\left(\frac{\pi}{4}\right)\sin\left(\frac{\pi}{6}\right) \\
&= \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} + \frac{1}{2} \\
&= \frac{\sqrt{6}+\sqrt{2}}{4}
\end{aligned}$$

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By setting $\beta = \alpha$ in the compound formulas, we have

$$\begin{aligned}\sin(2\alpha) &= 2\sin(\alpha)\cos(\alpha)\\ \cos(2\alpha) &= \cos^2(\alpha) - \sin^2(\alpha)\\ \tan(2\alpha) &= \frac{2\tan(\alpha)}{1-\tan^2(\alpha)}\end{aligned}$$

Using the Pythagorean identity, the cosine can also be written as

$$\begin{aligned}\cos(2\alpha) &= 1 - 2\sin^2(\alpha)\\ &= 2\cos^2(\alpha) - 1\end{aligned}$$

Notice that cosine has three different equivalent forms. This flexibility is extremely useful when solving identities and equations.

Example C.17. If $\sin(\theta) = \frac{3}{5}$ and θ is in Quadrant I, find $\cos(2\theta)$.

Solution. First find $\cos(\theta)$

$$\cos(\theta) = \frac{4}{5}$$

Now use

$$\begin{aligned}\cos(2\theta) &= \cos^2(\theta) - \sin^2(\theta)\\ &= \left(\frac{4}{5}\right)^2 - \left(\frac{3}{5}\right)^2\\ &= \frac{16}{25} - \frac{9}{25} = \frac{7}{25}\end{aligned}$$

Theorem C.18 (**Half-Angle Formulas**). *Using the double-angle identity for cosine, we solve for* $\sin^2(\theta)$ *and* $\cos^2(\theta)$ *to obtain*

$$\begin{aligned}\sin^2\left(\frac{\theta}{2}\right) &= \frac{1-\cos(\theta)}{2}\\ \cos^2\left(\frac{\theta}{2}\right) &= \frac{1+\cos(\theta)}{2}\end{aligned}$$

Thus,

$$\begin{aligned}\sin\left(\frac{\theta}{2}\right) &= \pm\sqrt{\frac{1-\cos(\theta)}{2}}\\ \cos\left(\frac{\theta}{2}\right) &= \pm\sqrt{\frac{1+\cos(\theta)}{2}}\end{aligned}$$

The sign depends on the quadrant of $\frac{\theta}{2}$. This is extremely important. Students often forget this.

Theorem C.19 (**Tangent Half-Angle Formula**).

$$\tan\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos(\theta)}{1+\cos(\theta)}} = \frac{\sin(\theta)}{1+\cos(\theta)} = \frac{1-\cos(\theta)}{\sin(\theta)}$$

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Example C.20. Find $\sin\left(\frac{\pi}{12}\right)$.

Solution. Since $\frac{\pi}{12} = \frac{1}{2} \cdot \frac{\pi}{6}$, using the half-angle formula,

$$\sin\left(\frac{\pi}{12}\right) = \sqrt{\frac{1-\cos\left(\frac{\pi}{6}\right)}{2}} = \sqrt{\frac{1-\frac{\sqrt{3}}{2}}{2}} = \sqrt{\frac{2-\sqrt{3}}{4}} = \frac{\sqrt{2-\sqrt{3}}}{2}$$

Since the angle is in Quadrant I, we take the positive root.

The following formulas convert products of sine and cosine into sums.

Theorem C.21 (**Product-to-Sum Identities**). *For any real numbers α and β*

$$\begin{aligned}
\sin(\alpha)\sin(\beta) &= \frac{1}{2}\left[\cos(\alpha-\beta) - \cos(\alpha+\beta)\right] \\
\cos(\alpha)\cos(\beta) &= \frac{1}{2}[\cos(\alpha-\beta) + \cos(\alpha+\beta)] \\
\sin(\alpha)\cos(\beta) &= \frac{1}{2}[\sin(\alpha+\beta) + \sin(\alpha-\beta)]
\end{aligned}$$

These identities are less commonly used in MHF4U problem-solving, but they are extremely important in areas including signal processing, Fourier analysis, advanced calculus, or physics. They are derived from the compound angle formulas.

Example C.22. Simplify $\sin(x)\cos(x)$.

Solution. Use product-to-sum

$$\frac{1}{2}\sin(2x)$$

Summary of Trigonometric Identities

For clarity and organization, we summarize the major identities studied.

Reciprocal Identities

$$\sin(\theta) = \frac{1}{\csc(\theta)}, \quad \cos(\theta) = \frac{1}{\sec(\theta)}, \quad \tan(\theta) = \frac{1}{\cot(\theta)}$$

Quotient Identities

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}, \quad \cot(\theta) = \frac{\cos(\theta)}{\sin(\theta)}$$

Pythagorean Identities

$$\begin{aligned}
\sin^2(\theta) + \cos^2(\theta) &= 1 \\
1 + \tan^2(\theta) &= \sec^2(\theta) \\
\cot^2(\theta) + 1 &= \csc^2(\theta)
\end{aligned}$$

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Even–Odd Identities

$$\sin(-\theta) = -\sin(\theta), \quad \cos(-\theta) = \cos(\theta), \quad \tan(-\theta) = -\tan(\theta)$$

Periodicity

$$\sin(\theta + 2\pi) = \sin(\theta), \quad \cos(\theta + 2\pi) = \cos(\theta), \quad \tan(\theta + \pi) = \tan(\theta)$$

Compound Angle Formulas

$$\begin{aligned}\sin(A \pm B) &= \sin(A)\cos(B) \pm \cos(A)\sin(B)\\ \cos(A \pm B) &= \cos(A)\cos(B) \mp \sin(A)\sin(B)\end{aligned}$$

Double Angle Formulas

$$\begin{aligned}\sin(2\theta) &= 2\sin(\theta)\cos(\theta)\\ \cos(2\theta) &= \cos^2(\theta) - \sin^2(\theta) = 1 - 2\sin^2(\theta) = 2\cos^2(\theta) - 1\end{aligned}$$

Half-Angle Formulas

$$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos(\theta)}{2}}, \quad \cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1+\cos(\theta)}{2}}$$

Sum-to-Product Formulas

$$\begin{aligned}\sin(A) + \sin(B) &= 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)\\ \sin(A) - \sin(B) &= 2\sin\left(\frac{A-B}{2}\right)\cos\left(\frac{A+B}{2}\right)\\ \cos(A) + \cos(B) &= 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)\\ \cos(A) - \cos(B) &= -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)\end{aligned}$$

Product-to-Sum Identities

$$\begin{aligned}\sin(A)\sin(B) &= \frac{1}{2}\left[\cos(A-B) - \cos(A+B)\right],\\ \cos(A)\cos(B) &= \frac{1}{2}\left[\cos(A-B) + \cos(A+B)\right],\\ \sin(A)\cos(B) &= \frac{1}{2}\left[\sin(A+B) + \sin(A-B)\right],\\ \cos(A)\sin(B) &= \frac{1}{2}\left[\sin(A+B) - \sin(A-B)\right].\end{aligned}$$

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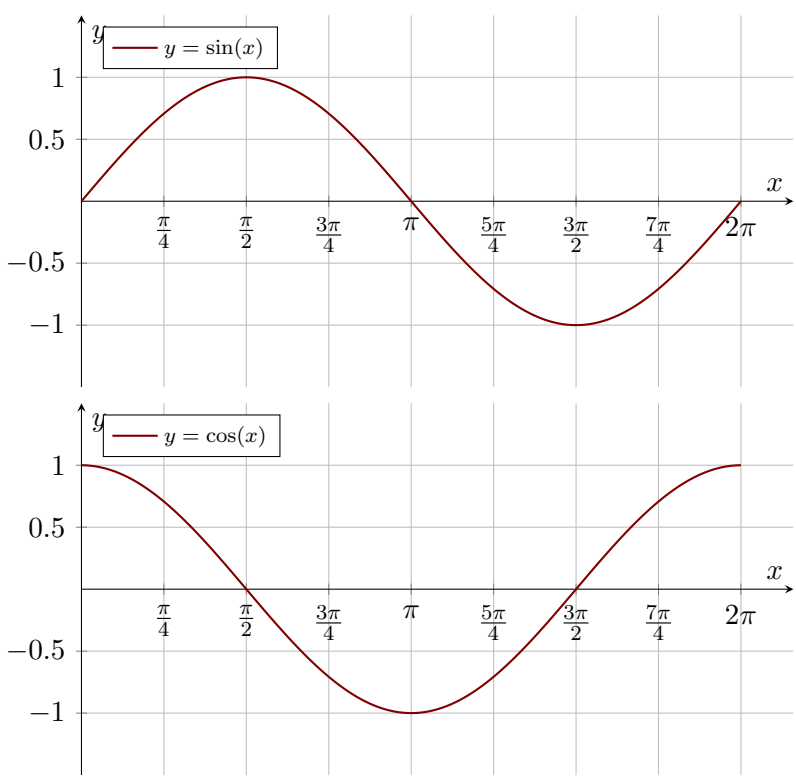
C.2 Trigonometric Functions

In Section C.1, we developed trigonometric functions using the unit circle and algebraic identities. In this section, we shift perspective and study trigonometric functions as graphs of periodic functions.

We first study the key properties of the primary trigonometric functions.

Property	**Sine**	**Cosine**	**Tangent**
Domain	$(-\infty, \infty)$	$(-\infty, \infty)$	$(-\infty, \infty) \setminus \left\{\frac{\pi}{2} + k\pi : k \in \mathbb{Z}\right\}$
Range	$[-1, 1]$	$[-1, 1]$	$(-\infty, \infty)$
Period	2π	2π	π
Zeros	$k\pi, k \in \mathbb{Z}$	$\frac{\pi}{2} + k\pi, k \in \mathbb{Z}$	$k\pi, k \in \mathbb{Z}$
Vertical Asymptotes	–	–	$\frac{\pi}{2} + k\pi, k \in \mathbb{Z}$
Maximum	1	1	–
Minimum	-1	-1	–

Below are the graphs of the three functions graphed on $[0, 2\pi]$.



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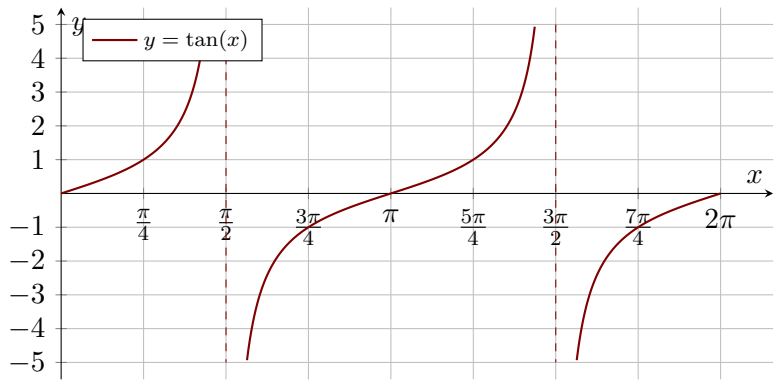

Notice that $\cos(x)$ is simply a horizontal shift of sine. That is,

$$\cos(x) = \sin\left(x + \frac{\pi}{2}\right)$$

Unlike sine and cosine, tangent is not bounded. Its asymptotic behaviour will later connect to limits in calculus.

Definition C.23. A general *sinusoidal function* takes the form

$$y = a\sin(b(x-c)) + d$$

where a is the *amplitude*, b is the *frequency*, c is the *phase shift*, and d is the *vertical shift*.

Definition C.24. The amplitude is the maximum vertical displacement from the midline.

Definition C.25. The period of a sinusoidal function is defined to be

$$\frac{2\pi}{|b|}$$

Theorem C.26. *For $y = a\sin(b(x-c)) + d$,*

- *Stretch/compress vertically by factor of a*
- *Compress horizontally by factor $\frac{1}{b}$*
- *Shift horizontally by c*
- *Shift vertically by d.*

Example C.27. Graph one cycle of $y = 2\sin(3x)$ on $[0, 2\pi]$ using transformations.

Solution. Here, we note that the amplitude is $|a| = 2$, and the period is $\frac{2\pi}{3}$. We use a table of values to graph the parent function $y = \sin(x)$ and $y = 2\sin(3x)$.

x	y	$\Rightarrow$	$\frac{1}{3}x$	$2y$
0	0		0	0
$\frac{\pi}{2}$	1		$\frac{\pi}{6}$	2
π	0		$\frac{\pi}{3}$	0
$\frac{3\pi}{2}$	-1		$\frac{\pi}{2}$	-2
2π	0		$\frac{2\pi}{3}$	0

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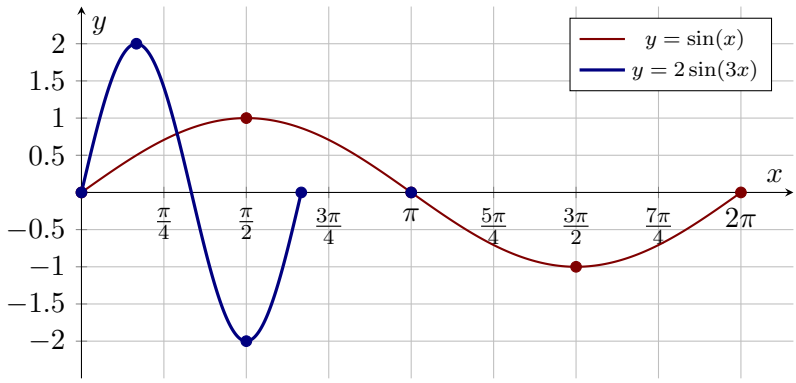
Thus, we obtain the following plot.



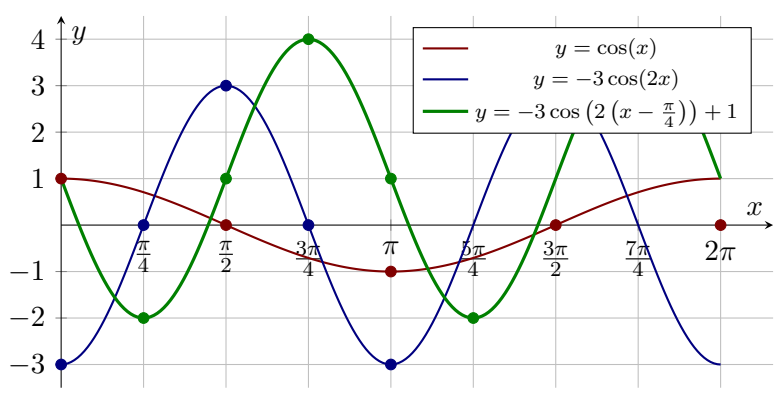
Increasing the value of b compresses the graph horizontally, and thus, causes the graph to oscillate faster.

Example C.28. Graph $y = -3\cos\left(2\left(x - \frac{\pi}{4}\right)\right) + 1$ using transformations.

Solution. Here, the amplitude is 3, the period is $\frac{2\pi}{2} = \pi$, the phase shift is $\frac{\pi}{4}$ to the right, and the vertical shift is $y = 1$. Furthermore, there is a reflection about the x-axis. Using a table of values

x	y	$\Rightarrow$	$\frac{1}{2}x$	$-3y$	$\Rightarrow$	$\frac{1}{2} + \frac{\pi}{4}$	$-3y + 1$
0	1		0	-3		$\frac{\pi}{4}$	-2
$\frac{\pi}{2}$	0		$\frac{\pi}{4}$	0		$\frac{\pi}{2}$	1
π	-1		$\frac{\pi}{2}$	3		$\frac{3\pi}{4}$	4
$\frac{3\pi}{2}$	0		$\frac{3\pi}{4}$	0		π	1
2π	1		π	-3		$\frac{5\pi}{4}$	-2

Thus, obtain the following plots



C.3 Solving Trigonometric Equations

In previous sections, we defined trigonometric functions using the unit circle and studied their graphical behaviour. In this section, we develop systematic methods for solving trigonometric equations.

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Unlike polynomial equations, trigonometric equations often have infinitely many solutions because trigonometric functions are periodic. Therefore, solutions must be expressed either

- As general solutions (using integer parameters), or
- As solutions restricted to a given interval (for example, $0 \leq x < 2\pi$).

Definition C.29. A *trigonometric equation* is an equation one or more trigonometric functions of a variable.

For example,

$$\sin(x) = \frac{1}{2}$$

is a trigonometric equation.

To solve trigonometric equations,

1. Isolate the trigonometric function.
2. Use the unit circle to find reference angles.
3. Determine all angles in the specified interval.
4. Write the general solution, if required.

Theorem C.30 (**General Solutions for Sine**). *If*

$$\sin(x) = \sin(\alpha)$$

then

$$x = \alpha + 2\pi n \quad \textit{or} \quad x = (\pi - \alpha) + 2\pi n$$

for $n \in \mathbb{Z}$.

Theorem C.31 (**General Solution for Cosine**). *If*

$$\cos(x) = \cos(\alpha)$$

then

$$x = \alpha + 2\pi n \quad \textit{or} \quad x = -\alpha + 2\pi n$$

for $n \in \mathbb{Z}$.

Example C.32. Solve

$$\sin(x) = \frac{1}{2}$$

on the interval $0 \leq x < 2\pi$.

Solution. Here, the reference angle is $\frac{\pi}{6}$. Since sine is positive in Quadrants I and II, the solutions are

$$x = \frac{\pi}{6} \quad \text{or} \quad x = \frac{5\pi}{6}$$

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If the interval was not specified in the above example, then the general solutions would be

$$x = \frac{\pi}{6} + 2\pi n \quad \text{or} \quad x = \frac{5\pi}{6} + 2\pi n$$

for $n \in \mathbb{Z}$.

Students often forget the second quadrant solution. Always use the ASTC rule.

Theorem C.33 (**General Solutions for Tangent**). *If*

$$\tan(x) = \tan(\alpha)$$

then

$$x = \alpha + \pi n$$

for $n \in \mathbb{Z}$.

Example C.34. Solve

$$\tan(x) = 1$$

Solution. Here, the reference angle is $\frac{\pi}{4}$, so the general solution is

$$x = \frac{\pi}{4} + \pi n$$

for $n \in \mathbb{Z}$.

There are other systematic techniques to solve trigonometric equations.

Example C.35. Solve $2\sin(x) - 1 = 0$ on the interval $0 \leq x < 2\pi$.

Solution. Rewrite the equation as

$$\begin{aligned} 2\sin(x) &= 1 \\ \sin(x) &= \frac{1}{2} \end{aligned}$$

and thus, we have found the solutions earlier, so

$$x = \frac{\pi}{6} \quad \text{or} \quad x = \frac{5\pi}{6}$$

Example C.36. Solve $2\cos^2(x) - 3\cos(x) + 1 = 0$.

Solution. Observe we have a quadratic equation in $\cos(x)$. To see this, let $u = \cos(x)$. We have

$$2u^2 - 3u + 1 = 0$$

Factoring the left side gives

$$(2u - 1)(u - 1) = 0$$

Thus, either

$$\cos(x) = \frac{1}{2} \quad \text{or} \quad \cos(x) = 1$$

First solving $\cos(x) = \frac{1}{2}$ gives

$$x = \frac{\pi}{3} \quad \text{or} \quad x = \frac{5\pi}{3}$$

Then solving $\cos(x) = 1$, we have

$$x = 0$$

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Example C.37. Solve $\sin(x)\cos(x) = 0$.

Solution. Using the Zero Product Property, we have either $\sin(x) = 0$ or $\cos(x) = 0$. Thus, we have

$$x = n\pi \quad \text{or} \quad x = \frac{\pi}{2} + n\pi$$

for $n \in \mathbb{Z}$.

Example C.38. Solve

$$\sin(2x) = \frac{\sqrt{3}}{2}$$

Solution. We first solve for $2x$

$$2x = \frac{\pi}{3} + 2\pi n \quad \text{or} \quad 2x = \frac{2\pi}{3} + 2\pi n$$

Dividing both sides by 2 gives

$$x = \frac{\pi}{6} + \pi n \quad \text{or} \quad x = \frac{\pi}{3} + \pi n$$

for $n \in \mathbb{Z}$.

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Appendix D

MHF4U Review: Exponential and Logarithmic Functions

Exponential and logarithmic functions model some of the most important phenomena in mathematics and the natural sciences: population growth, radioactive decay, compound interest, sound intensity, pH levels, and many other processes that change multiplicatively rather than additively. Unlike polynomial functions, which grow at rates determined by powers of x, exponential functions grow (or decay) at rates proportional to their current value. This fundamental difference gives them distinctive algebraic and graphical behaviour.

In this appendix, we study exponential functions first, focusing on their laws, transformations, and equations. We then introduce logarithmic functions as the inverse functions of exponentials, emphasizing their properties, domain restrictions, and algebraic rules. A careful understanding of logarithms is essential, as they provide the primary method for solving exponential equations that cannot be solved by rewriting with common bases.

D.1 Exponential Functions

In earlier algebra, we studied polynomial functions such as x^2 or x^3 where the variable appears in the base. In this section, we study a fundamentally different type of function: exponential functions, where the variable appears in the exponent.

This difference leads to dramatically different growth behaviour. Polynomial functions grow by repeated multiplication of the variable, while exponential functions grow (or decay) by repeated multiplication of a constant base. This produces rapid growth or decay that models many real-world phenomena, including population growth, compound interest, and radioactive decay.

Before introducing exponential functions, let us review laws of exponents.

Theorem D.1 (**Laws of Exponents**). *Let $a > 0$ and let $m, n \in \mathbb{R}$. Then*

1. $a^m \cdot a^n = a^{m+n}$.
2. $\frac{a^m}{a^n} = a^{m-n}$
3. $(a^m)^n = a^{mn}$

4. $(ab)^n = a^n b^n$

5. $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$

6. $a^0 = 1$

7. $a^{-n} = \frac{1}{a^n}$

8. $a^{\frac{m}{n}} = \sqrt[n]{a^m}$.

Exponent rules apply only when the bases are the same. For example,

$$2^x \cdot 3^x \neq 6^x$$

However

$$2^x \cdot 2^y = 2^{x+y}$$

Definition D.2. A function of the form $f(x) = a^x$ is called an *exponential function* where $a > 0$ and $a \neq 1$.

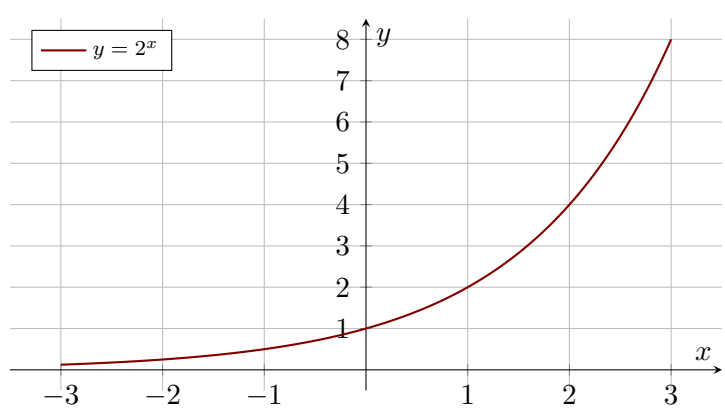
Note that if $a = 1$, then $1^x = 1$ is the constant function, and if $a \leq 0$, then the function is not defined for all real x.

Theorem D.3. *For $f(x) = a^x$, the domain is $(-\infty, \infty)$, the range is $(0, \infty)$, and the horizontal asymptote is at $y = 0$.*

Exponential functions are always positive. They never touch or cross the x-axis.

Definition D.4. For $f(x) = a^x$,

- If $a > 1$, the function represents an *exponential growth*.
- If $0 < a < 1$, the function represents *exponential decay*.



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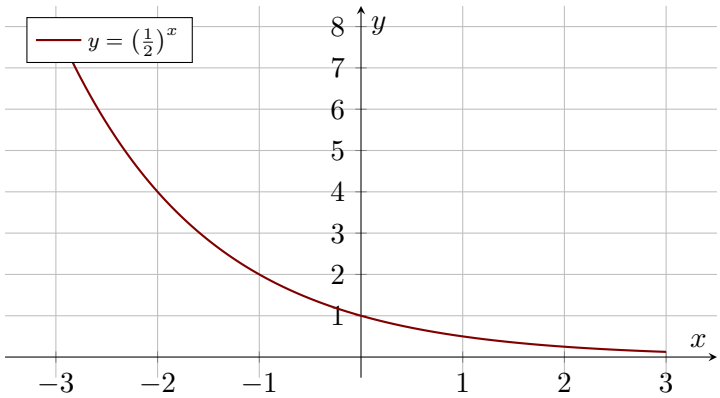

For growth, as $x \to \infty$, $f(x) \to \infty$, and for decay, as $x \to \infty$, $f(x) \to 0$.

Theorem D.5. *For all exponential functions,* $f(0) = 1$. *Thus, every exponential function passes through* $(0, 1)$.

Theorem D.6. *Exponential functions are one-to-one. Thus, they have inverses (known as logarithmic functions)*

This monotonic behaviour is crucial when solving equations later.

Definition D.7. The general form for transformations of exponential functions is given by

$$y = Aa^{B(x-C)} + D$$

where A is the vertical stretch/reflection, B is the horizontal compression/stretch, C is the horizontal shift, and D is the vertical shift. The horizontal asymptote of the transformed function is the line $y = D$.

Example D.8. Graph $y = 3(2^x) - 4$ using transformations.

Solution. Note that the parent function $y = 2^x$ is being vertically stretched by a factor of 3, and then translated 4 units down. Using a table of values

x	y	$\Rightarrow$	x	$3y$	$\Rightarrow$	x	$3y-4$
-2	$\frac{1}{4}$		-2	$\frac{3}{4}$		-2	$-\frac{13}{4}$
-1	$\frac{1}{2}$		-1	$\frac{3}{2}$		-1	$-\frac{5}{2}$
0	1		0	3		0	-1
1	2		1	6		1	2
2	4		2	12		2	8

Thus, we have the following plot shown below.

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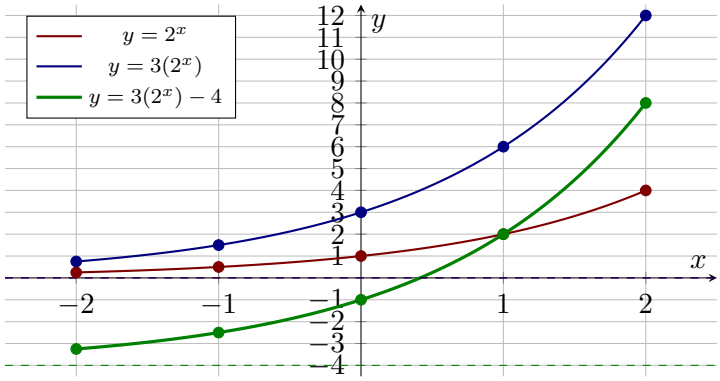

Definition D.9. The constant e is defined approximately as

$$e \approx 2.71828$$

Definition D.10. The natural exponential function is defined as

$$f(x) = e^x$$

The function e^x has unique properties in calculus. Generally, it models continuous growth, and also arises naturally in compound interest.

Definition D.11. The exponential growth formula is defined as

$$P(t) = P_0 a^t$$

where P_0 is the initial amount, and $a > 1$ is the growth factor.

Example D.12. The continuous growth model is generally defined by

$$P(t) = P_0 e^{kt}$$

where $k > 0$ is the growth rate.

Example D.13. A population starts at 500 and doubles every year. This can be modelled by

$$P(t) = 500(2^t)$$

Similar to exponential growth, we also have exponential decay.

Definition D.14. The exponential decay formula is defined as

$$P(t) = P_0 a^t$$

for $a \in (0, 1)$.

Example D.15. The continuous decay model is generally defined as

$$P(t) = P_0 e^{-kt}$$

where $k > 0$ is the decay rate.

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Example D.16. A radioactive substance decreases by 20% per year. The remaining fraction left over is 0.8. This can be modelled by

$$P(t) = P_0(0.8)^t$$

Theorem D.17. *For sufficiently large x, a^x grows faster than x^n for any fixed n.*

This theorem explains why exponential processes quickly outpace polynomial ones. This idea becomes fundamental in calculus and asymptotic analysis.

D.2 Solving Exponential Equations

In Section D.1, we defined exponential functions and studied their properties and models. We now turn to the problem of solving exponential equations, that is, equations in which the variable appears in the exponent.

Unlike polynomial equations, exponential equations often require

- Using exponent laws,
- Rewriting expressions with a common base,
- Applying inverse operations (logarithms).

A key structural idea is that exponential functions are one-to-one. This property allows us to equate exponents when the bases are equal and guarantees uniqueness of solutions.

In this section, we develop systematic methods for solving exponential equations in increasingly general settings.

Theorem D.18. *Let $a > 0$ with $a \neq 1$. If $a^x = a^y$, then $x = y$.*

This theorem justifies "dropping the base" when both sides have the same base. It only works when the bases are identical and positive. Whenever possible, rewrite both sides using powers of the same base.

Example D.19. Solve $2^{x+1} = 8$.

Solution. Rewrite 8 as 2^3, so then

$$2^{x+1} = 2^3$$

By Theorem D.18,

$$\begin{aligned} x + 1 &= 3 \\ x &= 2 \end{aligned}$$

Example D.20. Solve $9^x = 3^{2x-1}$.

Solution. Rewrite 9 as 3^2 so that

$$\begin{aligned} (3^2)^x &= 3^{2x-1} \\ 3^{2x} &= 3^{2x-1} \end{aligned}$$

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Thus,

$$2x = 2x - 1$$

This gives

$$0 = -1$$

which is impossible. Thus, there are no solutions.

Not all exponential equations have solutions. Always check algebra carefully.

Example D.21. Solve $5^{2x} = 5^{x+3}$.

Solution. Using Theorem D.18, since the bases are the same equating the exponents give

$$\begin{aligned} 2x &= x + 3 \\ x &= 3 \end{aligned}$$

Example D.22. Solve $4^x = 2^{x+1}$.

Solution. Rewrite $4 = 2^2$ so that

$$\begin{aligned} (2^2)^x &= 2^{x+1} \\ 2^{2x} &= 2^{x+1} \end{aligned}$$

Equate exponents

$$\begin{aligned} 2x &= x + 1 \\ x &= 1 \end{aligned}$$

Sometimes expressions cannot be written using the same base. For example,

$$2^x = 7$$

we cannot rewrite 7 as a power of 2. We need logarithms, which will be discussed in more detail in Section D.3.

Theorem D.23. *If $a^x = b$, then $x = \log_a(b)$.*

Using natural logarithms

$$\begin{aligned} a^x &= b \\ \ln(a^x) &= \ln(b) \\ x\ln(a) &= \ln(b) \\ x &= \frac{\ln(b)}{\ln(a)} \end{aligned}$$

Example D.24. Solve $2^x = 7$.

Solution. Take the natural logarithm

$$\begin{aligned} \ln(2^x) &= \ln(7) \\ x\ln(2) &= \ln(7) \\ x &= \frac{\ln(7)}{\ln(2)} \end{aligned}$$

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Logarithms allow us to "bring down" the exponent. This is why logs are the inverse of exponential functions.

Example D.25. Solve $3e^{2x} = 5$.

Solution. Isolate the exponential term

$$e^{2x} = \frac{5}{3}$$

and then take natural logarithm

$$\begin{aligned} \ln(e^{2x}) &= \ln\left(\frac{5}{3}\right) \\ 2x &= \ln\left(\frac{5}{3}\right) \\ x &= \frac{1}{2}\ln\left(\frac{5}{3}\right) \end{aligned}$$

Example D.26. Solve $2^{2x} - 5(2^x) + 6 = 0$.

Solution. Observe that this is a quadratic equation in 2^x. Indeed, let $u = 2^x$. Then

$$u^2 - 5u + 6 = 0$$

Factoring

$$(u-2)(u-3) = 0$$

Thus, either $2^x = 2$ or $2^x = 3$. Solving each, gives $x = 1$ or $x = \frac{\ln(3)}{\ln(2)}$.

D.3 Logarithmic Functions

In Section D.2, we solved exponential equations using logarithms. We now study logarithmic functions formally. Logarithms are defined as the inverse functions of exponential functions. They allow us to "undo" exponentiation and solve equations in which the variable appears in the exponent.

Logarithmic functions play a central role in algebra and calculus. In calculus, the natural logarithm $\ln(x)$ is especially important because of its unique derivative properties.

Definition D.27. Let $a > 0$ with $a \neq 1$. The logarithm base a of a positive number x is defined by

$$\log_a(x) = y \iff a^y = x$$

A logarithm answers the question: "To what power must we raise a to obtain x?"

Example D.28. $\log_2(8) = 3$ because $2^3 = 8$.

Example D.29. $\log_5(1) = 0$ because $5^0 = 1$.

Definition D.30. The *common logarithm* is the logarithm base 10, denoted by $\log(x)$.

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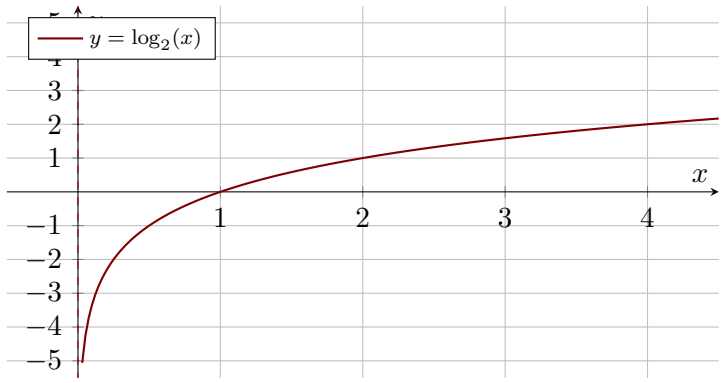
Definition D.31. The *natural logarithm* is the logarithm base e, denoted by $\ln(x)$.

The natural logarithm $\ln(x)$ is the most important logarithm in calculus.

Theorem D.32. *For $f(x) = \log_a(x)$, the domain is $(0, \infty)$, the range is $(-\infty, \infty)$, and there is a vertical asymptote at $x = 0$.*



Logarithmic functions increase slowly, cross at $(1, 0)$, and are undefined for non-positive x.

Theorem D.33 (Inverse Property). *For any $a > 0$ with $a \neq 1$,*

$$\log_a(a^x) = x, \quad a^{\log_a(x)} = x$$

Example D.34. $\ln(e^5) = 5$

These identities reflect the inverse relationship between exponentials and logarithms.

Theorem D.35 (Laws of Logarithm). *For any $x, y > 0$ and $a > 0$ with $a \neq 1$,*

1. $\log_a(xy) = \log_a(x) + \log_a(y)$
2. $\log_a\left(\frac{x}{y}\right) = \log_a(x) - \log_a(y)$
3. $\log_a(x^r) = r\log_a(x)$ *for any* $r \in \mathbb{R}$.

The power rule is especially powerful, as it allows us to "bring down" exponents.

Theorem D.36 (Change of Base). *For any positive bases*

$$\log_a(x) = \frac{\log_b(x)}{\log_b(a)}$$

In particular,

$$\log_a(x) = \frac{\ln(x)}{\ln(a)}$$

Example D.37. $\log_2(7) = \frac{\ln(7)}{\ln(2)}$.

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Example D.38. Expand $\log_3(9x^2)$.

Solution. Using the laws of logarithm,

$$\begin{aligned}\log_3(9x^2) &= \log_3(9) + \log_3(x^2)\\ &= 2 + 2\log_3(x)\end{aligned}$$

Example D.39. Condense

$$2\ln(x) - \ln(x-1)$$

Solution. Using the laws of logarithm,

$$\begin{aligned}2\ln(x) - \ln(x-1) &= \ln(x^2) - \ln(x-1)\\ &= \ln\left(\frac{x^2}{x-1}\right)\end{aligned}$$

Since the exponential function and logarithmic functions are inverses of each other, we can extend properties.

Proposition D.40. *For any $a > 0$ with $a \neq 1$ and $x \in \mathbb{R}$,*

$$a^x = e^{x\ln(a)}$$

D.4 Solving Logarithmic Equations

In Section D.3, we defined logarithmic functions and developed the laws of logarithms. We now turn to solving logarithmic equations, that is, equations in which the variable appears inside a logarithm.

Solving logarithmic equations requires careful attention to

- Domain restrictions
- The inverse relationship between exponentials and logarithms.
- Proper use of logarithmic laws.
- Checking for extraneous solutions.

Unlike exponential equations, logarithmic equations can easily produce invalid solutions if domain restrictions are ignored. Logical precision is therefore essential.

Theorem D.41. *Let $a > 0$ with $a \neq 1$. Then for any $x, y > 0$, if*

$$\log_a(x) = \log_a(y)$$

then

$$x = y$$

This theorem allows us to "drop the logarithm" when both sides have the same base. But only if both arguments are positive.

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Example D.42. Solve $\log_3(x) = 2$.

Solution. Rewrite in exponential form $3^2 = x$, so $x = 9$.

Example D.43. Solve $\ln(x) = 4$.

Solution. Rewrite $e^4 = x$.

Example D.44. Solve $\log_2(x-1) = \log_2(5)$.

Solution. Here, we note that we require $x - 1 > 0$, so $x > 1$. Dropping the logarithms and we get

$$\begin{aligned} x - 1 &= 5 \\ x &= 6 \end{aligned}$$

which is a valid solution.

Example D.45. Solve $\ln(x) + \ln(x-2) = 0$.

Solution. Note that we have $x > 0$ or $x - 2 > 0$, and thus $x > 2$. Using laws of logarithm,

$$\ln(x(x-2)) = 0$$

Converting to exponential form

$$\begin{aligned} x(x-2) &= e^0 \\ x(x-2) &= 1 \end{aligned}$$

Thus, solving the quadratic gives

$$\begin{aligned} x^2 - 2x - 1 &= 0 \\ x &= 1 \pm \sqrt{2} \end{aligned}$$

Note that $1 - \sqrt{2} < 0$, which is invalid, but $1 + \sqrt{2} > 2$, which is valid.

This example demonstrates why checking is essential. Logarithmic equations can generate extraneous solutions.

Example D.46. Solve $\log_3(x+4) = 2$.

Solution. Rewrite

$$\begin{aligned} 3^2 &= x + 4 \\ 9 &= x + 4 \\ x &= 5 \end{aligned}$$

Example D.47. Solve $\ln(x) = 2 - \ln(3)$.

Solution. Rewrite the right side as

$$2 = \ln(e^2)$$

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$$\ln(x) = \ln(e^2) - \ln(3)$$

Applying laws of logarithm

$$\ln(x) = \ln\left(\frac{e^2}{3}\right)$$

Therefore, dropping the logs,

$$x = \frac{e^2}{3}$$

as desired.

Example D.48. Solve $\ln(x-1) + \ln(x-3) = \ln(4)$.

Solution. Here, note that $x - 1 > 0$ or $x - 3 > 0$. Thus, we take $x > 3$. Combining logs give

$$\ln[(x-1)(x-3)] = \ln(4)$$

Dropping the logarithms

$$(x-1)(x-3) = 4$$

and expanding and solving

$$\begin{aligned} x^2 - 4x + 3 &= 4 \\ x^2 - 4x - 1 &= 0 \\ x &= 2 \pm \sqrt{5} \end{aligned}$$

Note $2 - \sqrt{5} < 3$, which is invalid, but $2 + \sqrt{5} > 3$, which is valid.

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